

Cloud Computing

Lab Manual

Submitted by : Inshal Nasir

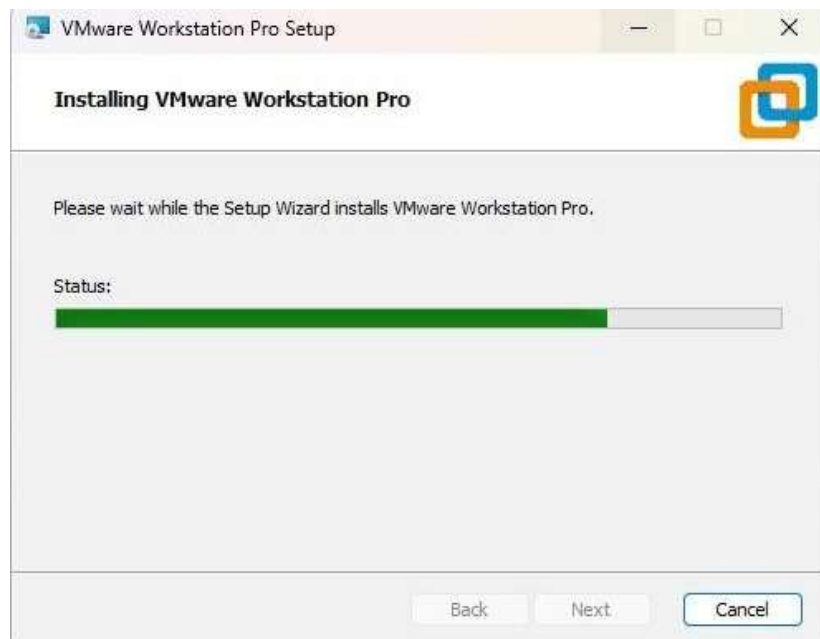
Section : A

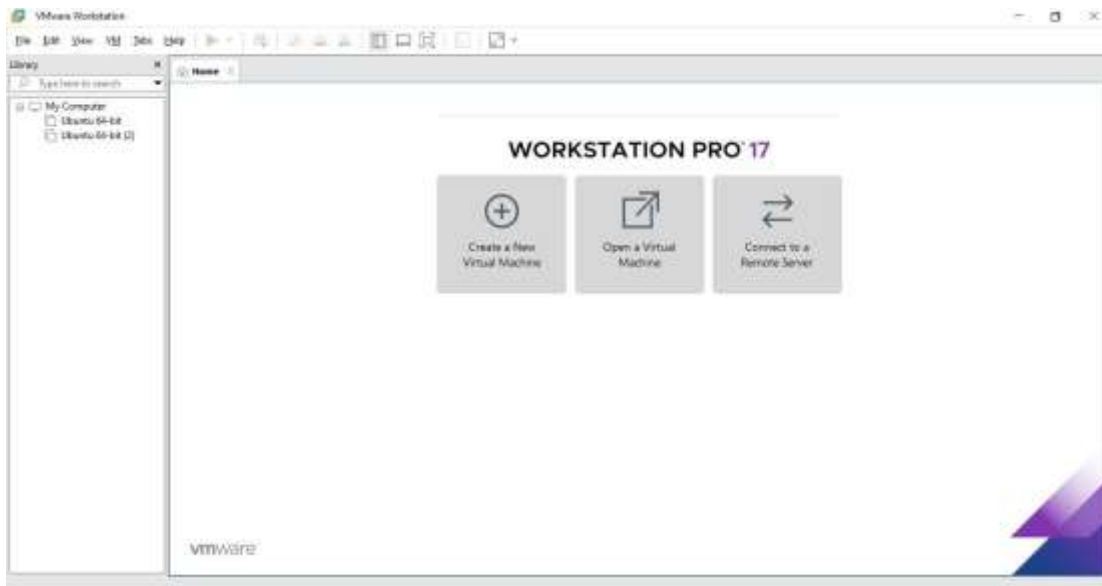
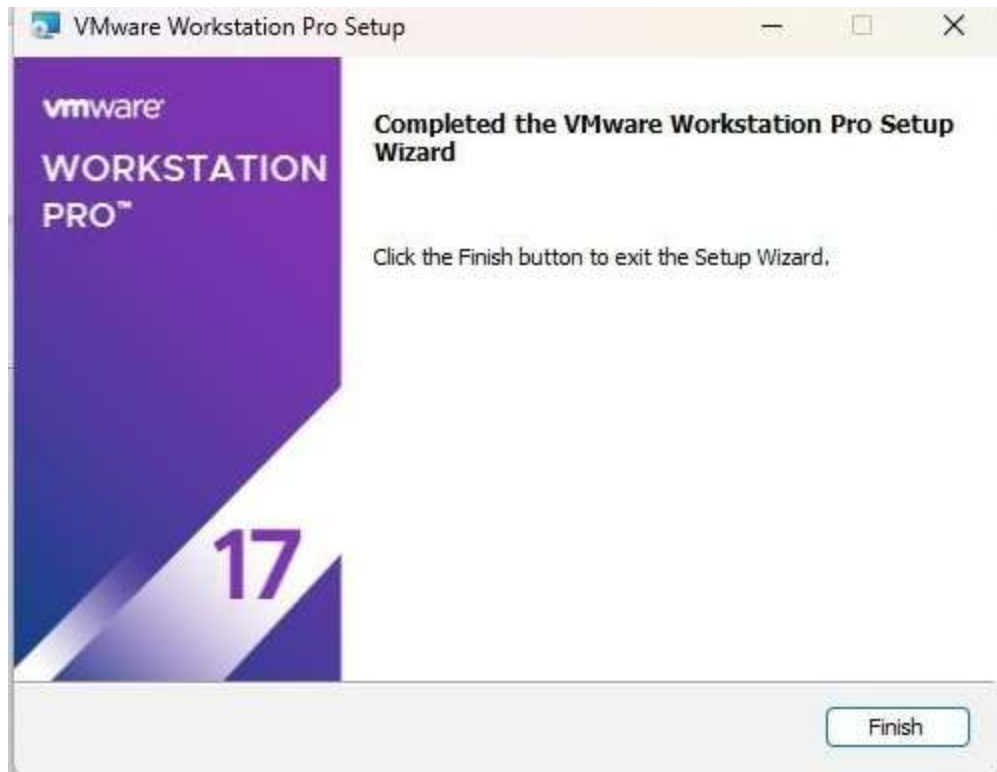
Roll No: 38

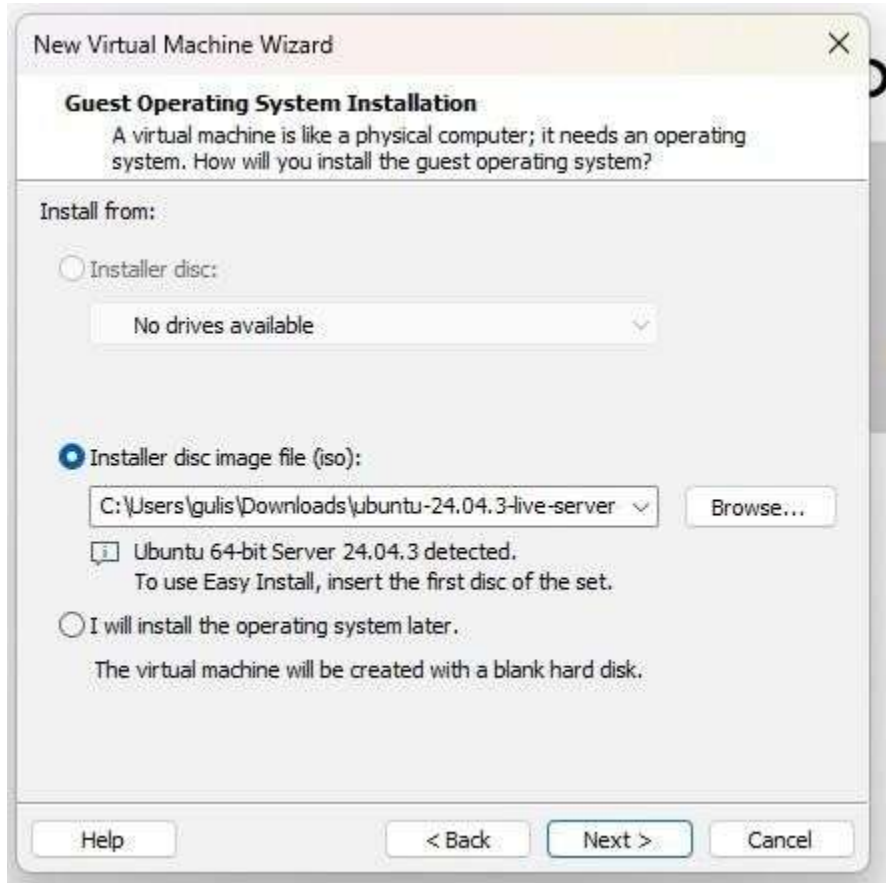
Submitted to: Sir Shoaib

Lab 1

INSTALLATION OF UBUNTU AND VMWARE WORKSTATION PRO CONFIGURING UBUNTU IN VMWARE







New Virtual Machine Wizard

Name the Virtual Machine

What name would you like to use for this virtual machine?

Virtual machine name:

Ubuntu 64-bit (3)

Location:

C:\Users\gulis\Documents\Virtual Machines\Ubuntu 64-bit (3)

Browse...

The default location can be changed at Edit > Preferences.

< Back

Next >

Cancel

New Virtual Machine Wizard

Ready to Create Virtual Machine

Click Finish to create the virtual machine and start installing Ubuntu 64-bit.

The virtual machine will be created with the following settings:

Name:Ubuntu 64-bit (3)

Location:C:\Users\gulis\Documents\Virtual Machines\Ubuntu 6...

Version:Workstation 17.5 or later

Operating System:Ubuntu 64-bit

Hard Disk:20 GB, Split

Memory:2048 MB

Network Adapter:NAT

Other Devices:2 CPU cores, CD/DVD, USB Controller, Sound Card

Customize Hardware...

☒ Power on this virtual machine after creation

< Back

Finish

Cancel

ubuntu archive mirror configuration

help

If you use an alternative mirror for Ubuntu, enter its details here.

mirror address:

You may provide an archive mirror to be used instead of the default.

The mirror location is being tested. ↵

```
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
```

Done

Back

Profile configuration

help

Enter the username and password you will use to log in to the system. You can configure SSB access on a later screen, but a password is still needed for sudo.

Your name:

Your server's name:

The name it uses when it talks to other computers.

Pick a username:

Choose a password:

Confirm your password:

Done

```

ubuntu 24.04.3 LTS follow tftg1
fatima login: fatima
Password:
welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-71-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/support

System information as of Fri 3 Oct 04:07:28 UTC 2025

System load:  0.8                Processes:    239
Usage of /:   22.8% of 19.51GB    Users logged in:  0
Memory usage: 12%                IPv4 address for ens33: 192.168.19.128
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

No updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo apt update

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo ".
See "man sudo_root" for details.

fatima@fatima:~$ _

```

```
System load: 0.0          Processes: 223
Usage of /: 22.2% of 19.5GB   Users logged in: 0
Memory usage: 17%          IPv4 address for ens33: 192.168.19.128
Swap usage: 0%

Expanded security maintenance for applications is not enabled.
@ updates can be applied immediately.

Enable ESM apps to receive additional future security updates.
See http://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo command",
see "man sudo_root" for details.

root@ubuntu:~#
```



FATIMA JINNAH WOMEN UNIVERSITY

Department of Software Engineering

Lab Title:	Experiment # :
Student Name:	Regd. #:
Experiment Name:	

Performance Indicator	Level of Achievements				
	Excellent (5)	Good (4)	Average (3)	Below Average (2)	Poor (1)
Understanding & implementation Problem					
Report and presentation					
Maximum Marks		10	Obtained Marks		
Lab Instructor:					

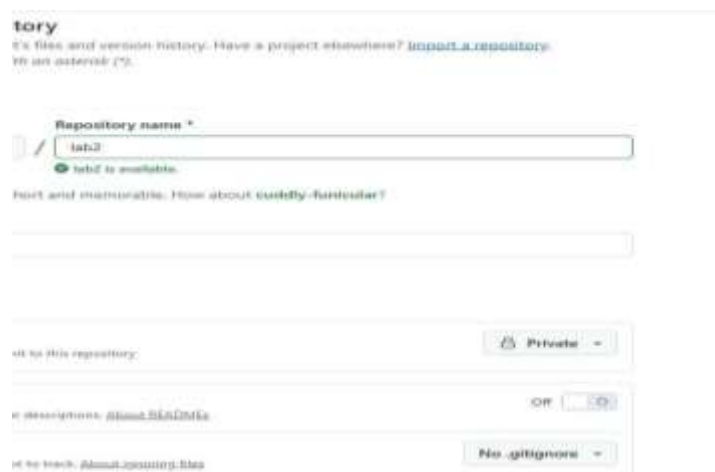
Lab 2

Install Git on laptop



Task #01

Create a new private repository named Lab2 on GitHub.

The image shows the 'Create a new repository' form on GitHub. The title is 'Create a new repository'. Below the title, it says 'It's free and version history. Have a project elsewhere? Import a repository in an asterisk (*).' The 'Repository name' field contains 'lab2' and a message below it says 'lab2 is available.' There is a hint: 'Short and memorable. How about cuddly-fantastic?'. Below the name field is a description field. At the bottom, there are two sections: 'Set up this repository' with a 'Private' dropdown menu, and 'OR' with 'Create repository from scratch' and a 'No .gitignore' dropdown menu.

Clone lab2 repo

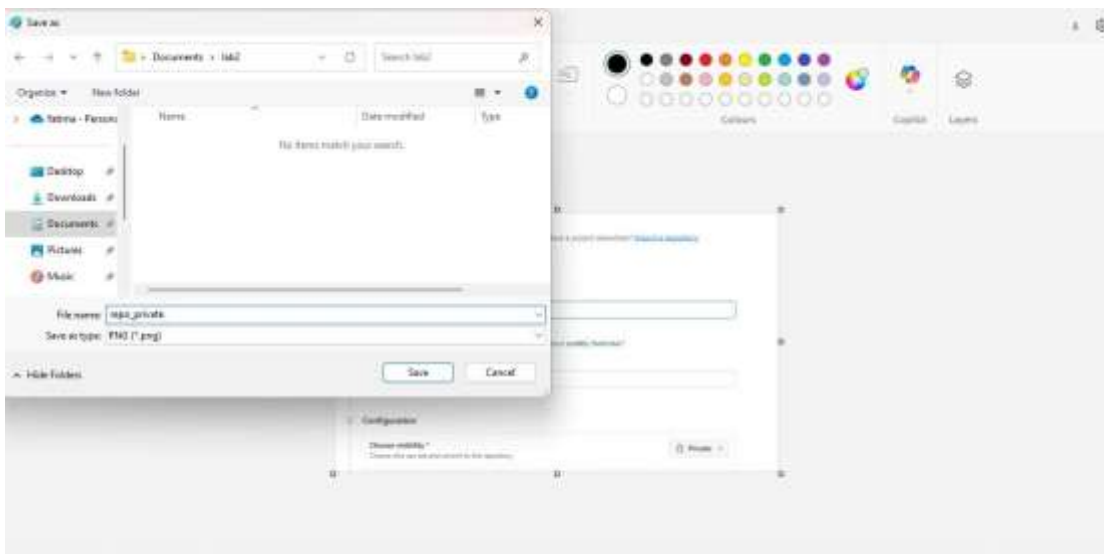
```

gulis@Laptop MINGW64 ~/Documents
$ git clone git@github.com:23-22411-018-png/lab2.git
Cloning into 'lab2'...
The authenticity of host 'github.com (20.207.73.82)' can't be established.
ED25519 key fingerprint is SHA256:+DiY3wvV6TuJJhbpZisF/zLDA0zPMSvHdkr4UvCOqU.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'github.com' (ED25519) to the list of known hosts.
git@github.com: Permission denied (publickey).
fatal: Could not read from remote repository.

Please make sure you have the correct access rights
and the repository exists.

```

Take a screenshot of your repo settings showing it's private. Save as repo_private.png in your Lab2 repo.



TASK # 02

Generate a new SSH key using PowerShell

```

gulis@Laptop MINGW64 ~/Documents
$ ssh-keygen -t ed25519 -C "fatimakashif897@gmail.com"
Generating public/private ed25519 key pair.
Enter file in which to save the key (/c:/Users/gulis/.ssh/id_ed25519):
Enter passphrase for "/c:/Users/gulis/.ssh/id_ed25519" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /c:/Users/gulis/.ssh/id_ed25519
Your public key has been saved in /c:/Users/gulis/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:7VMwTFuW1R7t6RIzltkV8K1p/IcmeE2Xh2XREIp7Io4 fatimakashif897@gmail.com
The key's randomart image is:
+--[ED25519 256]--+
  . ++.0+==
  =+00.+ .
  ..0. 00
  =.0.0.0
  oS=+0+0
  .000"++ .
  E..000 .
  +-----[SHA256]-----+

```

Add your SSH public key to GitHub (Settings > SSH and GPG keys). Take a screenshot of the added key as github_sshkey.png



Add new SSH Key

Title
key1

Key type
Authentication Key

Key
ssh-ed25519 AAAAC3NzaC1lZD11NTESAAAAI/ScipKPGN7h4h4xTL+vt1qG/5g9s2F9E60wJgQsNdd fatimakahf@gmail.com

Add SSH key

.Clone your Lab2 repo using SSH

```
gulis@Laptop MINGW64 ~/Documents/lab2 (main)
$ git clone git@github.com:23-22411-018-png/lab2.git
Cloning into 'lab2'...
warning: You appear to have cloned an empty repository.
```

TASK#03

Set up your Git identity (this ensures all commits are linked to you)

```
gulis@Laptop MINGW64 ~/Documents/lab2 (main)
$ git config --global user.name "fatima"

gulis@Laptop MINGW64 ~/Documents/lab2 (main)
$ git config --global user.email "fatima@gmail.com"
```

Verify your configuration:

```

gulis@Laptop MINGW64 ~/Documents/lab2 (main)
$ git config --list
diff.astextplain.textconv=astextplain
filter.lfs.clean=git-lfs clean -- %f
filter.lfs.smudge=git-lfs smudge -- %f
filter.lfs.process=git-lfs filter-process
filter.lfs.required=true
http.sslbackend=schannel
core.autocrlf=true
core.fscache=true
core.symlinks=false
pull.rebase=false
credential.helper=manager
credential.https://dev.azure.com.usehttppath=true
init.defaultbranch=master
user.name=fatima
user.email=fatima@gmail.com
core.repositoryformatversion=0
core.filemode=false
core.bare=false
core.logallrefupdates=true
core.symlinks=false
core.ignorecase=true
remote.origin.url=https://github.com/23-22411-018-png/lab2.git
remote.origin.fetch=+refs/heads/*:refs/remotes/origin/*
branch.main.remote=origin
branch.main.merge=refs/heads/main

```

TASK # 04

Navigate into your cloned repository folder.

```

gulis@Laptop MINGW64 ~/Documents/lab2 (main)
$ cd ~/Documents/lab2

```

Show hidden files and locate the .git directory.

```

gulis@Laptop MINGW64 ~/Documents/lab2 (main)
$ ls -a
./ ../ .git/ README.md git_config_list.png git_identity.png github_sshkey.png repo_private.png ssh_clone.png ssh_keygen.png

```

Explore whats inside

```

gulis@Laptop MINGW64 ~/Documents/lab2 (main)
$ ls -a .git
./ ../ COMMIT_EDITMSG HEAD config description hooks/ index info/ logs/ objects/ refs/

```

TASK#05

Delete the existing .git folder from your cloned repo using Git Bash

```

gulis@Laptop MINGW64 ~/Documents/lab2 (main)
$ rm -rf .git

```


Re-initialize the local git repository

```
gulis@Laptop MINGW64 ~/Documents/lab2
$ git init
Initialized empty Git repository in C:/Users/gulis/Documents/lab2/.git/
```

Add a file named README.md and commit it

```
gulis@Laptop MINGW64 ~/Documents/lab2 (main)
$ echo "# lab2" >> README.md

gulis@Laptop MINGW64 ~/Documents/lab2 (main)
$ git init
Reinitialized existing Git repository in C:/Users/gulis/Documents/lab2/.git/

gulis@Laptop MINGW64 ~/Documents/lab2 (main)
$ git add README.md
warning: in the working copy of 'README.md', LF will be replaced by CRLF the next time Git touches it

gulis@Laptop MINGW64 ~/Documents/lab2 (main)
$ git commit -m "first commit"
[main (root-commit) d610042] first commit
1 file changed, 1 insertion(+)
create mode 100644 README.md
```

Connect local repo to Github and push

```
gulis@Laptop MINGW64 ~/lab2 (master)
$ git remote add origin git@github.com:23-22411-018-png/lab2.git

gulis@Laptop MINGW64 ~/lab2 (master)
$ git push -u origin main
error: src refspec main does not match any
error: failed to push some refs to 'github.com:23-22411-018-png/lab2.git'

gulis@Laptop MINGW64 ~/lab2 (master)
$ git push -u origin master
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 233 bytes | 38.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'master' on GitHub by visiting:
remote:   https://github.com/23-22411-018-png/lab2/pull/new/master
remote:
To github.com:23-22411-018-png/lab2.git
 * [new branch]      master -> master
branch 'master' set up to track 'origin/master'.
```

TASK # 06

Create a new file notes.txt and write a note.

```
gulis@Laptop MINGW64 ~
$ echo "This is my first note." > notes.txt
```

Check status. Save screenshot as status1.png.

```
gulis@Laptop MINGW64 ~/Documents/lab2 (master)
$ git status
```

Stage and commit. Save screenshot as commit_notes.png.

```

gulis@Laptop MINGW64 ~/Documents/lab2 (master)
$ git add notes.txt
warning: in the working copy of 'notes.txt', LF will be replaced by CRLF the next time Git touches it

gulis@Laptop MINGW64 ~/Documents/lab2 (master)
$ git commit -m "Update notes.txt"
[master e5a9d46] Update notes.txt
1 file changed, 1 insertion(+)

```

Edit notes.txt and repeat status/commit steps.

```

gulis@Laptop MINGW64 ~/Documents/lab2 (master)
$ echo "This is an update to my notes." >> notes.txt

gulis@Laptop MINGW64 ~/Documents/lab2 (master)
$ git status
On branch master
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   notes.txt

Untracked files:
  (use "git add <files>..." to include in what will be committed)
        delete_git.png
        first_commit.png
        git_config_list.png
        git_folder.png
        git_identity.png
        github_sshkey.png
        hit_init.png
        repo_private.png
        ssh_clone.png
        ssh_keygen.png

no changes added to commit (use "git add" and/or "git commit -a")

gulis@Laptop MINGW64 ~/Documents/lab2 (master)
$ git add notes.txt
warning: in the working copy of 'notes.txt', LF will be replaced by CRLF the next time Git touches it

gulis@Laptop MINGW64 ~/Documents/lab2 (master)
$ git commit -m "Update notes.txt"
[master e5a9d46] Update notes.txt
1 file changed, 1 insertion(+)

```

TASK # 07

On GitHub (web interface), create a branch named bugfix/user-auth-error.



Pull the branch to your local repository to sync.

```

gulis@Laptop MINGW64 ~/lab2 (master)
$ git pull origin bugfix/user-auth-error
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 870 bytes | 48.00 KiB/s, done.
From github.com:23-22411-018-png/lab2
 * branch                bugfix/user-auth-error -> FETCH_HEAD
 * [new branch]          bugfix/user-auth-error -> origin/bugfix/user-auth-error
fatal: refusing to merge unrelated histories

```

TASK # 08

Create a branch named feature/db-connection using Git Bash. Push the branch to the remote repository.

```

gulis@Laptop MINGW64 ~/lab2 (master)
$ git checkout -b feature/db-connection
Switched to a new branch 'feature/db-connection'

gulis@Laptop MINGW64 ~/lab2 (feature/db-connection)
$ git push origin feature/db-connection
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'feature/db-connection' on GitHub by visiting:
remote:   https://github.com/23-22411-018-png/lab2/pull/new/feature/db-connection
remote:
To github.com:23-22411-018-png/lab2.git
 * [new branch]      feature/db-connection -> feature/db-connection

```

TASK # 09

Create and switch to a branch feature-1

```

gulis@Laptop MINGW64 ~/lab2 (feature/db-connection)
$ git checkout -b feature-1
Switched to a new branch 'feature-1'

```

Modify main.py (add a function) and commit

```

gulis@Laptop MINGW64 ~/lab2 (feature-1)
$ git add main.py

gulis@Laptop MINGW64 ~/lab2 (feature-1)
$ git commit -m "Add new function to main.py"
[feature-1 95cec15] Add new function to main.py
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 main.py

```

Switch back to main and merge

```

gulis@Laptop MINGW64 ~/lab2 (feature-1)
$ git checkout master
Switched to branch 'master'
Your branch is up to date with 'origin/master'.

gulis@Laptop MINGW64 ~/lab2 (master)
$ git merge feature-1
Updating 6c9f659..95cec15
Fast-forward
 main.py | 0
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 main.py

```

Push all branches

```

gulis@Laptop MINGW64 ~/lab2 (master)
$ git push origin master
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 284 bytes | 40.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:23-22411-018-png/lab2.git
6c9f659..95cec15 master -> master

gulis@Laptop MINGW64 ~/lab2 (master)
$ git push origin feature-1
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'feature-1' on GitHub by visiting:
remote:   https://github.com/23-22411-018-png/lab2/pull/new/feature-1
remote:
To github.com:23-22411-018-png/lab2.git
 * [new branch]      feature-1 -> feature-1

```

TASK # 12

Create the following branches to simulate a professional branching strategy

```

gulis@Laptop MINGW64 ~/lab2 (master)
$ git checkout master
Already on 'master'
Your branch is up to date with 'origin/master'.

gulis@Laptop MINGW64 ~/lab2 (master)
$ git branch develop

gulis@Laptop MINGW64 ~/lab2 (master)
$ git branch staging

gulis@Laptop MINGW64 ~/lab2 (master)
$ git branch feature/login

gulis@Laptop MINGW64 ~/lab2 (master)
$ git branch bugfix/typo

gulis@Laptop MINGW64 ~/lab2 (master)
$ git branch
  bugfix/typo
  develop
  feature-1
  feature/db-connection
  feature/login
* master
  staging

```

Documentation with Screenshots

```

gulis@Laptop MINGW64 ~/lab2 (master)
$ git checkout feature/login
Switched to branch 'feature/login'

gulis@Laptop MINGW64 ~/lab2 (feature/login)
$ git add main.py

gulis@Laptop MINGW64 ~/lab2 (feature/login)
$ git commit -m "Add login feature"
[feature/login bd3df14] Add login feature
1 file changed, 2 insertions(+)

gulis@Laptop MINGW64 ~/lab2 (feature/login)
$ git checkout develop
Switched to branch 'develop'

gulis@Laptop MINGW64 ~/lab2 (develop)
$ git merge feature/login
merge: feature/login - not something we can merge

gulis@Laptop MINGW64 ~/lab2 (develop)
$ git merge feature/login
Updating 95cec15..bd3df14
Fast-forward
 main.py | 2 ++
1 file changed, 2 insertions(+)

gulis@Laptop MINGW64 ~/lab2 (develop)
$ git checkout staging
Switched to branch 'staging'

gulis@Laptop MINGW64 ~/lab2 (staging)
$ git merge develop
Updating 95cec15..bd3df14
Fast-forward
 main.py | 2 ++
1 file changed, 2 insertions(+)

```

```

gulis@Laptop MINGW64 ~/lab2 (staging)
$ git checkout master
Switched to branch 'master'
Your branch is up to date with 'origin/master'.

gulis@Laptop MINGW64 ~/lab2 (master)
$ git merge staging
Updating 95cec15..bd3df14
Fast-forward
 main.py | 2 ++
1 file changed, 2 insertions(+)

```

TASK # 13

Delete Remote Branch After Merge

Your branches:

Branch	Updated	Check status	Behind / Ahead	Pull request
feature-1	41 minutes ago		1/2	
master	42 minutes ago		1/1	
feature/correction	Deleted now		1/1	
bugfix/user-auth-error	3 days ago		0/0	

Update Local Repository

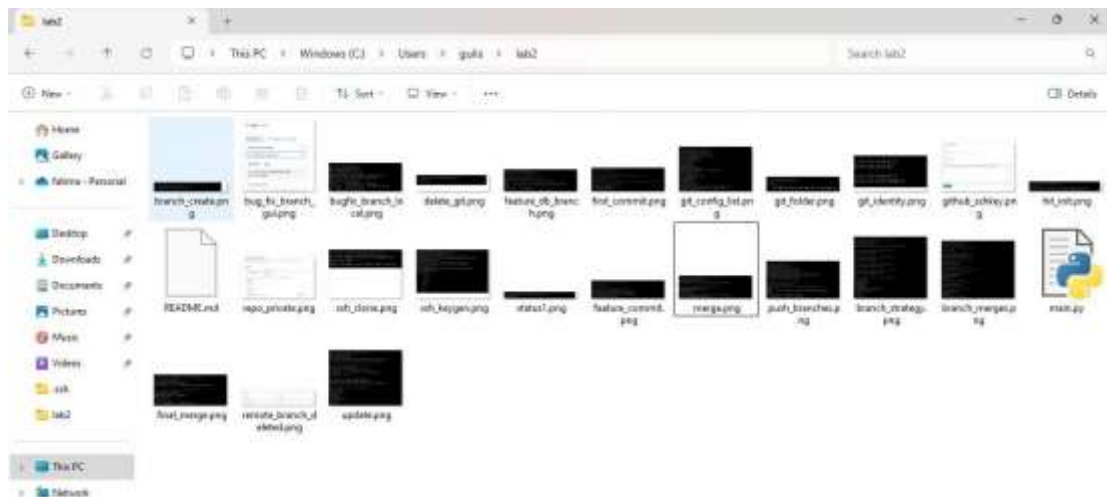
```
gulis@Laptop MINGW64 ~/lab2 (master)
$ git checkout master
Already on 'master'
Your branch is ahead of 'origin/master' by 1 commit.
(use "git push" to publish your local commits)
```

```
gulis@Laptop MINGW64 ~/lab2 (master)
$ git pull origin master
From github.com:23-22411-018-png/lab2
 * branch          master      -> FETCH_HEAD
Already up to date.
```

```
gulis@Laptop MINGW64 ~/lab2 (master)
$ git branch -d feature/db-connection
Deleted branch feature/db-connection (was 6c9f659).
```

```
gulis@Laptop MINGW64 ~/lab2 (master)
$ git branch
  bugfix/typo
  develop
  feature-1
  feature/login
* master
  staging
```

My lab2 document





FATIMA JINNAH WOMEN UNIVERSITY

Department of Software Engineering

Lab Title:	Experiment # :
Student Name:	Regd. #:
Experiment Name:	

Performance Indicator	Level of Achievements				
	Excellent (5)	Good (4)	Average (3)	Below Average (2)	Poor (1)
Understanding & implementation Problem					
Report and presentation					
Maximum Marks		10	Obtained Marks		
Lab Instructor:					

Lab 3

Task 1



```
HP@DESKTOP-JFEH5TK MINGW64 /d
$ cd "D:/FJWU/Semester 5/Cloud Computing/CC Lab-Engr. Shoaib/Lab3"

HP@DESKTOP-JFEH5TK MINGW64 /d/FJWU/Semester 5/Cloud Computing/CC Lab-Engr. Shoaib/Lab3 (main)
$ git clone https://github.com/hajrasarwar11/Lab3.git
Cloning into 'Lab3'...
remote: Enumerating objects: 7, done.
remote: Counting objects: 100% (7/7), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 7 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (7/7), done.

HP@DESKTOP-JFEH5TK MINGW64 /d/FJWU/Semester 5/Cloud Computing/CC Lab-Engr. Shoaib/Lab3 (main)
$ cd Lab3

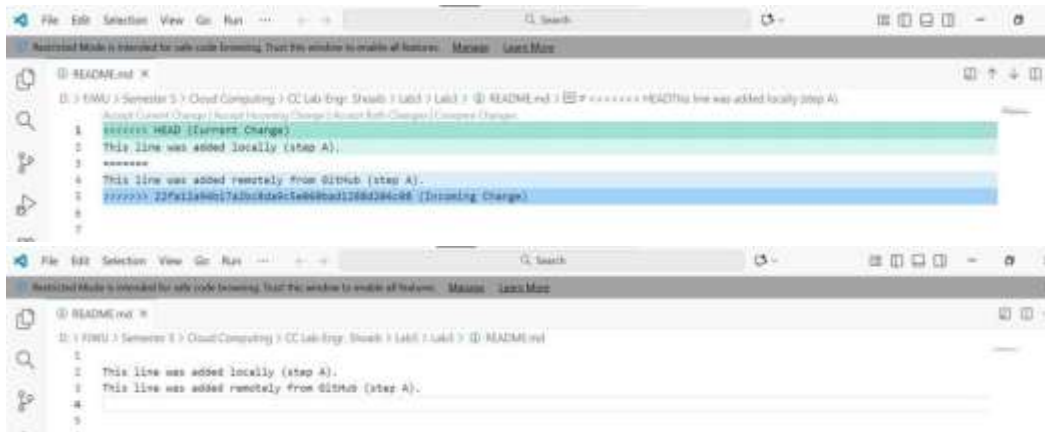
HP@DESKTOP-JFEH5TK MINGW64 /d/FJWU/Semester 5/Cloud Computing/CC Lab-Engr. Shoaib/Lab3/Lab3 (main)
$ git remote -v
origin https://github.com/hajrasarwar11/Lab3.git (fetch)
origin https://github.com/hajrasarwar11/Lab3.git (push)
```



```
HP@DESKTOP-JFEH5TK MINGW64 /d/FJWU/Semester 5/Cloud Computing/CC Lab-Engr. Shoaib/Lab3/Lab3 (main)
$ git add README.md

HP@DESKTOP-JFEH5TK MINGW64 /d/FJWU/Semester 5/Cloud Computing/CC Lab-Engr. Shoaib/Lab3/Lab3 (main)
$ git commit -m "Local update to README"
[main a2f594f] Local update to README
1 file changed, 1 insertion(+), 1 deletion(-)
```

```
HP@DESKTOP-JFEH5TK MINGW64 /d/FJWU/Semester 5/Cloud Computing/CC Lab-Engr. Shoaib/Lab3/Lab3 (main|MERGING)
$ git push -u origin main
To https://github.com/hajrasarwar11/Lab3.git
 ! [rejected]        main -> main (non-fast-forward)
error: failed to push some refs to 'https://github.com/hajrasarwar11/Lab3.git'
hint: Updates were rejected because the tip of your current branch is behind
hint: its remote counterpart. If you want to integrate the remote changes,
hint: use 'git pull' before pushing again.
hint: See the 'Note about fast-forwards' in 'git push --help' for details.
```

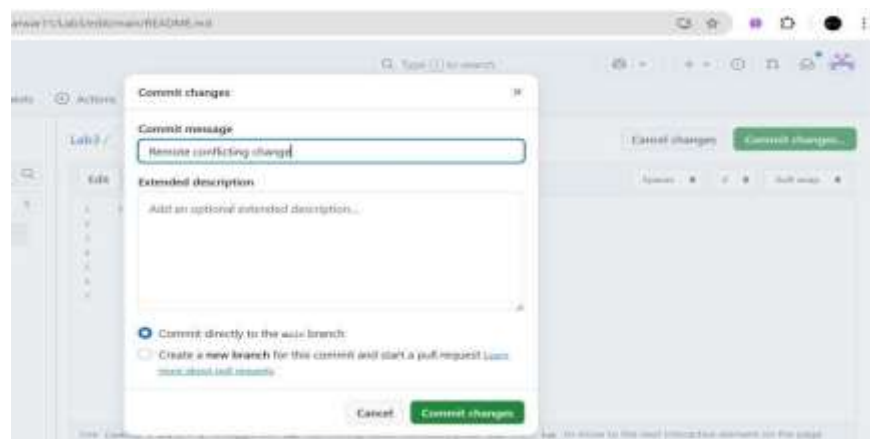
```
$ git pull --no-rebase origin main
remote: Enumerating objects: 5, done.
remote: Counting objects: 100% (5/5), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 959 bytes | 20.00 KiB/s, done.
From https://github.com/hajrasarwar11/Lab3
 * branch            main       -> FETCH_HEAD
    e2f594f..22fa12a  main       -> origin/main
Auto-merging README.md
CONFLICT (content): Merge conflict in README.md
Automatic merge failed; fix conflicts and then commit the result.
```

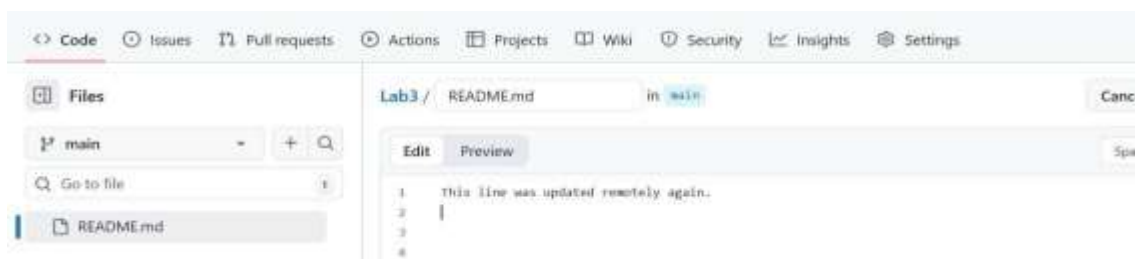
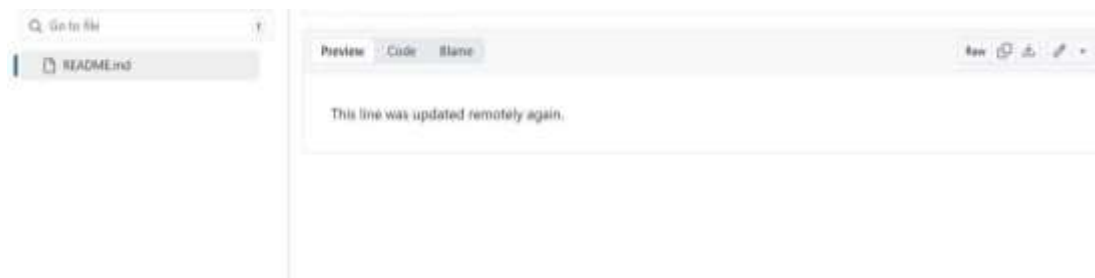
```
HP@DESKTOP-3FEH5TK MINGW64 /d/FJWU/Semester 5/Cloud Computing/CC Lab-Engr. Shoaib/Lab3/Lab3 (main)
$ git push origin main
Enumerating objects: 10, done.
Counting objects: 100% (10/10), done.
Delta compression using up to 8 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (6/6), 654 bytes | 50.00 KiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/hajrasarwar11/Lab3.git
    22fa12a..9339f5f  main -> main
```



```
HP@DESKTOP-3FEH5TK MINGW64 /d/FJWU/Semester 5/Cloud Computing/CC Lab-Engr. Shoaib/Lab3/Lab3 (main)
$ git push -u origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 339 bytes | 67.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/hajrasarwar11/Lab3.git
    df97ab4..ec04ce7  main -> main
branch 'main' set up to track 'origin/main'.
```

Task 2:



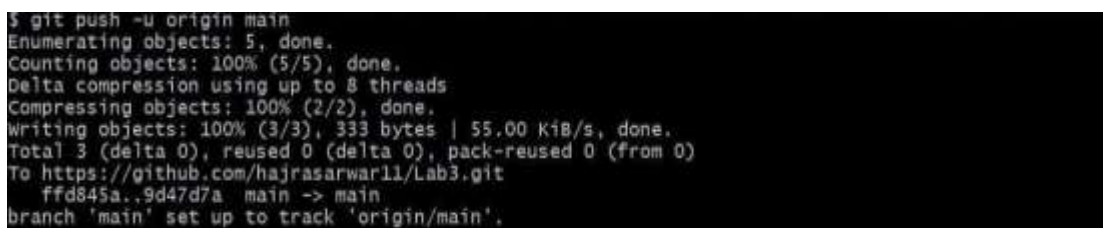
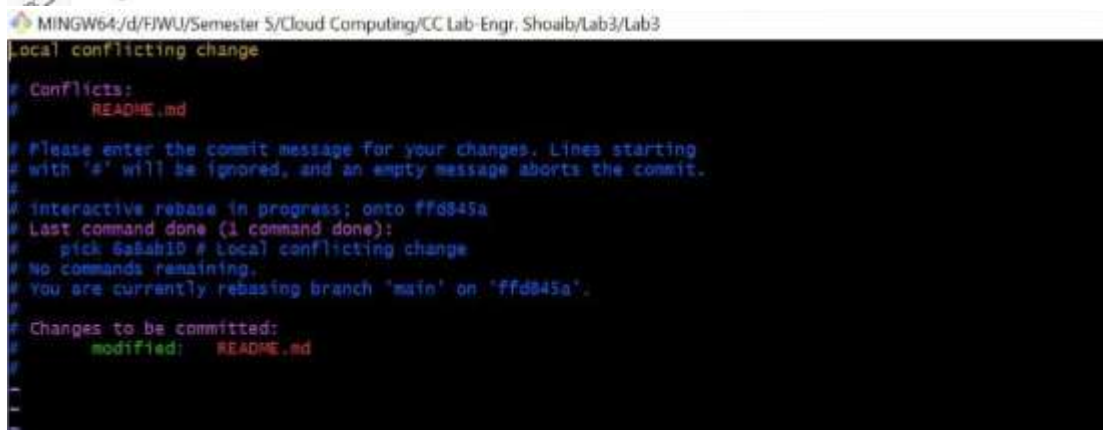


```
$ git add README.md
git commit -m "Local conflicting change"
[main 6a6ab10] Local conflicting change
1 file changed, 1 insertion(+), 2 deletions(-)
```

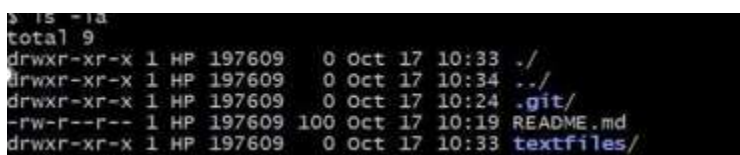
```
$ git push -u origin main
To https://github.com/hajrasarwar11/Lab3.git
! [rejected]        main -> main (fetch first)
error: failed to push some refs to 'https://github.com/hajrasarwar11/Lab3.git'
hint: Updates were rejected because the remote contains work that you do not
hint: have locally. This is usually caused by another repository pushing to
hint: the same ref. If you want to integrate the remote changes, use
hint: 'git pull' before pushing again.
hint: See the 'Note about fast-forwards' in 'git push --help' for details.
```



```
.git pull --rebase origin main
remote: Enumerating objects: 5, done.
remote: Counting objects: 100% (5/5), done.
remote: Compressing objects: 100% (1/1), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
unpacking objects: 100% (3/3), 940 bytes | 21.00 KiB/s, done.
from https://github.com/hajrasarwar11/Lab3
* branch      main      -> FETCH_HEAD
   ac04ce7..ffd845a  main  -> origin/main
Auto-merging README.md
CONFLICT (content): Merge conflict in README.md
error: could not apply 6a6ab10... Local conflicting change
hint: Resolve all conflicts manually, mark them as resolved with
hint: "git add/rm <conflicted_files>", then run "git rebase --continue".
hint: You can instead skip this commit: run "git rebase --skip".
hint: To abort and get back to the state before "git rebase", run "git rebase --abort".
hint: Disable this message with "git config set advice.mergeConflict false"
could not apply 6a6ab10... # Local conflicting change
```



Task 3 :



```
$ git add .
git commit -m "Added textfiles directory with three files"
git push origin main
warning: in the working copy of 'textfiles/a.txt', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'textfiles/b.txt', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'textfiles/c.txt', LF will be replaced by CRLF the next time Git touches it
[main 149c401] Added textfiles directory with three files
3 files changed, 3 insertions(+)
create mode 100644 textfiles/a.txt
create mode 100644 textfiles/b.txt
create mode 100644 textfiles/c.txt
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 8 threads.
Compressing objects: 100% (3/3), done.
Writing objects: 100% (6/6), 467 bytes | 66.00 KiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/hajrasarwar11/Lab3.git
   9d47d7a..149c401  main -> main
```

```
echo "File A content" > textfiles/a.txt
```

```
$ echo "File B content" > textfiles/b.txt
```

```
$ echo "File C content" > textfiles/c.txt
```

```
$ ls textfiles
a.txt b.txt c.txt
```

```
echo "textfiles/*" > .gitignore
```

```
$ cat .gitignore
textfiles/*
```

	gitignore	Added .gitignore to ignore textfiles directory	9 minutes ago
	README.md	Local conflicting change	34 minutes ago

Task 4 :

```
$ git checkout -b feature-branch
Switched to a new branch 'feature-branch'
```

```
$ git branch
feature-branch
main
```

```
$ git add .
git commit -m "Removed tracked textfiles directory"
git push origin main
[main fdd2375] Removed tracked textfiles directory
3 files changed, 3 deletions(-)
delete mode 100644 textfiles/a.txt
delete mode 100644 textfiles/b.txt
delete mode 100644 textfiles/c.txt
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 8 threads.
Compressing objects: 100% (2/2), done.
Writing objects: 100% (2/2), 244 bytes | 244.00 KiB/s, done.
Total 2 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/hajrasarwar11/Lab3.git
   b869d8a..fdd2375  main -> main
```

```
$ git add README.md
git commit -m "Changes the README.md file"
feature-branch c6fbafc] Changes the README.md file
1 file changed, 4 insertions(+)
```



```
git checkout main
error: Your local changes to the following files would be overwritten by checkout:
  README.md
Please commit your changes or stash them before you switch branches.
Aborting
```

```
git stash
Saved working directory and index state WIP on feature-branch: c6fbafe Changes the README.md file
```

```
git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
```

```
git checkout feature-branch
Switched to branch 'feature-branch'
```

```
git stash pop
On branch feature-branch
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   README.md

no changes added to commit (use "git add" and/or "git commit -a")
Dropped refs/stash@{0} (008ccee943c57456c18faaf52900f7d20bedad9f)
```

Task 5 :

```
$ git log --oneline
c6fbafe (HEAD -> feature-branch) Changes the README.md file
fdd2375 (origin/main, origin/HEAD, main) Removed tracked textfiles directory
bb69d8a Added .gitignore to ignore textfiles directory
149c401 Added textfiles directory with three files
9d47d7a Local conflicting change
ffd845a Remote conflicting change
ac04ce7 Local update to README (B)
df97ab4 Remote update to README (B)
c7ddeec Remote update to README (B)
9339f5f Resolve merge conflict in README.md (merge remote changes)
77e6bd2 Local update to README (A)
22fa12a Remote update to README (A)
e2f594f Local update to README
c1da5cc Update README.md
d121e68 Update README.md
28e9511 Initial commit
```



```

$ git checkout 28e9511
Note: switching to '28e9511'.

You are in 'detached HEAD' state. You can look around, make experimental
changes and commit them, and you can discard any commits you make in this
state without impacting any branches by switching back to a branch.

If you want to create a new branch to retain commits you create, you may
do so (now or later) by using -c with the switch command. Example:

  git switch -c <new-branch-name>

To undo this operation with:

  git switch -

Turn off this advice by setting config variable advice.detachedHead to false

HEAD is now at 28e9511 Initial commit

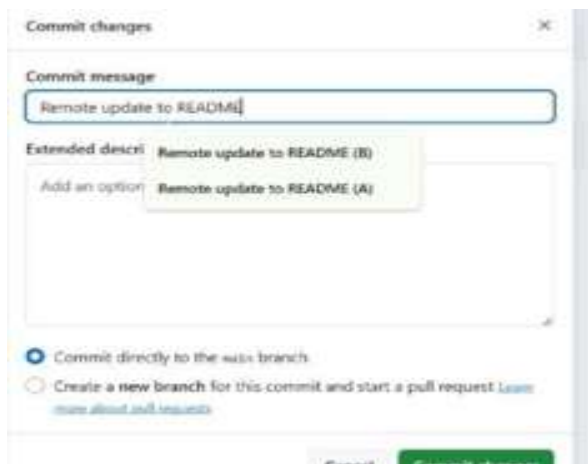
```

```

$ git checkout main
Previous HEAD position was 28e9511 Initial commit
Switched to branch 'main'
Your branch is up to date with 'origin/main'.

```

Task 6:



```

$ git commit -m "Added test line"
main 685e2a5] Added test line
1 file changed, 2 insertions(+)

```

```

$ git log --oneline
97772a1 (HEAD -> main) Second test commit
685e2a5 Added test line
fdd2375 (origin/main, origin/HEAD) Removed tracked textfiles directory
b869d8a Added .gitignore to ignore textfiles directory
149c401 Added textfiles directory with three files
9d47d7a Local conflicting change
ffd845a Remote conflicting change
ec04ce7 Local update to README (B)
df97ab4 Remote update to README (B)
c7ddeec Remote update to README (B)
9339f5f Resolve merge conflict in README.md (merge remote changes)
77e6bd2 Local update to README (A)
22fa12a Remote update to README (A)
e2f594f Local update to README
c1da5cc Update README.md
d121e68 Update README.md
8e9511 Initial commit

```

① README.md X

D: > FJWU > Semester 5 > Cloud Computing > CC Lab-Engr. Shoaib > Lab3 > Lab3 > ① README.md

```

1
2 This line was updated remotely again.
3 This line was updated locally at the same time.
4
5 this is my first commit.
6
7 this is my second commit.
8
9
10
11

```

```
git reset --soft HEAD~1
```

```

git log --oneline
85e2a5 (HEAD -> main) Added test line
dd2375 (origin/main, origin/HEAD) Removed tracked textfiles directory
869d8a Added .gitignore to ignore textfiles directory
49c401 Added textfiles directory with three files
d47d7a Local conflicting change
fd845a Remote conflicting change
c04ce7 Local update to README (B)
f97ab4 Remote update to README (B)
7ddeec Remote update to README (B)
339f5f Resolve merge conflict in README.md (merge remote changes)
7e6bd2 Local update to README (A)
2fa12a Remote update to README (A)
2f594f Local update to README
1da5cc Update README.md
121e68 Update README.md
8e9511 Initial commit

```

File Edit Selection View Go Run ... Search

Restricted Mode is intended for safe code browsing. Trust this window to enable all features. Manage Learn More

① README.md X

D: > FJWU > Semester 5 > Cloud Computing > CC Lab-Engr. Shoaib > Lab3 > Lab3 > ① README.md

```

1
2 This line was updated remotely again.
3 This line was updated locally at the same time.
4
5 this is my first commit.
6
7 this is my second commit.
8
9

```



```
git status
On branch main
Your branch is ahead of 'origin/main' by 1 commit.
  (use "git push" to publish your local commits)

nothing to commit, working tree clean
```

Task 7:

```
git commit -m "Fix log message"
[main 599f6f4] Fix log message
1 file changed, 3 insertions(+)
```



Task 8 :

```
echo "temporary text" >> temp.txt

$ git add .
git commit -m "Added temporary text file"
warning: in the working copy of 'temp.txt', LF will be replaced by CRLF the next time Git touches it
[main blaac4a] Added temporary text file
1 file changed, 1 insertion(+)
create mode 100644 temp.txt
```

```
git push origin main
To https://github.com/hajrasarwar11/Lab3.git
 ! [rejected]        main -> main (non-fast-forward)
error: failed to push some refs to 'https://github.com/hajrasarwar11/Lab3.git'
hint: Updates were rejected because a pushed branch tip is behind its remote
hint: counterpart. If you want to integrate the remote changes, use 'git pull'
hint: before pushing again.
hint: See the 'Note about fast-forwards' in 'git push --help' for details.
```



```

git push origin main
Enumerating objects: 11, done.
Counting objects: 100% (11/11), done.
Delta compression using up to 8 threads
Compressing objects: 100% (8/8), done.
Writing objects: 100% (9/9), 870 bytes | 72.00 KiB/s, done.
Total 9 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (3/3), completed with 1 local object.
To https://github.com/hajrasarwar11/Lab3.git
    d01a12f..ebf51d0  main -> main

```

```

git log --oneline
ebf51d0 (HEAD -> main, origin/main, origin/HEAD) Added temporary text file
e730e99 Fix log message
fd8fa7 Added test line
01a12f amending commit.
2cecf9 committing again
bfe4e6 Remote update to README
dd2375 Removed tracked textfiles directory
869d8a Added .gitignore to ignore textfiles directory
49c401 Added textfiles directory with three files
d47d7a Local conflicting change
fd845a Remote conflicting change
c04ce7 Local update to README (B)
f97ab4 Remote update to README (B)
7ddeec Remote update to README (B)
339f5f Resolve merge conflict in README.md (merge remote changes)
7e6bd2 Local update to README (A)
2fa12a Remote update to README (A)
2f594f Local update to README
1da5cc Update README.md
121e68 Update README.md

```

```

Revert "Added temporary text file"
This reverts commit ebf51d00caf1755f38123548020ae8344ecce10f.

Please enter the commit message for your changes. Lines starting
with '#' will be ignored, and an empty message aborts the commit.

On branch main
Your branch is up to date with 'origin/main'.

Changes to be committed:
  deleted:    temp.txt

```

```

$ git push origin main
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (2/2), 265 bytes | 132.00 KiB/s, done.
Total 2 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/hajrasarwar11/Lab3.git
    ebf51d0..20e9220  main -> main

```

Task 9

```

git checkout -b test-force
Switched to a new branch 'test-force'

```



```
git add .
```

```
$ git commit -m "Added test change for force push"
[test-force dbf2e32] Added test change for force push
1 file changed, 3 insertions(+)
```

```
git push origin test-force
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 8 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 345 bytes | 86.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
Remote: Resolving deltas: 100% (1/1), completed with 1 local object.
Remote:
Remote: Create a pull request for 'test-force' on GitHub by visiting:
Remote:   https://github.com/hajrasarwar11/Lab3/pull/new/test-force
Remote:
To https://github.com/hajrasarwar11/Lab3.git
 * [new branch]      test-force -> test-force
```

```
git reset --hard HEAD~1
HEAD is now at 20e9220 Revert "Added temporary text file"
```

```
git push origin test-force
To https://github.com/hajrasarwar11/Lab3.git
 ! [rejected]        test-force -> test-force (non-fast-forward)
error: failed to push some refs to 'https://github.com/hajrasarwar11/Lab3.git'
hint: Updates were rejected because the tip of your current branch is behind
hint: its remote counterpart. If you want to integrate the remote changes,
hint: use 'git pull' before pushing again.
hint: See the 'Note about fast-forwards' in 'git push --help' for details.
```

```
git push origin test-force --force
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/hajrasarwar11/Lab3.git
+ dbf2e32...20e9220 test-force -> test-force (forced update)
```

Task 10 :

Repository name *

/ ghos

ghos is available

the same as their upstream repository. You can customize the name to distinguish it.

less self-hosted all-in-one software development service, including Git hosting, code

only

enabled by adding your own branch [Learn more](#)

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Public
forked from go-gitea/gitea

[Pin](#)
[Watch](#)
[Fork](#)
[Star](#)

main
1 Branch
0 Tags

Add file
Code

This branch is up to date with [go-gitea/gitea](#).

Contribute
Sync fork

GiteaBot (skip ci) Updated translations via Crowdin
 etlilaf · 8 hours ago
39,944 Commits

	bump devcontainer to go 1.25 (go-gitea#35404)	last month
	Update demo site location from try.gitea.io => demo.gitea...	last year
	Bump actions/labeler to v6 (go-gitea#35681)	15 hours ago
	bump archiver/updater dep (go-gitea#25637)	5 days ago
	Update gofmt, add go.mod ignore directive (go-gitea#3...	last month

About

Git with a cup of tea! Painless self-hosted all-in-one software development service, including Git hosting, code review, team collaboration, package registry and CI/CD

[gitea.com](#)

[Readme](#)

[MIT license](#)

[Code of conduct](#)

[Contributing](#)

[Security policy](#)

[Activity](#)



Port	Forwarded Addr...	Running Process	Visibility	Origin
3000	https://supreme-s...	Process information unavailable	Private	User Forwarded

Port	Forwarded Addr...	Running Process	Visibility	Origin
3000	https://supreme-s...	Process information unavailable	Public	User Forwarded

[supreme-space:info-wv37xv5g35ppv:3000.app.github.dev](#)

Initial Configuration

If you run Gitea inside Docker, please read the [documentation](#) before changing any settings.

Database Settings

Gitea requires MySQL, PostgreSQL, MSSQL, SQLite3 or TiDB (MySQL protocol)

Database Type: MySQL

Host:

Username:

Password:

Database Name:

General Settings

Site Title:

You can enter your company name here.

supreme-space-orbit-wvj57jx5g35qqr-3000.app.github.dev

Gitea Base URL:
Base address for HTTP(S) clone URLs and email notifications.

Log Path:
Log files will be written to this directory.

☐ Enable Update Checker
Checks for new version releases periodically by connecting to gitea.io

Optional Settings

- Email Settings
- Server and Third-Party Service Settings
- Administrator Account Settings

These configuration options will be written into `/home/vscode/gitea/custom/conf/app.ini`

[Install Gitea](#)

Password:

Confirm Password:

[Register Account](#)

[Already have an account? Sign in now!](#)

Issues Pull Requests Milestones Explore

gitea

Account was successfully created. Welcome!

Repository Organisation

Repositories 0

Find a repository

All Sources Forks Stars Collaborators

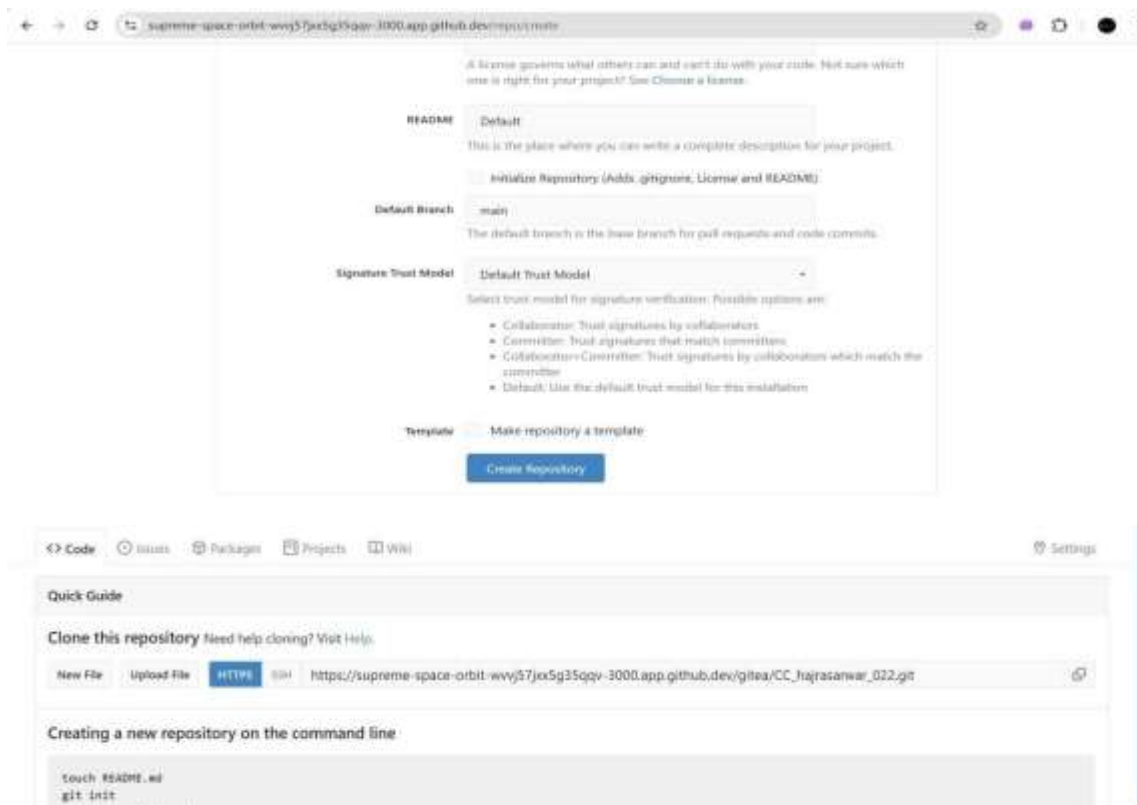
supreme-space-orbit-wvj57jx5g35qqr-3000.app.github.dev

Issues Pull Requests Milestones Explore

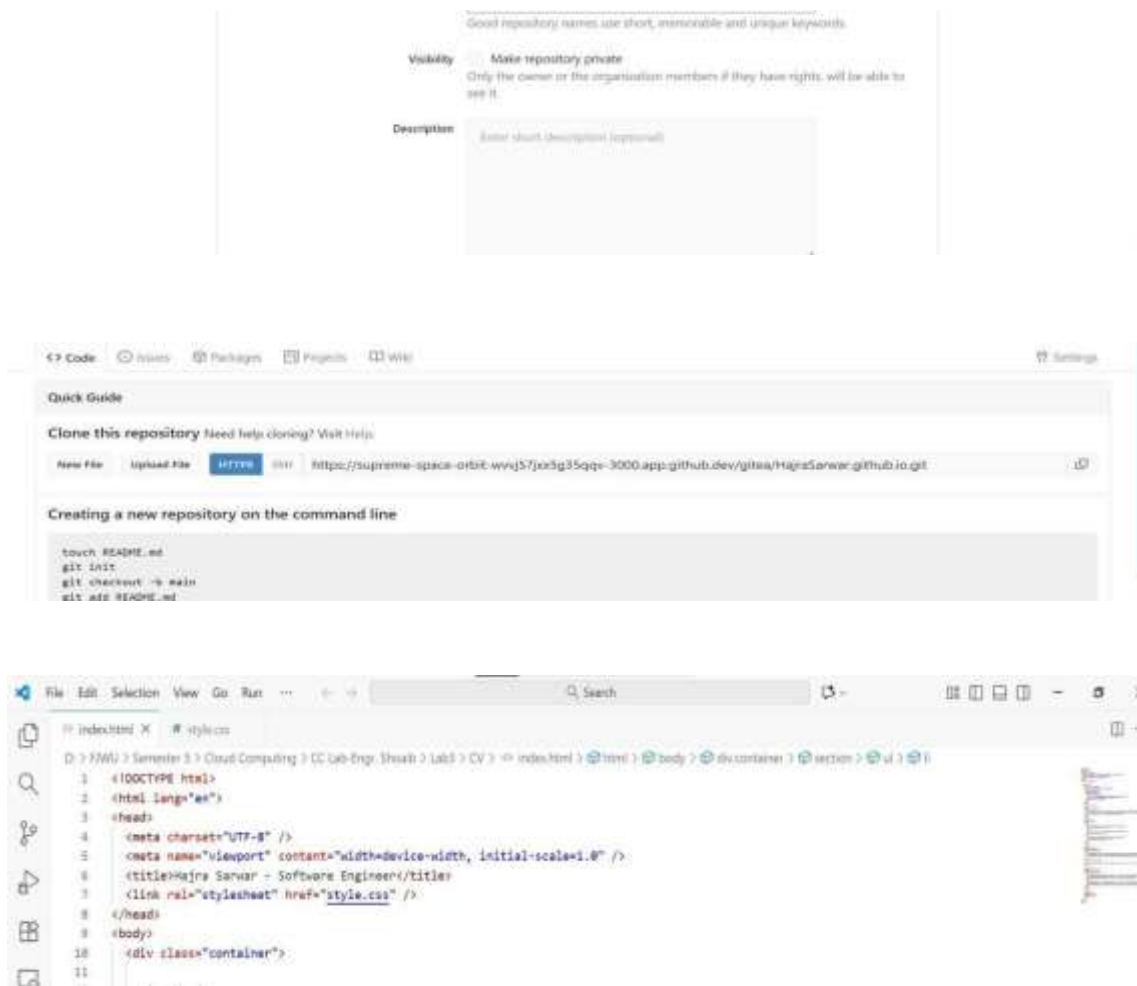
gitea

Repository

- + New Repository
- + New Migration
- + New Organization



Task 11:




```
cd "D:\FJWU\Semester 5\Cloud Computing\CC Lab-Engr. Shoaib\Lab3\CV"
```

```
git init
Initialized empty Git repository in D:\FJWU\Semester 5\Cloud Computing\CC Lab-Engr. Shoaib\Lab3\CV\.git/
```

```
git add .
git commit -m "Add portfolio site for Github Pages"
master (root-commit) 42055f3] Add portfolio site for Github Pages
1 files changed, 214 insertions(+)
create mode 100644 index.html
create mode 100644 style.css
```

```
git push -u origin main --force
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 8 threads
Compressing objects: 100% (7/7), done.
Writing objects: 100% (7/7), 2.77 KiB | 142.00 KiB/s, done.
Total 7 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/hajrasarwar11/HajraSarwar.github.io.git
+ 3a75a43...2066045 main -> main (forced update)
Branch 'main' set up to track 'origin/main'.
```

inshal nasir

3D software engineering | Github: 2020-03-08

[Home](#) [Skills](#) [Projects](#) [Contact](#)

About Me

I am a Computer Science student passionate about web development, networking, and software engineering. I enjoy learning new technologies and building creative digital projects.

Technical Skills

- Python, C++, JavaScript
- Python (Django)
- Git & GitHub
- Networking (Cisco Packet Tracer)

Projects

Library Management System (Cisco Packet Tracer)

Designed a network-based library using routers, switches, and servers to manage library operations efficiently.

Personal Portfolio Website

Created a responsive portfolio website using Python and CSS hosted on GitHub Pages.

Contact Me

Email: inshaln@outlook.com
GitHub: <https://github.com/inshaln>

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FATIMA JINNAH WOMEN UNIVERSITY

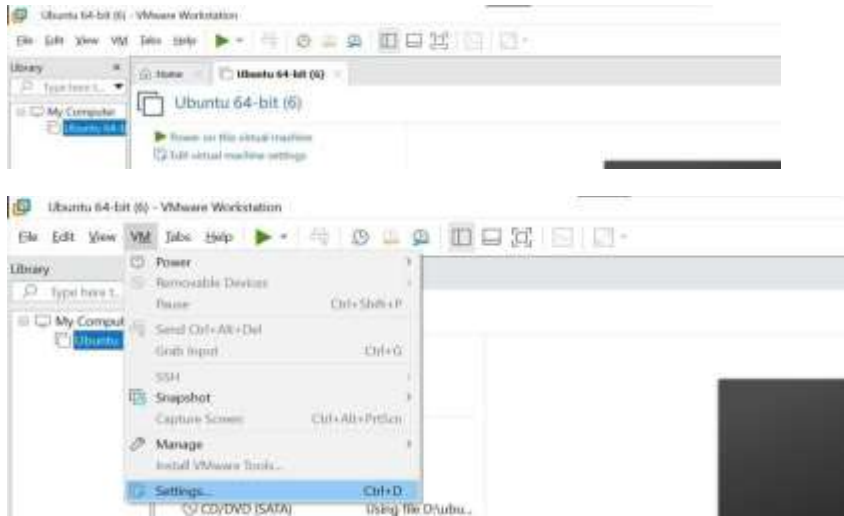
Department of Software Engineering

Lab Title:	Experiment # :
Student Name:	Regd. #:
Experiment Name:	

Performance Indicator	Level of Achievements				
	Excellent (5)	Good (4)	Average (3)	Below Average (2)	Poor (1)
Understanding & implementation					
Problem					
Report and presentation					
Maximum Marks		10	Obtained Marks		
Lab Instructor:					

Lab 4

Task 1;



Task 2:



```
ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
link/ether 08:0c:29:6a:f6:07 brd ff:ff:ff:ff:ff:ff
aliasname enp2s1
inet 192.168.161.129/24 metric 100 brd 192.168.161.255 scope global dynamic ens33
    valid_lft 1082sec preferred_lft 1082sec
inet6 fe80::20c:29ff:fe6a:f607/64 scope link
    valid_lft forever preferred_lft forever
```

```
PS C:\Users\HP> ping 192.168.161.129

Pinging 192.168.161.129 with 32 bytes of data:
Reply from 192.168.161.129: bytes=32 time=1ms TTL=64
Reply from 192.168.161.129: bytes=32 time<1ms TTL=64
Reply from 192.168.161.129: bytes=32 time<1ms TTL=64
Reply from 192.168.161.129: bytes=32 time=3ms TTL=64

Ping statistics for 192.168.161.129:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 3ms, Average = 1ms
```

```

Warning: Permanently added '192.168.161.129' (fd25519) to the list of known hosts.
hajra@192.168.161.129's password:
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-84-generic x86_64)

```

```

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

```

System information as of Sat 27 Sep 10:28:45 UTC 2025

```

System load:  1.46      Processes:            276
Usage of /:   70.2% of 11.21GB   Users logged in:    1
Memory usage: 57%      IPv4 address for ens33: 192.168.161.129
Swap usage:   9%

```

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See <https://ubuntu.com/esm> or run: `sudo pro status`

```

ls -la
total 39593
-rwxr-xr-x 1 Alli 197609 0 Nov 15 09:10 ./
-rwxr-xr-x 1 Alli 197609 0 Oct 16 00:59 ../
-rw-r--r-- 1 Alli 197609 11439 Nov 20 01:26 .bash_history
-rwxr-xr-x 1 Alli 197609 0 Dec 10 2024 .dotnet/
-rwxr-xr-x 1 Alli 197609 0 Oct 18 03:51 .git/
-rw-r--r-- 1 Alli 197609 301 Oct 18 02:07 .gitconfig
-rwxr-xr-x 1 Alli 197609 0 Dec 5 2024 .gk/
-rwxr-xr-x 1 Alli 197609 0 Dec 5 2024 .group/
-rw-r--r-- 1 Alli 197609 20 Oct 18 03:48 .ssh/

```

```

ls -la /bin
total 90304
-rwxr-xr-x 1 Alli 197609 0 Oct 8 20:18 ./
-rwxr-xr-x 1 Alli 197609 0 Oct 8 20:19 ../

```

```

Alli@DESKTOP-KQE4P80 MINGW64 - (main)
$ whoami
Alli
Alli@DESKTOP-KQE4P80 MINGW64 - (main)
$ /home/Alli
bash: /home/Alli: No such file or directory
Alli@DESKTOP-KQE4P80 MINGW64 - (main)
$ pwd
/c/Users/Alli
Alli@DESKTOP-KQE4P80 MINGW64 - (main)
$

```

Task 3:

```

total 96
drwxr-xr-x 12 root root 4096 Aug 5 16:54 .
drwxr-xr-x 23 root root 4096 Sep 26 21:15 ..
drwxr-xr-x 2 root root 36864 Oct 21 17:19 .git
drwxr-xr-x 2 root root 4096 Sep 26 21:15 .gitignore
drwxr-xr-x 25 root root 4096 Oct 14 05:07 .idea
drwxr-xr-x 23 root root 4096 Oct 21 17:20 .vs
drwxr-xr-x 2 root root 4096 Sep 26 21:18 .vscode
drwxr-xr-x 13 root root 4096 Oct 14 05:06 .vscode-test
drwxr-xr-x 10 root root 4096 Aug 5 16:54 .vscode-test
drwxr-xr-x 2 root root 20480 Oct 21 17:18 .vscode-test
drwxr-xr-x 128 root root 4096 Oct 14 05:07 .vscode-test
drwxr-xr-x 7 root root 4096 Oct 21 17:20 .vscode-test

```

```

total 16
drwxr-xr-x 4 root root 4096 Sep 26 21:51 .
drwxr-xr-x 23 root root 4096 Sep 26 21:15 ..
drwxr-xr-x 2 root root 4096 Sep 26 21:51 .git
drwxr-xr-x 4 root root 4096 Sep 26 21:51 .gitignore

```

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```
lab4/workspace/python_project$ cp README.md README.md.copy.md
lab4/workspace/python_project$
```

```
hain@buntu-lab:~/lab4/workspace/python_project$
```

Task 5

```
~/7lab4/workspace/python_project$ uname -a
Linux 5.8.0-85-generic #85-Ubuntu SMP PREEMPT_DYNAMIC Thu Sep 10 15:26:59 UTC 2025 x86_64 x86_64 GNU/Linux
```

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```

```

haiba@ubuntu-lab:~/lab4/workspace/python_project$ free -h
              total        used        free      shared  buff/cache   available
Mem:           1.9Gi       1.2Gi       132Mi       12Mi       734Mi       667Mi
Swap:          2.0Gi       223Mi       1.8Gi
haiba@ubuntu-lab:~/lab4/workspace/python_project$

```

Task 6

```

info: Adding user 'lab4user' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group 'lab4user' (1002) ...
info: Adding new user 'lab4user' (1002) with group 'lab4user' (1002) ...
info: Creating home directory '/home/lab4user' ...
info: Copying files from '/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for lab4user
Enter the new value, or press ENTER for the default
  Full Name []: hajira
    Room Number []: twenty two
    Work Phone []: 032wo7-5007883
    Home Phone []: 032wo7-5007883
    Other []: nil
Is the information correct? [Y/n] Y
info: adding new user 'lab4user' to supplemental / extra groups 'users' ...
info: adding user 'lab4user' to group 'users' ...

```

```
lab4user:x:1002:1002:hajra,twenty two,03two7-5007883,03two7-5007883,nil:/home/lab4user:/bin/bash
hajra@ubuntu-lab:~/lab4/workspace/python_projects$
```

```
info: Looking for files to backup/remove ...
info: Removing files ...
info: Removing crontab ...
info: Removing user 'lab4user' ...
haira@ubuntu-lab:~/lab4/workspace/python projects$
```



```

info: Adding user `lab4user' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `lab4user' (1002) ...
info: Adding new user `lab4user' (1002) with group `lab4user (1002)' ...
info: Creating home directory `/home/lab4user' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for lab4user
Enter the new value, or press ENTER for the default
  Full Name []: hajra
    Room Number []: twenty two
    Work Phone []: 03two7-5007883
    Home Phone []: 03two7-5007883
    Other []: nil
Is the information correct? [Y/n] Y
info: Adding new user `lab4user' to supplemental / extra groups `users' ...
info: Adding user `lab4user' to group `users' ...

```

```

hajra@ubuntu:~$ cd /lab4user; pwd; cd; pwd; cd /lab4user;
Password:
lab4user@ubuntu-lab:~$ _

```



FATIMA JINNAH WOMEN UNIVERSITY

Department of Software Engineering

Lab Title:	Experiment # :
Student Name:	Regd. #:
Experiment Name:	

Performance Indicator	Level of Achievements				
	Excellent (5)	Good (4)	Average (3)	Below Average (2)	Poor (1)
Understanding & implementation Problem					
Report and presentation					
Maximum Marks		10	Obtained Marks		
Lab Instructor:					

Lab 5

Task 1:

```
Command 'java' not found, but can be installed with:
sudo apt install openjdk-17-jre-headless # version 17.0.16+8~us1-0ubuntu1~24.04.1, or
sudo apt install openjdk-21-jre-headless # version 21.0.8+9~us1-0ubuntu1~24.04.1
sudo apt install default-jre # version 2:1.17-75
sudo apt install openjdk-11-jre-headless # version 11.0.28+6-1ubuntu1~24.04.1
sudo apt install openjdk-25-jre-headless # version 25+36-1~24.04.2
sudo apt install openjdk-8-jre-headless # version 8u462-ga~us1-0ubuntu2~24.04.2
sudo apt install openjdk-19-jre-headless # version 19.0.2+7-4
sudo apt install openjdk-20-jre-headless # version 20.0.2+9-1
sudo apt install openjdk-22-jre-headless # version 22~22ea-1
hsjra@ubuntu-lab:~$
```

```
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  alsa-topology-conf alsa-ucm-conf ca-certificates-java java-common libasound2-data libasound2t64 libpcsc-lite1t2
Suggested packages:
  default-jre alsa-utils libasound2-plugins xosro libasound2-plugins fontconfig fontconfig-glyphs fontconfig-urw fontconfig-misc fontconfig-urw-misc
1 fonts-ubuntu fontconfig
The following NEW packages will be installed:
  alsa-topology-conf alsa-ucm-conf ca-certificates-java java-common libasound2-data libasound2t64 libpcsc-lite1t2 openjdk-17-jre-headless
0 upgraded, 8 newly installed, 0 to remove and 0 not upgraded.
Need to get 46.5 MB of archives.
After this operation, 133 MB of additional disk space will be used.
Do you want to continue? [Y/n] Y
Get:1 http://archive.ubuntu.com/ubuntu noble/main amd64 alsa-topology-conf all 1.2.5.1-2 [15.5 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble/main amd64 alsa-ucm-conf all 1.2.11-1ubuntu1 [121.1 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libasound2t64 amd64 1.2.11-1ubuntu1 [199 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 alsa-ucm-conf all 1.2.11-1ubuntu1 [66.4 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble/main amd64 ca-certificates-java all 20240419 [11.8 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble/main amd64 java-common all 0.17esset [6,796 B]
Get:7 http://archive.ubuntu.com/ubuntu noble/main amd64 libpcsc-lite1t2 amd64 1.9.3-1build1 [21.4 kB]
Get:8 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 openjdk-17-jre-headless amd64 17.0.16+8~us1-0ubuntu1~24.04.1 [46.8 MB]
Fetched 46.5 MB in 3s (16.7 MB/s)
Selecting previously unselected package alsa-topology-conf.
Preparing database ... Done
```

```
openjdk 17.0.16 2025-07-15
OpenJDK Runtime Environment (build 17.0.16+8-Ubuntu-0ubuntu124.04.1)
OpenJDK 64-Bit Server VM (build 17.0.16+8-Ubuntu-0ubuntu124.04.1, mixed mode, sharing)
hsjra@ubuntu-lab:~$
```

```
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  alsa-topology-conf alsa-ucm-conf ca-certificates-java java-common libasound2-data libasound2t64 libpcsc-lite1t2
Use 'sudo apt autoremove' to remove them.
The following packages will be REMOVED:
  openjdk-17-jre-headless
0 upgraded, 0 newly installed, 1 to remove and 20 not upgraded.
After this operation, 133 MB disk space will be freed.
(Reading database ... 143550 files and directories currently installed.)
Removing openjdk-17-jre-headless:amd64 (17.0.16+8~us1-0ubuntu1~24.04.1) ...
hsjra@ubuntu-lab:~$
```

```
hsjra@ubuntu-lab:~$ java
-bash: /usr/bin/java: No such file or directory
hsjra@ubuntu-lab:~$
```

```
Command 'java' not found, but can be installed with:
sudo apt install openjdk-17-jre-headless # version 17.0.16+8~us1-0ubuntu1~24.04.1, or
sudo apt install openjdk-21-jre-headless # version 21.0.8+9~us1-0ubuntu1~24.04.1
sudo apt install default-jre # version 2:1.17-75
sudo apt install openjdk-11-jre-headless # version 11.0.28+6-1ubuntu1~24.04.1
sudo apt install openjdk-25-jre-headless # version 25+36-1~24.04.2
sudo apt install openjdk-8-jre-headless # version 8u462-ga~us1-0ubuntu2~24.04.2
sudo apt install openjdk-19-jre-headless # version 19.0.2+7-4
sudo apt install openjdk-20-jre-headless # version 20.0.2+9-1
sudo apt install openjdk-22-jre-headless # version 22~22ea-1
hsjra@ubuntu-lab:~$
```

Task 2

```

Building package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  default-jre-headless fonts-dejavu-extra libatk-wrapper-java libatk-wrapper-java-jni libgit7 libicb-shape0 libitm2 libxv libxvfbkgd openjdk-21-jre
openjdk-21-jre-headless x11-utils
Suggested packages:
  libasound2 fonts-lob3font-gothic fonts-lob3font-silcho fonts-way-microhei | fonts-way-zerhei fonts-lob3 debconf-utilities
Recommended packages:
  libxv
The following new packages will be installed:
  default-jre-headless fonts-dejavu-extra libatk-wrapper-java libatk-wrapper-java-jni libgit7 libicb-shape0 libitm2 libxv libxvfbkgd
openjdk-21-jre openjdk-21-jre-headless x11-utils
0 upgraded, 13 newly installed, 0 to remove and 20 not upgraded.
Need to get 45.0 MB of archives.
After this operation, 212 MB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 openjdk-21-jre-headless amd64 21.0.8-9*~ubuntu24.04.1 [46.4 MB]
Get:2 http://archive.ubuntu.com/ubuntu noble/main amd64 default-jre-headless amd64 2:1.21-75~exp1 [3,094 B]
Get:3 http://archive.ubuntu.com/ubuntu noble/main amd64 libgit7 amd64 5.2.8-1ubuntu1 [125.0 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 openjdk-21-jre amd64 21.0.8-9*~ubuntu24.04.1 [228 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libxv amd64 2:1.0.11-1ubuntu1 [116.7 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble/main amd64 fonts-dejavu-extra all 2.37-3 [1.54 kB]
Get:7 http://archive.ubuntu.com/ubuntu noble/main amd64 libicb-shape0 amd64 1.15-1ubuntu2 [6,100 B]
Get:8 http://archive.ubuntu.com/ubuntu noble/main amd64 libxvfbkgd amd64 2.0.6-1ubuntu1 [345.9 kB]
Get:9 http://archive.ubuntu.com/ubuntu noble/main amd64 libxv amd64 2:1.0.11-1ubuntu1 [116.7 kB]
Get:10 http://archive.ubuntu.com/ubuntu noble/main amd64 x11-utils amd64 1:1.5-4ubuntu1 [111.6 kB]
Get:11 http://archive.ubuntu.com/ubuntu noble/main amd64 x11-utils amd64 7.74-0ubuntu1 [189 kB]
Get:12 http://archive.ubuntu.com/ubuntu noble/main amd64 libatk-wrapper-java all 0.40.0-2ubuntu1 [54.0 kB]
Get:13 http://archive.ubuntu.com/ubuntu noble/main amd64 libatk-wrapper-java-jni amd64 0.40.0-2ubuntu1 [46.4 kB]
Needed 45.0 MB in 6s (0.240 MB/s)
Selecting previously unselected package openjdk-21-jre-headless:amd64.
(Reading database ... 153265 files and directories currently installed.)
Preparing to unpack .../00-openjdk-21-jre-headless_21.0.8-9*~ubuntu24.04.1_amd64.deb ...
Unpacking openjdk-21-jre-headless:amd64 (21.0.8-9*~ubuntu24.04.1) ...

```

```
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  alsa-topology-conf alsa-ucm-conf ca-certificates-java default-jre-headless fonts-dejavu-extra java-common libasound2-data libasound2t64 libatk-wrapper-java
  libatk-wrapper-java-gtk libxfont2 libxcursor-dev libxtst-dev libxv libxfixes0gpgal openjdk-8-jre openjdk-8-jre-headless x11-utils
Use 'dpkg --get-selections' to configure them.
The following packages will be REMOVED:
  default-jre
0 upgraded, 0 newly installed, 1 to remove and 76 not upgraded.
After this operation, 6,144 B disk space will be freed.
(Reading database ... 143707 files and directories currently installed.)
Removing default-jre (2:1.8_75-expl) ...
dpkg-query -f='${Package} ${Version} ${Architecture}\n'
```

Task 3

[sudo] password for najra:

NAME	MAJ:MIN	RM	SIZE	RO	TYPE	MOUNTPOINTS
loop0	7:0	0	52.2M	1	loop	/snap/aws-cli/1668
loop1	7:1	0	52.2M	1	loop	/snap/aws-cli/1672
loop2	7:2	0	55.5M	1	loop	/snap/core18/2952
loop3	7:3	0	63.8M	1	loop	/snap/core20/2669
loop4	7:4	0	73.9M	1	loop	/snap/core22/2133
loop5	7:5	0	66.8M	1	loop	/snap/core24/1196
loop6	7:6	0	73.9M	1	loop	/snap/core22/2139
loop7	7:7	0	140.6M	1	loop	/snap/docker/3265
loop8	7:8	0	66.8M	1	loop	/snap/core24/1225
loop9	7:9	0	44.1M	1	loop	/snap/doctl/2263
loop10	7:10	0	420.3M	1	loop	/snap/google-cloud-sdk/608
loop11	7:11	0	44.1M	1	loop	/snap/doctl/2275
loop12	7:12	0	164.3M	1	loop	/snap/microk8s/8511
loop13	7:13	0	118.3M	1	loop	/snap/lxd/35624
loop14	7:14	0	22.7M	1	loop	/snap/keepalived/3093
loop15	7:15	0	118.4M	1	loop	/snap/lxd/36153
loop16	7:16	0	95.7M	1	loop	/snap/kata-containers/2446
loop17	7:17	0	336M	1	loop	/snap/nextcloud/50464
loop18	7:18	0	76.6M	1	loop	/snap/powershell/313
loop19	7:19	0	164.2M	1	loop	/snap/microk8s/8355
loop20	7:20	0	3.1M	1	loop	/snap/mosquitto/1006
loop21	7:21	0	50.9M	1	loop	/snap/snapd/25577
loop22	7:22	0	76.6M	1	loop	/snap/powershell/316
loop23	7:23	0	13.9M	1	loop	/snap/slcli/2957
loop24	7:24	0	50.8M	1	loop	/snap/snapd/25702
loop25	7:25	0	23.2M	1	loop	/snap/wormhole/509
sda	8:0	0	35G	0	disk	
└─sda1	8:1	0	1M	0	part	
└─sda2	8:2	0	2G	0	part	/boot
└─sda3	8:3	0	23G	0	part	
└─ubuntu--vg-ubuntu--lv	252:0	0	11.5G	0	lvm	/
sr0	11:0	1	1024M	0	rom	

```
$ nano ~/apt_update_vs_upgrade.md
```

One common issue apt update encounters is the former available packages from online repositories. It does not install or remove anything; it just updates the local index. On the other hand, `apt-get upgrade` actually installs the new versions of already-installed packages using the updated index.

Earlier, I faced issues where updates failed due to low disk space, and I had to extend the partition (`/dev/sda1`) using `growpart` to fix it. After fixing the partition, both `apt update` and `apt upgrade` ran successfully.

Without running `apt update` first, `apt upgrade` would not know about the latest versions available.

This process requires the system stays up-to-date and secure.

Task 4

Name	Version	Rev	Tracking	Publisher	Notes
code	7d842fb8	211	latest/stable	vscode	classic

```
code 7d842fb8 from Visual Studio Code (vscode) installed
```

```
1.105.1  
7d842fb85a0275a4a8e4d7e040d2625abbf7f084  
x64
```

```
name: code  
summary: Code editing. Redefined.  
publisher: Visual Studio Code (vscode)  
store-url: https://snapcraft.io/code  
contact: https://twitter.com/code  
license: unset  
description: |  
  Visual Studio Code is a new choice of tool that combines the  
  simplicity of a code editor with what developers need for the core  
  edit-build-debug cycle.  
commands:  
  - code  
  - code-uri-handler  
snap-id: Htmaw6Zufm0rBuhaw7J8z2paysk281u  
tracking: latest/stable  
refresh-date: today at 20:41 UTC  
channels:  
  latest/stable: 7d842fb8-2025-10-15 (211) 360MB classic  
  latest/candidate: ?  
  latest/beta: ?  
  latest/edge: ?  
installed: 7d842fb8 (211) 360MB classic
```

Task 5

```
The authenticity of host '192.168.161.129 (192.168.161.129)' can't be established.  
ED25519 key fingerprint is SHA256:M5unwK7+jxUhr6KM3bbMPRkghlGd+Kq5usPTQZ5Up3w.  
This key is not known by any other names.  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
```

```
Warning: Permanently added '192.168.161.129' (ED25519) to the list of known hosts.  
hajrag@192.168.161.129's password:  
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-84-generic x86_64)
```

```
* Documentation:  https://help.ubuntu.com  
* Management:    https://landscape.canonical.com  
* Support:        https://ubuntu.com/pro
```

System information as of Sat 27 Sep 10:28:45 UTC 2025

System load:	1.46	Processes:	276
Usage of /:	70.2% of 11.21GB	Users logged in:	1
Memory usage:	52%	IPv4 address for ens33:	192.168.161.129
Swap usage:	9%		

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See <https://ubuntu.com/esm> or run: `sudo pro status`

```
Hit:1 http://security.ubuntu.com/ubuntu noble-security InRelease  
Hit:2 http://archive.ubuntu.com/ubuntu noble InRelease  
Hit:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease  
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
All packages are up to date.  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
Calculating upgrade... Done  
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
```


[illegible]

```
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
 libfuse2764 libpipewire-0.3-modules erp pipewire-module-erp songbird
Suggested packages:
 pipewire-bin gstreamer
The following new packages will be installed:
 libfuse2764 libpipewire-0.3-modules erp pipewire-module-erp songbird xrdp
0 upgraded, 5 newly installed, 0 to remove and 0 not upgraded.
Service restarts being deferred:
/etc/passwd start /usr/sbin/sshd.service
systemctl restart system-logind.service
systemctl restart unattended-upgrades.service
... containers need to be restarted.
```

```

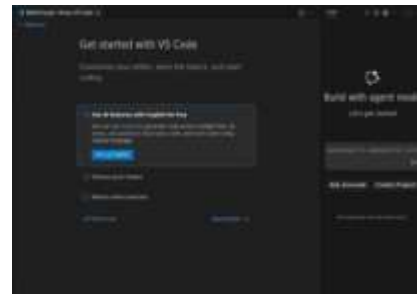
xrdp.service - xrdp daemon
Loaded: loaded (/usr/lib/systemd/system/xrdp.service; enabled; preset: enabled)
Active: active (running) since Fri 2025-10-24 21:03:34 UTC; 3min 27s ago
Docs: man:xrdp(8)
      man:xrdp.ini(5)
Main PID: 464810 (xrdp)
Tasks: 1 (limit: 2210)
Memory: 936.0K (peak: 1.0M)
CPU: 72ms
CGroup: /system.slice/xrdp.service
        └─464810 /usr/sbin/xrdp

Oct 24 21:05:33 ubuntu-lab systemd[1]: Starting xrdp.service - xrdp daemon...
Oct 24 21:05:33 ubuntu-lab xrdp[464810]: [INFO] address [0.0.0.0] port [3389] mode 1
Oct 24 21:05:33 ubuntu-lab xrdp[464810]: [INFO] listening to port 3389 on 0.0.0.0
Oct 24 21:05:33 ubuntu-lab xrdp[464810]: [INFO] xrdp_listen up done
Oct 24 21:05:33 ubuntu-lab systemd[1]: xrdp.service: Can't open PID file /run/xrdp/xrdp.pid (yet?) after start: no such file or directory
Oct 24 21:05:34 ubuntu-lab systemd[1]: Started xrdp.service - xrdp daemon.
Oct 24 21:05:35 ubuntu-lab xrdp[464810]: [INFO] starting xrdp with pid 464810
Oct 24 21:05:35 ubuntu-lab xrdp[464810]: [INFO] address [0.0.0.0] port [3389] mode 1
Oct 24 21:05:35 ubuntu-lab xrdp[464810]: [INFO] listening to port 3389 on 0.0.0.0
Oct 24 21:05:35 ubuntu-lab xrdp[464810]: [INFO] xrdp_listen up done

```

```
$ echo xfce4-session > ~/.xsession
$
```





```

Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
lightdm is already the newest version (1.30.0-0ubuntu14).
lightdm set to manually installed.
The following additional packages will be installed:
  gnome-accessibility-themes gnome-themes-extra gnome-themes-extra-data gtk2-engines-pixbuf
The following NEW packages will be installed:
  gnome-accessibility-themes gnome-themes-extra gnome-themes-extra-data gtk2-engines-pixbuf lightdm-gtk-greeter
0 upgraded, 5 newly installed, 0 to remove and 0 not upgraded.

```

synchronizing state of lightdm.service with your service script with `sudo systemctl daemon-reload`.
 Executing: `sudo systemctl daemon-reload`
 The unit files have no installation config (WantedBy=, Requires=, Upstream=, Alias=, or AliasSettings) in the [Install] section, and default instances for complete units). This means they are not meant to be enabled or disabled using `systemctl`.

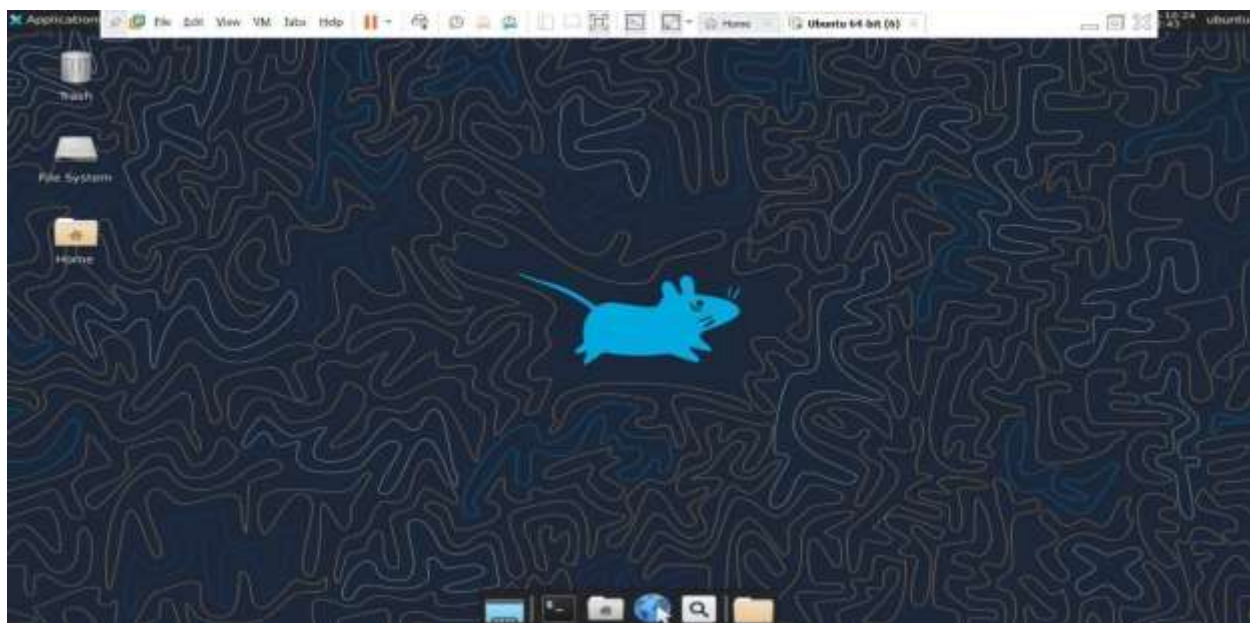
Possible reasons for having these kinds of unit files:

- A unit may be statically enabled by being symlinked from another unit's `WantedBy`, `Requires`, or `AliasSettings` directory.
- A unit's purpose may be to act as a helper for some other unit which has a `Requires=` dependency on it.
- A unit may be started when needed via activation (triggers, path, timer, etc), some `systemd` unit(s) call it.
- In case of template units, the unit is meant to be enabled with some instance name specified.

```

systemctl cat $ service unit set default graphical target
Created symlink /usr/systemd/system/default.target → /usr/lib/systemd/system/graphical.target.
Unit group (id) 0 =

```



```

ubuntu@24.04.3 LTS: ubuntu-lab $ curl
ubuntu-lab login: haifa
Password:
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-86-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

System information as of Wed 22 Oct 12:09:52 UTC 2024

System load:  0.96               Processes:    270
Usage of /:   36.9% of 11.71GB   Users logged in: 1
Memory usage: 57%              IPv4 address for enp3s1: 192.168.161.129
Swap usage:   0%

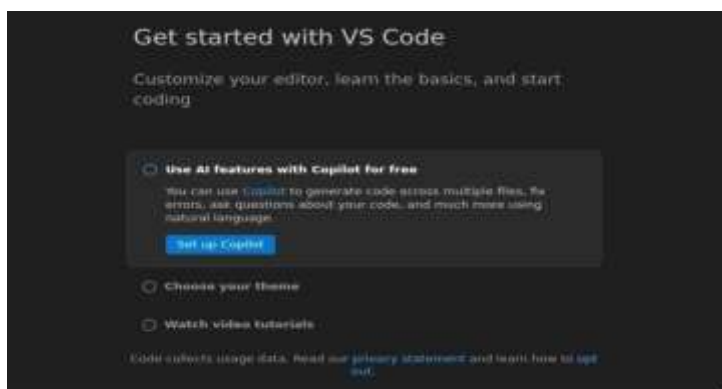
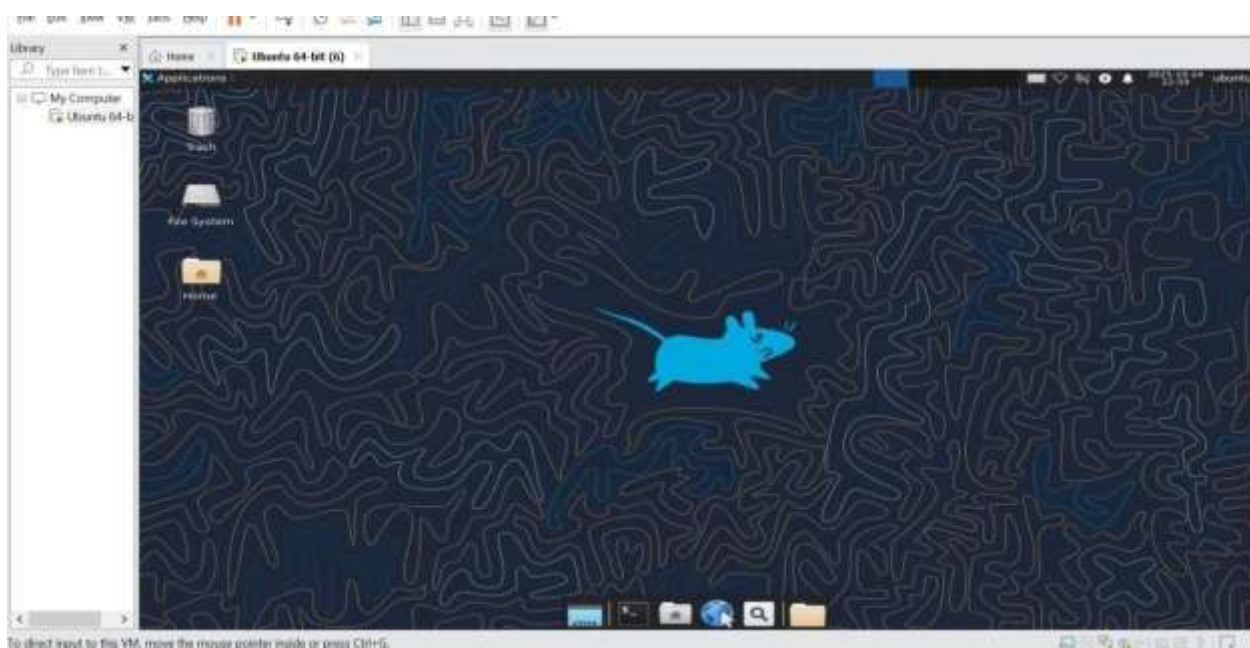
≡ / is using 36.9% of 11.71GB

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
   just raised the bar for easy, resilient and secure K8s cluster deployment.
   https://ubuntu.com/engage/secure-kubernetes-at-the-edge

Expanded security maintenance for Applications is not enabled.
e updates can be applied immediately.
n additional security updates can be applied with COS Apps.
Learn more about enabling COS Apps service at https://ubuntu.com/cos

```

```
sudo systemctl stop lightdm
```



Task 7

```
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
E: Unable to locate package google-chrome-stable
```

```
total 48
drwxr-xr-x  9 root root 4096 Sep 26 23:06
drwxr-xr-x 147 root root 12288 Oct 24 21:25
drwxr-xr-x  2 root root 4096 Oct 24 20:19
drwxr-xr-x  2 root root 4096 Mar 31 2024
drwxr-xr-x  2 root root 4096 Mar 31 2024
drwxr-xr-x  2 root root 4096 Aug 5 17:14
drwxr-xr-x  2 root root 4096 Aug 5 17:14
drwxr-xr-x  2 root root 4096 Aug 5 17:14
drwxr-xr-x  1 root root  78 Sep 26 23:24
drwxr-xr-x  2 root root 4096 Sep 26 23:24
drwxr-xr-x  2 root root 4096 Aug 5 17:01
```

```
types: deb
URIs: http://archive.ubuntu.com/ubuntu
Suites: noble noble-updates noble-backports
Components: main restricted universe multiverse
Signed-By: /usr/share/keyrings/ubuntu-archive-keyring.gpg
```

```
types: deb
URIs: http://security.ubuntu.com/ubuntu/
Suites: noble-security
Components: main restricted universe multiverse
Signed-By: /usr/share/keyrings/ubuntu-archive-keyring.gpg
```

```

Pages: 404
Info:
  url: http://www.mirrorservice.org/ip/ftp/pub/openssh/openssh-4.7p1.tar.gz
  comment: sshd restricted access build for ip
  signed-by: /usr/share/keyrings/debian-archive-keyring.gpg

```

```
types: deb
file: /usr/share/doc/iptables/iptables_1.4.21-1ubuntu1_all.deb
sha256: 7000000000000000000000000000000000000000000000000000000000000000
signature: 7000000000000000000000000000000000000000000000000000000000000000
signed-by: /usr/share/keyrings/ubuntu-archive-keyring.gpg
```

```

Access: pub
URL: https://www.googleapis.com/auth/cloud-platform
Scopes: stable
Components: stable
Architecture: amd64
Signed by: /etc/apt/keyrings/google.gpg

```

Help write Out where To Cut Execute Location Undo Set Mark To Bracket Previous Back
Exit Read File Replace Paste Testify No To Line Root Copy where was Next Forward

```

Get:11 http://dl.google.com/linux/chrome/deb/stable/main amd64 InRelease [1,825 B]
Get:12 http://security.ubuntu.com/ubuntu noble-security/inRelease [126 B]
Get:13 http://archive.ubuntu.com/ubuntu noble-inRelease [126 B]
Get:14 http://archive.ubuntu.com/ubuntu noble-updates/inRelease [126 B]
Get:15 http://dl.google.com/linux/chrome/deb/stable/main amd64 Packages [1,337 B]
Get:16 http://archive.ubuntu.com/ubuntu noble-backports/inRelease [126 B]
Get:17 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.0 B]
Get:18 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Components [212 B]
Get:19 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [52.3 B]
Get:20 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Components [212 B]
Get:21 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1,341 KB]
Get:22 http://archive.ubuntu.com/ubuntu noble-updates/main translation-en [189 KB]
Get:23 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [175 KB]
Get:24 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [2,170 KB]
Get:25 http://archive.ubuntu.com/ubuntu noble-updates/restricted translation-en [489 KB]
Get:26 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Components [212 B]
Get:27 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [377 KB]
Get:28 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [348 B]
Get:29 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [7,128 B]
Get:30 http://archive.ubuntu.com/ubuntu noble-backports/restricted amd64 Components [212 B]
Get:31 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [31.8 KB]
Get:32 http://archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Components [212 B]
Fetched 3,531 kB in 3s (1,004 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.

Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  fonts-liberation fonts-liberation-sans-narrow
The following NEW packages will be installed:
  fonts-liberation fonts-liberation-sans-narrow google-chrome-stable
0 upgraded, 3 newly installed, 0 to remove and 0 not upgraded.
Need to get 122 MB of archives.
After this operation, 401 MB of additional disk space will be used.
Get:1 http://dl.google.com/linux/chrome/deb/stable/main amd64 google-chrome-stable amd64 141.0.7390.122-1 [128 MB]
Get:2 http://archive.ubuntu.com/ubuntu noble/main amd64 fonts-liberation all 1:1.15.3 [1,083 KB]
Get:3 http://archive.ubuntu.com/ubuntu noble/main amd64 fonts-liberation-sans-narrow all 1:1.07.6-4 [107 KB]
Fetched 122 MB in 11s (11,723 kB/s)
Processing triggers for base-files (0.0+22.06.20220217ubuntu5) ...
Rebuilding /usr/share/applications/bamf-2.index...
Processing triggers for desktop-file-utils (0.27-2build1) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
~/graphonuts lab1 $

```

Task 8

```

vim - vi Improved

version 9.1.007
by Bram Moolenaar et al.
Modified by IamvishnuPrakash@Debian.org
vim is open source and freely distributable

Type :help sponsor for information
Type :ru to exit
Type :help or :h to view help
Type :help version for version info

$ vim k8s-sample.yaml

apiVersion: v1
kind: Pod
metadata:
  name: nginx-pod
spec:
  containers:
    - name: nginx
      image: nginx:1.19
      ports:
        - containerPort: 80

```

Task 9

```
cd ~/Lab5
vim k8s-sample.yaml
```

```
apiVersion: v1
kind: Pod
metadata:
  annotations:
    lab: lesson11
```

```
apiVersion: v1
kind: Pod
metadata:
  annotations:
    lab: lesson11
  name: nginx-pod
spec:
  containers:
    - name: nginx
      image: nginx:1.19
      ports:
        - containerPort: 80
      restartPolicy: Always
```

```
apiVersion: v1
kind: Pod
metadata:
  annotations:
    lab: lesson11
  name: nginx-pod
spec:
  containers:
    - name: nginx
      image: nginx:1.19
      ports:
        - containerPort: 80
      restartPolicy: Always
```

```
apiVersion: v1
kind: Pod
metadata:
  annotations:
    lab: lesson11
  name: nginx-pod
spec:
  containers:
    - name: nginx
      image: nginx:1.19
      ports:
        - containerPort: 80
      restartPolicy: Always
  # temp: do-not-keep
```

```
vim k8s-sample.yaml
```

```
apiVersion: v1
kind: Pod
metadata:
  annotations:
    lab: lesson11
  name: nginx-pod
spec:
  containers:
    - name: nginx
      image: nginx:1.19
      ports:
        - containerPort: 80
      restartPolicy: Always
  # temp: do-not-keep
```

Task 10

```
$ cd ~/Lab5
$ vim k8s-sample.yaml
```

```
- name: nginx
  image: nginx:1.19
  ports:
```

```

- name: nginx
  ports:
    - containerPort: 80
    - containerPort: 443
- name: nginx
  image: nginx:1.19
  ports:
    - containerPort: 80
    - containerPort: 443
apiVersion: v1
kind: Pod
metadata:
  annotations:
    lab: lesson11
  name: nginx-pod
  namespace: default
spec:
  containers:
    - name: nginx
      image: nginx:1.19
      ports:
        - containerPort: 80
        - containerPort: 443
      restartPolicy: Always
      # temp: do-not-keep
vim k8s-sample.yaml

```

Task 11:

```

$ cd ~/Lab5
$ vim k8s-sample.yaml

```

```

- name: nginx
  image: nginx:1.19
  ports:
    - containerPort: 80
    - containerPort: 443
  restartPolicy: Always
  # temp: do-not-keep
  # nginx added for testing
  # # another nginx line

```

```

apiVersion: v1
kind: Pod
metadata:
  annotations:
    lab: lesson11
  name: nginx-pod
  namespace: default
spec:
  containers:
    - name: nginx
      image: nginx:1.19
      ports:
        - containerPort: 80
        - containerPort: 443
      restartPolicy: Always
      # temp: do-not-keep
vim k8s-sample.yaml

```

Docker desktop Installation:


```

Get:1 http://dl.google.com/linux/chrome/deb/stable InRelease [1,625 B]
Hit:2 http://archive.ubuntu.com/ubuntu noble InRelease
Get:3 http://dl.google.com/linux/chrome/deb/stable/main amd64 Packages [1,235 B]
Get:4 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Hit:5 http://ppa.launchpadcontent.net/obsproject/obs-studio/ubuntu noble InRelease
Get:6 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Hit:7 https://ppa.launchpadcontent.net/ubuntuhandbook1/audacity/ubuntu noble InRelease
Hit:8 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.5 kB]
Hit:9 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Components [208 B]
Get:10 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [52.2 kB]
Get:11 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Components [200 B]
Get:12 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:13 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1,542 kB]
Get:14 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [170 kB]
Get:15 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Components [212 B]
Get:16 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [377 kB]
Get:17 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Get:18 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [7,124 B]
Get:19 http://archive.ubuntu.com/ubuntu noble-backports/restricted amd64 Components [212 B]
Get:20 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [11.0 kB]
Get:21 http://archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Components [212 B]
Fetched 3,569 kB in 33s (79.0 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.

```

```

sudo mkdir -p /etc/apt/keyrings
curl -fsSL https://download.docker.com/linux/ubuntu/gpg | sudo gpg --dearmor -o /etc/apt/keyrings/docker.gpg

```

```

Get:1 https://download.docker.com/linux/ubuntu noble InRelease [48.5 kB]
Hit:2 http://security.ubuntu.com/ubuntu noble-security InRelease
Get:3 https://download.docker.com/linux/ubuntu noble/stable amd64 Packages [33.3 kB]
Hit:4 http://dl.google.com/linux/chrome/deb/stable InRelease
Hit:5 http://archive.ubuntu.com/ubuntu noble InRelease
Get:6 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Hit:7 https://ppa.launchpadcontent.net/obsproject/obs-studio/ubuntu noble InRelease
Hit:8 https://ppa.launchpadcontent.net/ubuntuhandbook1/audacity/ubuntu noble InRelease
Hit:9 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Get:10 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1,541 kB]
Fetched 1,749 kB in 12s (143 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.

```

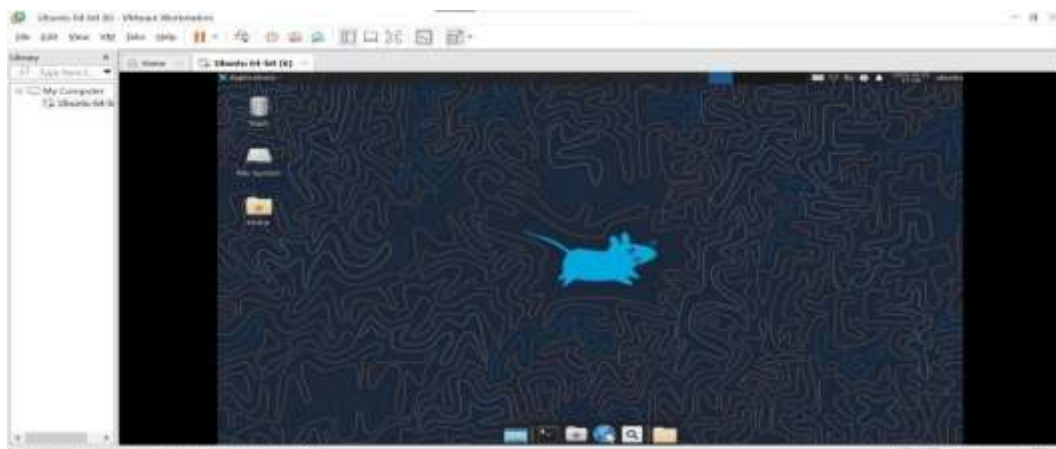
```

Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  docker-ce-rootless-extras libltdl7e0 sigs-killpatterns
Suggested packages:
  (groupfs-mount) | cgroup-lite docker-model-plugin
The following new packages will be installed:
  containerd.io docker-buildx-plugin docker-ce-cli docker-ce-rootless-extras docker-compose-plugin libltdl7e0 pigs-killpatterns
0 upgraded, 8 newly installed, 0 to remove and 0 not upgraded.
Need to get 365 MB of archives.
After this operation, 637 MB of additional disk space will be used.
Get:1 https://download.docker.com/linux/ubuntu noble/stable amd64 containerd.io amd64 1.7.28-1-ubuntu.24.04-ubuntu [31.9 MB]
Get:2 http://archive.ubuntu.com/ubuntu noble/universe amd64 sigs-killpatterns 2.0-1 [25.0 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble/main amd64 libltdl7e0 amd64 2.0.0-ubuntu [63.0 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble/universe amd64 killpatterns amd64 1.2.1-1build2 [34.9 kB]
Get:5 https://download.docker.com/linux/ubuntu noble/stable amd64 docker-ce-cli amd64 5:28.5.3-1-ubuntu.24.04-ubuntu [16.5 MB]
Get:6 https://download.docker.com/linux/ubuntu noble/stable amd64 docker-ce amd64 5:28.5.3-1-ubuntu.24.04-ubuntu [19.7 MB]
Get:7 https://download.docker.com/linux/ubuntu noble/stable amd64 docker-buildx-plugin amd64 0.20.1-1-ubuntu.24.04-ubuntu [25.9 MB]
Get:8 https://download.docker.com/linux/ubuntu noble/stable amd64 docker-ce-rootless-extras amd64 5:28.5.3-1-ubuntu.24.04-ubuntu [8,401 MB]
Get:9 https://download.docker.com/linux/ubuntu noble/stable amd64 docker-compose-plugin amd64 2.40.2-1-ubuntu.24.04-ubuntu [14.3 MB]
Fetched 365 MB in 47s (2,228 kB/s)
Selecting previously unselected package containerd.io.
(Reading database ... 18903 files and directories currently installed.)
Preparing to unpack .../6-containerd.io_1.7.28-1-ubuntu.24.04-ubuntu_amd64.deb ...
Unpacking containerd.io (1.7.28-1-ubuntu.24.04-ubuntu) ...
Selecting previously unselected package docker-ce-cli.
Preparing to unpack .../1-docker-ce-cli_5:28.5.3-1-ubuntu.24.04-ubuntu_amd64.deb ...
Unpacking docker-ce-cli (5:28.5.3-1-ubuntu.24.04-ubuntu) ...
Selecting previously unselected package docker-ce.
Preparing to unpack .../2-docker-ce_5:28.5.3-1-ubuntu.24.04-ubuntu_amd64.deb ...
Unpacking docker-ce (5:28.5.3-1-ubuntu.24.04-ubuntu) ...
Selecting previously unselected package pigs.
Preparing to unpack .../3-pigs_2.0-1_amd64.deb ...
Unpacking pigs (2.0-1) ...
Selecting previously unselected package docker-buildx-plugin.
Preparing to unpack .../4-docker-buildx-plugin_0.20.1-1-ubuntu.24.04-ubuntu_amd64.deb ...
Unpacking docker-buildx-plugin (0.20.1-1-ubuntu.24.04-ubuntu) ...
Selecting previously unselected package docker-ce-rootless-extras.
Preparing to unpack .../5-docker-ce-rootless-extras_5:28.5.3-1-ubuntu.24.04-ubuntu_amd64.deb ...
Unpacking docker-ce-rootless-extras (5:28.5.3-1-ubuntu.24.04-ubuntu) ...
Selecting previously unselected package docker-compose-plugin.
Preparing to unpack .../8-docker-compose-plugin_2.40.2-1-ubuntu.24.04-ubuntu_amd64.deb ...
Unpacking docker-compose-plugin (2.40.2-1-ubuntu.24.04-ubuntu) ...

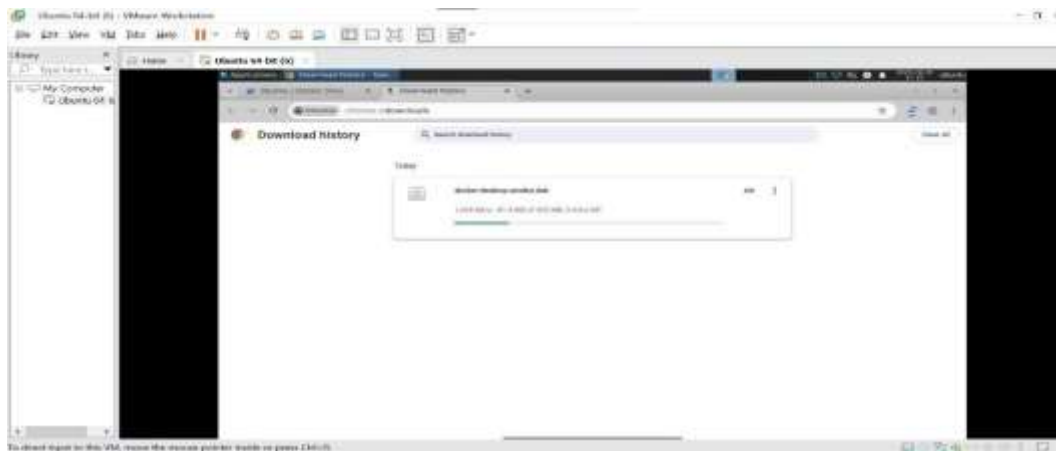
```

```
systemctl --user start docker-desktop
```

[illegible]



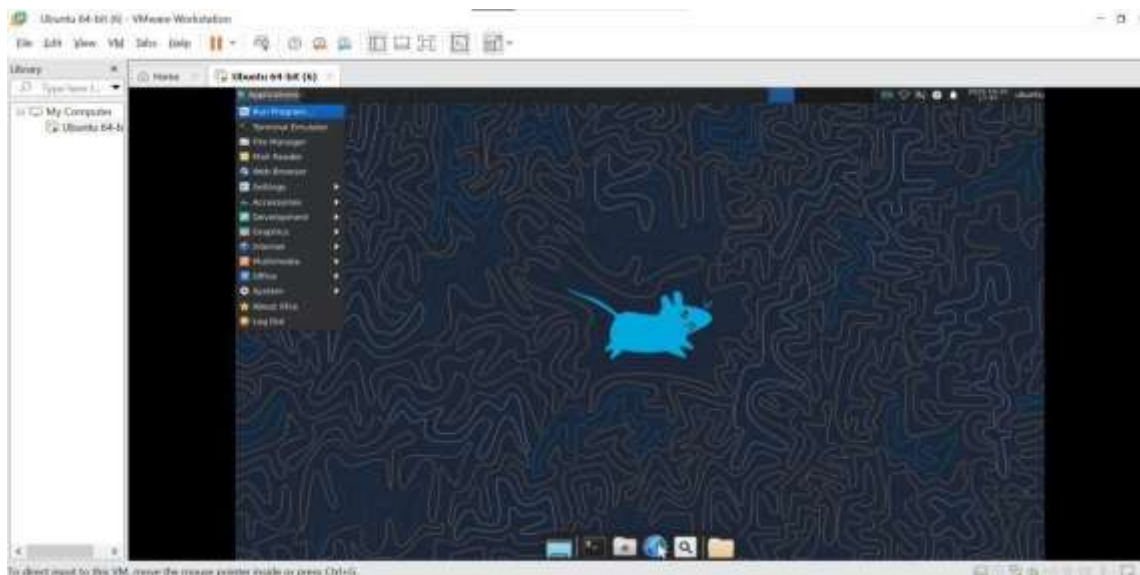
To direct input to this VM, move the mouse pointer inside or press Ctrl+Q.

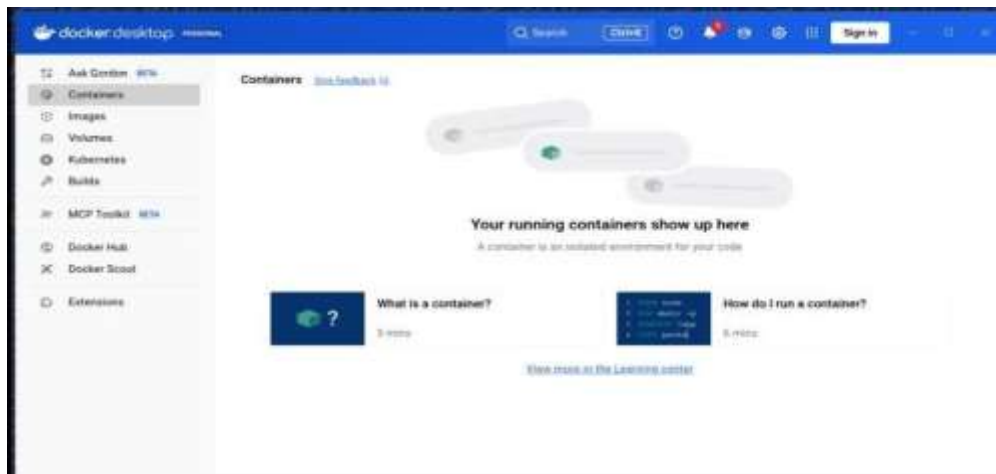
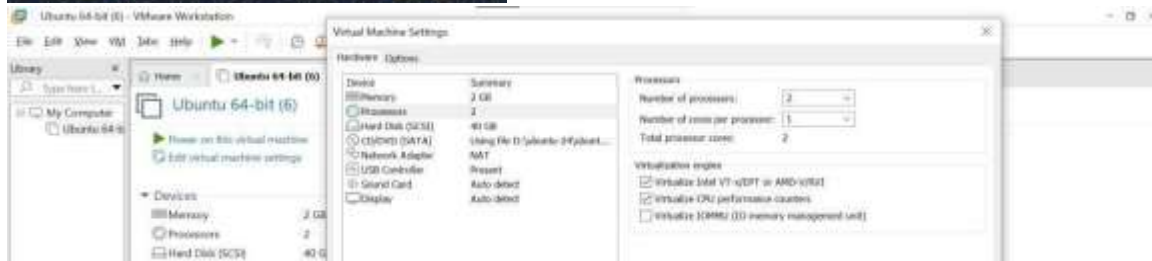


```

Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Note, selecting 'docker-desktop' instead of './docker-desktop-amd64.deb'
docker-desktop is already the newest version (4.49.0-208700).
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.

```







FATIMA JINNAH WOMEN UNIVERSITY

Department of Software Engineering

Lab Title:	Experiment # :
Student Name:	Regd. #:
Experiment Name:	

Performance Indicator	Level of Achievements				
	Excellent (5)	Good (4)	Average (3)	Below Average (2)	Poor (1)
Understanding & implementation Problem					
Report and presentation					
Maximum Marks		10	Obtained Marks		
Lab Instructor:					

Lab 6

Task 1

```
inshal@inshal:~$ sudo passwd root
[sudo] password for inshal:
New password:
Retype new password:
passwd: password updated successfully
inshal@inshal:~$
```

```
inshal@inshal:~$ su -
Password:
root@inshal:~# id
uid=0(root) gid=0(root) groups=0(root)
root@inshal:~#
```

```
root@inshal:~# exit
logout
inshal@inshal:~$ whoami
inshal
inshal@inshal:~$
```

Task 2

```

inshal@inshal:~$ sudo adduser tom
info: Adding user `tom' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `tom' (1001) ...
info: Adding new user `tom' (1001) with group `tom (1001)' ...
info: Creating home directory `/home/tom' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for tom
Enter the new value, or press ENTER for the default
    Full Name []: inshal
    Room Number []:
    Work Phone []:
    Home Phone []:
    Other []:
Is the information correct? [Y/n]
info: Adding new user `tom' to supplemental / extra groups `users' ...
info: Adding user `tom' to group `users' ...

```

```

inshal@inshal:~$ cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534::/nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:998:998:systemd Network Management:/:/usr/sbin/nologin
systemd-timesync:x:997:997:systemd Time Synchronization:/:/usr/sbin/nologin
dhcpcd:x:100:65534:DHCP Client Daemon,,,:/usr/lib/dhcpcd:/bin/false
messagebus:x:101:102::/nonexistent:/usr/sbin/nologin
systemd-resolve:x:992:992:systemd Resolver:/:/usr/sbin/nologin
pollinate:x:102:1::/var/cache/pollinate:/bin/false
polkitd:x:991:991:User for polkitd:/:/usr/sbin/nologin
syslog:x:103:104::/nonexistent:/usr/sbin/nologin
uidd:x:104:105::/run/uidd:/usr/sbin/nologin
tcpdump:x:105:107::/nonexistent:/usr/sbin/nologin
tss:x:106:108:TPM software stack,,,:/var/lib/tpm:/bin/false
landscape:x:107:109::/var/lib/landscape:/usr/sbin/nologin
fwupd-refresh:x:989:989:Firmware update daemon:/var/lib/fwupd:/usr/sbin/nologin
usbmux:x:108:46:usbmux daemon,,,:/var/lib/usbmux:/usr/sbin/nologin

```

```
inshal@inshal:~$ cat /etc/group
root:x:0:
daemon:x:1:
bin:x:2:
sys:x:3:
adm:x:4:syslog,inshal
tty:x:5:
disk:x:6:
lp:x:7:
mail:x:8:
news:x:9:
uucp:x:10:
man:x:12:
proxy:x:13:
kmem:x:15:
dialout:x:20:
fax:x:21:
voice:x:22:
cdrom:x:24:inshal
floppy:x:25:
tape:x:26:
sudo:x:27:inshal
audio:x:29:
dip:x:30:inshal
www-data:x:33:
backup:x:34:
operator:x:37:
list:x:38:
irc:x:39:
src:x:40:
```

```
inshal@inshal:~$ sudo cat /etc/shadow
root:$y$j9T$ENoprgu8qsk6uBwNRF3SD1$0BNfxCjx4.6QuLVk733Wd8NT00LxdR9h0rA230tLQ12:20399:0:99999:7:
daemon:*:20305:0:99999:7:::
bin:*:20305:0:99999:7:::
sys:*:20305:0:99999:7:::
sync:*:20305:0:99999:7:::
games:*:20305:0:99999:7:::
man:*:20305:0:99999:7:::
lp:*:20305:0:99999:7:::
mail:*:20305:0:99999:7:::
news:*:20305:0:99999:7:::
uucp:*:20305:0:99999:7:::
proxy:*:20305:0:99999:7:::
www-data:*:20305:0:99999:7:::
backup:*:20305:0:99999:7:::
list:*:20305:0:99999:7:::
irc:*:20305:0:99999:7:::
_apt:*:20305:0:99999:7:::
nobody:*:20305:0:99999:7:::
systemd-network:!*:20305::::::
systemd-timesync:!*:20305::::::
dhcpcd:!:20305::::::
messagebus:!:20305::::::
systemd-resolve:!*:20305::::::
pollinate:!:20305::::::
polkitd:!*:20305::::::
syslog:!:20305::::::
uuid:!:20305::::::
tcpdump:!:20305::::::
tss:!:20305::::::
landscape:!:20305::::::
fwupd-refresh:!*:20305::::::
```

Task 3

```

groupadd: group 'developer' already exists
inshal@inshal:~$ sudo addgroup designer
info: Selecting GID from range 1000 to 59999 ...
info: Adding group 'designer' (GID 1003) ...
inshal@inshal:~$ sudo addgroup design
info: Selecting GID from range 1000 to 59999 ...
info: Adding group 'design' (GID 1004) ...
inshal@inshal:~$ cat /etc/groups
cat: /etc/groups: No such file or directory
inshal@inshal:~$ cat /etc/group
root:x:0:
daemon:x:1:
bin:x:2:
sys:x:3:
adm:x:4:syslog,inshal
tty:x:5:
disk:x:6:
lp:x:7:
mail:x:8:
news:x:9:
uucp:x:10:
man:x:12:
proxy:x:13:
kmem:x:15:
dialout:x:20:
fax:x:21:
voice:x:22:
cdrom:x:24:inshal
floppy:x:25:
tape:x:26:
sudo:x:27:inshal
audio:x:29:
dip:x:30:inshal

```

```

inshal@inshal:~$ sudo usermod -g designer tom
inshal@inshal:~$ id tom
uid=1001(tom) gid=1003(designer) groups=1003(designer),100(users)
inshal@inshal:~$

```

```

inshal@inshal:~$ sudo usermod -aG designer,design tom
inshal@inshal:~$ id tom
uid=1001(tom) gid=1003(designer) groups=1003(designer),100(users),1004(design)
inshal@inshal:~$

```

Task 4

```

inshal@inshal:~$ sudo adduser Jerry
err: Please enter a username matching the regular expression
      configured via the NAME_REGEX configuration variable. Use the
      '--allow-bad-names' option to relax this check or reconfigure
      NAME_REGEX in configuration.
inshal@inshal:~$

```

```

inshal@inshal:~$ su - scooby
Password:

```



```
scooby@inshal:~$ sudo passwd scooby
[sudo] password for scooby:
New password:
Retype new password:
passwd: password updated successfully
scooby@inshal:~$
```

```
scooby@inshal:~$ sudo mkdir -p /home/scooby
scooby@inshal:~$ sudo chown scooby:scooby /home/scooby
scooby@inshal:~$ sudo chmod 750 /home/scooby
scooby@inshal:~$ ls -ld /home/scooby
drwxr-x--- 2 scooby scooby 4096 Nov  7 05:33 /home/scooby
scooby@inshal:~$
```

```
scooby@inshal:~$ ls -ld /home/scooby
drwxr-x--- 2 scooby scooby 4096 Nov  7 05:33 /home/scooby
scooby@inshal:~$
```

```
scooby@inshal:~$ su - Scooby
pwd
ls -la
su: user Scooby does not exist or the user entry does not contain all the required fields
/home/scooby
total 24
drwxr-x--- 2 scooby scooby 4096 Nov  7 05:33 .
drwxr-xr-x 6 root    root   4096 Nov  7 05:36 ..
-rw-r----- 1 scooby scooby   5 Nov  7 05:33 .bash_history
-rw-r--r-- 1 scooby scooby  220 Nov  7 05:22 .bash_logout
-rw-r--r-- 1 scooby scooby 3771 Nov  7 05:22 .bashrc
-rw-r--r-- 1 scooby scooby  807 Nov  7 05:22 .profile
-rw-r--r-- 1 scooby scooby   0 Nov  7 05:32 .sudo_as_admin_successful
scooby@inshal:~$
```

```
scooby@inshal:~$ exit
cat /etc/passwd
logout
```

```
uuid:x:104:105:./run/uuid:/usr/sbin/nologin
tcpdump:x:105:107:./nonexistent:/usr/sbin/nologin
tss:x:106:108:TPM software stack,,,:/var/lib/tpm:/bin/false
landscape:x:107:109:./var/lib/landscape:/usr/sbin/nologin
fwupd-refresh:x:989:989:Firmware update daemon:/var/lib/fwupd:/usr/sbin/nologin
usbmux:x:108:46:usbmux daemon,,,:/var/lib/usbmux:/usr/sbin/nologin
sshd:x:109:65534:./run/sshd:/usr/sbin/nologin
fatima:x:1000:1000:fatima:/home/fatima:/bin/bash
tom:x:1001:1004:,,,:/home/tom:/bin/bash
Scooby:x:1002:1005:./home/Scooby:/bin/sh
```

Task 5


```

Password:
$ touch file1
$ mkdir -p dir1
$ touch dir1/file2
$ ls -l
total 4
drwxrwxr-x 2 student student 4096 Nov  6 15:17 dir1
-rw-rw-r-- 1 student student   0 Nov  6 15:16 file1

$ sudo chgrp devops file1
$ ls -l file1
-rw-rw-r-- 1 tom devops 0 Nov  6 15:16 file1

```

```

$ ls -l file1
-rw-rw-r-- 1 tom devops 0 Nov  6 15:16 file1
$ ls -l
total 4
drwxrwxr-x 2 student student 4096 Nov  6 15:17 dir1
-rw-rw-r-- 1 tom devops 0 Nov  6 15:16 file1
$ ls -l dir1
total 0
-rw-rw-r-- 1 student student 0 Nov  6 15:17 file2
$ ls il /dev/null
ls: cannot access 'il': No such file or directory
/dev/null
$ ls -l /dev/null
crw-rw-rw- 1 root root 1, 3 Nov  6 11:44 /dev/null
$ file file1 dir1 /dev/null
file1:      empty
dir1:       directory
/dev/null:  character special (1/3)

```

```
$ exit
```

Task 6

```

Password:
$ cd ~
$ ls -l file1
-rw-rw-r-- 1 tom devops 0 Nov  6 15:16 file1

```

```

$ sudo chmod -rwx file1
[sudo] password for student:
chmod: file1: new permissions are ---w---, not -----
$ ls -l file1
---w--- 1 tom devops 0 Nov  6 15:16 file1

```

```

$ sudo chmod +r file1
$ ls -l file1
-r--rw-r-- 1 tom devops 0 Nov  6 15:16 file1

```

```

$ sudo chmod u+x file1
$ ls -l file1
-r-xrw-r-- 1 tom devops 0 Nov  6 15:16 file1

```

Task 7

```
fatima@fatima:~$ su - student
Password:
$ cd ~
$ ls -l file1
----- 1 tom devops 0 Nov  6 15:16 file1
$
```

```
$ sudo chmod ugo=rwx file1
$ ls -l file1
-rwxrwxrwx 1 tom devops 0 Nov  6 15:16 file1
$
```

```
$ sudo chmod go=rw file1
$ ls -l file1
-rwxrw-rw- 1 tom devops 0 Nov  6 15:16 file1
$
```

```
$ sudo chmod ugo= file1
$ ls -l file1
----- 1 tom devops 0 Nov  6 15:16 file1
$
```

Task 8

```
Password:
$ cd ~
$ ls -l file1
----- 1 tom devops 0 Nov  6 15:16 file1
$
```

```
$ sudo chmod 777 file1
$ ls -l file1
-rwxrwxrwx 1 tom devops 0 Nov  6 15:16 file1
$
```

```
$ sudo chmod 700 file1
$ ls -l file1
-rwx----- 1 tom devops 0 Nov  6 15:16 file1
$
```

```
$ sudo chmod 744 file1
$ ls -l file1
-rwxr--r-- 1 tom devops 0 Nov  6 15:16 file1
$
```

Task 9

```
$ sudo cat /var/log/syslog | less
$
```

```
$ sudo cat /var/log/syslog | more
2025-11-06T11:45:41.588489+00:00 fatima systemd-modules-load[463]: Inserted module 'msr'
2025-11-06T11:45:41.589229+00:00 fatima systemd-modules-load[463]: Inserted module 'dm_multipath'
2025-11-06T11:45:41.589256+00:00 fatima lvm[446]: 1 logical volume(s) in volume group "ubuntu-vg" monitored
2025-11-06T11:45:41.589269+00:00 fatima systemd[1]: Starting systemd-journal-flush.service - Flush Journal to Persistent Storage...
```

```
$ sudo grep -E 'fail|error' /var/log/syslog | head
2025-11-06T11:45:41.589296+00:00 fatima multipathd[481]: sda: failed to get udev uid: No data available
2025-11-06T11:45:41.589488+00:00 fatima multipath: sda: failed to get sysfs uid: No such file or directory
2025-11-06T11:45:41.589496+00:00 fatima multipath: sda: failed to get sgio uid: No such file or directory
2025-11-06T11:45:41.589587+00:00 fatima multipathd[481]: sda: failed to get udev uid: No data available
2025-11-06T11:45:41.589518+00:00 fatima multipathd[481]: sda: failed to get path uid
2025-11-06T11:45:41.589548+00:00 fatima multipathd[481]: uevent trigger error
2025-11-06T11:45:41.590711+00:00 fatima multipath: sda: failed to get sysfs uid: No such file or directory
2025-11-06T11:45:41.590734+00:00 fatima multipath: sda: failed to get sgio uid: No such file or directory
2025-11-06T11:45:41.590746+00:00 fatima multipathd[481]: sda: failed to get udev uid: No data available
2025-11-06T11:45:41.590756+00:00 fatima multipathd[481]: sda: failed to get path uid
```

```
$ sudo grep -i systemd /var/log/syslog > ~/syslog_systemd.txt
#
```

```
$ sudo journalctl | less
$ sudo journalctl -u systemd | grep -i error > ~/journal_errors.txt
#
```

Task 10

```
#!/bin/bash
# Define and show var1
var1="Hollo from Lab 6"
echo "var1: $var1"
```

```
# Directory check
if [ -d "dir1" ]; then
    echo "Directory dir1 exists."
else
    echo "Directory dir1 does not exist. Creating..."
    mkdir -p "dir1"
    echo "Directory dir1 created."
fi
```

Task 11

```
#!/bin/bash
num=$1
str=$2
```

```
cd ~
vim setup.sh
chmod +x setup.sh
./setup.sh 10 student
```

```
if [ "$num" -eq 10 ]; then
    echo "$num is equal to 10 (-eq)."
else
    echo "$num is NOT equal to 10 (-eq)."
fi
```

```

if [ -z "$str" ]; then
    echo "Second argument is empty (zero-length)."
else
    echo "Second argument is not empty."
fi

```

Task 12

```

#!/bin/bash
# Script to demonstrate printing all user-entered arguments using $*

```

```

vim setup.sh
./setup.sh

```

```

# Print all arguments using $*
echo "Printing all arguments using \${*}:"
for arg in $*; do
    echo "Argument: $arg"
done

```

Task 13


UbuntuMachine
Public
Watch 0

main
1 Branch
0 Tags

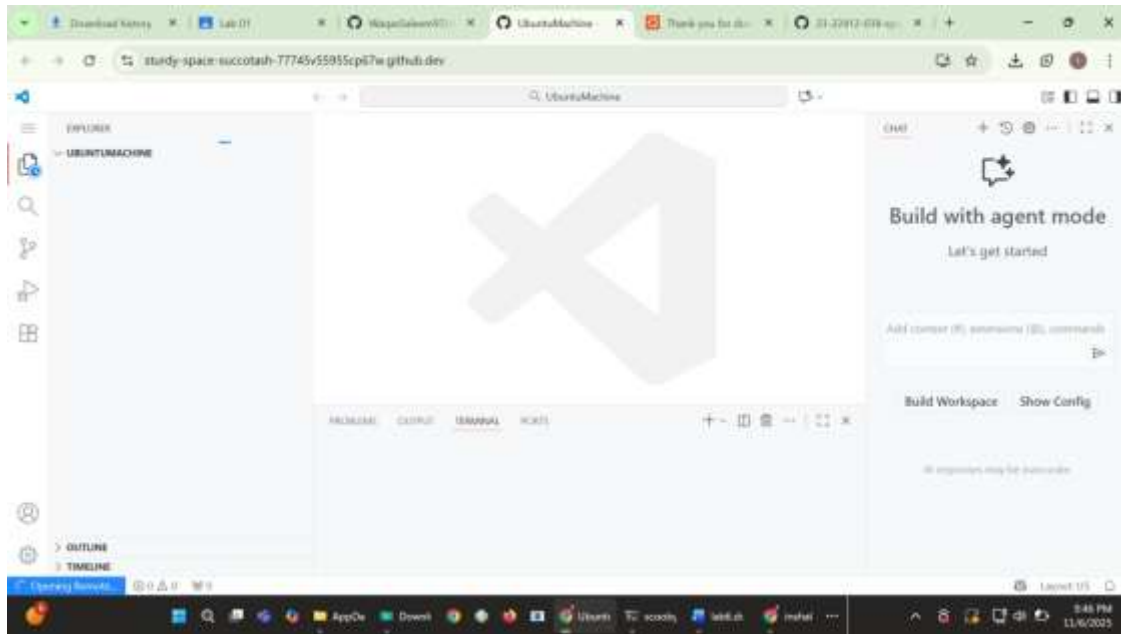
Add file
Code

Commit	Message	Author	Date
240438	Adding the stop script	5a1f77d	last week
	initial commit		2 weeks ago
	Adding the stop script		last week
	Adding the stop script		last week

README

UbuntuMachine

Ubuntu Machine



```
@23-22412-038-sys →/workspaces/UbuntuMachine (main) $ ls -l start-desktop.sh stop-desktop.sh
```

```
-rwxrwxrwx 1 codespace root 1333 Nov  7 05:45 start-desktop.sh
-rwxrwxrwx 1 codespace root  428 Nov  7 05:45 stop-desktop.sh
```

```
[*] Updating system and installing dependencies...
```

```
Get:1 https://dl.yarnpkg.com/debian stable InRelease
```

```
Get:2 https://repo.anaconda.com/pkg/misc/debrepo/conda stable InRelease [3961 B]
```

```
Get:3 https://repo.anaconda.com/pkg/misc/debrepo/conda stable/main amd64 Packages [4557 B]
```

```
Get:4 http://archive.ubuntu.com/ubuntu noble InRelease [256 kB]
```

```
Get:5 https://dl.yarnpkg.com/debian stable/main all Packages [11.8 kB]
```

```
Get:6 https://packages.microsoft.com/repos/microsoft-ubuntu-noble-prod noble InRelease [364 B]
```

```
Get:7 https://packages.microsoft.com/repos/microsoft-ubuntu-noble-prod noble/main all Packages [643 B]
```

```
Get:8 https://packages.microsoft.com/repos/microsoft-ubuntu-noble-prod noble/main amd64 Packages [66.5 kB]
```

```
Get:9 https://dl.yarnpkg.com/debian stable/main amd64 Packages [11.8 kB]
```

```
Get:10 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
```

```
Get:11 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
```

```
Get:12 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
```

```
Get:13 http://archive.ubuntu.com/ubuntu noble/restricted amd64 Packages [117 kB]
```

```
Get:14 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [1170 kB]
```

```
Get:15 http://archive.ubuntu.com/ubuntu noble/multiverse amd64 Packages [331 kB]
```

```
Get:16 http://archive.ubuntu.com/ubuntu noble/main amd64 Packages [1808 kB]
```

```
Get:17 http://archive.ubuntu.com/ubuntu noble/universe amd64 Packages [19.3 MB]
```

```
Get:18 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [2703 kB]
```

```
Get:19 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Packages [33.1 kB]
```

```
Get:20 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1639 kB]
```

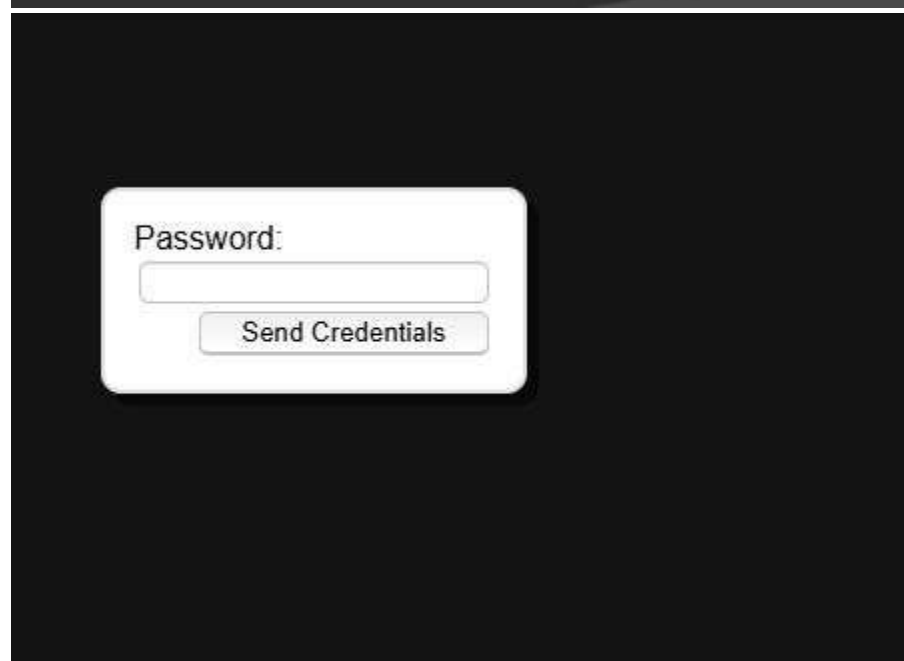
```
Get:21 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1939 kB]
```

```
Get:22 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [2005 kB]
```

```
0
```


PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS 3

Port	Forwarded Add...	Running Process	Visibility	Origin
5900	https://upgraded...	x11vnc -display :1 -rfbauth /h...	Private	Auto Forwarded
5901	https://upgraded...	x11vnc -display :1 -rfbauth /h...	Private	Auto Forwarded
6080	https://upgraded...	/usr/bin/python3 /usr/bin/we...	Private	Auto Forwarded
Add Port				





```
[*] Stopping noVNC server...
```

```
Terminating WebSockets proxy (23988)
```

```
[*] Stopping x11vnc server...
```

```
In exit
```

```
caught signal: 15
```

```
06/11/2025 20:10:35 deleted 60 tile_row polling images.
```

```
[*] Stopping XFCE desktop session...
```

```
[*] Stopping virtual X server (Xvfb)...
```

```
[✓] All services stopped.
```

```
(xfce4-power-manager:23817): libxfce4ui-WARNING **: 20:10:35.131: ICE I/O Error
```

```
(xfce4-power-manager:23817): libxfce4ui-WARNING **: 20:10:35.131: Disconnected from session manager.
```

```
(xfce4-panel:23789): libxfce4ui-WARNING **: 20:10:35.132: ICE I/O Error
```

```
(xfce4-panel:23789): libxfce4ui-WARNING **: 20:10:35.135: Disconnected from session manager.
```



FATIMA JINNAH WOMEN UNIVERSITY

Department of Software Engineering

Lab Title:	Experiment # :
Student Name:	Regd. #:
Experiment Name:	

Performance Indicator	Level of Achievements				
	Excellent (5)	Good (4)	Average (3)	Below Average (2)	Poor (1)
Understanding & implementation Problem					
Report and presentation					
Maximum Marks		10	Obtained Marks		
Lab Instructor:					

Lab 7

Task 1

```
Alli@DESKTOP-KQE4P80 MINGW64 ~ (main)
$ printenv
ProgramFiles(x86)=C:\Program Files (x86)
!::=:
CommonProgramFiles(x86)=C:\Program Files (x86)\Common Files
PyCharm Community Edition=C:\Program Files\JetBrains\PyCharm Community Edition
024.2.4\bin;
SHELL=/usr/bin/bash
NUMBER_OF_PROCESSORS=8
FPS_BROWSER_USER_PROFILE_STRING=Default
PROCESSOR_LEVEL=6
TERM_PROGRAM_VERSION=3.8.1
MINGW_PREFIX=/mingw64
PKG_CONFIG_PATH=/mingw64/lib/pkgconfig:/mingw64/share/pkgconfig
USERDOMAIN_ROAMINGPROFILE=DESKTOP-KQE4P80
HOSTNAME=DESKTOP-KQE4P80
PROGRAMFILES=C:\Program Files
MSYSTEM=MINGW64
PATHEXT=.COM;.EXE;.BAT;.CMD;.VBS;.VBE;.JS;.JSE;.WSF;.WSH;.MSC
ORIGINAL_TEMP=/tmp
MINGW_CHOST=x86_64-w64-mingw32
```

```
Alli@DESKTOP-KQE4P80 MINGW64 ~ (main)
$ printenv | grep SHELL
SHELL=/usr/bin/bash

Alli@DESKTOP-KQE4P80 MINGW64 ~ (main)
$ printenv | grep Home

Alli@DESKTOP-KQE4P80 MINGW64 ~ (main)
$ printenv | grep USER
FPS_BROWSER_USER_PROFILE_STRING=Default
USERDOMAIN_ROAMINGPROFILE=DESKTOP-KQE4P80
USERDOMAIN=DESKTOP-KQE4P80
USERPROFILE=C:\Users\Alli
ALLUSERSPROFILE=C:\ProgramData
USERNAME=Alli
```

Task 2

```

A11i@DESKTOP-KQE4P80 MINGW64 ~ (main)
$ export DB_URL="postgres://db.example.local:5432/mydb"
export DB_USER="labuser"
export DB_PASSWORD="labpass123"

A11i@DESKTOP-KQE4P80 MINGW64 ~ (main)
$ echo "$DB_URL"
echo "$DB_USER"
echo "$DB_PASSWORD"
postgres://db.example.local:5432/mydb
labuser
labpass123

A11i@DESKTOP-KQE4P80 MINGW64 ~ (main)
$ ^[[200~printenv | grep '^DB_~
bash: $'\E[200~printenv': command not found

A11i@DESKTOP-KQE4P80 MINGW64 ~ (main)
$ printenv | grep '^DB_'
DB_PASSWORD=labpass123
DB_USER=labuser
DB_URL=postgres://db.example.local:5432/mydb

```

Task 3

```

A11i@DESKTOP-KQE4P80 MINGW64 ~ (main)
$ vim ~/.bashrc
# add at the end:
# Lab 7 persistent DB variables
export DB_URL="postgres://db.example.local:5432/mydb"
export DB_USER="labuser"
export DB_PASSWORD="labpass123"

A11i@DESKTOP-KQE4P80 MINGW64 ~ (main)
$ source ~/.bashrc
echo "$DB_URL"
echo "$DB_USER"
echo "$DB_PASSWORD"
printenv | grep '^DB_'
bash: /c/Users/A11i/.bashrc: No such file or directory
postgres://db.example.local:5432/mydb
labuser
labpass123
DB_PASSWORD=labpass123
DB_USER=labuser
DB_URL=postgres://db.example.local:5432/mydb

A11i@DESKTOP-KQE4P80 MINGW64 ~ (main)
$ echo "$DB_URL"
printenv | grep '^DB_'
postgres://db.example.local:5432/mydb
DB_PASSWORD=labpass123
DB_USER=labuser
DB_URL=postgres://db.example.local:5432/mydb

```

Task 4

```
PATH="/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin"
```

```
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin
```

```
PATH="/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin"  
class="CC-SA"
```

```
PATH="/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin"  
class="CC-SA"
```

```
> #!/bin/bash  
> echo "Welcome to Cloud Computing $USER"  
> EOF
```

```
PATH=$PATH:~
```

```
-- INSERT --
```

Task 5

```
Command may disrupt existing ssh connections. Proceed with operation (y|n)? y  
Firewall is active and enabled on system startup
```

```
Status: active  
Logging: on (low)  
Default: deny (incoming), allow (outgoing), deny (routed)  
New profiles: skip
```

```
logout
Connection to 192.168.161.129 closed.
PS C:\Users\HP> ssh hajra@192.168.161.129
ssh: connect to host 192.168.161.129 port 22: Connection timed out
PS C:\Users\HP>
```

[illegible][illegible][illegible]

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1997, 1998, 1999, 2000, 2001, 2002, 2003, 2004, 2005, 2006, 2007, 2008, 2009, 2010, 2011, 2012, 2013, 2014, 2015, 2016, 2017, 2018, 2019, 2020, 2021, 2022, 2023, 2024, 2025, 2026, 2027, 2028, 2029, 2030, 2031, 2032, 2033, 2034, 2035, 2036, 2037, 2038, 2039, 2040, 2041, 2042, 2043, 2044, 2045, 2046, 2047, 2048, 2049, 2050, 2051, 2052, 2053, 2054, 2055, 2056, 2057, 2058, 2059, 2060, 2061, 2062, 2063, 2064, 2065, 2066, 2067, 2068, 2069, 2070, 2071, 2072, 2073, 2074, 2075, 2076, 2077, 2078, 2079, 2080, 2081, 2082, 2083, 2084, 2085, 2086, 2087, 2088, 2089, 2090, 2091, 2092, 2093, 2094, 2095, 2096, 2097, 2098, 2099, 2100, 2101, 2102, 2103, 2104, 2105, 2106, 2107, 2108, 2109, 2110, 2111, 2112, 2113, 2114, 2115, 2116, 2117, 2118, 2119, 2120, 2121, 2122, 2123, 2124, 2125, 2126, 2127, 2128, 2129, 2130, 2131, 2132, 2133, 2134, 2135, 2136, 2137, 2138, 2139, 2140, 2141, 2142, 2143, 2144, 2145, 2146, 2147, 2148, 2149, 2150, 2151, 2152, 2153, 2154, 2155, 2156, 2157, 2158, 2159, 2160, 2161, 2162, 2163, 2164, 2165, 2166, 2167, 2168, 2169, 2170, 2171, 2172, 2173, 2174, 2175, 2176, 2177, 2178, 2179, 2180, 2181, 2182, 2183, 2184, 2185, 2186, 2187, 2188, 2189, 2190, 2191, 2192, 2193, 2194, 2195, 2196, 2197, 2198, 2199, 2200, 2201, 2202, 2203, 2204, 2205, 2206, 2207, 2208, 2209, 2210, 2211, 2212, 2213, 2214, 2215, 2216, 2217, 2218, 2219, 2220, 2221, 2222, 2223, 2224, 2225, 2226, 2227, 2228, 2229, 2230, 2231, 2232, 2233, 2234, 2235, 2236, 2237, 2238, 2239, 2240, 2241, 2242, 2243, 2244, 2245, 2246, 2247, 2248, 2249, 2250, 2251, 2252, 2253, 2254, 2255, 2256, 2257, 2258, 2259, 2260, 2261, 2262, 2263, 2264, 2265, 2266, 2267, 2268, 2269, 2270, 2271, 2272, 2273, 2274, 2275, 2276, 2277, 2278, 2279, 2280, 2281, 2282, 2283, 2284, 2285, 2286, 2287, 2288, 2289, 2290, 2291, 2292, 2293, 2294, 2295, 2296, 2297, 2298, 2299, 2300, 2301, 2302, 2303, 2304, 2305, 2306, 2307, 2308, 2309, 2310, 2311, 2312, 2313, 2314, 2315, 2316, 2317, 2318, 2319, 2320, 2321, 2322, 2323, 2324, 2325, 2326, 2327, 2328, 2329, 2330, 2331, 2332, 2333, 2334, 2335, 2336, 2337, 2338, 2339, 2340, 2341, 2342, 2343, 2344, 2345, 2346, 2347, 2348, 2349, 2350, 2351, 2352, 2353, 2354, 2355, 2356, 2357, 2358, 2359, 2360, 2361, 2362, 2363, 2364, 2365, 2366, 2367, 2368, 2369, 2370, 2371, 2372, 2373, 2374, 2375, 2376, 2377, 2378, 2379, 2380, 2381, 2382, 2383, 2384, 2385, 2386, 2387, 2388, 2389, 2390, 2391, 2392, 2393, 2394, 2395, 2396, 2397, 2398, 2399, 2400, 2401, 2402, 2403, 2404, 2405, 2406, 2407, 2408, 2409, 2410, 2411, 2412, 2413, 2414, 2415, 2416, 2417, 2418, 2419, 2420, 2421, 2422, 2423, 2424, 2425, 2426, 2427, 2428, 2429, 2430, 2431, 2432, 2433, 2434, 2435, 2436, 2437, 2438, 2439, 2440, 2441, 2442, 2443, 2444, 2445, 2446, 2447, 2448, 2449, 2450, 2451, 2452, 2453, 2454, 2455, 2456, 2457, 2458, 2459, 2460, 2461, 2462, 2463, 2464, 2465, 2466, 2467, 2468, 2469, 2470, 2471, 2472, 2473, 2474, 2475, 2476, 2477, 2478, 2479, 2480, 2481, 2482, 2483, 2484, 2485, 2486, 2487, 2488, 2489, 2490, 2491, 2492, 2493, 2494, 2495, 2496, 2497, 2498, 2499, 2500, 2501, 2502, 2503, 2504, 2505, 2506, 2507, 2508, 2509, 2510, 2511, 2512, 2513, 2514, 2515, 2516, 2517, 2518, 2519, 2520, 2521, 2522, 2523, 2524, 2525, 2526, 2527, 2528, 2529, 2530, 2531, 2532, 2533, 2534, 2535, 2536, 2537, 2538, 2539, 2540, 2541, 2542, 2543, 2544, 2545, 2546, 2547, 2548, 2549, 2550, 2551, 2552, 2553, 2554, 2555, 2556, 2557, 2558, 2559, 2560, 2561, 2562, 2563, 2564, 2565, 2566, 2567, 2568, 2569, 2570, 2571, 2572, 2573, 2574, 2575, 2576, 2577, 2578, 2579, 2580, 2581, 2582, 2583, 2584, 2585, 2586, 2587, 2588, 2589, 2590, 2591, 2592, 2593, 2594, 2595, 2596, 2597, 2598, 2599, 2600, 2601, 2602, 2603, 2604, 2605, 2606, 2607, 2608, 2609, 2610, 2611, 2612, 2613, 2614, 2615, 2616, 2617, 2618, 2619, 2620, 2621, 2622, 2623, 2624, 2625, 2626, 2627, 2628, 2629, 2630, 2631, 2632, 2633, 2634, 2635, 2636, 2637, 2638, 2639, 2640, 2641, 2642, 2643, 2644, 2645, 2646, 2647, 2648, 2649, 2650, 2651, 2652, 2653, 2654, 2655, 2656, 2657, 2658, 2659, 2660, 2661, 2662, 2663, 2664, 2665, 2666, 2667, 2668, 2669, 2670, 2671, 2672, 2673, 2674, 2675, 2676, 2677, 2678, 26

Q1

```

Alli@DESKTOP-KQE4P80 MINGW64 ~ (main)
$ printenv
ProgramFiles(x86)=C:\Program Files (x86)
!::=:\
CommonProgramFiles(x86)=C:\Program Files (x86)\Common Files
PyCharm Community Edition=C:\Program Files\JetBrains\PyCharm Community Edition
024.2.4\bin;
SHELL=/usr/bin/bash
NUMBER_OF_PROCESSORS=8
FPS_BROWSER_USER_PROFILE_STRING=Default
PROCESSOR_LEVEL=6
TERM_PROGRAM_VERSION=3.8.1
MINGW_PREFIX=/mingw64
PKG_CONFIG_PATH=/mingw64/lib/pkgconfig:/mingw64/share/pkgconfig
USERDOMAIN_ROAMINGPROFILE=DESKTOP-KQE4P80
HOSTNAME=DESKTOP-KQE4P80
PROGRAMFILES=C:\Program Files
MSYSTEM=MINGW64
PATHEXT=.COM;.EXE;.BAT;.CMD;.VBS;.VBE;.JS;.JSE;.WSF;.WSH;.MSC
ORIGINAL_TEMP=/tmp
MINGW_CHOST=x86_64-w64-mingw32

```

Q2

```

$ export STUDENT_NAME="John Doe"
$ export STUDENT_ROLL_NUMBER="12345"
$ export STUDENT_SEMESTER="3"

```

```

STUDENT_NAME=John Doe
STUDENT_SEMESTER=3
STUDENT_ROLL_NUMBER=12345

```

```

STUDENT_NAME=John Doe
STUDENT_SEMESTER=3
STUDENT_ROLL_NUMBER=12345

```

Q3

```

export STUDENT_NAME="John Doe"
export STUDENT_ROLL_NUMBER="12345"
export STUDENT_SEMESTER="3"
-- INSERT --

```

```

ho "$STUDENT_ROLL_NUMBER"
echo "$STUDENT_SEMESTER"
printenv | grep '^STUDENT_'

```

Q4

Status: active

To	Action	From
--	-----	----
22/tcp	ALLOW	Anywhere
Anywhere on vxlan.calico	ALLOW	Anywhere
Anywhere on cali+	ALLOW	Anywhere
22/tcp (v6)	ALLOW	Anywhere (v6)
Anywhere (v6) on vxlan.calico	ALLOW	Anywhere (v6)
Anywhere (v6) on cali+	ALLOW	Anywhere (v6)
Anywhere	ALLOW OUT	Anywhere on vxlan.calico
Anywhere	ALLOW OUT	Anywhere on cali+
Anywhere (v6)	ALLOW OUT	Anywhere (v6) on vxlan.calico
Anywhere (v6)	ALLOW OUT	Anywhere (v6) on cali+

Status: active

To	Action	From
--	-----	----
[1] 22/tcp	ALLOW IN	Anywhere
[2] Anywhere on vxlan.calico	ALLOW IN	Anywhere
[3] Anywhere	ALLOW OUT	Anywhere on vxlan.calico (out)
[4] Anywhere on cali+	ALLOW IN	Anywhere
[5] Anywhere	ALLOW OUT	Anywhere on cali+ (out)
[6] 22/tcp (v6)	ALLOW IN	Anywhere (v6)
[7] Anywhere (v6) on vxlan.calico	ALLOW IN	Anywhere (v6)
[8] Anywhere (v6)	ALLOW OUT	Anywhere (v6) on vxlan.calico (out)
[9] Anywhere (v6) on cali+	ALLOW IN	Anywhere (v6)
[10] Anywhere (v6)	ALLOW OUT	Anywhere (v6) on cali+ (out)

PS C:\Users\HP> ping 192.168.161.129

Pinging 192.168.161.129 with 32 bytes of data:

Request timed out.

Request timed out.

Request timed out.

Request timed out.

Ping statistics for 192.168.161.129:

Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

PS C:\Users\HP>

PS C:\Users\HP> ping 192.168.161.129

Pinging 192.168.161.129 with 32 bytes of data:

Reply from 192.168.161.129: bytes=32 time<1ms TTL=64

Reply from 192.168.161.129: bytes=32 time<1ms TTL=64

Reply from 192.168.161.129: bytes=32 time<1ms TTL=64

Reply from 192.168.161.129: bytes=32 time<1ms TTL=64

Ping statistics for 192.168.161.129:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 0ms, Maximum = 1ms, Average = 0ms

PS C:\Users\HP>



FATIMA JINNAH WOMEN UNIVERSITY

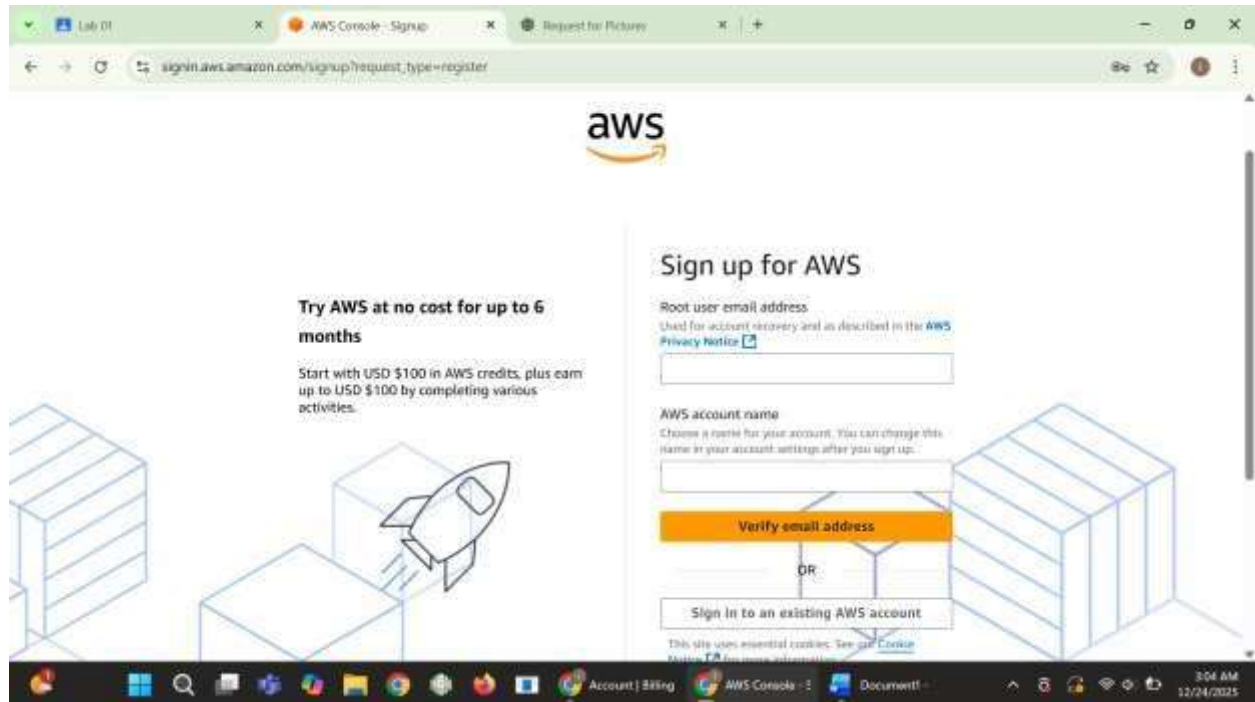
Department of Software Engineering

Lab Title:	Experiment # :
Student Name:	Regd. #:
Experiment Name:	

Performance Indicator	Level of Achievements				
	Excellent (5)	Good (4)	Average (3)	Below Average (2)	Poor (1)
Understanding & implementation Problem					
Report and presentation					
Maximum Marks	10		Obtained Marks		
Lab Instructor:					

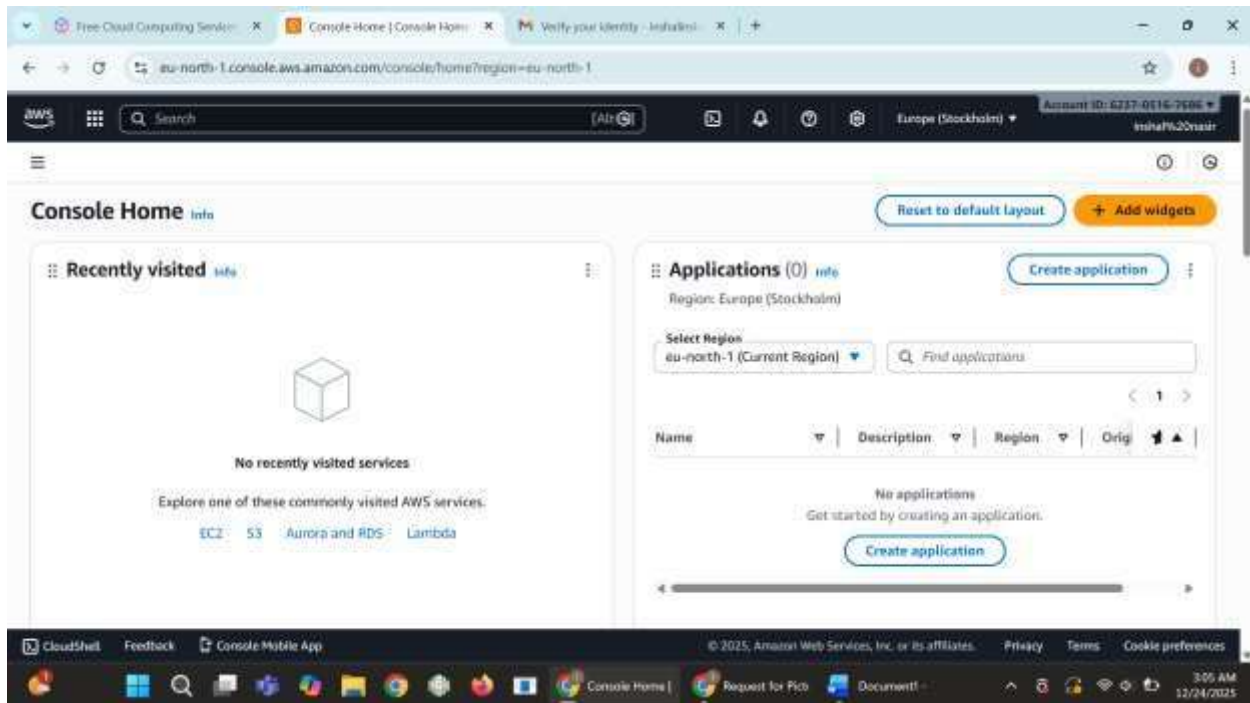
Lab 8

Task 1

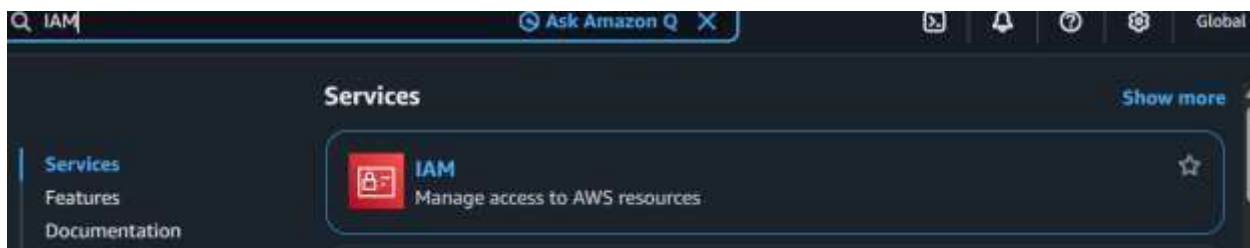


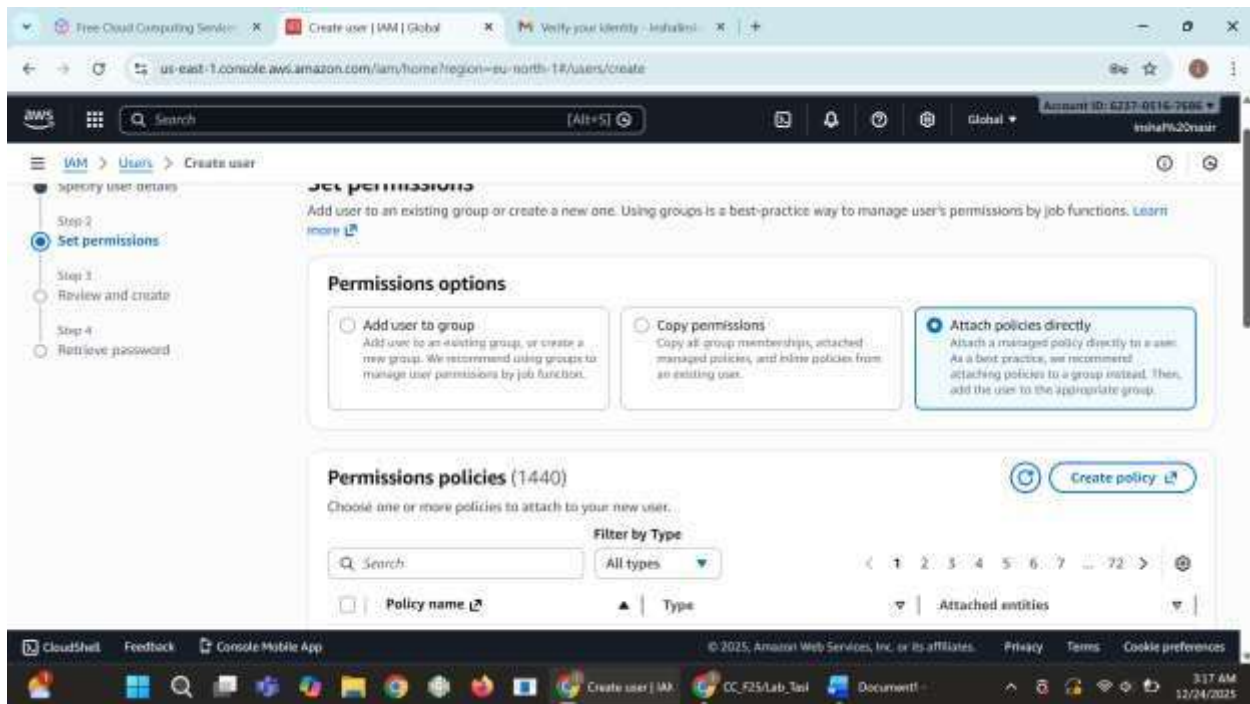
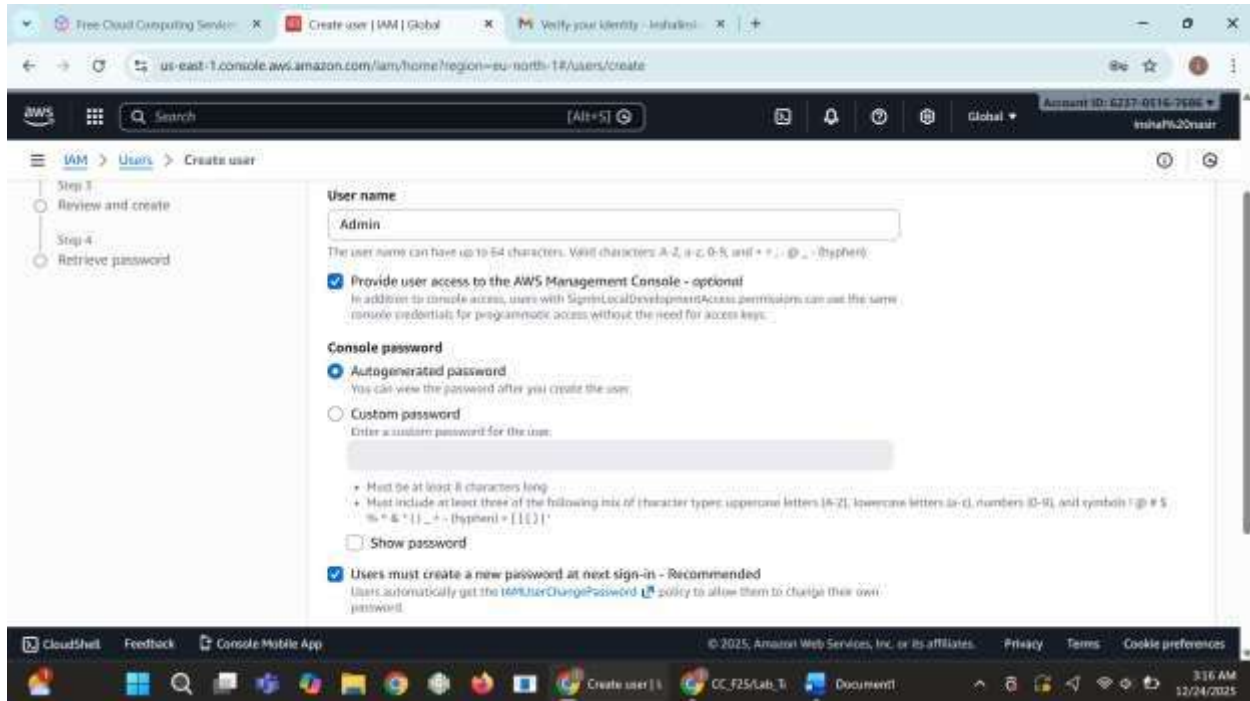
The screenshot shows the AWS Sign Up page in a web browser. The browser's address bar displays the URL `signin.aws.amazon.com/signup/request_type=register`. The page features the AWS logo at the top center. On the left, a promotional message states: "Try AWS at no cost for up to 6 months. Start with USD \$100 in AWS credits, plus earn up to USD \$100 by completing various activities." Below this text is an illustration of a rocket launching from a cube. On the right, the "Sign up for AWS" form is visible. It includes a "Root user email address" field with a "Privacy Notice" link, an "AWS account name" field, and a prominent orange "Verify email address" button. Below the button is a "OR" separator and a "Sign in to an existing AWS account" link. At the bottom of the form, a small disclaimer states: "This site uses essential cookies. See our Cookie Notice." The Windows taskbar at the bottom shows the Start button, search icon, and several open applications including "Account | Billing", "AWS Console", and "Document". The system clock in the bottom right corner indicates the time is 10:04 AM on 12/24/2025.





Task 2





Free Cloud Computing Services | Create user | IAM | Global | Verify your identity - Inhabitat

us-east-1.console.aws.amazon.com/iam/home?region=us-north-1#/users/create

Search [Alt+S] Global Account ID: 6237-0116-7686 Inhabitat@20nash

IAM > Users > Create user

Filter by Type

Search All types

	Policy name	Type	Attached entities
☐	AccessAnalyzerServiceRole	AWS managed	0
☐	AccountManagementFro...	AWS managed	0
☒	AdministratorAccess	AWS managed - job function	0
☐	AdministratorAccess-Amp...	AWS managed	0
☐	AdministratorAccess-AWS...	AWS managed	0
☐	AIOpsAssistantIncidentRe...	AWS managed	0
☐	AIOpsAssistantPolicy	AWS managed	0
☐	AIOpsConsoleAdminPolicy	AWS managed	0
☐	AIOpsOperatorAccess	AWS managed	0

CloudShell Feedback Console Mobile App

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Create user | IAM CC_F25/Lab_Tail Document1

3:18 AM 12/24/2025

Free Cloud Computing Services | Create user | IAM | Global | Verify your identity - Inhabitat

us-east-1.console.aws.amazon.com/iam/home?region=us-north-1#/users/create

Search [Alt+S] Global Account ID: 6237-0116-7686 Inhabitat@20nash

IAM > Users > Create user

Step 1: Review and create

Step 2: Retrieve password

User details

User name	Admin
Console password type	Autogenerated
Require password reset	Yes

Permissions summary

Name	Type	Used as
AdministratorAccess	AWS managed - job function	Permissions policy
IAMUserChangePassword	AWS managed	Permissions policy

Tags - optional

Tags are key-value pairs you can add to AWS resources to help identify, organize, or search for resources. Choose any tags you want to associate with this user.

No tags associated with the resource.

CloudShell Feedback Console Mobile App

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Create user | IAM CC_F25/Lab_Tail Document1

3:18 AM 12/24/2025

Free Cloud Computing Services | Create user | IAM | Global | Verify your identity - Initial...

us-east-1.console.aws.amazon.com/iam/home?region=us-north-1#/users/create

Search [Alt+S]

Global Account ID: 6237-0516-7686

iam > Users > Create user

Step 1: Specify user details
Step 2: Set permissions
Step 3: Review and create
Step 4: Retrieve password

Retrieve password

You can view and download the user's password below or email users instructions for signing in to the AWS Management Console. This is the only time you can view and download this password.

Console sign-in details

[Email sign-in instructions](#)

Console sign-in URL
<https://623705167686.signin.aws.amazon.com/console>

User name
[Admin](#)

Console password
***** [Show](#)

Cancel [Download .csv file](#) [Return to users list](#)

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Free Cloud Computing Services | Users | IAM | Global | Verify your identity - Initial...

us-east-1.console.aws.amazon.com/iam/home?region=us-north-1#/users

Search [Alt+S]

Global Account ID: 6237-0516-7686

iam > Users

Identity and Access Management (IAM)

Search IAM

Dashboard

Access management

- User groups
- Users**
- Roles
- Policies
- Identity providers
- Account settings
- Root access management
- Temporary delegation requests

Users (1)

An IAM user is an identity with long-term credentials that is used to interact with AWS in an account.

Search

1

☐	User name	Path	Group	Last activity	MFA	Password age
☐	Admin	/	0			5 minutes

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Today

Admin_credentials.csv	12/24/2025 3:20 AM	Microsoft Excel Com...	1 KB
-----------------------	--------------------	------------------------	------

4.4 GB Backup

Free Cloud Computing x AWS Management Console x Verify your identity (info) x AWS Management Console x Console Home (Console) x

eu-north-1.console.aws.amazon.com/console/home?region=eu-north-1

Search [Alt+S] Europe (Stockholm) Account ID: 6237-0116-7686 Admin

Console Home Info

[Reset to default layout](#) [+ Add widgets](#)

Recently visited Info

No recently visited services

Explore one of these commonly visited AWS services:

[EC2](#) [S3](#) [Aurora and RDS](#) [Lambda](#)

Applications (0) Info

Region: Europe (Stockholm)

Select Region: eu-north-1 (Current Region)

Name	Description	Region	Orig
No applications			
Get started by creating an application.			

[Create application](#)

CloudShell Feedback Console Mobile App

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3:43 AM 12/24/2025

Free Cloud Computing x AWS Management Console x Verify your identity (info) x AWS Management Console x Lab8user | IAM | Global x

us-east-1.console.aws.amazon.com/iam/home?region=eu-north-1#/users/details/Lab8user/section=permissions

Search [Alt+S] Global Account ID: 6237-0116-7686 Admin

IAM > Users > Lab8user

Identity and Access Management (IAM)

Search IAM

Dashboard

Access management

- User groups
- Users**
- Roles
- Policies
- Identity providers
- Account settings
- Root access management
- Temporary delegation requests

User created successfully

You can view and download the user's password and email instructions for signing in to the AWS Management Console.

[View user](#)

[Delete](#)

Lab8user Info

Summary

ARN <code>arn:aws:iam::623705167686:user/Lab8user</code>	Console access Enabled without MFA	Access key 1 Create access key
Created December 24, 2025, 03:46 (UTC-08:00)	Last console sign-in Never	

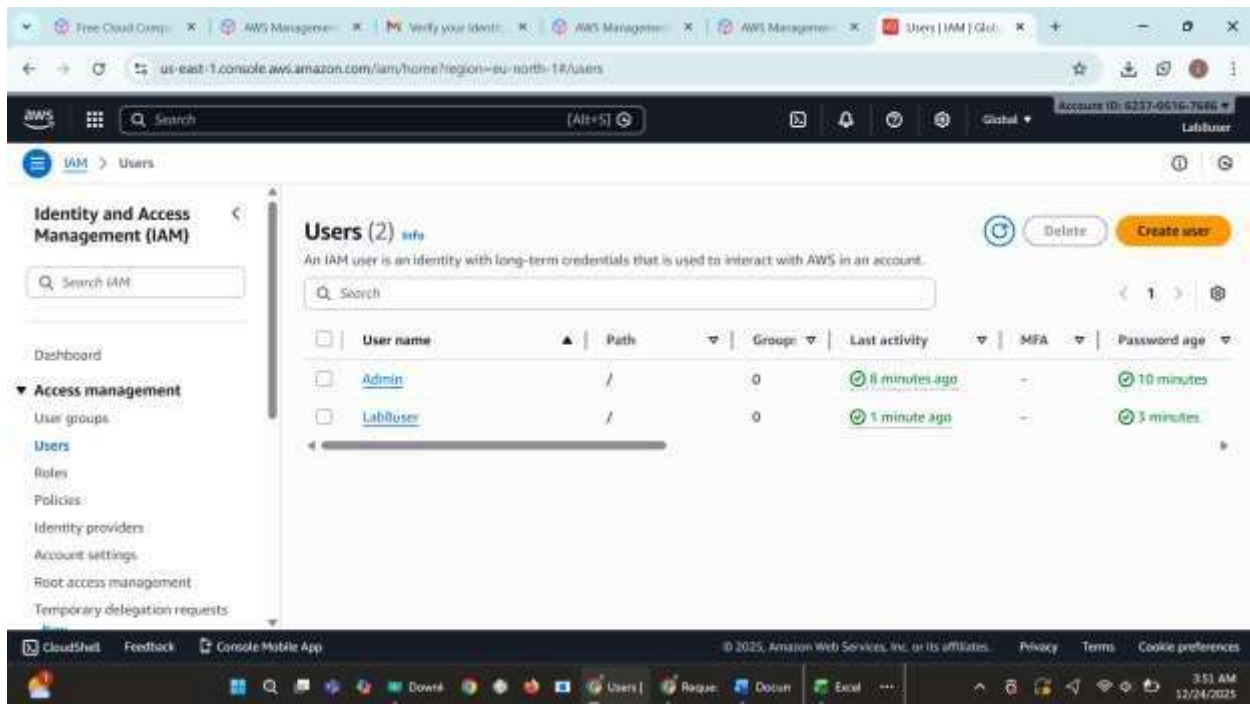
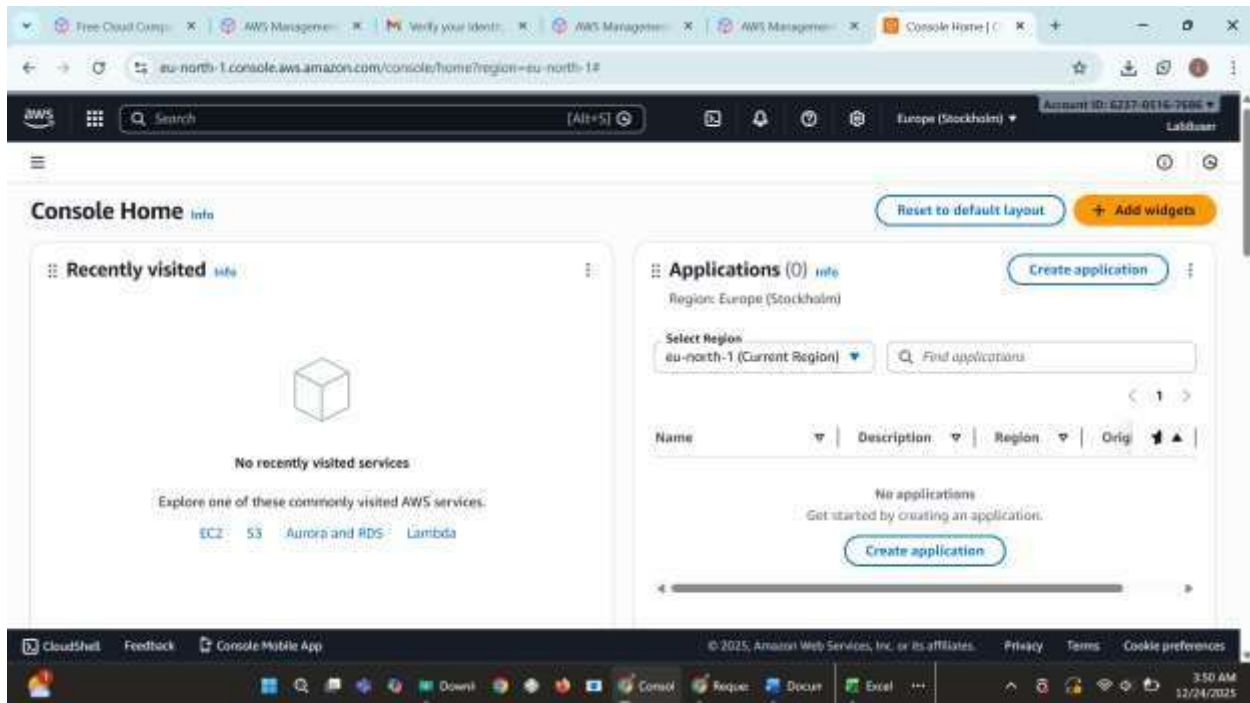
[Permissions](#) [Groups](#) [Tags](#) [Security credentials](#) [Last Accessed](#)

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3:46 AM 12/24/2025

Lab8user_credentials.csv	12/24/2025 3:47 AM	Microsoft Excel Com...	1 KB
--------------------------	--------------------	------------------------	------



Task 3

Free Class x AWS Man... x VPC Cons... x Verify you... x AWS Man... x AWS Man... x VPC | eu... x New Tab x +

me-central-1.console.aws.amazon.com/vpcconsole/home?region=me-central-1#Home

Search [Alt+S] Mobile East (UAE) Account ID: 6237-0516-7686 Initial name

VPC dashboard

AWS Global View Filter by VPC

Virtual private cloud

- Your VPCs
- Subnets
- Route tables
- Internet gateways
- Egress-only internet gateways
- DHCP option sets
- Elastic IPs
- Managed prefix lists
- Endpoints
- Endpoint services

Resources by Region

You are using the following Amazon VPC resources

Refresh Resources

VPCs ▶ See all regions	UAE 1	Endpoint Services ▶ See all regions	UAE 0
Subnets ▶ See all regions	UAE 3	NAT Gateways ▶ See all regions	UAE 0
Route Tables ▶ See all regions	UAE 1	VPC Peering Connections ▶ See all regions	UAE 0
Internet Gateways ▶ See all regions	UAE 1	Network ACLs ▶ See all regions	UAE 1

Settings

- [Block Public Access](#)
- [Zones](#)
- [Console Experiments](#)

Additional Information

- [VPC Documentation](#)
- [All VPC Resources](#)
- [Forums](#)
- [Report an issue](#)

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3:58 AM 12/24/2025

Free Class x AWS Man... x vpc | VPC x Verify you... x AWS Man... x AWS Man... x VPC | eu... x New Tab x +

me-central-1.console.aws.amazon.com/vpcconsole/home?region=me-central-1#vpcs

Search [Alt+S] Mobile East (UAE) Account ID: 6237-0516-7686 Initial name

VPC dashboard

AWS Global View Filter by VPC

Virtual private cloud

- Your VPCs
- Subnets
- Route tables
- Internet gateways
- Egress-only internet gateways
- DHCP option sets
- Elastic IPs
- Managed prefix lists
- Endpoints
- Endpoint services

Your VPCs

VPCs VPC encryption controls

Last updated less than a minute ago Actions Create VPC

Find VPCs by attribute or tag

☐	Name	VPC ID	State	Encryption c...	Encryption contr
☐	-	vpc-079a25ed8b9cb32f2	Available	-	-

Select a VPC above

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3:58 AM 12/24/2025

me-central-1.console.aws.amazon.com/vpcconsole/home?region=me-central-1#subnets

VPC dashboard

Subnets (3)

Find subnets by attribute or tag

Name	Subnet ID	State	VPC
-	subnet-03c38a2593aaf6920	Available	vpc-079a26ed9b9cb32f2
-	subnet-08a73f8aa1dee72d1	Available	vpc-079a26ed9b9cb32f2
-	subnet-05e4d0f087e63c8e	Available	vpc-079a26ed9b9cb32f2

Select a subnet

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3:59 AM 12/24/2025

me-central-1.console.aws.amazon.com/vpcconsole/home?region=me-central-1#RouteTables

VPC dashboard

Route tables (1)

Find route tables by attribute or tag

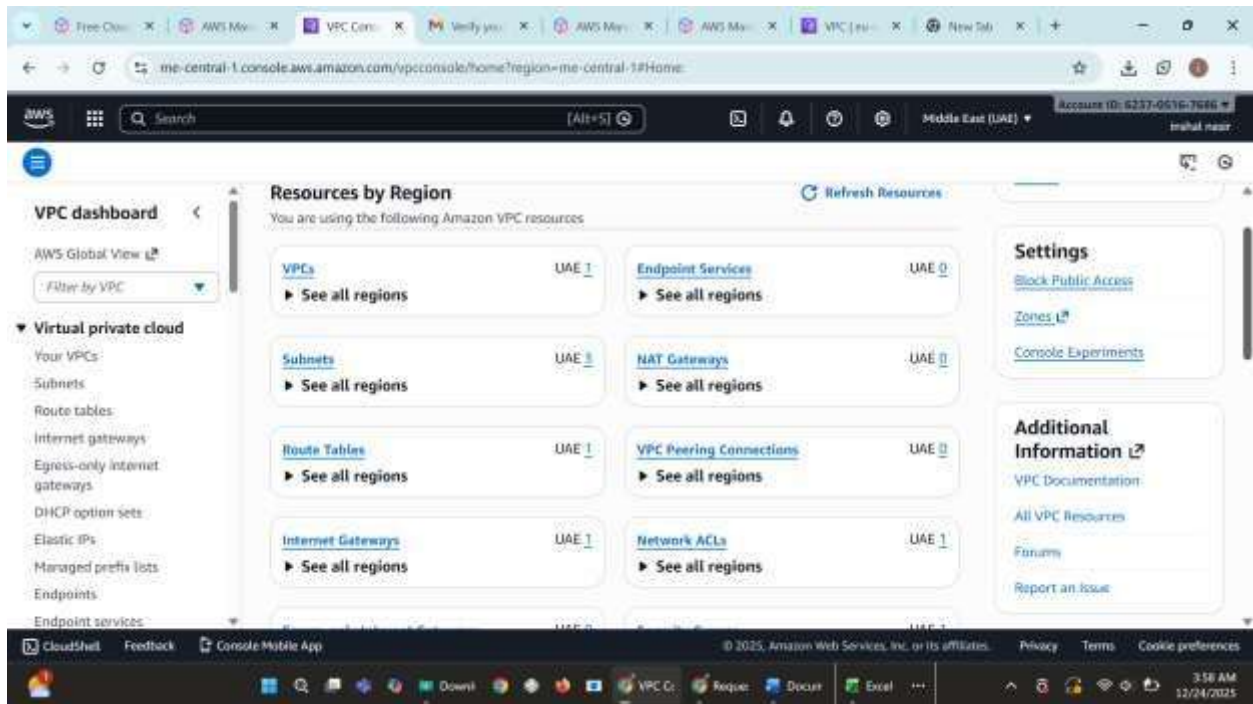
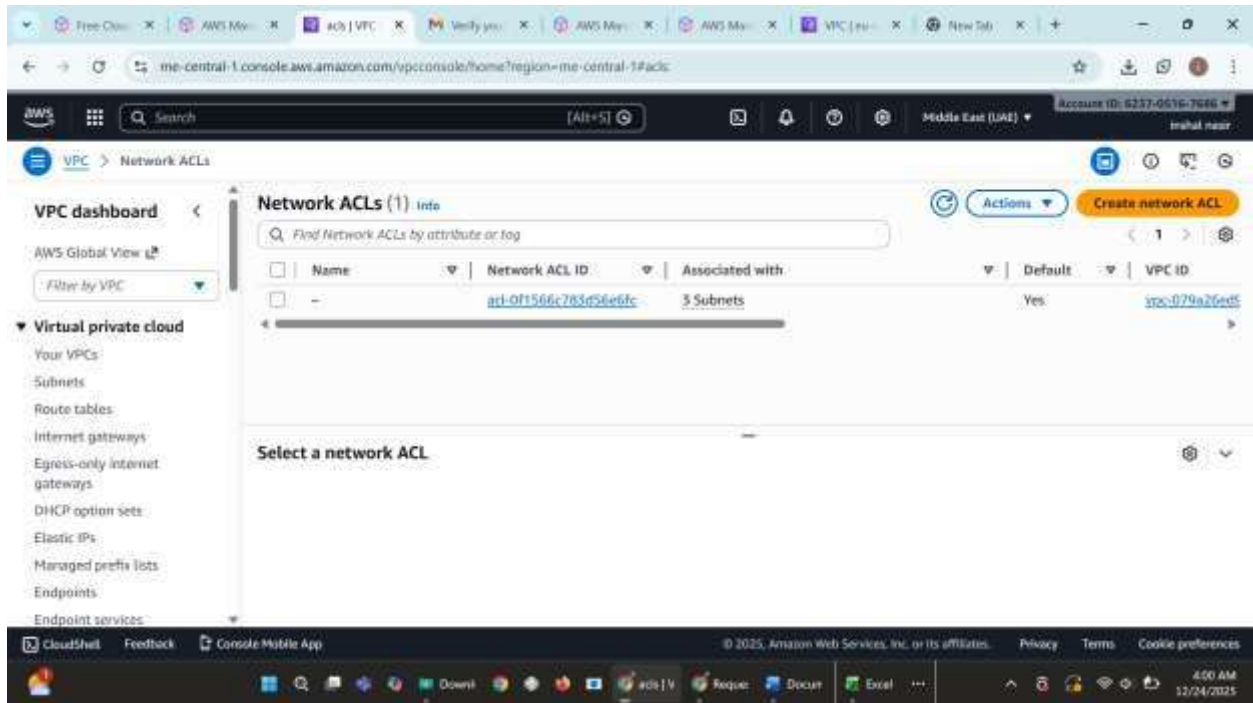
Name	Route table ID	Explicit subnet associ...	Edge associations	Main
-	rtb-08fard96c7ana3f9d1	-	-	Yes

Select a route table

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3:59 AM 12/24/2025



Task 4

Free Class x AWS Man... x Instances x Verify you x AWS Man... x AWS Man... x VPC | eu... x New Tab x +

me-central-1.console.aws.amazon.com/ec2/home?region=me-central-1#Instances

Search [Alt+S] Middle East (UAE) Account ID: 6237-0516-7686 Initial name

EC2 > Instances

Instances (1) Info Connect Instance state Actions Launch instances

Find instance by attribute or tag (case-sensitive) All states < 1 >

☐	Name	Instance ID	Instance state	Instance type	Status check	Alarm status
☐	Lab8Machine	i-05ed5f211b5974265	Running	t3.micro	2/2 checks passed	View alarms +

Select an instance

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Lab8key.pem

12/24/2025 4:2

Free Class x AWS Man... x Instances x Verify you x AWS Man... x AWS Man... x VPC | eu... x New Tab x +

me-central-1.console.aws.amazon.com/ec2/home?region=me-central-1#Instances

Search [Alt+S] Middle East (UAE) Account ID: 6237-0516-7686 Initial name

EC2 > Instances

Instances (1/1) Info Connect Instance state Actions Launch instances

Find instance by attribute or tag (case-sensitive) All states < 1 >

☒	Name	Instance ID	Instance state	Instance type	Status check	Alarm status
☒	Lab8Machine	i-05ed5f211b5974265	Running	t3.micro	3/3 checks passed	View alarms +

i-05ed5f211b5974265 (Lab8Machine)

Instance ID: i-05ed5f211b5974265

Public IPv4 address: 3.28.181.174 [open address](#)

Private IPv4 addresses: 172.31.10.218

Public DNS: ec2-3-28-181-174.me-central-

Instance state: Running

IPv6 address: -

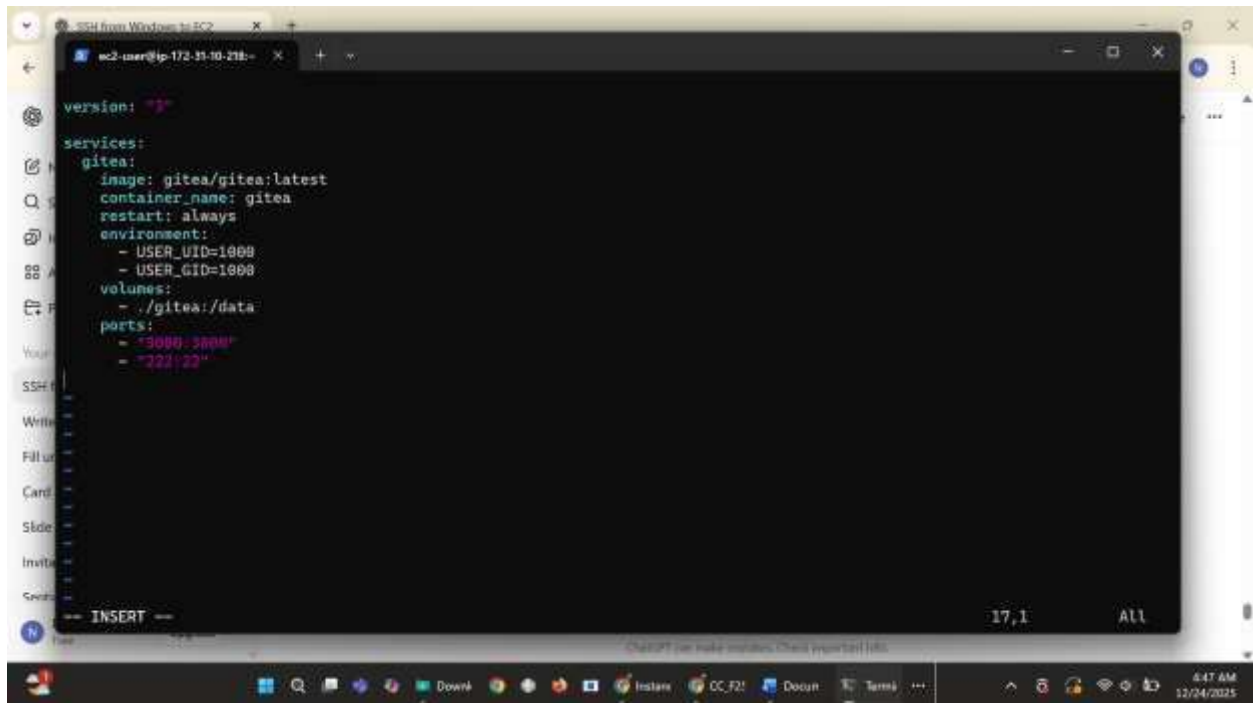
CloudShell Feedback Console Mobile App © 2025, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences 4:24 AM 12/24/2025


```
ec2-user@ip-172-31-10-218:~$ sudo yum install docker
Install 11 Packages
Total download size: 74 M
Installed size: 288 M
Downloading Packages:
(1/11): container-selinux-2.242.0-1.amzn2023.noarch.rpm 1.4 MB/s | 58 kB 00:00
(2/11): iptables-libs-1.8.8-3.amzn2023.0.2.x86_64.rpm 12 MB/s | 481 kB 00:00
(3/11): iptables-nft-1.8.8-3.amzn2023.0.2.x86_64.rpm 6.4 MB/s | 183 kB 00:00
(4/11): libcgroupp-3.0-1.amzn2023.0.1.x86_64.rpm 2.4 MB/s | 75 kB 00:00
(5/11): libnetfilter_conntrack-1.0.8-2.amzn2023.0.2.x86_64.rpm 2.4 MB/s | 58 kB 00:00
(6/11): libnftnl-1.0.1-19.amzn2023.0.2.x86_64.rpm 1.3 MB/s | 38 kB 00:00
(7/11): libnftnl-1.2.2-2.amzn2023.0.2.x86_64.rpm 2.6 MB/s | 84 kB 00:00
(8/11): pigz-2.5-1.amzn2023.0.3.x86_64.rpm 3.1 MB/s | 83 kB 00:00
(9/11): containerd-2.1.5-1.amzn2023.0.1.x86_64.rpm 66 MB/s | 23 MB 00:00
(10/11): runc-1.3.3-2.amzn2023.0.1.x86_64.rpm 26 MB/s | 3.9 MB 00:00
(11/11): docker-25.0.13-1.amzn2023.0.2.x86_64.rpm 68 MB/s | 46 MB 00:00
-----
Total 105 MB/s | 74 MB 00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
Preparing :
Installing : runc-1.3.3-2.amzn2023.0.1.x86_64 1/11
Installing : containerd-2.1.5-1.amzn2023.0.1.x86_64 2/11
Running scriptlet: containerd-2.1.5-1.amzn2023.0.1.x86_64 2/11
Installing : pigz-2.5-1.amzn2023.0.3.x86_64 3/11
Installing : libnftnl-1.2.2-2.amzn2023.0.2.x86_64 4/11
```

```
ec2-user@ip-172-31-10-218:~$ sudo yum install docker
Running scriptlet: container-selinux-4:2.242.0-1.amzn2023.noarch 11/11
Running scriptlet: docker-25.0.13-1.amzn2023.0.2.x86_64 11/11
Verifying : container-selinux-4:2.242.0-1.amzn2023.noarch 1/11
Verifying : containerd-2.1.5-1.amzn2023.0.1.x86_64 2/11
Verifying : docker-25.0.13-1.amzn2023.0.2.x86_64 3/11
Verifying : iptables-libs-1.8.8-3.amzn2023.0.2.x86_64 4/11
Verifying : iptables-nft-1.8.8-3.amzn2023.0.2.x86_64 5/11
Verifying : libcgroupp-3.0-1.amzn2023.0.1.x86_64 6/11
Verifying : libnetfilter_conntrack-1.0.8-2.amzn2023.0.2.x86_64 7/11
Verifying : libnftnl-1.0.1-19.amzn2023.0.2.x86_64 8/11
Verifying : libnftnl-1.2.2-2.amzn2023.0.2.x86_64 9/11
Verifying : pigz-2.5-1.amzn2023.0.3.x86_64 10/11
Verifying : runc-1.3.3-2.amzn2023.0.1.x86_64 11/11

Installed:
container-selinux-4:2.242.0-1.amzn2023.noarch
docker-25.0.13-1.amzn2023.0.2.x86_64
iptables-libs-1.8.8-3.amzn2023.0.2.x86_64
libnetfilter_conntrack-1.0.8-2.amzn2023.0.2.x86_64
libnftnl-1.2.2-2.amzn2023.0.2.x86_64
runc-1.3.3-2.amzn2023.0.1.x86_64
containerd-2.1.5-1.amzn2023.0.1.x86_64
iptables-nft-1.8.8-3.amzn2023.0.2.x86_64
libcgroupp-3.0-1.amzn2023.0.1.x86_64
libnftnl-1.0.1-19.amzn2023.0.2.x86_64
pigz-2.5-1.amzn2023.0.3.x86_64

Complete!
% Total % Received % Xferd Average Speed Time Time Time Current
Dload Upload Total Spent Left Speed
0 0 0 0 0 0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0 0 0 0 0 0
100 29.8M 100 29.8M 0 0 35.7M 0 0 0 0 0 35.7M
[ec2-user@ip-172-31-10-218 ~]$
```



The screenshot shows an SSH terminal window titled "SSH from Windows to EC2". The terminal displays a Docker Compose file for a service named "gitea". The file content is as follows:

```
version: "3"

services:
  gitea:
    image: gitea/gitea:latest
    container_name: gitea
    restart: always
    environment:
      - USER_UID=1000
      - USER_GID=1000
    volumes:
      - ./gitea:/data
    ports:
      - "3000:3000"
      - "22:22"
```

The terminal window has a sidebar on the left with various icons and a status bar at the bottom showing "17,1" and "All".

```
[ec2-user@ip-172-31-10-218 ~]$ ls -l
total 4
-rw-r--r--. 1 root root 255 Dec 24 12:48 compose.yaml
[ec2-user@ip-172-31-10-218 ~]$
```

```
[ec2-user@ip-172-31-10-218 ~]$ sudo vim compose.yaml
[ec2-user@ip-172-31-10-218 ~]$ ls -l
total 4
-rw-r--r--. 1 root root 255 Dec 24 12:48 compose.yaml
[ec2-user@ip-172-31-10-218 ~]$ groups
ec2-user adm wheel systemd-journal
[ec2-user@ip-172-31-10-218 ~]$ sudo usermod -aG docker $USER
[ec2-user@ip-172-31-10-218 ~]$ groups
ec2-user adm wheel systemd-journal
[ec2-user@ip-172-31-10-218 ~]$ exit
logout
Connection to 3.28.181.174 closed.
```



```
#
# ##### Amazon Linux 2023
### \#####\
### \####\
### \#\
### V~' '~> https://aws.amazon.com/linux/amazon-linux-2023
#####
##### /
##### /w/'

Last login: Wed Dec 24 12:48:31 2025 from 154.192.2.99
[ec2-user@ip-172-31-10-218 ~]$ groups
ec2-user adm wheel systemd-journal docker
[ec2-user@ip-172-31-10-218 ~]$ docker

Usage:  docker [OPTIONS] COMMAND

A self-sufficient runtime for containers

Common Commands:
run      Create and run a new container from an image
exec     Executes a command in a running container
ps       List containers
build    Build an image from a Dockerfile
pull     Download an image from a registry
push     Upload an image to a registry
images   List images
login    Log in to a registry
logout   Log out from a registry
```

The screenshot shows the AWS Management Console interface for configuring a security group. A green notification banner at the top states: "Inbound security group rules successfully modified on security group [sg-0e0224478c86fcd89 | Lab8SecurityGroup]". Below this, the "Inbound rules" tab is selected, displaying a table of two rules:

Name	Security group rule ID	IP version	Type	Protocol
-	sgr-007171b215c3d88bf	IPv4	Custom TCP	TCP
-	sgr-035e14f649f077a5a	IPv4	SSH	TCP

The console also shows the "EC2" sidebar with options like Dashboard, Global View, and Instances. The top navigation bar includes the AWS logo, a search bar, and the account ID: 6237-0616-7666.

Lab 01 CC/F25/Lab_Tasks/Lab... Request for Pictures Gitex: Git with a cup of... Register - Gitex: Git with...

Not secure 3.28.181.174:3000/user/sign_up

Explore Help Register Sign In

Register

Username *

Email Address *

Password *

Confirm Password *

[Register Account](#)

or

 [Sign in with OpenID](#)

9:06 AM 12/24/2025

Lab 01 CC/F25/Lab_Tasks/Lab... Request for Pictures Gitex: Git with a cup of... admin - Dashboard - Git...

Not secure 3.28.181.174:3000

Issues Pull Requests Milestones Explore

admin

Account was successfully created. Welcome!



No Activity

You are currently not following any repositories or users, so there is no content to display. You can explore repositories or users of interest from the links below.

[Explore repositories](#) - [Explore users](#)

Repository Organization

Repositories

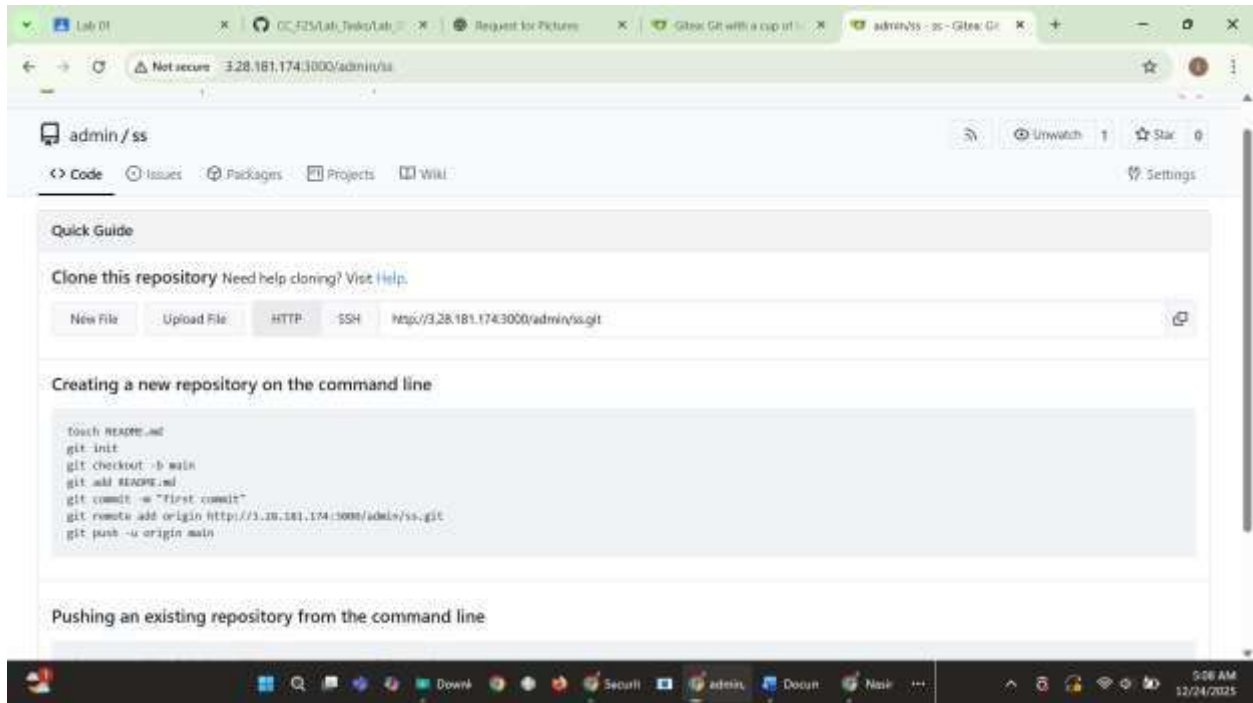


There are no repositories yet.

Powered by Gitexa Version: 1.25.3 Page: 14ms Template: 12ms

English Licenses API

9:07 AM 12/24/2025



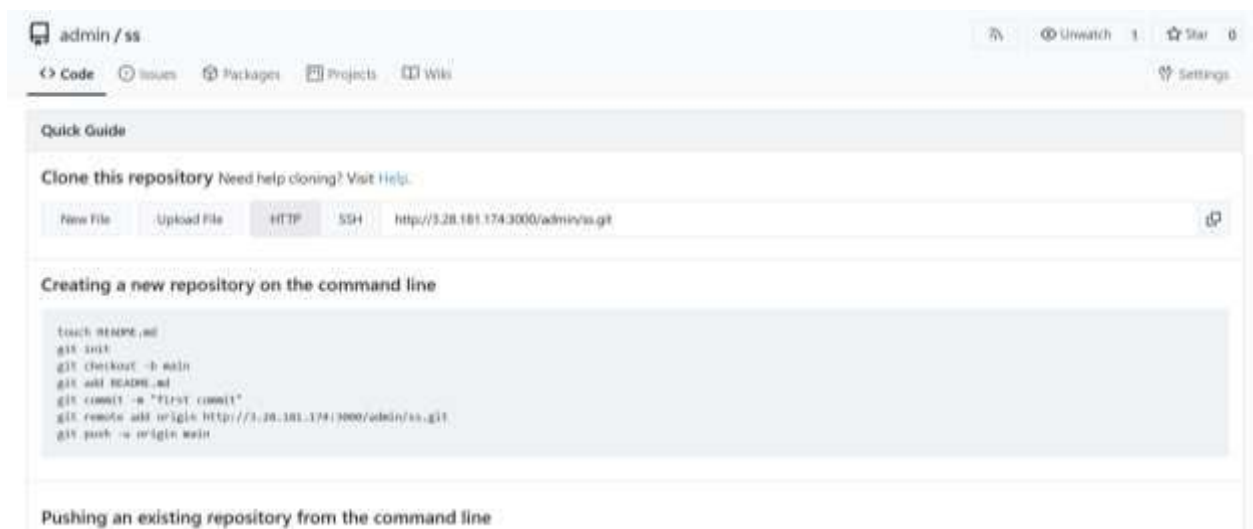
☐	Name	Instance ID	Instance state	Instance type	Status check	Alarm status
☐	LabBMachine	i-05ed5f211b5974265	Running	t3.micro	3/3 checks passed	View alarms +

▼ Network & Security

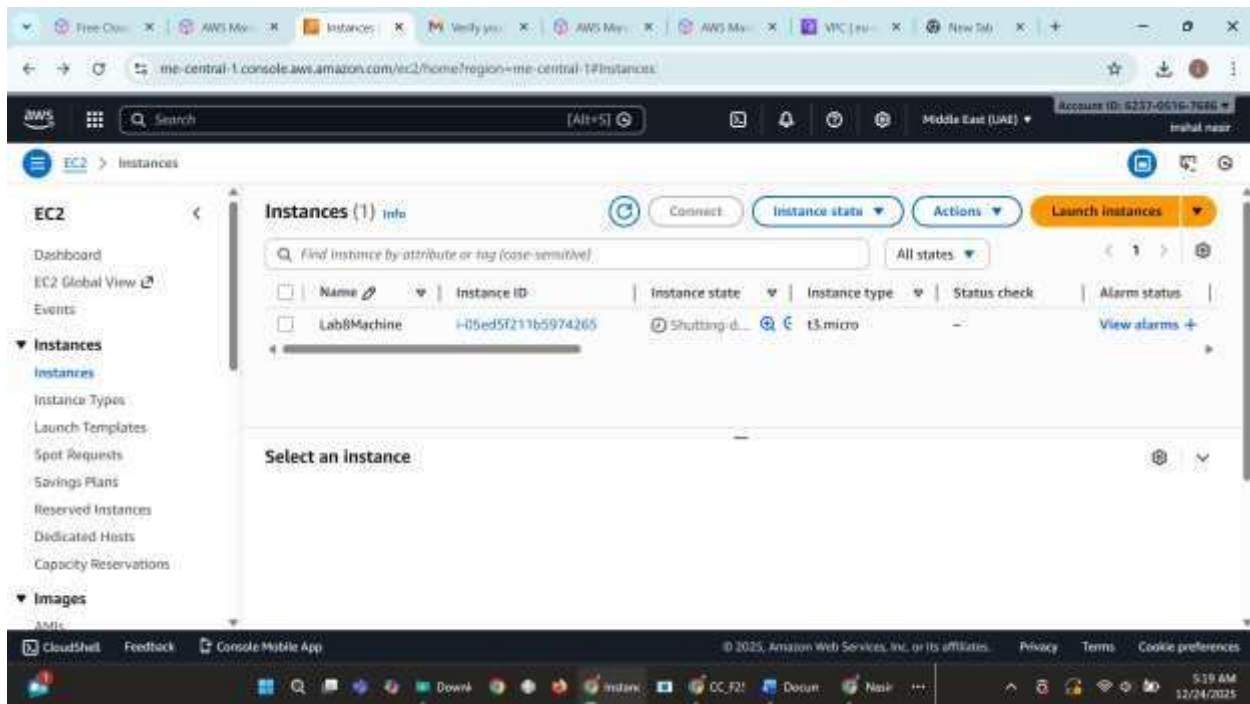
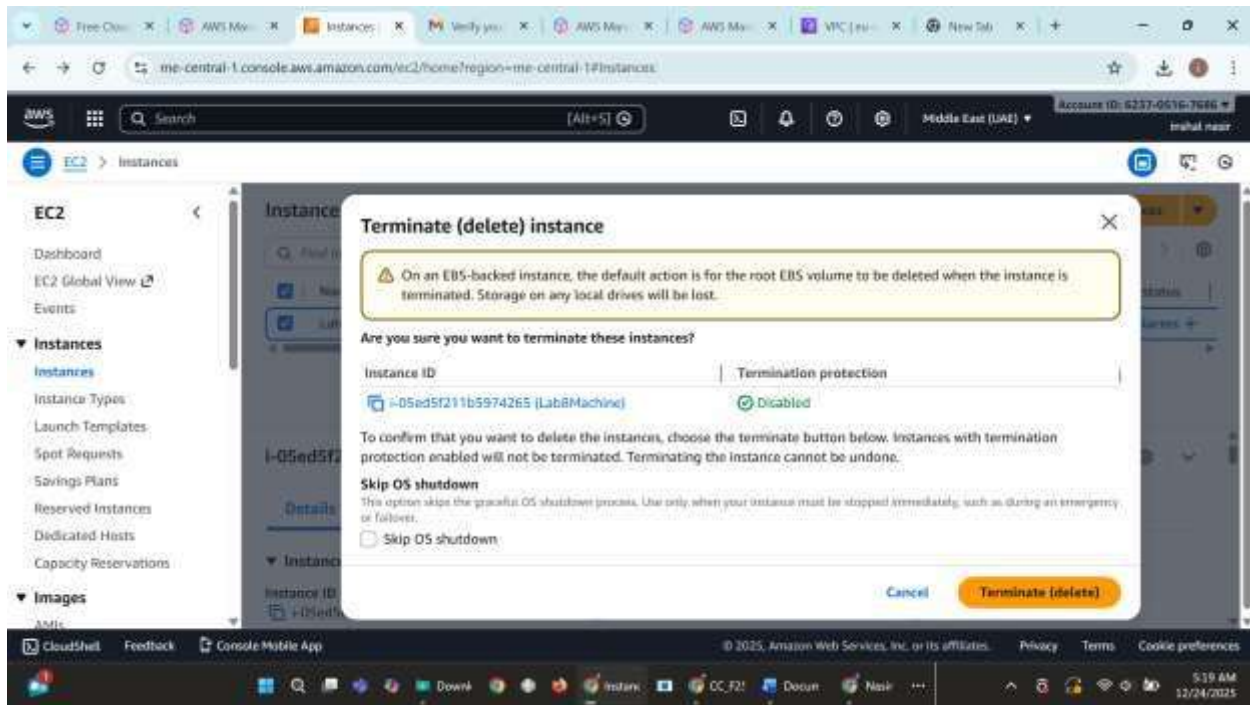
- Security Groups
- Elastic IPs
- Placement Groups
- Key Pairs
- Network Interfaces

▼ Load Balancing

Inbound rules (2)						
Search						
☐	Name	Security group rule ID	IP version	Type	Protocol	
☐	-	sgr-007171b2f5c3d88bf	IPv4	Custom TCP	TCP	
☐	-	sgr-035e14f649f077a5a	IPv4	SSH	TCP	



Remove



Free Class AWS Main Volumes Verify you AWS Main AWS Main VPC | eu New Tab

me-central-1.console.aws.amazon.com/ec2/home?region=me-central-1#Volumes

Search [Alt+S] Mobile East (UAE) Account ID: 6237-0516-7686 Initial name

Volumes

Info Last updated 1 minute ago Recycle Bin Actions Create volume

Saved filter sets Choose filter set Search

Name	Volume ID	Type	Size	IOPS	Throughput	Snapshot ID
You currently have no volumes in this region.						

Fault tolerance for all volumes in this Region

Snapshot summary

Recently backed up volumes / Total # volumes

0 / 0

Last updated on Wed, Dec 24, 2025, 05:21:03 AM (GMT-08:00)

Data Lifecycle Manager default policy for EBS Snapshots status
No default policy set up | Create policy

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Free Class AWS Main Security Verify you AWS Main AWS Main VPC | eu New Tab

me-central-1.console.aws.amazon.com/ec2/home?region=me-central-1#SecurityGroups

Search [Alt+S] Mobile East (UAE) Account ID: 6237-0516-7686 Initial name

Security Groups (1/2)

Info Actions Export security groups to CSV Create security group

Find security groups by attribute or tag

Name	Security group ID	VPC ID
sg-0e0224478c86fdb9	sg-0e0224478c86fdb9	vpc-079a26ed9b9cb32f2
sg-0cb37394bf16ab2a5	sg-0cb37394bf16ab2a5	vpc-079a26ed9b9cb32f2

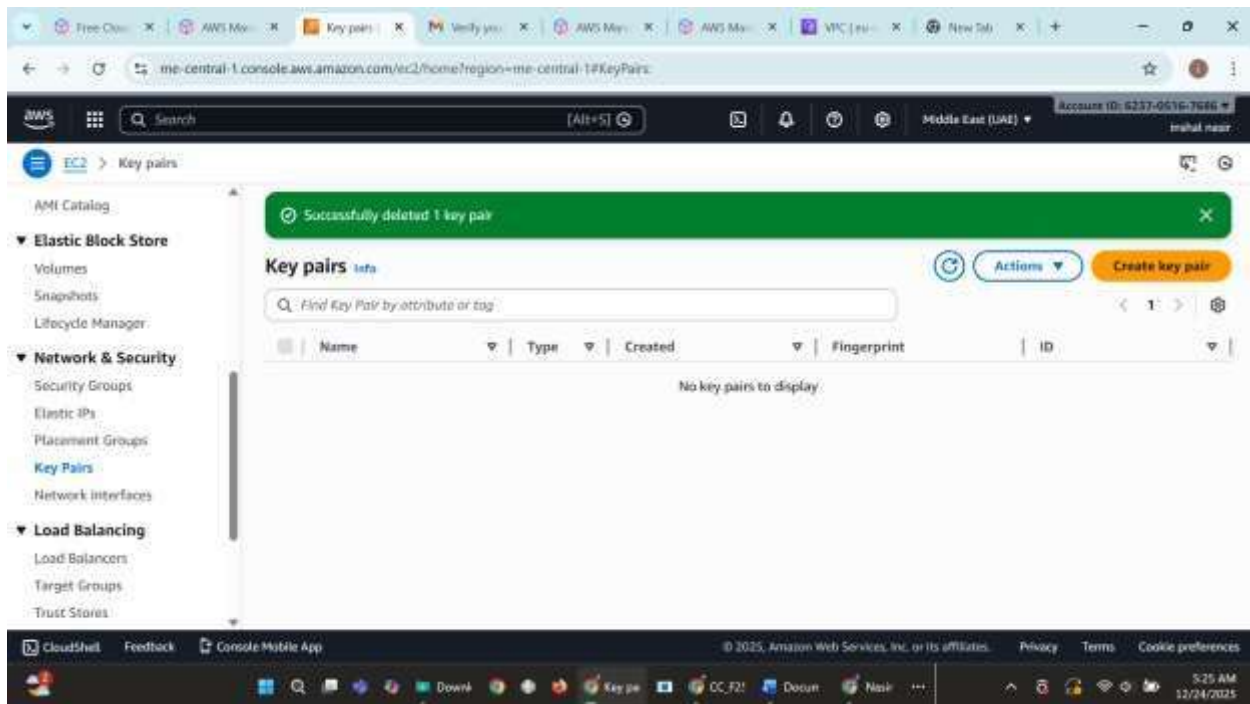
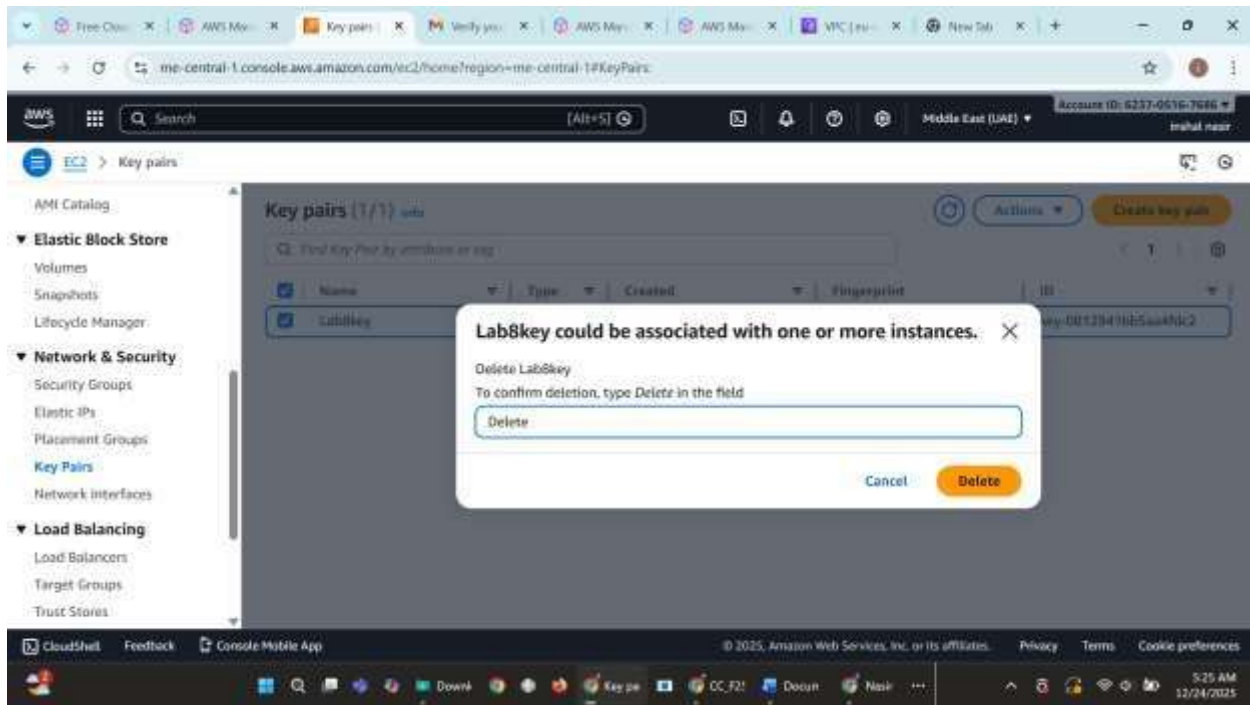
sg-0e0224478c86fdb9 - Lab8SecurityGroup

Details Inbound rules Outbound rules Sharing VPC associations Tags

Details

Security group name	Security group ID	Description	VPC ID
Lab8SecurityGroup	sg-0e0224478c86fdb9	Launch wizard 1 created 20	vpc-079a26ed9b9cb32f2

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Free Class AWS M... SecurityC... Verify you... AWS M... AWS M... VPC | eu... New Tab

me-central-1.console.aws.amazon.com/ec2/home?region=me-central-1#SecurityGroups

Search [Alt+S] Mobile East (UAE) Account ID: 6237-0516-7686 initial.nasir

EC2 > Security Groups

AMI Catalog

▼ Elastic Block Store

- Volumes
- Snapshots
- Lifecycle Manager

▼ Network & Security

- Security Groups
- Elastic IPs
- Placement Groups
- Key Pairs
- Network interfaces

▼ Load Balancing

- Load Balancers
- Target Groups
- Trust Stores

Security Groups (1) info

Find security groups by attribute or tag

Actions Export security groups to CSV Create security group

☐	Name	Security group ID	Security group name	VPC ID
☐	-	sg-0cb37394bf16ab2e5	default	vpc-079a26ed9b9cb52f2

Select a security group

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5:25 AM 12/24/2025

Free Class AWS M... Users | IAM Verify you... AWS M... AWS M... VPC | eu... New Tab

us-east-1.console.aws.amazon.com/iam/home?region=me-central-1#/users

Search IAM

IAM > Users

Identity and Access Management (IAM)

Search IAM

Dashboard

▼ Access management

- User groups
- Users
- Roles
- Policies
- Identity providers
- Account settings
- Account access management
- Temporary delegation requests

Delete Lab8user?

Delete Lab8user permanently? This will also delete all its user data, security credentials and inline policies.

User name	Last activity
Lab8user	1 hour ago

Note: Recent activity usually appears within 4 hours. Data is stored for a maximum of 365 days, depending when your region began supporting this feature. [Learn more](#)

To avoid accidental deletions, we ask you to provide additional written consent.

To confirm this deletion, type "confirm".

confirm

Cancel Delete user

CloudShell Feedback Console Mobile App

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5:26 AM 12/24/2025

us-east-1.console.aws.amazon.com/iam/home?region=me-central-1#/users

Identity and Access Management (IAM)

Users (1)

An IAM user is an identity with long-term credentials that is used to interact with AWS in an account.

User name	Path	Groups	Last activity	MFA	Password age
Admin	/	0	1 hour ago	-	1 hour

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5:27 AM 12/24/2025

us-east-1.console.aws.amazon.com/costmanagement/home?region=me-central-1#/cost-explorer?chartStyle=STACK&costAggregate=unBlendedCost&en...

Billing and Cost Management

Cost Explorer

New cost and usage report

Introducing Enhanced Forecasts with Explanations

Forecast up to 18 months ahead and understand how predictions are calculated with AI-powered explanations. Configure your forecast in Report parameters. For better accuracy, enable 36-month historical data in settings.

Don't show this again

New cost and usage report

Report parameters

Record reports

Save to report library

Cost and usage overview

Total cost	Average monthly cost	Service count
\$0.00	\$0.00	0

Cost and usage graph

Costs (\$)

Report parameters

Time

Standard Compare

Date Range

2025-06-01 — 2025-11-30

Displaying last 6 months

Granularity

Monthly

Group by

Dimension

Service

Filters

Applied filters (0)

Service

Linked account

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4:18 AM 12/27/2025



FATIMA JINNAH WOMEN UNIVERSITY

Department of Software Engineering

Lab Title:	Experiment # :
Student Name:	Regd. #:
Experiment Name:	

Performance Indicator	Level of Achievements				
	Excellent (5)	Good (4)	Average (3)	Below Average (2)	Poor (1)
Understanding & implementation Problem					
Report and presentation					
Maximum Marks		10	Obtained Marks		
Lab Instructor:					

Lab 9

Task 1

```
Ali1@DESKTOP-KQE4P80 MINGW64 ~ (main)
$ winget install --id GitHub.cli
Found an existing package already installed. Trying to upgrade the installed package...

Ali1@DESKTOP-KQE4P80 MINGW64 ~ (main)
$ gh auth login -s codespace
Where do you use GitHub? GitHub.com
What is your preferred protocol for Git operations on this host? HTTPS
Authenticate Git with your GitHub credentials? Yes
How would you like to authenticate GitHub CLI? Login with a web browser

First copy your one-time code: B36D-2706
Press Enter to open https://github.com/login/device in your browser...
Authentication complete.
$ gh config set -h github.com git_protocol https
Configured git protocol
Logged in as 23-22412-038-sys
You were already logged in to this account

Ali1@DESKTOP-KQE4P80 MINGW64 ~ (main)
$ gh codespace list


| NAME           | DISPLAY NAME   | REPOSITORY     | BRANCH | STATE    | CREATED AT     |
|----------------|----------------|----------------|--------|----------|----------------|
| sturdy-para... | sturdy para... | 23-22412-03... | main   | Shutdown | about 12 da... |
| obscure-zeb... | obscure zebra  | 23-22412-03... | main*  | Shutdown | about 12 da... |



? Choose codespace: 23-22412-038-sys/-inshal-nasir-38- [main*]: obscure zebra
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-1030-azure x86_64)

 * Documentation: https://help.ubuntu.com
 * Management:   https://landscape.canonical.com
 * Support:      https://ubuntu.com/pro
Last login: Sun Dec 14 16:05:23 2025 from ::1
@23-22412-038-sys → /workspaces/-inshal-nasir-38- (main) $
```

Task 2

```
@23-22412-038-sys → /workspaces/-inshal-nasir-38- (main) $ aws --version
aws-cli/2.32.16 Python/3.13.11 Linux/6.8.0-1030-azure exe/x86_64.ubuntu.24
```

```
@23-22412-038-sys → /workspaces/-inshal-nasir-38- (main) $ aws configure
AWS Access Key ID [None]: AKIAZCN5ZS5DFNEY4OHR
AWS Secret Access Key [None]: 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
Default region name [None]: me-central-1
Default output format [None]: json
```

```
@23-22412-038-sys → /workspaces/-inshal-nasir-38- (main) $ cat ~/.aws/credentials
cat ~/.aws/config
[default]
aws_access_key_id = AKIAZCN5ZS5DFNEY4OHR
aws_secret_access_key = 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
[default]
region = me-central-1
output = json
```

```
@23-22412-038-sys → /workspaces/-inshal-nasir-38- (main) $ aws sts get-caller-identity
{
  "UserId": "AIDAZCN5ZS5DNPM5O4PRX",
  "Account": "623705167686",
  "Arn": "arn:aws:iam::623705167686:user/zzzz"
}
```

Task 3


```
@23-22412-038-sys + /workspaces/-inshal-nasir-38- (main) $ aws ec2 create-security-group \
--group-name MySecurityGroup \
--description "My Security Group" \
--vpc-id vpc-079a26ed9b9cb32f2
{
  "GroupId": "sg-05ccb0c9bde8724e8",
  "SecurityGroupArn": "arn:aws:ec2:me-central-1:623705167686:security-group/sg-05ccb0c9bde8724e8"
}
```

```
@23-22412-038-sys + /workspaces/-inshal-nasir-38- (main) $ aws ec2 describe-security-groups --group-ids sg-05ccb0c9bde8724e8
{
  "SecurityGroups": [
    {
      "GroupId": "sg-05ccb0c9bde8724e8",
      "IpPermissionsEgress": [
        {
          "IpProtocol": "-1",
          "UserIdGroupPairs": [],
          "IpRanges": [
            {
              "CidrIp": "0.0.0.0/0"
            }
          ],
          "Ipv6Ranges": [],
          "PrefixListIds": []
        }
      ],
      "VpcId": "vpc-079a26ed9b9cb32f2",
      "SecurityGroupArn": "arn:aws:ec2:me-central-1:623705167686:security-group/sg-05ccb0c9bde8724e8",
      "OwnerId": "623705167686",
      "GroupName": "MySecurityGroup",
      "Description": "My Security Group",
      "IpPermissions": []
    }
  ]
}
```

```
@23-22412-038-sys + /workspaces/-inshal-nasir-38- (main) $ curl icanhazip.com
4.240.18.227
```

```
$ aws ec2 authorize-security-group-ingress \
--group-id sg-05ccb0c9bde8724e8 \
--protocol tcp \
--port 22 \
--cidr 4.240.18.227/32
{
  "Return": true,
  "SecurityGroupRules": [
    {
      "SecurityGroupRuleId": "sgr-03c9e24c44966c9f3",
      "GroupId": "sg-05ccb0c9bde8724e8",
      "GroupOwnerId": "623705167686",
      "IsEgress": false,
      "IpProtocol": "tcp",
      "FromPort": 22,
      "ToPort": 22,
      "CidrIpv4": "4.240.18.227/32",
      "SecurityGroupRuleArn": "arn:aws:ec2:me-central-1:623705167686:security-group-rule/sgr-03c9e24c44966c9f3"
    }
  ]
}
```



```
$ aws ec2 authorize-security-group-ingress \
--group-id sg-05ccb0c9bde8724e8 \
--ip-permissions '[{"FromPort":80,"ToPort":80,"IpProtocol":"tcp","IpRanges":[{"CidrIp":"4.240.18.227/32"}]}'
{
  "Return": true,
  "SecurityGroupRules": [
    {
      "SecurityGroupRuleId": "sgr-04e16199952c72ef0",
      "GroupId": "sg-05ccb0c9bde8724e8",
      "GroupOwnerId": "623705167686",
      "IsEgress": false,
      "IpProtocol": "tcp",
      "FromPort": 80,
      "ToPort": 80,
      "CidrIpv4": "4.240.18.227/32",
      "SecurityGroupRuleArn": "arn:aws:ec2:me-central-1:623705167686:security-group-rule/sgr-04e16199952c72ef0"
    }
  ]
}
```

```
923-22412-038-sys → /workspaces/-inshai-nasir-38- (main) $ aws ec2 describe-security-groups --group-ids sg-05ccb0c9bde8724e8
{
  "SecurityGroups": [
    {
      "GroupId": "sg-05ccb0c9bde8724e8",
      "IpPermissionsEgress": [
        {
          "IpProtocol": "-1",
          "UserIdGroupPairs": [],
          "IpRanges": [
            {
              "CidrIp": "0.0.0.0/0"
            }
          ],
          "Ipv6Ranges": [],
          "PrefixListIds": []
        }
      ],
      "VpcId": "vpc-079a26ed9b9cb32f2",
      "SecurityGroupArn": "arn:aws:ec2:me-central-1:623705167686:security-group/sg-05ccb0c9bde8724e8",
      "OwnerId": "623705167686",
      "GroupName": "MySecurityGroup",
      "Description": "My Security Group",
      "IpPermissions": [
        {
          "IpProtocol": "tcp",
          "FromPort": 80,
          "ToPort": 80,
          "UserIdGroupPairs": [],
          "IpRanges": [
            {
              "CidrIp": "4.240.18.227/32"
            }
          ],
          "Ipv6Ranges": [],
          "PrefixListIds": []
        },
        {
          "IpProtocol": "tcp",
          "FromPort": 22,
          "ToPort": 22,
          "UserIdGroupPairs": [],
          "IpRanges": [
            {
              "CidrIp": "4.240.18.227/32"
            }
          ],
          "Ipv6Ranges": [],
          "PrefixListIds": []
        }
      ]
    }
  ]
}
```

Task 4

```
--key-name MyED25519Key \
--key-type ed25519 \
--key-format pem \
--query 'KeyMaterial' \
--output text > MyED25519Key.pem

-rw-rw-rw- 1 MyED25519Key.pem
v-rw-rw- 1 codespace codespace 388 Dec 27 14:10 MyED25519Key.pem
```

```

@23-22412-038-sys → /workspaces/-inshal-nasir-38- (main) $ aws ec2 describe-key-pairs
{
  "KeyPairs": [
    {
      "KeyPairId": "key-0a17c60c74d86b8c8",
      "KeyType": "ed25519",
      "Tags": [],
      "CreateTime": "2025-12-27T14:10:18.282000+00:00",
      "KeyName": "MyED25519Key",
      "KeyFingerprint": "6y4AWdtINiWwR6vWoGF4hqc1l+x3qt3H1lNggXMDwzw="
    }
  ]
}

```

```

@23-22412-038-sys → /workspaces/-inshal-nasir-38- (main) $ aws ec2 run-instances \
  --image-id ami-05e66df2bafcb7dea \
  --count 1 \
  --instance-type t3.micro \
  --key-name MyED25519Key \
  --security-group-ids sg-05ccb0c9bde8724e8 \
  --subnet-id subnet-03c38a2393aaf6920 \
  --tag-specifications "ResourceType=instance,Tags=[{Key=Name,Value=MyServer}]"
{
  "ReservationId": "r-0ce6ef4ae6b3739c4",
  "OwnerId": "623705167686",
  "Groups": [],
  "Instances": [
    {
      "Architecture": "x86_64",
      "BlockDeviceMappings": [],
      "ClientToken": "ca979562-fea4-44de-8aaf-f289077ea081",
      "EbsOptimized": false,
      "EnaSupport": true,
      "Hypervisor": "xen",
      "NetworkInterfaces": [
        {
          "Attachment": {
            "AttachTime": "2025-12-27T14:16:49+00:00",
            "AttachmentId": "eni-attach-0042831c73ba683d8",
            "DeleteOnTermination": true,
            "DeviceIndex": 0,
            "Status": "attaching",
            "NetworkCardIndex": 0
          },
          "Description": "",
          "Groups": [
            {
              "GroupId": "sg-05ccb0c9bde8724e8",
              "GroupName": "MySecurityGroup"
            }
          ],
          "Ipv6Addresses": [],
          "MacAddress": "0e:15:50:a5:4b:13",
          "NetworkInterfaceId": "eni-0e457fc8c20348c4f",

```

```

@23-22412-038-sys → /workspaces/-inshal-nasir-38- (main) $ aws ec2 describe-instances \
  --filters "Name=tag:Name,Values=MyServer" \
  --query "Reservations[*].Instances[*].[InstanceId,PublicIpAddress]" \
  --output table

```

DescribeInstances	
i-051db8825b44630bb	3.28.200.73

Please type 'yes', 'no' or the fingerprint: yes
 Warning: Permanently added '3.28.200.73' (ED25519) to the list of known hosts.

```
@23-22412-038-sys → /workspaces/~inshal-nasir-38- (main) $ # Stop instance
aws ec2 stop-instances --instance-ids i-051db8825b44630bb

# Start instance
aws ec2 start-instances --instance-ids i-051db8825b44630bb

# Terminate instance (to avoid charges)
aws ec2 terminate-instances --instance-ids i-051db8825b44630bb
{
  "StoppingInstances": [
    {
      "InstanceId": "i-051db8825b44630bb",
      "CurrentState": {
        "Code": 64,
        "Name": "stopping"
      },
      "PreviousState": {
        "Code": 16,
        "Name": "running"
      }
    }
  ]
}
```

```
{
  "TerminatingInstances": [
    {
      "InstanceId": "i-051db8825b44630bb",
      "CurrentState": {
        "Code": 32,
        "Name": "shutting-down"
      },
      "PreviousState": {
        "Code": 64,
        "Name": "stopping"
      }
    }
  ]
}
```

Task 5

```

aws ec2 describe-vpcs
aws ec2 describe-subnets
aws ec2 describe-instances
aws ec2 describe-regions
aws ec2 describe-availability-zones
{
  "SecurityGroups": [
    {
      "GroupId": "sg-05ccb0c9bde8724e8",
      "IpPermissionsEgress": [
        {
          "IpProtocol": "-1",
          "UserIdGroupPairs": [],
          "IpRanges": [
            {
              "CidrIp": "0.0.0.0/0"
            }
          ],
          "Ipv6Ranges": [],
          "PrefixListIds": []
        }
      ],
      "VpcId": "vpc-079a26ed9b9cb32f2",
      "SecurityGroupArn": "arn:aws:ec2:me-central-1:623705167686:security-group/sg-05ccb0c9bde8724e8",
      "OwnerId": "623705167686",
      "GroupName": "MySecurityGroup",
      "Description": "My Security group",
      "IpPermissions": [
        {
          "IpProtocol": "tcp",
          "FromPort": 80,
          "ToPort": 80,
          "UserIdGroupPairs": [],
          "IpRanges": [
            {
              "CidrIp": "4.240.18.227/32"
            }
          ],
          "Ipv6Ranges": [],
          "PrefixListIds": []
        }
      ],
      {
        "IpProtocol": "tcp",

```

```

    "IpPermissionsEgress": [
      {
        "IpProtocol": "-1",
        "UserIdGroupPairs": [],
        "IpRanges": [
          {
            "CidrIp": "0.0.0.0/0"
          }
        ],
        "Ipv6Ranges": [],
        "PrefixListIds": []
      }
    ],
    "VpcId": "vpc-079a26ed9b9cb32f2",
    "SecurityGroupArn": "arn:aws:ec2:me-central-1:623705167686:security-group/sg-0cb37394bf16ab2a5",
    "OwnerId": "623705167686",
    "GroupName": "default",
    "Description": "default VPC security group",
    "IpPermissions": [
      {
        "IpProtocol": "tcp",
        "FromPort": 80,
        "ToPort": 80,
        "UserIdGroupPairs": [],
        "IpRanges": [
          {
            "CidrIp": "154.192.1.59/32"
          }
        ],
        "Ipv6Ranges": [],
        "PrefixListIds": []
      },
      {
        "IpProtocol": "tcp",
        "FromPort": 22,
        "ToPort": 22,
        "UserIdGroupPairs": [],
        "IpRanges": [
          {
            "CidrIp": "206.84.147.78/32"
          }
        ],
        "Ipv6Ranges": [],
        "PrefixListIds": []
      }
    ]
  }
}

```

```

{
  "Vpcs": [
    {
      "OwnerId": "623705167686",
      "InstanceTenancy": "default",
      "CidrBlockAssociationSet": [
        {
          "AssociationId": "vpc-cidr-assoc-0dbcc892062f89bef",
          "CidrBlock": "172.31.0.0/16",
          "CidrBlockState": {
            "State": "associated"
          }
        }
      ],
      "IsDefault": true,
      "BlockPublicAccessStates": {
        "InternetGatewayBlockMode": "off"
      },
      "VpcId": "vpc-079a26ed9b9cb32f2",
      "State": "available",
      "CidrBlock": "172.31.0.0/16",
      "DhcpOptionsId": "dopt-02dd4ed017908b331"
    }
  ]
}

```

```

"Subnets": [
  {
    "AvailabilityZoneId": "mec1-az3",
    "MapCustomerOwnedIpOnLaunch": false,
    "OwnerId": "623705167686",
    "AssignIpv6AddressOnCreation": false,
    "Ipv6CidrBlockAssociationSet": [],
    "SubnetArn": "arn:aws:ec2:me-central-1:623705167686:subnet/subnet-03c38a2393aaf6920",
    "EnabledDns64": false,
    "Ipv6Native": false,
    "PrivateDnsNameOptionsOnLaunch": {
      "HostnameType": "ip-name",
      "EnableResourceNameDnsARecord": false,
      "EnableResourceNameDnsAAAARecord": false
    },
    "BlockPublicAccessStates": {
      "InternetGatewayBlockMode": "off"
    },
    "SubnetId": "subnet-03c38a2393aaf6920",
    "State": "available",
    "VpcId": "vpc-079a26ed9b9cb32f2",
    "CidrBlock": "172.31.0.0/20",
    "AvailableIpAddressCount": 4091,
    "AvailabilityZone": "me-central-1c",
    "DefaultForAz": true,
    "MapPublicIpOnLaunch": true
  },
  {
    "AvailabilityZoneId": "mec1-az1",
    "MapCustomerOwnedIpOnLaunch": false,
    "OwnerId": "623705167686",
    "AssignIpv6AddressOnCreation": false,
    "Ipv6CidrBlockAssociationSet": [],
    "SubnetArn": "arn:aws:ec2:me-central-1:623705167686:subnet/subnet-08a73f8aa1dee72d3",
    "EnabledDns64": false,
    "Ipv6Native": false,
    "PrivateDnsNameOptionsOnLaunch": {
      "HostnameType": "ip-name",
      "EnableResourceNameDnsARecord": false,
      "EnableResourceNameDnsAAAARecord": false
    },
    "BlockPublicAccessStates": {
      "InternetGatewayBlockMode": "off"
    },
    "SubnetId": "subnet-08a73f8aa1dee72d3",
    "State": "available",
  }
]

```



```
"Reservations": [
  {
    "ReservationId": "r-0ce6ef4ae6b3739c4",
    "OwnerId": "623705167686",
    "Groups": [],
    "Instances": [
      {
        "Architecture": "x86_64",
        "BlockDeviceMappings": [],
        "ClientToken": "ca979562-fea4-44de-8aaf-f289077ea081",
        "EbsOptimized": false,
        "EnaSupport": true,
        "Hypervisor": "xen",
        "NetworkInterfaces": [],
        "RootDeviceName": "/dev/xvda",
        "RootDeviceType": "ebs",
        "SecurityGroups": [],
        "StateReason": {
          "Code": "Client.UserInitiatedShutdown",
          "Message": "Client.UserInitiatedShutdown: User initiated shutdown"
        },
        "Tags": [
          {
            "Key": "Name",
            "Value": "MyServer"
          }
        ],
        "VirtualizationType": "hvm",
        "CpuOptions": {
          "CoreCount": 1,
          "ThreadsPerCore": 2
        },
        "CapacityReservationSpecification": {
          "CapacityReservationPreference": "open"
        },
        "HibernationOptions": {
          "Configured": false
        },
        "MetadataOptions": {
          "State": "pending",
          "HttpTokens": "required",
          "HttpPutResponseHopLimit": 2,
          "HttpEndpoint": "enabled",
          "HttpProtocolIpv6": "disabled",
          "InstanceMetadataTags": "disabled"
        }
      }
    ]
  }
]
```

```
"Regions": [  
  {  
    "OptInStatus": "opt-in-not-required",  
    "RegionName": "ap-south-1",  
    "Endpoint": "ec2.ap-south-1.amazonaws.com"  
  },  
  {  
    "OptInStatus": "opt-in-not-required",  
    "RegionName": "eu-north-1",  
    "Endpoint": "ec2.eu-north-1.amazonaws.com"  
  },  
  {  
    "OptInStatus": "opt-in-not-required",  
    "RegionName": "eu-west-3",  
    "Endpoint": "ec2.eu-west-3.amazonaws.com"  
  },  
  {  
    "OptInStatus": "opt-in-not-required",  
    "RegionName": "eu-west-2",  
    "Endpoint": "ec2.eu-west-2.amazonaws.com"  
  },  
  {  
    "OptInStatus": "opt-in-not-required",  
    "RegionName": "eu-west-1",  
    "Endpoint": "ec2.eu-west-1.amazonaws.com"  
  },  
  {  
    "OptInStatus": "opt-in-not-required",  
    "RegionName": "ap-northeast-3",  
    "Endpoint": "ec2.ap-northeast-3.amazonaws.com"  
  },  
  {  
    "OptInStatus": "opt-in-not-required",  
    "RegionName": "ap-northeast-2",  
    "Endpoint": "ec2.ap-northeast-2.amazonaws.com"  
  },  
  {  
    "OptInStatus": "opt-in-not-required",  
    "RegionName": "ap-northeast-1",  
    "Endpoint": "ec2.ap-northeast-1.amazonaws.com"  
  },  
],
```

```

"AvailabilityZones": [
  {
    "OptInStatus": "opt-in-not-required",
    "Messages": [],
    "RegionName": "me-central-1",
    "ZoneName": "me-central-1a",
    "ZoneId": "mec1-az1",
    "GroupName": "me-central-1-zg-1",
    "NetworkBorderGroup": "me-central-1",
    "ZoneType": "availability-zone",
    "GroupLongName": "Middle East (UAE) 1",
    "State": "available"
  },
  {
    "OptInStatus": "opt-in-not-required",
    "Messages": [],
    "RegionName": "me-central-1",
    "ZoneName": "me-central-1b",
    "ZoneId": "mec1-az2",
    "GroupName": "me-central-1-zg-1",
    "NetworkBorderGroup": "me-central-1",
    "ZoneType": "availability-zone",
    "GroupLongName": "Middle East (UAE) 1",
    "State": "available"
  },
  {
    "OptInStatus": "opt-in-not-required",
    "Messages": [],
    "RegionName": "me-central-1",
    "ZoneName": "me-central-1c",
    "ZoneId": "mec1-az3",
    "GroupName": "me-central-1-zg-1",
    "NetworkBorderGroup": "me-central-1",
    "ZoneType": "availability-zone",
    "GroupLongName": "Middle East (UAE) 1",
    "State": "available"
  }
]

```

Task 6

```

023-22412-038-sys → /workspaces/-inshal-nasir-38- (main) $ aws iam create-group --group-name MyGroupCli
aws iam get-group --group-name MyGroupCli
aws iam create-user --user-name MyUserCli
aws iam get-user --user-name MyUserCli
{
  "Group": {
    "Path": "/",
    "GroupName": "MyGroupCli",
    "GroupId": "AGPAZCN5ZS5DHF5F2RFNY",
    "Arn": "arn:aws:iam::623705167686:group/MyGroupCli",
    "CreateDate": "2025-12-27T17:10:33+00:00"
  }
}
{
  "Users": [],
  "Group": {
    "Path": "/",
    "GroupName": "MyGroupCli",
    "GroupId": "AGPAZCN5ZS5DHF5F2RFNY",
    "Arn": "arn:aws:iam::623705167686:group/MyGroupCli",
    "CreateDate": "2025-12-27T17:10:33+00:00"
  }
}
{
  "User": {
    "Path": "/",
    "UserName": "MyUserCli",
    "UserId": "AIDAZCN5ZS5DCV6JAI6E5",
    "Arn": "arn:aws:iam::623705167686:user/MyUserCli",
    "CreateDate": "2025-12-27T17:10:36+00:00"
  }
}
{
  "User": {
    "Path": "/",
    "UserName": "MyUserCli",
    "UserId": "AIDAZCN5ZS5DCV6JAI6E5",
    "Arn": "arn:aws:iam::623705167686:user/MyUserCli",
    "CreateDate": "2025-12-27T17:10:36+00:00"
  }
}

```

```

023-22412-038-sys → /workspaces/-inshal-nasir-38- (main) $ aws iam add-user-to-group --user-name MyUserCli --group-name MyGroupCli
aws iam get-group --group-name MyGroupCli
{
  "Users": [
    {
      "Path": "/",
      "UserName": "MyUserCli",
      "UserId": "AIDAZCN5ZS5DCV6JAI6E5",
      "Arn": "arn:aws:iam::623705167686:user/MyUserCli",
      "CreateDate": "2025-12-27T17:10:36+00:00"
    }
  ],
  "Group": {
    "Path": "/",
    "GroupName": "MyGroupCli",
    "GroupId": "AGPAZCN5ZS5DHF5F2RFNY",
    "Arn": "arn:aws:iam::623705167686:group/MyGroupCli",
    "CreateDate": "2025-12-27T17:10:33+00:00"
  }
}

```

```

@23-22412-038-sys → /workspaces/~inshal-nasir-38- (main) $ aws iam list-policies \
--query "Policies[?contains(PolicyName, 'EC2')].PolicyName" \
--output text

aws iam attach-group-policy \
--group-name MyGroupCli \
--policy-arn arn:aws:iam::aws:policy/AmazonEC2FullAccess

aws iam list-attached-group-policies --group-name MyGroupCli
AmazonEC2FullAccess AmazonEC2ReadOnlyAccess AmazonElasticMapReduceforEC2Role AmazonEC2RoleforDataPipelineRole
onEC2ContainerServiceRole
AmazonEC2RoleforAWSCodeDeploy AmazonEC2RoleforSSM CloudwatchActionsEC2Access AmazonEC2ContainerRegistryReadOnly
onEC2ContainerRegistryFullAccess AmazonEC2ContainerServiceAutoscaleRole AmazonEC2SpotFleetAutoscaleRole
AWSLambdaElasticBeanstalkCustomPlatformforEC2Role AmazonEC2ContainerServiceEventsRole AmazonEC2SpotFleetTaggingRole Aw
ScheduledInstances AWSSEC2SpotFleetServiceRolePolicy AWSApplicationAutoscalingEC2SpotFleetRequestPolicy
AWSSEC2FleetServiceRolePolicy AWSAutoScalingPlansEC2AutoScalingPolicy
EC2InstanceConnect
AmazonEC2RolePolicyforLaunchWizard EC2InstanceProfileForImageBuilder EC2FleetTimeShifttableServiceRolePolicy
AmazonEC2RoleforAWSCodeDeploylimited
EC2InstanceProfileForImageBuilderECRContainerBuilds AWSApplicationMigrationEC2Access
AWSSEC2CapacityReservationFleetRolePolicy EC2FastLaunchServiceRolePolicy
AmazonSSMManagedEC2InstanceDefaultPolicy AWSFaultInjectionSimulatorEC2Access
EC2ImageBuilderLifecycleExecutionPolicy AWSSEC2VssSnapshotPolicy EC2FastLaunchFullAccess
AmazonEC2ContainerRegistryPullOnly DeclarativePoliciesEC2Report
AmazonEC2ImageReferencesAccessPolicy AWSSEC2CapacityManagerServiceRolePolicy
AWSSEC2SqlHaServiceRolePolicy AWSSEC2SqlHaInstancePolicy AWSLambdaManagedEC2ResourceOperator
{
  "AttachedPolicies": [
    {
      "PolicyName": "AmazonEC2FullAccess",
      "PolicyArn": "arn:aws:iam::aws:policy/AmazonEC2FullAccess"
    }
  ]
}

```

```

@23-22412-038-sys → /workspaces/~inshal-nasir-38- (main) $ aws iam create-login-profile \
--user-name MyUserCli \
--password TempPassword@123 \
--password-reset-required

{
  "LoginProfile": {
    "UserName": "MyUserCli",
    "CreateDate": "2025-12-27T17:14:28+00:00",
    "PasswordResetRequired": true
  }
}

```

```

@23-22412-038-sys → /workspaces/~inshal-nasir-38- (main) $ aws iam create-login-profile \
--user-name MyUserCli \
--password TempPassword@123 \
--password-reset-required

{
  "LoginProfile": {
    "UserName": "MyUserCli",
    "CreateDate": "2025-12-27T17:14:28+00:00",
    "PasswordResetRequired": true
  }
}

@23-22412-038-sys → /workspaces/~inshal-nasir-38- (main) $ aws iam attach-group-policy \
--group-name MyGroupCli \
--policy-arn arn:aws:iam::aws:policy/IAMUserChangePassword

@23-22412-038-sys → /workspaces/~inshal-nasir-38- (main) $ aws iam detach-group-policy \
--group-name MyGroupCli \
--policy-arn arn:aws:iam::aws:policy/IAMUserChangePassword

```



```

@23-22412-038-sys → /workspaces/-inshal-nasir-38- (main) $ aws iam create-access-key --user-name MyUserCli
aws iam list-access-keys --user-name MyUserCli
{
  "AccessKey": {
    "UserName": "MyUserCli",
    "AccessKeyId": "AKIAZCN5ZS5DIQQZWZMM",
    "Status": "Active",
    "SecretAccessKey": "O Ae9Am18XCPdpV4EBnGmooGsFN170cIN98Hrove+",
    "CreateDate": "2025-12-27T17:16:40+00:00"
  }
}
{
  "AccessKeyMetadata": [
    {
      "UserName": "MyUserCli",
      "AccessKeyId": "AKIAZCN5ZS5DIQQZWZMM",
      "Status": "Active",
      "CreateDate": "2025-12-27T17:16:40+00:00"
    }
  ]
}

```

```

@23-22412-038-sys → /workspaces/-inshal-nasir-38- (main) $ export AWS_ACCESS_KEY_ID=AKIAZCN5ZS5DIQQZWZMM
export AWS_SECRET_ACCESS_KEY=O Ae9Am18XCPdpV4EBnGmooGsFN170cIN98Hrove+
@23-22412-038-sys → /workspaces/-inshal-nasir-38- (main) $ printenv | grep AWS_
AWS_SECRET_ACCESS_KEY=O Ae9Am18XCPdpV4EBnGmooGsFN170cIN98Hrove+
AWS_ACCESS_KEY_ID=AKIAZCN5ZS5DIQQZWZMM

```

Task 7

```

@23-22412-038-sys → /workspaces/-inshal-nasir-38- (main) $ aws ec2 describe-instances \
--filters "Name=instance-type,Values=t3.micro" \
--query "Reservations[].Instances[].InstanceId" \
--output table
@23-22412-038-sys → /workspaces/-inshal-nasir-38- (main) $ aws ec2 describe-instances \
--filters "Name=instance-type,Values=t3.micro" \
--query "Reservations[].Instances[].InstanceId" \
--output table

```

Task 8

```

@23-22412-038-sys → /workspaces/-inshal-nasir-38- (main) $ aws ec2 describe-instances \
--query "Reservations[*].Instances[*].InstanceId,State,Name" \
--output table

```

Cleanup

```

@23-22412-038-sys → /workspaces/-inshal-nasir-38- (main) $ aws ec2 delete-security-group --group-id sg-05ccb0c9bde8724e8
{
  "Return": true,
  "GroupId": "sg-05ccb0c9bde8724e8"
}

```

```

@23-22412-038-sys → /workspaces/-inshal-nasir-38- (main) $ aws ec2 delete-key-pair --key-name MyED25519Key
{
  "Return": true,
  "KeyPairId": "key-0a17c60c74d86b8c8"
}

```




FATIMA JINNAH WOMEN UNIVERSITY

Department of Software Engineering

Lab Title:	Experiment # :
Student Name:	Regd. #:
Experiment Name:	

Performance Indicator	Level of Achievements				
	Excellent (5)	Good (4)	Average (3)	Below Average (2)	Poor (1)
Understanding & implementation Problem					
Report and presentation					
Maximum Marks		10	Obtained Marks		
Lab Instructor:					

Lab 10

Task 1

```
ti@DESKTOP-KQE4P80 MINGW64 ~ (main)
winget install --id GitHub.cli
```

```
: First copy your one-time code: 0A98-C194
Press Enter to open https://github.com/login/device in your browser
✓ Authentication complete.
- gh config set -h github.com git_protocol https
✓ Configured git protocol
✓ Logged in as 23-22412-038-sys
! You were already logged in to this account
```

```
$ gh codespace list
```

NAME	DISPLAY NAME
sturdy-parake...	sturdy par...
obscure-zebra...	obscure zebra
reimagined-ba...	reimagined...
ominous-waffl...	ominous wa...

```
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-1030-azure x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.
```

Task 2

```

inflating: aws/dist/prompts/lookup-3.0.31.dist-info/licenses/LICENSE
inflating: aws/dist/wheel-0.45.1.dist-info/direct_url.json
inflating: aws/dist/wheel-0.45.1.dist-info/METADATA
inflating: aws/dist/wheel-0.45.1.dist-info/INSTALLER
inflating: aws/dist/wheel-0.45.1.dist-info/entry_points.txt
inflating: aws/dist/wheel-0.45.1.dist-info/WHEEL
inflating: aws/dist/wheel-0.45.1.dist-info/LICENSE.txt
inflating: aws/dist/wheel-0.45.1.dist-info/RECORD
inflating: aws/dist/wheel-0.45.1.dist-info/REQUESTED
You can now run: /usr/local/bin/aws --version
aws-cli/2.32.24 Python/3.13.11 Linux/6.8.0-1030-azure exe/x86_64.ubuntu.24

```

```

cat ~/.aws/config
[default]
aws_access_key_id = AKIAZCN5ZS5DATBPGP50
aws_secret_access_key = N5eC7zbrdor1vNf3KxUHFMc150p/d7VP7Q9s3Vgy
[default]
region = me-central-1
output = json

```

```

{
  "UserId": "623705167686",
  "Account": "623705167686",
  "Arn": "arn:aws:iam::623705167686:root"
}

```

```

spring.pgk
echo "deb [arch=$(dpkg --print-architecture) signed-by=/usr/share/keyrings/hashicorp-archive-keyring.gpg] \
https://apt.releases.hashicorp.com $(cat /dev/null) main" | \
sudo tee /etc/apt/sources.list.d/hashicorp.list

sudo apt update
sudo apt install terraform
which terraform
terraform --version
2025-12-28 13:00:41-- https://apt.releases.hashicorp.com/gpg
Resolving apt.releases.hashicorp.com (apt.releases.hashicorp.com)... 18.172.78.80, 18.172.78.65, 18.172.78.12, ...
Connecting to apt.releases.hashicorp.com (apt.releases.hashicorp.com)[18.172.78.80]:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 3980 (3.9K) [binary/octet-stream]
Saving to: 'STDOUT'

- 100%[=====] 3.98K --48.9/s in 0s

2025-12-28 13:00:41 (938 MB/s) = written to stdout [3980/3980]

deb [arch=amd64 signed-by=/usr/share/keyrings/hashicorp-archive-keyring.gpg] https://apt.releases.hashicorp.com noble main
Get:1 https://apt.releases.hashicorp.com noble InRelease [12.9 kB]
Get:2 https://apt.releases.hashicorp.com noble/main amd64 Packages [166 kB]
Get:3 https://dl.yarnpkg.com/debian stable InRelease
Get:4 https://repo.anaconda.com/pkgs/misc/debrapa/conda stable InRelease [3961 B]
Get:5 https://packages.microsoft.com/repos/microsoft-ubuntu-noble-prod noble InRelease [3600 B]
Get:6 https://repo.anaconda.com/pkgs/misc/debrapa/conda stable/main amd64 Packages [4557 B]
Get:7 https://packages.microsoft.com/repos/microsoft-ubuntu-noble-prod noble/main all Packages [643 B]
Get:8 https://packages.microsoft.com/repos/microsoft-ubuntu-noble-prod noble/main amd64 Packages [77.3 kB]
Get:9 https://dl.yarnpkg.com/debian stable/main amd64 Packages [11.6 kB]
Get:10 https://dl.yarnpkg.com/debian stable/main all Packages [11.6 kB]
Get:11 http://archive.ubuntu.com/ubuntu noble InRelease [256 kB]
Get:12 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:13 http://archive.ubuntu.com/ubuntu noble-updates InRelease [116 kB]
Get:14 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1712 kB]
Get:15 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:16 http://archive.ubuntu.com/ubuntu noble/universe amd64 Packages [331 kB]
Get:17 http://archive.ubuntu.com/ubuntu noble/universe amd64 Packages [39.3 MB]
Get:18 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [33.1 kB]
Get:19 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [1183 kB]

```

```

Get:19 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [1183 kB]
Get:20 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [2898 kB]
Get:21 http://archive.ubuntu.com/ubuntu noble/main amd64 Packages [1808 kB]
Get:22 http://archive.ubuntu.com/ubuntu noble/restricted amd64 Packages [117 kB]
Get:23 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [3059 kB]
Get:24 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1950 kB]
Get:25 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [2130 kB]
Get:26 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Packages [35.9 kB]
Get:27 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 Packages [49.5 kB]
Get:28 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Packages [34.6 kB]
Fetched 35.7 MB in 6s (6034 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
51 packages can be upgraded. Run 'apt list --upgradable' to see them.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  terraform
0 upgraded, 1 newly installed, 0 to remove and 51 not upgraded.
Need to get 30.6 MB of archives.
After this operation, 101 MB of additional disk space will be used.
Get:1 https://apt.releases.hashicorp.com noble/main amd64 terraform amd64 1.14.3-1 [30.6 MB]
Fetched 30.6 MB in 0s (84.4 MB/s)
Selecting previously unselected package terraform.
(Reading database ... 58629 files and directories currently installed.)
Preparing to unpack .../terraform_1.14.3-1_amd64.deb ...
Unpacking terraform (1.14.3-1) ...
Setting up terraform (1.14.3-1) ...
/usr/bin/terraform
Terraform v1.14.3
on linux_amd64

```

```

codespace@codespaces-76b767: /bin/mkdir
provider "aws" {
  shared_config_files    = ["~/.aws/config"]
  shared_credentials_files = ["~/.aws/credentials"]
}

```


Initializing the backend...

Initializing provider plugins...

- Finding latest version of hashicorp/aws...

- Installing hashicorp/aws v6.27.0...

- Installed hashicorp/aws v6.27.0 (signed by HashiCorp)

Terraform has created a lock file **.terraform.lock.hcl** to record the provider selections it made above. Include this file in your version control repository so that Terraform can guarantee to make the same selections by default when you run "terraform init" in the future.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see any changes that are required for your infrastructure. All Terraform commands should now work.

If you ever set or change modules or backend configuration for Terraform, rerun this command to reinitialize your working directory. If you forget, other commands will detect it and remind you to do so if necessary.

```
ls .terraform/
# This file is maintained automatically by "terraform init".
# Manual edits may be lost in future updates.

provider "registry.terraform.io/hashicorp/aws" {
  version = "6.27.0"
  hashes = [
    "h1:bixp2PSsP5ZGbczGCxcbsDn6lF5QFlUXlNroq9cdab4=",
    "zh:177a24b806c72e8484b5cabc93b2b38e3d770ae6f745a998b54d6619fd0e8129",
    "zh:4ac4a85c14fb868a3306b542e6a56c10bd6c6d5a67bc0c9b8f6a9060cf5f3be7",
    "zh:552652185bc85c8balda1d65dea47c454728a5c6839c458b6dcd3ce71c19ccfc",
    "zh:60284b8172d09aee91eae0856f09855eaf040ce3a58d6933602ae17c53f8ed04",
    "zh:6be38d156756ca61fb8e7c752cc5d769cd709686700ac4b230f40a6e95b5dbc9",
    "zh:7a409138fae4ef42e3a637e37cb9efedf96459e28a3c764fc4e855e8db9a7485",
    "zh:8070cf5224ed1ed3a3e9a59f7c30ff88bf071c7567165275d477c1738a56c064",
    "zh:894439ef340a9a79f69cd759e27ad11c7826adeca27be1b1ca82b3c9702fa300",
    "zh:89d035eebf08a97c89374ff06040955ddc09f275ecca609d0c9d58d149bef5cf",
    "zh:985b1145d724fc1f38369099e4a5087141885740fd6c0b1dbc492171e73c2e49",
    "zh:9b12af85486a96aedd8d7984b0ff811a4b42e3d88dad1a3fb4c0b580d04fa425",
    "zh:a80b47ae8d1475201c86bd94a5dcb9dd4da5e8b73102a90820b68b66b76d50fd",
    "zh:d3395be1556210f82199b9166a6b2e677cee9c4b67e96e63f6c3a98325ad7ab0",
    "zh:db0b869d09657f6f1e4110b56093c5fcdcf9dbdd97c020db1e577b239c0adcbee",
    "zh:ffc72e680370ae7c21f9bd3082c6317730df805c6797427839a6b6b7e9a26a01",
  ]
}
```

Task 3

```

provider "aws" {
  shared_config_files    = ["~/.aws/config"]
  shared_credentials_files = ["~/.aws/credentials"]
}

resource "aws_vpc" "development_vpc" {
  cidr_block = "10.0.0.0/16"
}

resource "aws_subnet" "dev_subnet_1" {
  vpc_id            = aws_vpc.development_vpc.id
  cidr_block        = "10.0.10.0/24"
  availability_zone = "me-central-1a"
}

```

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:

- + create

Terraform will perform the following actions:

```

# aws_subnet.dev_subnet_1 will be created
+ resource "aws_subnet" "dev_subnet_1" {
  + arn                                = (known after apply)
  + assign_ipv6_address_on_creation    = false
  + availability_zone                  = "me-central-1a"
  + availability_zone_id                = (known after apply)
  + cidr_block                         = "10.0.10.0/24"
  + enable_dns64                       = false
  + enable_resource_name_dns_a_record_on_launch = false
  + enable_resource_name_dns_aaaa_record_on_launch = false
  + id                                 = (known after apply)
  + ipv6_cidr_block_association_id     = (known after apply)
  + ipv6_native                        = false
  + map_public_ip_on_launch            = false
  + owner_id                           = (known after apply)
  + private_dns_hostname_type_on_launch = (known after apply)
  + region                             = "me-central-1"
  + tags_all                           = (known after apply)
  + vpc_id                             = (known after apply)
}

# aws_vpc.development_vpc will be created
+ resource "aws_vpc" "development_vpc" {
  + arn                = (known after apply)
  + cidr_block          = "10.0.0.0/16"
  + default_network_acl_id = (known after apply)
  + default_route_table_id = (known after apply)
  + default_security_group_id = (known after apply)
}

```

Enter a value: yes

```

aws_vpc.development_vpc: Creating...
aws_vpc.development_vpc: Creation complete after 2s [id=vpc-00e832dc4b5877ee2]
aws_subnet.dev_subnet_1: Creating...
aws_subnet.dev_subnet_1: Creation complete after 0s [id=subnet-0132ca84323b35596]

```

Apply complete! Resources: 2 added, 0 changed, 0 destroyed.


```
{
  "Subnets": [
    {
      "AvailabilityZoneId": "mec1-az1",
      "MapCustomerOwnedIpOnLaunch": false,
      "OwnerId": "623705167686",
      "AssignIpv6AddressOnCreation": false,
      "Ipv6CidrBlockAssociationSet": [],
      "SubnetArn": "arn:aws:ec2:me-central-1:623705167686:subnet/subnet-0132ca84323b35596",
      "EnableDns64": false,
      "Ipv6Native": false,
      "PrivateDnsNameOptionsOnLaunch": {
        "HostnameType": "ip-name",
        "EnableResourceNameDnsARecord": false,
        "EnableResourceNameDnsAAAARecord": false
      },
      "BlockPublicAccessStates": {
        "InternetGatewayBlockMode": "off"
      },
      "SubnetId": "subnet-0132ca84323b35596",
      "State": "available",
      "VpcId": "vpc-00e832dc4b5877ee2",
      "CidrBlock": "10.0.10.0/24",
      "AvailableIpAddressCount": 251,
      "AvailabilityZone": "me-central-1a",
      "DefaultForAz": false,
      "MapPublicIpOnLaunch": false
    }
  ]
}
```

```
{
  "Vpcs": [
    {
      "OwnerId": "623705167686",
      "InstanceTenancy": "default",
      "CidrBlockAssociationSet": [
        {
          "AssociationId": "vpc-cidr-assoc-0c45e37cf75768225",
          "CidrBlock": "10.0.0.0/16",
          "CidrBlockState": {
            "State": "associated"
          }
        }
      ],
      "IsDefault": false,
      "BlockPublicAccessStates": {
        "InternetGatewayBlockMode": "off"
      },
      "VpcId": "vpc-00e832dc4b5877ee2",
      "State": "available",
      "CidrBlock": "10.0.0.0/16",
      "DhcpOptionsId": "dopt-02dd4ed017908b331"
    }
  ]
}
```

Task 4

```

provider "aws" {
  shared_config_files    = ["~/.aws/config"]
  shared_credentials_files = ["~/.aws/credentials"]
}

resource "aws_vpc" "development_vpc" {
  cidr_block = "10.0.0.0/16"
}

resource "aws_subnet" "dev_subnet_1" {
  vpc_id            = aws_vpc.development_vpc.id
  cidr_block        = "10.0.10.0/24"
  availability_zone = "me-central-1a"
}

data "aws_vpc" "existing_vpc" {
  default = true
}

resource "aws_subnet" "dev_subnet_1_existing" {
  vpc_id            = data.aws_vpc.existing_vpc.id
  cidr_block        = "172.31.48.0/24"
  availability_zone = "me-central-1a"
}

```

```

data.aws_vpc.existing_vpc: Reading...
aws_vpc.development_vpc: Refreshing state... [id=vpc-0be832dc4b5877ee2]
data.aws_vpc.existing_vpc: Read complete after 0s [id=vpc-079a26ed9b9cb32f2]
aws_subnet.dev_subnet_1: Refreshing state... [id=subnet-0132ca84323b35596]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
+ Create

Terraform will perform the following actions:

# aws_subnet.dev_subnet_1_existing will be created
+ resource "aws_subnet" "dev_subnet_1_existing" {
  + arn                                     = (known after apply)
  + assign_ipv6_address_on_creation        = false
  + availability_zone                      = "me-central-1a"
  + availability_zone_id                   = (known after apply)
  + cidr_block                            = "172.31.48.0/24"
  + enable_dns64                          = false
  + enable_resource_name_dns_a_record_on_launch = false
  + enable_resource_name_dns_aaaa_record_on_launch = false
  + id                                     = (known after apply)
  + ipv6_cidr_block_association_id         = (known after apply)
  + ipv6_native                           = false
  + map_public_ip_on_launch               = false
  + owner_id                              = (known after apply)
  + private_dns_hostname_type_on_launch   = (known after apply)
  + region                                = "me-central-1"
  + tags_all                              = (known after apply)
  + vpc_id                                = "vpc-079a26ed9b9cb32f2"
}

Plan: 1 to add, 0 to change, 0 to destroy.

Do you want to perform these actions?
  Terraform will perform the actions described above.
  Only 'yes' will be accepted to approve.

Enter a value: yes

aws_subnet.dev_subnet_1_existing: Creating...
aws_subnet.dev_subnet_1_existing: Creation complete after 1s [id=subnet-07f2eeb9f45745f76]

Apply complete! Resources: 1 added, 0 changed, 0 destroyed.

```

```

terraform refresh
terraform apply
terraform destroy
terraform plan
terraform apply
data.aws_vpc.existing_vpc: Reading...
data.aws_vpc.existing_vpc: Read complete after 0s [id=vpc-079a26ed9b9cb32f2]
aws_subnet.dev_subnet_1_existing: Refreshing state... [id=subnet-07f2eeb9f45745f76]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
- destroy

Terraform will perform the following actions:

# aws_subnet.dev_subnet_1_existing will be destroyed
- resource "aws_subnet" "dev_subnet_1_existing" {
  + arn                                = "arn:aws:ec2:me-central-1:623705167686:subnet/subnet-07f2eeb9f45745f76" -> null
  + assign_ipv6_address_on_creation    = false -> null
  + availability_zone                  = "me-central-1a" -> null
  + availability_zone_id                = "mecl-a21" -> null
  + cidr_block                         = "172.31.48.0/24" -> null
  + enable_dns64                       = false -> null
  + enable_ini_at_device_index         = 0 -> null
  + enable_resource_name_dns_a_record_on_launch = false -> null
  + enable_resource_name_dns_aaaa_record_on_launch = false -> null
  + id                                 = "subnet-07f2eeb9f45745f76" -> null
  + ipv6_native                        = false -> null
  + map_customer_owned_ip_on_launch    = false -> null
  + map_public_ip_on_launch            = false -> null
  + owner_id                           = "623705167686" -> null
  + private_dns_hostname_type_on_launch = "ip-name" -> null
  + region                             = "me-central-1" -> null
  + tags                               = {} -> null
  + tags_all                           = {} -> null
}

```

```

Destroy complete! Resources: 1 destroyed.
data.aws_vpc.existing_vpc: Reading...
aws_vpc.development_vpc: Refreshing state... [id=vpc-00e832dc4b5877ee2]
data.aws_vpc.existing_vpc: Read complete after 1s [id=vpc-079a26ed9b9cb32f2]
aws_subnet.dev_subnet_1: Refreshing state... [id=subnet-0132ca84323b35596]
data.aws_vpc.existing_vpc: Reading...
aws_vpc.development_vpc: Refreshing state... [id=vpc-00e832dc4b5877ee2]
data.aws_vpc.existing_vpc: Read complete after 1s [id=vpc-079a26ed9b9cb32f2]
aws_subnet.dev_subnet_1: Refreshing state... [id=subnet-0132ca84323b35596]

```

```

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
+ create

Terraform will perform the following actions:

# aws_subnet.dev_subnet_1_existing will be created
+ resource "aws_subnet" "dev_subnet_1_existing" {
  + arn                                = (known after apply)
  + assign_ipv6_address_on_creation    = false
  + availability_zone                  = "me-central-1a"
  + availability_zone_id                = (known after apply)
  + cidr_block                         = "172.31.48.0/24"
  + enable_dns64                       = false
  + enable_resource_name_dns_a_record_on_launch = false
  + enable_resource_name_dns_aaaa_record_on_launch = false
  + id                                 = (known after apply)
  + ipv6_cidr_block_association_id     = (known after apply)
  + ipv6_native                        = false
  + map_customer_owned_ip_on_launch    = false
  + map_public_ip_on_launch            = (known after apply)
  + owner_id                           = (known after apply)
  + private_dns_hostname_type_on_launch = (known after apply)
  + region                             = "me-central-1"
  + tags                               = (known after apply)
  + tags_all                           = (known after apply)
  + vpc_id                             = "vpc-079a26ed9b9cb32f2"
}

Plan: 1 to add, 0 to change, 0 to destroy.

Do you want to perform these actions?
  Terraform will perform the actions described above.
  Only 'yes' will be accepted to approve.

Enter a value: yes

aws_subnet.dev_subnet_1_existing: Creating...
aws_subnet.dev_subnet_1_existing: Creation complete after 0s [id=subnet-03a58fana503e156a]

Apply complete! Resources: 1 added, 0 changed, 0 destroyed.
data.aws_vpc.existing_vpc: Reading...

```

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
- destroy

Terraform will perform the following actions:

```
# aws_subnet.dev_subnet_1 will be destroyed
- resource "aws_subnet" "dev_subnet_1" {
  - arn                                = "arn:aws:ec2:me-central-1:623705167686:subnet/subnet-0132ca84323b35596" -> null
  - assign_ipv6_address_on_creation    = false -> null
  - availability_zone                  = "me-central-1a" -> null
  - availability_zone_id               = "mecl-a21" -> null
  - cidr_block                         = "10.0.10.0/24" -> null
  - enable_dns64                      = false -> null
  - enable_ini_at_device_index        = 0 -> null
  - enable_resource_name_dns_a_record_on_launch = false -> null
  - enable_resource_name_dns_aaaa_record_on_launch = false -> null
  - id                                = "subnet-0132ca84323b35596" -> null
  - ipv6_native                       = false -> null
  - map_customer_owned_ip_on_launch   = false -> null
  - map_public_ip_on_launch           = false -> null
  - owner_id                          = "623705167686" -> null
  - private_dns_hostnames_type_on_launch = "ip-name" -> null
  - region                            = "me-central-1" -> null
  - tags                              = {} -> null
  - tags_all                          = {} -> null
  - vpc_id                            = "vpc-00e832dc4b5877ee2" -> null
  # (4 unchanged attributes hidden)
}

# aws_subnet.dev_subnet_1_existing will be destroyed
- resource "aws_subnet" "dev_subnet_1_existing" {
  - arn                                = "arn:aws:ec2:me-central-1:623705167686:subnet/subnet-03a88faa503e156a" -> null
  - assign_ipv6_address_on_creation    = false -> null
  - availability_zone                  = "me-central-1a" -> null
  - availability_zone_id               = "mecl-a21" -> null
  - cidr_block                         = "172.31.48.0/24" -> null
  - enable_dns64                      = false -> null
  - enable_ini_at_device_index        = 0 -> null
  - enable_resource_name_dns_a_record_on_launch = false -> null
  - enable_resource_name_dns_aaaa_record_on_launch = false -> null
  - id                                = "subnet-03a88faa503e156a" -> null
}
```

```
# aws_vpc.development_vpc will be destroyed
- resource "aws_vpc" "development_vpc" {
  - arn                                = "arn:aws:ec2:me-central-1:623705167686:vpc/vpc-00e832dc4b5877ee2" -> null
  - assign_generated_ipv6_cidr_block   = false -> null
  - cidr_block                         = "10.0.0.0/16" -> null
  - default_network_acl_id             = "acl-00182c8af491379a9" -> null
  - default_route_table_id             = "rtb-0fface4094c2af403" -> null
  - default_security_group_id          = "sg-0a7f615fdebcac670" -> null
  - dhcp_options_id                   = "dopt-02dd4ed017908b331" -> null
  - enable_dns_hostnames               = false -> null
  - enable_dns_support                 = true -> null
  - enable_network_address_usage_metrics = false -> null
  - id                                = "vpc-00e832dc4b5877ee2" -> null
  - instance_tenancy                   = "default" -> null
  - ipv6_netmask_length                = 0 -> null
  - main_route_table_id                = "rtb-0fface4094c2af403" -> null
  - owner_id                          = "623705167686" -> null
  - region                            = "me-central-1" -> null
  - tags                              = {} -> null
  - tags_all                          = {} -> null
  # (4 unchanged attributes hidden)
}
```

Plan: 0 to add, 0 to change, 3 to destroy.

Do you really want to destroy all resources?

Terraform will destroy all your managed infrastructure, as shown above.
There is no undo. Only 'yes' will be accepted to confirm.

Enter a value: yes

```
aws_subnet.dev_subnet_1_existing: Destroying... [id=subnet-03a88faa503e156a]
aws_subnet.dev_subnet_1: Destroying... [id=subnet-0132ca84323b35596]
aws_subnet.dev_subnet_1_existing: Destruction complete after 1s
aws_subnet.dev_subnet_1: Destruction complete after 1s
aws_vpc.development_vpc: Destroying... [id=vpc-00e832dc4b5877ee2]
aws_vpc.development_vpc: Destruction complete after 1s
```

Destroy complete! Resources: 3 destroyed.

data.aws_vpc.existing_vpc: Reading...

data.aws_vpc.existing_vpc: Read complete after 0s [id=vpc-079a26ed9b9cb32f2]


```

Destroy complete! Resources: 3 destroyed.
data.aws_vpc.existing_vpc: Reading...
data.aws_vpc.existing_vpc: Read complete after 0s [id=vpc-079a26ed9b9cb32f2]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
+ create

```

Terraform will perform the following actions:

```

# aws_subnet.dev_subnet_1 will be created
+ resource "aws_subnet" "dev_subnet_1" {
  + arn                                = (known after apply)
  + assign_ipv6_address_on_creation    = false
  + availability_zone                  = "me-central-1a"
  + availability_zone_id               = (known after apply)
  + cidr_block                         = "10.0.10.0/24"
  + enable_dns64                       = false
  + enable_resource_name_dns_a_record_on_launch = false
  + enable_resource_name_dns_aaaa_record_on_launch = false
  + id                                 = (known after apply)
  + ipv6_cidr_block_association_id     = (known after apply)
  + ipv6_native                        = false
  + map_public_ip_on_launch           = false
  + owner_id                           = (known after apply)
  + private_dns_hostname_type_on_launch = (known after apply)
  + region                             = "me-central-1"
  + tags_all                           = (known after apply)
  + vpc_id                             = (known after apply)
}

# aws_subnet.dev_subnet_1_existing will be created
+ resource "aws_subnet" "dev_subnet_1_existing" {
  + arn                                = (known after apply)
  + assign_ipv6_address_on_creation    = false
  + availability_zone                  = "me-central-1a"
  + availability_zone_id               = (known after apply)
  + cidr_block                         = "172.31.48.0/24"
  + enable_dns64                       = false
  + enable_resource_name_dns_a_record_on_launch = false
  + enable_resource_name_dns_aaaa_record_on_launch = false
  + id                                 = (known after apply)
}

```

```

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
+ create

```

Terraform will perform the following actions:

```

# aws_subnet.dev_subnet_1 will be created
+ resource "aws_subnet" "dev_subnet_1" {
  + arn                                = (known after apply)
  + assign_ipv6_address_on_creation    = false
  + availability_zone                  = "me-central-1a"
  + availability_zone_id               = (known after apply)
  + cidr_block                         = "10.0.10.0/24"
  + enable_dns64                       = false
  + enable_resource_name_dns_a_record_on_launch = false
  + enable_resource_name_dns_aaaa_record_on_launch = false
  + id                                 = (known after apply)
  + ipv6_cidr_block_association_id     = (known after apply)
  + ipv6_native                        = false
  + map_public_ip_on_launch           = false
  + owner_id                           = (known after apply)
  + private_dns_hostname_type_on_launch = (known after apply)
  + region                             = "me-central-1"
  + tags_all                           = (known after apply)
  + vpc_id                             = (known after apply)
}

# aws_subnet.dev_subnet_1_existing will be created
+ resource "aws_subnet" "dev_subnet_1_existing" {
  + arn                                = (known after apply)
  + assign_ipv6_address_on_creation    = false
  + availability_zone                  = "me-central-1a"
  + availability_zone_id               = (known after apply)
  + cidr_block                         = "172.31.48.0/24"
  + enable_dns64                       = false
  + enable_resource_name_dns_a_record_on_launch = false
  + enable_resource_name_dns_aaaa_record_on_launch = false
  + id                                 = (known after apply)
  + ipv6_cidr_block_association_id     = (known after apply)
  + ipv6_native                        = false
  + map_public_ip_on_launch           = false
  + owner_id                           = (known after apply)
  + private_dns_hostname_type_on_launch = (known after apply)
  + region                             = "me-central-1"
}

```

Plan: 3 to add, 0 to change, 0 to destroy.

Do you want to perform these actions?

Terraform will perform the actions described above.
Only 'yes' will be accepted to approve.

Enter a value: yes

```
aws_vpc.development_vpc: Creating...
aws_subnet.dev_subnet_1_existing: Creating...
aws_subnet.dev_subnet_1_existing: Creation complete after 1s [id=subnet-056d280c40cb85779]
aws_vpc.development_vpc: Creation complete after 1s [id=vpc-08bae3f872bd21535]
aws_subnet.dev_subnet_1: Creating...
aws_subnet.dev_subnet_1: Creation complete after 1s [id=subnet-02a9b9c4a3211c350]

Apply complete! Resources: 3 added, 0 changed, 0 destroyed.
```

```
provider "aws" {
  shared_config_files   = ["~/.aws/config"]
  shared_credentials_files = ["~/.aws/credentials"]
}

resource "aws_vpc" "development_vpc" {
  cidr_block = "10.0.0.0/16"
  tags = {
    Name      = "development"
    vpc_env   = "dev"
  }
}
```

```
terraform apply -auto-approve
aws_vpc.development_vpc: Refreshing state... [id=vpc-08bae3f872bd21535]
aws_subnet.dev_subnet_1: Refreshing state... [id=subnet-02a9b9c4a3211c350]
aws_subnet.dev_subnet_1_existing: Refreshing state... [id=subnet-056d280c40cb85779]
aws_vpc.development_vpc: Refreshing state... [id=vpc-08bae3f872bd21535]
aws_subnet.dev_subnet_1_existing: Refreshing state... [id=subnet-056d280c40cb85779]
aws_subnet.dev_subnet_1: Refreshing state... [id=subnet-02a9b9c4a3211c350]
```

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:

- update in-place
- destroy

Terraform will perform the following actions:

```
# aws_subnet.dev_subnet_1 will be destroyed
# (because aws_subnet.dev_subnet_1 is not in configuration)
- resource "aws_subnet" "dev_subnet_1" {
  - arn                                     = "arn:aws:ec2:me-central-1:623705167686:subnet/subnet-02a9b9c4a3211c350" -> null
  - assign_ipv6_address_on_creation        = false -> null
  - availability_zone                     = "me-central-1a" -> null
  - availability_zone_id                  = "mecl-azi" -> null
  - cidr_block                           = "10.0.10.0/24" -> null
  - enable_dns64                         = false -> null
  - enable_ni_at_device_index            = 0 -> null
  - enable_resource_name_dns_a_record_on_launch = false -> null
  - enable_resource_name_dns_aaaa_record_on_launch = false -> null
  - id                                    = "subnet-02a9b9c4a3211c350" -> null
  - ipv6_native                          = false -> null
  - map_customer_owned_ip_on_launch      = false -> null
  - map_public_ip_on_launch              = false -> null
  - owner_id                             = "623705167686" -> null
  - private_dns_hostname_type_on_launch  = "ip-name" -> null
  - region                               = "me-central-1" -> null
  - tags                                 = {} -> null
  - tags_all                             = {} -> null
  - vpc_id                               = "vpc-08bae3f872bd21535" -> null
  # (4 unchanged attributes hidden)
}
```

```
# aws_subnet.dev_subnet_1_existing will be destroyed
# (because aws_subnet.dev_subnet_1_existing is not in configuration)
```



```

# (because aws_subnet.dev_subnet_1_existing is not in configuration)
- resource "aws_subnet" "dev_subnet_1_existing" {
  - arn = "arn:aws:ec2:me-central-1:623705167686:subnets/subnet-056d280c40cb85779"
  - assign_ipv6_address_on_creation = false -> null
  - availability_zone = "me-central-1a" -> null
  - availability_zone_id = "mec1-az1" -> null
  - cidr_block = "172.31.48.0/24" -> null
  - enable_dns64 = false -> null
  - enable_lni_at_device_index = 0 -> null
  - enable_resource_name_dns_a_record_on_launch = false -> null
  - enable_resource_name_dns_aaaa_record_on_launch = false -> null
  - id = "subnet-056d280c40cb85779" -> null
  - ipv6_native = false -> null
  - map_customer_owned_ip_on_launch = false -> null
  - map_public_ip_on_launch = false -> null
  - owner_id = "623705167686" -> null
  - private_dns_hostname_type_on_launch = "ip-name" -> null
  - region = "me-central-1" -> null
  - tags = {} -> null
  - tags_all = {} -> null
  - vpc_id = "vpc-079a26ed9b9cb32f2" -> null
  # (4 unchanged attributes hidden)
}

# aws_vpc.development_vpc will be updated in-place
- resource "aws_vpc" "development_vpc" {
  id = "vpc-08bae3f872bd21535"
  - tags = {
    + "Name" = "development"
    + "vpc_env" = "dev"
  }
  - tags_all = {
    + "Name" = "development"
    + "vpc_env" = "dev"
  }
  # (19 unchanged attributes hidden)
}

```

Plan: 0 to add, 1 to change, 2 to destroy.

aws_subnet.dev_subnet_1_existing: Destroying... [id=subnet-056d280c40cb85779]

aws_subnet.dev_subnet_1: Destroying... [id=subnet-02a9b9c4a3211c350]

aws_subnet.dev_subnet_1_existing: Destruction complete after 1s

aws_subnet.dev_subnet_1: Destruction complete after 1s

aws_vpc.development_vpc: Modifying... [id=vpc-08bae3f872bd21535]

aws_vpc.development_vpc: Modifications complete after 1s [id=vpc-08bae3f872bd21535]

Apply complete! Resources: 0 added, 1 changed, 2 destroyed.

```

terraform apply
aws_vpc.development_vpc: Refreshing state... [id=vpc-08bae3f872bd21535]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
~ update in-place

Terraform will perform the following actions:

# aws_vpc.development_vpc will be updated in-place
~ resource "aws_vpc" "development_vpc" {
  id          = "vpc-08bae3f872bd21535"
  ~ tags      = {
    "Name"      = "development"
    "vpc_env"   = "dev" -> null
  }
  ~ tags_all  = {
    "vpc_env"   = "dev" -> null
    # (1 unchanged element hidden)
  }
  # (19 unchanged attributes hidden)
}

Plan: 0 to add, 1 to change, 0 to destroy.

Note: You didn't use the -out option to save this plan, so Terraform can't guarantee to take exactly these actions if you run "terraform apply" now.
aws_vpc.development_vpc: Refreshing state... [id=vpc-08bae3f872bd21535]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
~ update in-place

Terraform will perform the following actions:

# aws_vpc.development_vpc will be updated in-place
~ resource "aws_vpc" "development_vpc" {
  id          = "vpc-08bae3f872bd21535"
  ~ tags      = {
    "Name"      = "development"
    "vpc_env"   = "dev" -> null
  }
  ~ tags_all  = {
    "vpc_env"   = "dev" -> null
    # (1 unchanged element hidden)
  }
  # (19 unchanged attributes hidden)
}

Note: You didn't use the -out option to save this plan, so Terraform can't guarantee to take exactly these actions if you run "terraform apply" now.
aws_vpc.development_vpc: Refreshing state... [id=vpc-08bae3f872bd21535]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
~ update in-place

Terraform will perform the following actions:

# aws_vpc.development_vpc will be updated in-place
~ resource "aws_vpc" "development_vpc" {
  id          = "vpc-08bae3f872bd21535"
  ~ tags      = {
    "Name"      = "development"
    "vpc_env"   = "dev" -> null
  }
  ~ tags_all  = {
    "vpc_env"   = "dev" -> null
    # (1 unchanged element hidden)
  }
  # (19 unchanged attributes hidden)
}

Plan: 0 to add, 1 to change, 0 to destroy.

Do you want to perform these actions?
  Terraform will perform the actions described above.
  Only 'yes' will be accepted to approve.

Enter a value: yes

aws_vpc.development_vpc: Modifying... [id=vpc-08bae3f872bd21535]
aws_vpc.development_vpc: Modifications complete after 0s [id=vpc-08bae3f872bd21535]

Apply complete! Resources: 0 added, 1 changed, 0 destroyed

```

Task 5

```

cat terraform.tfstate
cat terraform.tfstate.backup
terraform apply
cat terraform.tfstate
cat terraform.tfstate.backup
terraform state list
terraform state show aws_vpc.development_vpc
aws_vpc.development_vpc: Refreshing state... [id=vpc-08bae3f872bd21535]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
- destroy

Terraform will perform the following actions:

# aws_vpc.development_vpc will be destroyed
- resource "aws_vpc" "development_vpc" {
  - arn                                = "arn:aws:ec2:me-central-1:623705167686:vpc/vpc-08bae3f872bd21535" -> null
  - assign_generated_ipv6_cidr_block   = false -> null
  - cidr_block                         = "10.0.0.0/16" -> null
  - default_network_acl_id            = "acl-03c0a6ba5c9b37a31" -> null
  - default_route_table_id            = "rtb-01326304ae881fa22" -> null
  - default_security_group_id         = "sg-03094d41a61ebc8ef" -> null
  - dhcp_options_id                   = "dopt-02dd4ed017908b331" -> null
  - enable_dns_hostnames               = false -> null
  - enable_dns_support                 = true -> null
  - enable_network_address_usage_metrics = false -> null
  - id                                = "vpc-08bae3f872bd21535" -> null
  - instance_tenancy                   = "default" -> null
  - ipv6_netmask_length                = 0 -> null
  - main_route_table_id                = "rtb-01326304ae881fa22" -> null
  - owner_id                           = "623705167686" -> null
  - region                             = "me-central-1" -> null
  - tags                               = {
    - "Name" = "development"
  } -> null
  - tags_all                           = {
    - "Name" = "development"
  } -> null
  # (4 unchanged attributes hidden)
}

Plan: 0 to add, 0 to change, 1 to destroy.

Do you really want to destroy all resources?
  Terraform will destroy all your managed infrastructure, as shown above.
  There is no undo. Only 'yes' will be accepted to confirm.

```

```

Do you really want to destroy all resources?
  Terraform will destroy all your managed infrastructure, as shown above.
  There is no undo. Only 'yes' will be accepted to confirm.

```

Enter a value: yes

```

aws_vpc.development_vpc: Destroying... [id=vpc-08bae3f872bd21535]
aws_vpc.development_vpc: Destruction complete after 0s

```

Destroy complete! Resources: 1 destroyed.

```

{
  "version": 4,
  "terraform_version": "1.14.3",
  "serial": 27,
  "lineage": "3b3c750a-2506-d918-7465-edae5f5918f3",
  "outputs": {},
  "resources": [],
  "check_results": null
}

{
  "version": 4,
  "terraform_version": "1.14.3",
  "serial": 25,
  "lineage": "3b3c750a-2506-d918-7465-edae5f5918f3",
  "outputs": {},
  "resources": [
    {
      "mode": "managed",
      "type": "aws_vpc",
      "name": "development_vpc",
      "provider": "provider[\"registry.terraform.io/hashicorp/aws\"]",
      "instances": [
        {
          "schema_version": 1,
          "attributes": {
            "arn": "arn:aws:ec2:me-central-1:623705167686:vpc/vpc-08bae3f872bd21535",
            "assign_generated_ipv6_cidr_block": false,
            "cidr_block": "10.0.0.0/16",
            "default_network_acl_id": "acl-03c0a6ba5c9b37a31",
            "default_route_table_id": "rtb-01326304ae881fa22",
            "default_security_group_id": "sg-03094d41a61ebc8ef",
            "dhcp_options_id": "dopt-02dd4ed017908b331",
            "enable_dns_hostnames": false,
            "enable_dns_support": true,
            "enable_network_address_usage_metrics": false,
            "id": "vpc-08bae3f872bd21535",
            "instance_tenancy": "default",

```

```

    "ipv6_netmask_length": null,
    "ipv6_association_id": "",
    "ipv6_cidr_block": "",
    "ipv6_cidr_block_network_border_group": "",
    "ipv6_ipam_pool_id": "",
    "ipv6_netmask_length": 0,
    "main_route_table_id": "rtb-01326304ae881fa22",
    "owner_id": "623705167686",
    "region": "us-central-1",
    "tags": {
      "Name": "development"
    },
    "tags_all": {
      "Name": "development"
    }
  },
  "sensitive_attributes": [],
  "identity_schema_version": 0,
  "identity": {
    "account_id": "623705167686",
    "id": "vpc-08bba3f872bd21535",
    "region": "us-central-1"
  },
  "private": "eyJ2Y2h1bWVyc2lvdjE6IjEtfQ=="
}
}
}
"check_results": null

```

terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:

- + create

terraform will perform the following actions:

```

# aws_vpc.development_vpc will be created
+ resource "aws_vpc" "development_vpc" {
  + arn                               = (known after apply)
  + cidr_block                        = "10.0.0.0/16"
  + default_network_acl_id           = (known after apply)
  + default_route_table_id           = (known after apply)
  + default_security_group_id        = (known after apply)
  + dhcp_options_id                  = (known after apply)
  + enable_dns_hostnames              = (known after apply)
  + enable_dns_support                = true
  + enable_network_address_usage_metrics = (known after apply)
  + id                               = (known after apply)

```

```

+ ipv6_cidr_block_network_border_group = (known after apply)
+ main_route_table_id                  = (known after apply)
+ owner_id                             = (known after apply)
+ region                               = "me-central-1"
+ tags                                 = {
+   + "Name" = "development"
+ }
+ tags_all                             = {
+   + "Name" = "development"
+ }
}

```

Plan: 1 to add, 0 to change, 0 to destroy.

Do you want to perform these actions?

Terraform will perform the actions described above.
Only 'yes' will be accepted to approve.

Enter a value: yes

aws_vpc.development_vpc: Creating...

aws_vpc.development_vpc: Creation complete after 2s [id=vpc-064ef4080514a164e]

Apply complete! Resources: 1 added, 0 changed, 0 destroyed.

```

{
  "version": 4,
  "terraform_version": "1.14.3",
  "serial": 29,
  "lineage": "3b3c750a-2506-d918-7465-edae5f5918f3",
  "outputs": {},
  "resources": [
    {
      "mode": "managed",
      "type": "aws_vpc",
      "name": "development_vpc",
      "provider": "provider[\"registry.terraform.io/hashicorp/aws\"]",
      "instances": [
        {
          "schema_version": 1,
          "attributes": {
            "arn": "arn:aws:ec2:me-central-1:623705167686:vpc/vpc-064ef4080514a164e",
            "assign_generated_ipv6_cidr_block": false,
            "cidr_block": "10.0.0.0/16",
            "default_network_acl_id": "acl-0d19d823cb0f44c17",
            "default_route_table_id": "rtb-04b6a67794e4345da",
            "default_security_group_id": "sg-0a22cbdcaadb4ba55",
            "dhcp_options_id": "dopt-02dd4ed017908b331",
            "enable_dns_hostnames": false,

```


Task 6

Outputs:

```
dev-subnet-cidr_block = "10.0.10.0/24"
dev-subnet-region = "me-central-1a"
dev-subnet-tags_all = tomap({
  "Name" = "subnet-1-dev"
})
dev-subnet-tags_name = "subnet-1-dev"
dev-vpc-cidr_block = "10.0.0.0/16"
dev-vpc-region = "me-central-1"
dev-vpc-tags_all = tomap({
  "Name" = "development"
})
dev-vpc-tags_name = "development"
```

Clean up

```
cat terraform.tfstate
cat terraform.tfstate.backup
aws_vpc.development_vpc: Refreshing state... [id=vpc-064ef4080514a164e]
aws_subnet.dev_subnet_1: Refreshing state... [id=subnet-0af87dafdb1dc4f4]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
  ~ destroy

Terraform will perform the following actions:

# aws_subnet.dev_subnet_1 will be destroyed
+ resource "aws_subnet" "dev_subnet_1" {
  ~ arn                                = "arn:aws:ec2:me-central-1:623705167686:subnet/subnet-0af87dafdb1dc4f4" -> null
  ~ assign_ipv6_address_on_creation    = false -> null
  ~ availability_zone                  = "me-central-1a" -> null
  ~ availability_zone_id               = "ec2-ez1" -> null
  ~ cidr_block                         = "10.0.10.0/24" -> null
  ~ enable_dns64                       = false -> null
  ~ enable_internet_gateway_attachment = 0 -> null
  ~ enable_resource_name_dns_a_record_on_launch = false -> null
  ~ enable_resource_name_dns_saaa_record_on_launch = false -> null
  ~ id                                = "subnet-0af87dafdb1dc4f4" -> null
  ~ ipv6_native                        = false -> null
  ~ map_customer_owned_ip_on_launch    = false -> null
  ~ map_public_ip_on_launch            = false -> null
  ~ owner_id                           = "623705167686" -> null
  ~ private_dns_hostname_type_on_launch = "ip-name" -> null
  ~ region                             = "me-central-1" -> null
  ~ tags                               = {
    ~ "Name" = "subnet-1-dev"
  } -> null
  ~ tags_all                           = {
    ~ "Name" = "subnet-1-dev"
  } -> null
  ~ vpc_id                             = "vpc-064ef4080514a164e" -> null
  ~ # 4 unchanged attributes hidden
}

# aws_vpc.development_vpc will be destroyed
```

```
# aws_vpc.development_vpc will be destroyed
- resource "aws_vpc" "development_vpc" {
  - arn = "arn:aws:ec2:me-central-1:623705167686:vpc/vpc-064ef4080514a164e" -> null
  - assign_generated_ipv6_cidr_block = false -> null
  - cidr_block = "10.0.0.0/16" -> null
  - default_network_acl_id = "acl-0d19d823cb0f44e17" -> null
  - default_route_table_id = "rtb-04b6a67794e4345da" -> null
  - default_security_group_id = "sg-0a22cbdcadd4ba55" -> null
  - dhcp_options_id = "dopt-02dd4ed017908b331" -> null
  - enable_dns_hostnames = false -> null
  - enable_dns_support = true -> null
  - enable_network_address_usage_metrics = false -> null
  - id = "vpc-064ef4080514a164e" -> null
  - instance_tenancy = "default" -> null
  - ipv6_netmask_length = 0 -> null
  - main_route_table_id = "rtb-04b6a67794e4345da" -> null
  - owner_id = "623705167686" -> null
  - region = "me-central-1" -> null
  - tags = {
    - "Name" = "development"
  } -> null
  - tags_all = {
    - "Name" = "development"
  } -> null
  # (4 unchanged attributes hidden)
}
```

Plan: 0 to add, 0 to change, 2 to destroy.

Changes to Outputs:

```
- dev-subnet-cidr_block = "10.0.10.0/24" -> null
- dev-subnet-region = "me-central-1a" -> null
- dev-subnet-tags_all = {
  - Name = "subnet-1-dev"
} -> null
- dev-subnet-tags_name = "subnet-1-dev" -> null
- dev-vpc-cidr_block = "10.0.0.0/16" -> null
- dev-vpc-region = "me-central-1" -> null
- dev-vpc-tags_all = {
  - Name = "development"
} -> null
- dev-vpc-tags_name = "development" -> null
```

Do you really want to destroy all resources?

Terraform will destroy all your managed infrastructure, as shown above.
There is no undo. Only 'yes' will be accepted to confirm.

Enter a value: yes

```
aws_subnet.dev_subnet_1: Destroying... [id=subnet-0af87dafd1b1dc4f4]
aws_subnet.dev_subnet_1: Destruction complete after 1s
aws_vpc.development_vpc: Destroying... [id=vpc-064ef4080514a164e]
aws_vpc.development_vpc: Destruction complete after 1s
```

Destroy complete! Resources: 2 destroyed.

```
{
  "version": 4,
  "terraform_version": "1.14.3",
  "serial": 34,
  "lineage": "3b3c750a-2506-d918-7465-edae5f5918f3",
  "outputs": {},
  "resources": [],
  "check_results": null
}
{
  "version": 4,
  "terraform_version": "1.14.3",
  "serial": 31,
  "lineage": "3b3c750a-2506-d918-7465-edae5f5918f3",
  "outputs": {
    "dev-subnet-cidr_block": {
      "value": "10.0.10.0/24",
      "type": "string"
    },
    "dev-subnet-region": {
      "value": "me-central-1a",
      "type": "string"
    },
    "dev-subnet-tags_all": {
      "value": {
        "Name": "subnet-1-dev"
      },
      "type": [
        "map",
        "string"
      ]
    },
    "dev-subnet-tags_name": {
      "value": "subnet-1-dev",
      "type": "string"
    },
    "dev-vpc-cidr_block": {
      "value": "10.0.0.0/16",
      "type": "string"
    },
    "dev-vpc-region": {
      "value": "me-central-1",

```



```

    "type": "aws_vpc",
    "name": "development_vpc",
    "provider": "provider[\"registry.terraform.io/hashicorp/aws\"]",
    "instances": [
      {
        "schema_version": 1,
        "attributes": {
          "arn": "arn:aws:ec2:me-central-1:623705167686:vpc/vpc-064ef4080514a164e",
          "assign_generated_ipv6_cidr_block": false,
          "cidr_block": "10.0.0.0/16",
          "default_network_acl_id": "acl-0d19d823cb0f44c17",
          "default_route_table_id": "rtb-04b6a67794e4345da",
          "default_security_group_id": "sg-0a22cbdcaadb4ba55",
          "dhcp_options_id": "dopt-02dd4ed017908b331",
          "enable_dns_hostnames": false,
          "enable_dns_support": true,
          "enable_network_address_usage_metrics": false,
          "id": "vpc-064ef4080514a164e",
          "instance_tenancy": "default",
          "ipv4_ipam_pool_id": null,
          "ipv4_netmask_length": null,
          "ipv6_association_id": "",
          "ipv6_cidr_block": "",
          "ipv6_cidr_block_network_border_group": "",
          "ipv6_ipam_pool_id": "",
          "ipv6_netmask_length": 0,
          "main_route_table_id": "rtb-04b6a67794e4345da",
          "owner_id": "623705167686",
          "region": "me-central-1",
          "tags": {
            "Name": "development"
          },
          "tags_all": {
            "Name": "development"
          }
        },
        "sensitive_attributes": [],
        "identity_schema_version": 0,
        "identity": {
          "account_id": "623705167686",
          "id": "vpc-064ef4080514a164e",
          "region": "me-central-1"
        },
        "private": "eyJzY2h1bWFFdmVyc2lubiI6IjEifQ=="
      }
    ]
  },
  "check_results": null
}

```

```

    },
    "sensitive_attributes": [],
    "identity_schema_version": 0,
    "identity": {
      "account_id": "623705167686",
      "id": "vpc-064ef4080514a164e",
      "region": "me-central-1"
    },
    "private": "eyJzY2h1bWFFdmVyc2lubiI6IjEifQ=="
  }
}
],
"check_results": null
}

```



FATIMA JINNAH WOMEN UNIVERSITY

Department of Software Engineering

Lab Title:	Experiment # :
Student Name:	Regd. #:
Experiment Name:	

Performance Indicator	Level of Achievements				
	Excellent (5)	Good (4)	Average (3)	Below Average (2)	Poor (1)
Understanding & implementation Problem					
Report and presentation					
Maximum Marks		10	Obtained Marks		
Lab Instructor:					

Lab 11

Task 0:

```
gh codespace create --repo 23-22412-038-sys/Lab11
  / Codespaces usage for this repository is paid for by 23-22412-038-sys
error getting devcontainer.json paths: HTTP 400: The 'ref' provided was invalid.
Please specify a valid branch name or commit SHA (https://api.github.com/repos/
itories/1125688696/codespaces/devcontainers?per_page=100&ref=main)

Alli@DESKTOP-KQE4P80 MINGW64 ~ (main)
$ gh codespace list


| NAME           | DISPLAY NAME   | REPOSITORY     | BRANCH | STATE    | CREATED AT     |
|----------------|----------------|----------------|--------|----------|----------------|
| sturdy-para... | sturdy para... | 23-22412-03... | main   | Shutdown | about 16 da... |
| obscure-zeb... | obscure zebra  | 23-22412-03... | main*  | Shutdown | about 16 da... |
| reimagined-... | reimagined ... | 23-22412-03... | main   | Shutdown | about 2 day... |
| ominous-waf... | ominous waffle | 23-22412-03... | main*  | Shutdown | about 2 day... |



Alli@DESKTOP-KQE4P80 MINGW64 ~ (main)
$ gh repo delete 23-22412-038-sys/Lab11 --yes
HTTP 403: Must have admin rights to Repository. (https://api.github.com/repos/23
-22412-038-sys/Lab11)
This API operation needs the "delete_repo" scope. To request it, run: gh auth r
efresh -h github.com -s delete_repo

Alli@DESKTOP-KQE4P80 MINGW64 ~ (main)
$ gh repo create Lab11 --public --add-readme
HTTP 422: Repository creation failed. (https://api.github.com/user/repos)
? Choose codespace: 23-22412-038-sys/Lab11 [main]: didactic invention
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-1030-azure x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

@23-22412-038-sys → /workspaces/Lab11 (main) $
```

Task 1

```
@23-22412-038-sys → /workspaces/Lab11 (main) $
@23-22412-038-sys → /workspaces/Lab11 (main) $ touch main.tf
@23-22412-038-sys → /workspaces/Lab11 (main) $
```

```
code@code: /workspaces/79c51b: nano main.tf
GNU nano 2.9.3
provider "aws" {
  region                  = "ae-central-1" # Middle East (UAE)
  shared_config_files     = ["~/.aws/config"]
  shared_credentials_files = ["~/.aws/credentials"]
}
```

```
@23-22412-038-sys → /workspaces/Lab11 (main) $ terraform init
Initializing the backend...
Initializing provider plugins...
- Finding latest version of hashicorp/aws...
- Installing hashicorp/aws v6.27.0...
- Installed hashicorp/aws v6.27.0 (signed by HashiCorp)
Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
```

```
code-server: /root/.ssh: nano main.tf
GNU nano 2.9.3
provider "aws" {
  shared_config_files = ["~/.aws/config"]
  shared_credentials_files = ["~/.aws/credentials"]
}
variable "subnet_cidr_block" {
  type = string
}

output "subnet_cidr_block_output" {
  value = var.subnet_cidr_block
}
```

File Name to write: main.tf

Help Cancel

Mac OS Format Mac Format

Append Prepend

Backup File Show

Download Lab 11 localhost results XAMPP OC F2! Git Ba

11:29 PM 12/30/2025

```
#21-22412-038-855 ~ /workspaces/Lab11 (main) $ terraform apply -auto-approve
var.subnet_cidr_block
Enter a value: 1

Changes to Outputs:
  + subnet_cidr_block_output = "1"

You can apply this plan to save these new output values to the Terraform state, without changing any real infrastructure.

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.

Outputs:
subnet_cidr_block_output = "1"
```

```
code-server: /workspace: 7951b: nano main.tf
GNU nano 2.2
provider "aws" {
  shared_config_files = ["~/aws/config"]
  shared_credentials_files = ["~/aws/credentials"]
}

variable "subnet_cidr_block" {
  type = string
  default = "10.0.0.0/24"
}

output "subnet_cidr_block_output" {
  value = var.subnet_cidr_block
}
```

```
k23-22412-038-sys + /workspaces/Lab11 (main) $ terraform apply -auto-approve

Changes to Outputs:
  - subnet_cidr_block_output = "1" -> "10.0.0.0/24"

You can apply this plan to save these new output values to the Terraform state, without changing any real infrastructure.

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.

Outputs:
subnet_cidr_block_output = "10.0.0.0/24"
```

```
k23-22412-038-sys + /workspaces/Lab11 (main) $ export TF_VAR_subnet_cidr_block=10.0.20.0/24
terraform apply -auto-approve

Changes to Outputs:
  - subnet_cidr_block_output = "10.0.0.0/24" -> "10.0.20.0/24"

You can apply this plan to save these new output values to the Terraform state, without changing any real infrastructure.

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.

Outputs:
subnet_cidr_block_output = "10.0.20.0/24"
```

```
k23-22412-038-sys + /workspaces/Lab11 (main) $ terraform apply -auto-approve

Changes to Outputs:
  - subnet_cidr_block_output = "10.0.20.0/24" -> "10.0.30.0/24"

You can apply this plan to save these new output values to the Terraform state, without changing any real infrastructure.

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.

Outputs:
subnet_cidr_block_output = "10.0.30.0/24"
```

```
k23-22412-038-sys + /workspaces/Lab11 (main) $ terraform apply -auto-approve -var "subnet_cidr_block=10.0.40.0/24"

Changes to Outputs:
  - subnet_cidr_block_output = "10.0.30.0/24" -> "10.0.40.0/24"

You can apply this plan to save these new output values to the Terraform state, without changing any real infrastructure.

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.

Outputs:
subnet_cidr_block_output = "10.0.40.0/24"
```

```
@23-22412-038-sys → /workspaces/Lab11 (main) $ printenv | grep TF_VAR_
unset TF_VAR_subnet_cidr_block
printenv | grep TF_VAR_
TF_VAR_subnet_cidr_block=10.0.20.0/24
```

Task 2

```
@23-22412-038-sys → /workspaces/Lab11 (main) $ terraform apply -auto-approve -var "subnet_cidr_block=10.0.0.0/24"

Changes to Outputs:
- subnet_cidr_block_output = "10.0.40.0/24" -> "10.0.0.0/24"

You can apply this plan to save these new output values to the Terraform state, without changing any real infrastructure.

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.

Outputs:
subnet_cidr_block_output = "10.0.0.0/24"
```

```
@23-22412-038-sys → /workspaces/Lab11 (main) $ terraform apply -auto-approve -var "api_session_token=my_API_session_Token_12345"

Changes to Outputs:
- api_session_token_output = (sensitive value)
- subnet_cidr_block_output = "10.0.0.0/24" -> "10.0.30.0/24"

You can apply this plan to save these new output values to the Terraform state, without changing any real infrastructure.

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.

Outputs:
api_session_token_output = <sensitive>
subnet_cidr_block_output = "10.0.30.0/24"
```

```

"Version": 4,
"terraform_version": "1.14.3",
"serial": 7,
"lineage": "0d7ac30c-5970-354f-956a-9b5b08f74f33",
"outputs": {
  "api_session_token_output": {
    "value": "my_API_session_Token_12345",
    "type": "string",
    "sensitive": true
  },
  "subnet_cidr_block_output": {
    "value": "10.0.30.0/24",
    "type": "string"
  }
},
"resources": [],
"check_results": [
  {
    "object_kind": "var",
    "config_addr": "var.api_session_token",
    "status": "pass",
    "objects": [
      {
        "object_addr": "var.api_session_token",
        "status": "pass"
      }
    ]
  },
  {
    "object_kind": "var",
    "config_addr": "var.subnet_cidr_block",
    "status": "pass",
    "objects": [
      {
        "object_addr": "var.subnet_cidr_block",
        "status": "pass"
      }
    ]
  }
]

```

```
@23-22412-038-sys → /workspaces/Lab11 (main) $ terraform apply -auto-approve -var "api_session_token=my_API_session_Token_123456"

Changes to Outputs:
- api_session_token_output = (sensitive value)
- subnet_cidr_block_output = "10.0.0.0/24" -> "10.0.30.0/24"

You can apply this plan to save these new output values to the Terraform state, without changing any real infrastructure.
```

```
@23-22412-038-sys → /workspaces/Lab11 (main) $ cat terraform.tfstate | grep api_session_token -n
7:   "api_session_token_output": {
21:     "config_addr": "var.api_session_token",
25:     "object_addr": "var.api_session_token",
```

Task 3


```
comopet@codepages:7515b: nano main.tf
GNU nano 2.2 main.tf
provider "aws" {
  shared_config_files = ["~/aws/config"]
  shared_credentials_files = ["~/aws/credentials"]
}

variable "subnet_cidr_block" {
  type = string
  default = ""
  description = "CIDR block to assign to the application subnet"
  sensitive = false
  nullable = false
  ephemeral = false

  validation {
    condition = can(regex("^[0-9]{1,3}\\.[0-9]{1,3}/[0-9]{1,3}$", var.subnet_cidr_block))
    error_message = "The subnet_cidr_block must be a valid CIDR notation string."
  }
}

output "subnet_cidr_block_output" {
  value = var.subnet_cidr_block
}

variable "api_session_token" {
  type = string
  default = ""
  sensitive = true
  nullable = false
  ephemeral = false

  validation {
    condition = can(regex("^[A-Za-z0-9-]{20,}$", var.api_session_token))
    error_message = "Token must be 20+ characters."
  }
}

output "api_session_token_output" {
  value = var.api_session_token
  sensitive = true
}

variable "environment" {}
variable "project_name" {}
variable "primary_subnet_id" {}
variable "subnet_count" {}
variable "monitoring" {}

Help Exit Write Out Where Is Cut Paste Execute Duplicat Location Go To Line Undo Redo Set Mark Copy To Bracket Where Was Previous Next
```

```
@23-22412-038-sys → /workspaces/Lab11 (main) $ aws ec2 describe-subnets \
--filters "Name=availability-zone,Values=me-central-1a" \
--query "Subnets[].SubnetId" \
--output text
subnet-08a73f8aa1dee72d3
```

```
@23-22412-038-sys → /workspaces/Lab11 (main) $ cat terraform.tfvars
subnet_cidr_block = "10.0.30.0/24"
environment       = "dev"
project_name      = "lab_work"
primary_subnet_id = "subnet-08a73f8aa1dee72d3"
"
subnet_count      = 3
monitoring        = true
@23-22412-038-sys → /workspaces/Lab11 (main) $ !
```



```
codeops@codeops-79c1b: nano local.tf
GNU nano 2.2 local.tf
locals {
  resource_name    = "${var.project_name}-${var.environment}"
  primary_public_subnet = var.primary_subnet_id
  subnet_count     = var.subnet_count
  is_production    = var.environment == "prod"
  monitoring_enabled = var.monitoring || !local.is_production
}
```

```
codeops@codeops-79c1b: nano main.tf
GNU nano 2.2 main.tf
value = var.subnet_cidr_block
}
variable "api_session_token" {
  type        = string
  default     = ""
  sensitive   = true
  nullable    = false
  ephemeral   = false
  validation {
    condition = can(regex("^[A-Za-z0-9-]{20,}$", var.api_session_token))
    error_message = "Token must be 20+ characters."
  }
}
output "api_session_token_output" {
  value     = var.api_session_token
  sensitive = true
}
variable "environment" {}
variable "project_name" {}
variable "primary_subnet_id" {}
variable "subnet_count" {}
variable "monitoring" {}
output "resource_name" {
  value = local.resource_name
}
output "primary_public_subnet" {
  value = local.primary_public_subnet
}
output "subnet_count" {
  value = local.subnet_count
}
output "is_production" {
  value = local.is_production
}
output "monitoring_enabled" {
  value = local.monitoring_enabled
}
```

```
023-22412-036-sys → /workspaces/Lab11 (main) $ terraform apply -auto-approve
```

Changes to Outputs:

- api_session_token_output = (sensitive value)
- + is_production = false
- + monitoring_enabled = true
- + primary_public_subnet = "subnet-08a73f8aa1dee72d3"
- + resource_name = "lab_work-dev"
- subnet_cidr_block_output = "10.0.30.0/24" -> "10.0.0.0/24"
- + subnet_count = 3

You can apply this plan to save these new output values to the Terraform state, without changing any real infrastructure.

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.

Outputs:

```
api_session_token_output = <sensitive>
is_production = false
monitoring_enabled = true
primary_public_subnet = "subnet-08a73f8aa1dee72d3"
resource_name = "lab_work-dev"
subnet_cidr_block_output = "10.0.0.0/24"
subnet_count = 3
```

Task 4

```
023-22412-036-sys → /workspaces/Lab11 (main) $ terraform apply -auto-approve
```

Changes to Outputs:

- + tags = {
 - Environment = "dev"
 - Owner = "platform-team"
 - Project = "sample-app"
 }

You can apply this plan to save these new output values to the Terraform state, without changing any real infrastructure.

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.

Outputs:

```
api_session_token_output = <sensitive>
is_production = false
monitoring_enabled = true
primary_public_subnet = "subnet-08a73f8aa1dee72d3"
resource_name = "lab_work-dev"
subnet_cidr_block_output = "10.0.0.0/24"
subnet_count = 3
tags = tomap({
  "Environment" = "dev"
  "Owner" = "platform-team"
  "Project" = "sample-app"
})
```

```
023-22412-036-sys → /workspaces/Lab11 (main) $ terraform apply -auto-approve
```

Changes to Outputs:

- + server_config = {
 - backup_enabled = false
 - instance_type = "t3.micro"
 - monitoring = true
 - name = "web-server"
 - storage_gb = 20
 }

You can apply this plan to save these new output values to the Terraform state, without changing any real infrastructure.

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.

Outputs:

```
api_session_token_output = <sensitive>
is_production = false
monitoring_enabled = true
primary_public_subnet = "subnet-08a73f8aa1dee72d3"
resource_name = "lab_work-dev"
server_config = {
  "backup_enabled" = false
  "instance_type" = "t3.micro"
  "monitoring" = true
  "name" = "web-server"
  "storage_gb" = 20
}
subnet_cidr_block_output = "10.0.0.0/24"
subnet_count = 3
tags = tomap({
  "Environment" = "dev"
  "Owner" = "platform-team"
  "Project" = "sample-app"
})
```

```
023-22412-036-sys → /workspaces/Lab11 (main) $
```

Task 5

```
codepaces@codepaces-79c1b: nano main.tf
GNU nano 2.9.3 main.tf

variable "server_config" {
  type = object({
    name           = string
    instance_type  = string
    monitoring     = bool
    storage_gb     = number
    backup_enabled = bool
  })
  description = "Server configuration object"
}

output "server_config" {
  value = var.server_config
}

variable "server_names" {
  type = list(string)
  default = ["web-2", "web-1", "web-2"]
  description = "List of server names (duplicates allowed)"
}

variable "server_metadata" {
  type = tuple(string, number, bool)
  default = ["web-1", 4, true]
  description = "Tuple of server info (immutable, mixed types)"
}

variable "availability_zones" {
  type = set(string)
  default = ["us-central-1a", "us-central-1b"]
  description = "Set of availability zones (unique, unordered)"
}

output "compare_collections" {
  value = {
    list_example = var.server_names
    tuple_example = var.server_metadata
    set_example = var.availability_zones
  }
}
```

Outputs:

```
api_session_token_output = <sensitive>
compare_collections = {
  "list_example" = tolist([
    "web-2",
    "web-1",
    "web-2",
  ])
  "set_example" = toset([
    "us-central-1a",
    "us-central-1b",
  ])
  "tuple_example" = [
    "web-1",
    4,
    true,
  ]
}
is_production = false
monitoring_enabled = true
primary_public_subnet = "subnet-08a73f8aa1dee72d3"
resource_name = "lab_work-dev"
server_config = {
  "backup_enabled" = false
  "instance_type" = "t3.micro"
  "monitoring" = true
  "name" = "web-server"
  "storage_gb" = 20
}
subnet_cidr_block_output = "10.0.0.0/24"
subnet_count = 3
tags = tomap({
  "Environment" = "dev"
  "Owner" = "platform-team"
  "Project" = "sample-app"
})
```

```
codopace@codopace:79c31b: nano locals.tf
GNU nano 2.2.1 locals.tf
locals {
  # Task 2: locals
  resource_name = "${var.project_name}-${var.environment}"
  primary_public_subnet = var.primary_subnet_id
  subnet_count = var.subnet_count
  is_production = var.environment == "prod"
  monitoring_enabled = var.monitoring || !local.is_production

  # Task 3: mutation locals
  mutated_list = setunion(var.server_names, ["web-3"])
  mutated_tuple = setunion(var.server_metadata, ["web-3"])
  mutated_set = setunion(var.availability_zones, ["me-central-1c"])
}

Help Exit Write Out Read File Where Is Replace Cut Paste Execute Justify Location Go To Line Undo Redo Set Mark Copy To Bracket Where Was Previous Next
```

```
codopace@codopace:79c31b: nano main.tf
GNU nano 2.2.1 main.tf
name = string
instance_type = string
monitoring = bool
storage_gb = number
backup_enabled = bool
}
description = "Server configuration object"

output "server_config" {
  value = var.server_config
}

variable "server_names" {
  type = list(string)
  default = ["web-2", "web-1", "web-2"]
  description = "List of server names (duplicates allowed)"
}

variable "server_metadata" {
  type = tuple([string, number, bool])
  default = ["web-1", 4, true]
  description = "Tuple of server info (immutable, mixed types)"
}

variable "availability_zones" {
  type = set(string)
  default = ["me-central-1b", "me-central-1a", "me-central-1c"]
  description = "Set of availability zones (unique, unordered)"
}

output "compare_collections" {
  value = {
    list_example = var.server_names
    tuple_example = var.server_metadata
    set_example = var.availability_zones
  }
}

output "mutation_comparison" {
  value = {
    original_tuple = var.server_metadata
    mutated_tuple = local.mutated_tuple
  }
}

Help Exit Write Out Read File Where Is Replace Cut Paste Execute Justify Location Go To Line Undo Redo Set Mark Copy To Bracket Where Was Previous Next
```

```
423-22412-036-sys + /workspaces/Lab11 (main) $ terraform apply -auto-approve
```

Changes to Outputs:

```
+ mutation_comparison = {  
  + mutated_tuple = [  
    + "4",  
    + "true",  
    + "web-1",  
    + "web-2",  
  ]  
+ original_tuple = [  
  + "web-1",  
  + 4,  
  + true,  
+ ]  
}
```

You can apply this plan to save these new output values to the Terraform state, without changing any real infrastructure.

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.

Outputs:

```
api_session_token_output = <sensitive>  
compare_collections = {  
  "list_example" = tolist([  
    "web-2",  
    "web-1",  
    "web-2",  
  ])  
  "set_example" = toset([  
    "me-central-1a",  
    "me-central-1b",  
  ])  
  "tuple_example" = [  
    "web-1",  
    4,  
    true,  
  ]  
}  
is_production = false  
monitoring_enabled = true  
mutation_comparison = {  
  "mutated_tuple" = toset([  
    "4",  
    "true",  
    "web-1",  
    "web-2",  
  ])  
  "original_tuple" = [  
    "web-1",  
    4,  
    true,  
  ]  
}
```

```

    })
    "set_example" = toset([
        "me-central-1a",
        "me-central-1b",
    ])
    "tuple_example" = [
        "web-1",
        4,
        true,
    ]
}
is_production = false
monitoring_enabled = true
mutation_comparison = {
    "mutated_tuple" = toset([
        "4",
        "true",
        "web-1",
        "web-2",
    ])
    "original_tuple" = [
        "web-1",
        4,
        true,
    ]
}
primary_public_subnet = "subnet-08a73f8aa1dee72d3"
resource_name = "lab_work-dev"
server_config = {
    "backup_enabled" = false
    "instance_type" = "t3.micro"
    "monitoring" = true
    "name" = "web-server"
    "storage_gb" = 20
}
subnet_cidr_block_output = "10.0.0.0/24"
subnet_count = 3
tags = tomap({
    "Environment" = "dev"
    "Owner" = "platform-team"
    "Project" = "sample-app"
}))

```

Task 6


```
codeops@codeops:79c1b: nano locals.tf
GNU nano 2.2
locals.tf
locals {
  # Task 1: locals
  resource_name = "${var.project_name}-${var.environment}"
  primary_public_subnet = var.primary_subnet_id
  subnet_count = var.subnet_count
  is_production = var.environment == "prod"
  monitoring_enabled = var.monitoring || local.is_production

  # Task 2: mutation locals
  mutated_list = setunion(var.server_names, ["web-3"])
  mutated_tuple = setunion(var.server_metadata, ["web-3"])
  mutated_set = setunion(var.availability_zones, ["us-central-1c"])

  # Task 3: null/dynamic locals
  server_tag = merge(
    { name = "web-server" },
    var.optional_tag != null ? { "Custom" = var.optional_tag } : {}
  )
}

Help Exit Write Out Read File Where Is Replace Cut Paste Execute Justify Location Go To Line Undo Redo Set Mark Copy To Bracket Where Was Previous Next
```

```
codeops@codeops:79c1b: nano main.tf
GNU nano 2.2
main.tf
type = tuple(string, number, bool)
default = ["web-1", 4, true]
description = "Tuple of server info (immutable, mixed types)"

variable "availability_zones" {
  type = set(string)
  default = ["us-central-1b", "us-central-1a", "us-central-1c"]
  description = "Set of availability zones (unique, unordered)"
}

output "compare_collections" {
  value = {
    list_example = var.server_names
    tuple_example = var.server_metadata
    set_example = var.availability_zones
  }
}

output "mutation_comparison" {
  value = {
    original_tuple = var.server_metadata
    mutated_tuple = local.mutated_tuple
  }
}

variable "optional_tag" {
  type = string
  description = "A tag that may or may not be provided"
  default = null
}

variable "dynamic_value" {
  type = any
  description = "A variable that can accept any data type"
  default = null
}

output "optional_tag" {
  value = local.server_tags
}

output "value_received" {
  value = var.dynamic_value
}
```

```
codopac@codopac:79c17b:/dev/bash
$ cd /workspaces/Lab11 (main) $ nano main.tf
$ cd /workspaces/Lab11 (main) $ nano locals.tf
$ cd /workspaces/Lab11 (main) $ nano main.tf
$ cd /workspaces/Lab11 (main) $ terraform apply -auto-approve

Changes to Outputs:
  + optional_tag = [
    - name = "web-server"
  ]

you can apply this plan to save these new output values to the Terraform state, without changing any real infrastructure.

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.

Outputs:
api_session_token_output = <sensitive>
compare_collections = {
  "list_example" = tolist([
    "web-1",
    "web-1",
    "web-2",
  ])
  "set_example" = toset([
    "ne-central-1a",
    "ne-central-1b",
  ])
  "tuple_example" = [
    "web-1",
    4,
    true,
  ]
}
is_production = false
monitoring_enabled = true
mutation_comparison = {
  "mutated_tuple" = toset([
    "4",
    "true",
    "web-1",
    "web-2",
  ])
  "original_tuple" = [
    "web-1",
    4,
    true,
  ]
}
```

```
codopac@codopac:79c17b:/dev/bash
"web-1",
"web-2",
})
"set_example" = toset([
  "ne-central-1a",
  "ne-central-1b",
])
"tuple_example" = [
  "web-1",
  4,
  true,
]
}
is_production = false
monitoring_enabled = true
mutation_comparison = {
  "mutated_tuple" = toset([
    "4",
    "true",
    "web-1",
    "web-2",
  ])
  "original_tuple" = [
    "web-1",
    4,
    true,
  ]
}
optional_tag = {
  name = "web-server"
}
primary_public_subnet = "subnet-08a73f8a5d4ee72d3"
resource_name = "lab_work-dev"
server_config = {
  backup_enabled = false
  instance_type = "t3.micro"
  monitoring = true
  name = "web-server"
  storage_gp = 20
}
subnet_cidr_block_output = "10.0.0.0/24"
subnet_count = 3
tags = tomap([
  Environment = "dev"
  Owner = "platform-team"
  Project = "sample-app"
])
$ cd /workspaces/Lab11 (main) $
```

```
codeps@codeps:7931b:/nv$ cat
})
set_example = toset([
    "ne-central-1a",
    "ne-central-1b",
])
tuple_example = [
    "web-1",
    4,
    true,
]
}
}
is_production = false
monitoring_enabled = true
mutation_comparison = {
    mutated_tuple = toset([
        "a",
        "true",
        "web-1",
        "web-2",
    ])
    original_tuple = [
        "web-1",
        4,
        true,
    ]
}
}
optional_tag = {
    "Custom" = "dev"
    "Name" = "web-server"
}
primary_public_subnet = "subnet-08a73f8a1dee72d3"
resource_name = "lab_work-dev"
server_config = {
    "backup_enabled" = false
    "instance_type" = "t3.micro"
    "monitoring" = true
    "name" = "web-server"
    "storage_gb" = 20
}
subnet_cidr_block_output = "10.0.0.0/24"
subnet_count = 3
tags = tomap([
    "Environment" = "dev"
    "Owner" = "platform-team"
    "Project" = "sample-app"
])
value_received = "hello"
$? 22412 033-ys /workspaces/Lab11 (main) $
```

```
codeps@codeps:7931b:/nv$ cat
})
set_example = toset([
    "ne-central-1a",
    "ne-central-1b",
])
tuple_example = [
    "web-1",
    4,
    true,
]
}
}
is_production = false
monitoring_enabled = true
mutation_comparison = {
    mutated_tuple = toset([
        "a",
        "true",
        "web-1",
        "web-2",
    ])
    original_tuple = [
        "web-1",
        4,
        true,
    ]
}
}
optional_tag = {
    "Custom" = "dev"
    "Name" = "web-server"
}
primary_public_subnet = "subnet-08a73f8a1dee72d3"
resource_name = "lab_work-dev"
server_config = {
    "backup_enabled" = false
    "instance_type" = "t3.micro"
    "monitoring" = true
    "name" = "web-server"
    "storage_gb" = 20
}
subnet_cidr_block_output = "10.0.0.0/24"
subnet_count = 3
tags = tomap([
    "Environment" = "dev"
    "Owner" = "platform-team"
    "Project" = "sample-app"
])
value_received = "42"
$? 22412 033-ys /workspaces/Lab11 (main) $
```

```
codepaces@codepaces-79c17b: /dev/bash
}}
tuple_example = [
  "web-1",
  4,
  true,
]
}
is_production = false
monitoring_enabled = true
mutation_comparison = {
  "mutated_tuple" = toset([
    "4",
    "true",
    "web-1",
    "web-2",
  ])
  "original_tuple" = [
    "web-1",
    4,
    true,
  ]
}
optional_tag = {
  "Custom" = "dev"
  "Name" = "web-server"
}
primary_public_subnet = "subnet-08a7f96a1dee72d3"
resource_name = "lab_work-dev"
server_config = {
  "backup_enabled" = false
  "instance_type" = "t3.micro"
  "monitoring" = true
  "name" = "web-server"
  "storage_gb" = 20
}
subnet_cidr_block_output = "10.0.0.0/24"
subnet_count = 3
tags = tomap([
  "Environment" = "dev"
  "Owner" = "platform-team"
  "Project" = "sample-app"
])
value_received = [
  "a",
  "b",
  "c",
]
code-21A22-038-aps - /workspaces/Lab11 (wsl) $
```

```
codepaces@codepaces-79c17b: /dev/bash
}}
tuple_example = [
  "web-1",
  4,
  true,
]
}
is_production = false
monitoring_enabled = true
mutation_comparison = {
  "mutated_tuple" = toset([
    "4",
    "true",
    "web-1",
    "web-2",
  ])
  "original_tuple" = [
    "web-1",
    4,
    true,
  ]
}
optional_tag = {
  "Custom" = "dev"
  "Name" = "web-server"
}
primary_public_subnet = "subnet-08a7f96a1dee72d3"
resource_name = "lab_work-dev"
server_config = {
  "backup_enabled" = false
  "instance_type" = "t3.micro"
  "monitoring" = true
  "name" = "web-server"
  "storage_gb" = 20
}
subnet_cidr_block_output = "10.0.0.0/24"
subnet_count = 3
tags = tomap([
  "Environment" = "dev"
  "Owner" = "platform-team"
  "Project" = "sample-app"
])
value_received = {
  "cpu" = 4
  "name" = "server"
}
code-21A22-038-aps - /workspaces/Lab11 (wsl) $
```

```

codepaces@codepaces-79c3b: /workspaces
}
name = "web-server"
storage_gb = 20
}
subnet_cidr_block_output = "10.0.0.0/24"
subnet_count = 3
tags = tomap([
  "Environment" = "dev",
  "Owner" = "platform-team",
  "Project" = "sample-app"
])
value_received = {
  "cpu" = 4
  "name" = "server"
}
@23-22412-038-sys + /workspaces/Lab11 (main) $ nano terraform.tfvars
@23-22412-038-sys + /workspaces/Lab11 (main) $ terraform apply -auto-approve

Error: Missing attribute value

on terraform.tfvars line 27:
27: dynamic_value = { "null"
28:   name = "server"

Expected an attribute value, introduced by an equals sign (=).

@23-22412-038-sys + /workspaces/Lab11 (main) $
@23-22412-038-sys + /workspaces/Lab11 (main) $
@23-22412-038-sys + /workspaces/Lab11 (main) $ nano terraform.tfvars
@23-22412-038-sys + /workspaces/Lab11 (main) $ terraform apply -auto-approve

changes to outputs:
- value_received = {
  "cpu" = 4
  "name" = "server"
} -> null

You can apply this plan to save these new output values to the Terraform state, without changing any real infrastructure.
Apply complete! Resources: 0 added, 0 changed, 0 destroyed.

OUTPUTS:
api_session_token_output = <sensitive>
compare_collections = {
  "list_example" = tolist([
    "web-2",
    "web-1",
    "web-2",
  ])
}

```

Task 7

```

@23-22412-038-sys → /workspaces/Lab11 (main) $ touch .gitignore
@23-22412-038-sys → /workspaces/Lab11 (main) $ nano .gitignore
@23-22412-038-sys → /workspaces/Lab11 (main) $ cat .gitignore
.terraform/*
*.tfstate
*.tfstate.*
*.tfvars
*.pem
@23-22412-038-sys + /workspaces/Lab11 (main) $

```

Task 8

```

codepaces@codepaces-79c3b: /workspaces
@23-22412-038-sys + /workspaces/Lab11 (main) $ nano main.tf
@23-22412-038-sys + /workspaces/Lab11 (main) $ nano main.tf
@23-22412-038-sys + /workspaces/Lab11 (main) $ nano terraform.tfvars
@23-22412-038-sys + /workspaces/Lab11 (main) $ terraform init

terraform init
terraform apply -auto-approve
Initializing the backend...
Initializing provider plugins...
- Reusing previously-installed hashicorp/aws from the dependency lock file
- Using previously-installed hashicorp/cons v0.27.0.

Terraform has been successfully initialized!

You may now begin working with Terraform. Run "terraform plan" to see
the changes that are planned for your infrastructure. All Terraform commands
should now work.

If you ever see a warning about a "provider" or "plugin" in Terraform,
re-run this command to fetch the latest versions of the plugin, or use "terraform
upgrade" to do so. If you see this message, that means the provider or plugin
you are using is out of date.

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
  ~ create
  ~ destroy
  ~ update

Terraform will perform the following actions:

# aws_internet_gateway.igw will be created
+ resource "aws_internet_gateway" "igw" {
  + arn = (known after apply)
  + id = (known after apply)
  + owner_id = (known after apply)
  + region = "us-east-1"
  + tags = {
    "Environment" = "dev"
    "Name" = "myinternet-igw"
    "Owner" = "platform"
  }
  + tags_all = {
    "Environment" = "dev"
    "Name" = "myinternet-igw"
    "Owner" = "platform"
  }
  + vpc_id = (known after apply)
}

# aws_route_table.main_rt will be created
+ resource "aws_route_table" "main_rt" {

```

```
codepaces@codepaces-79c17b: /bin/bash
Note: You didn't use the -out option to save this plan, so Terraform can't guarantee to take exactly these actions if you run "terraform apply" now.
Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
  + create
Terraform will perform the following actions:

# aws_internet_gateway.igw will be created
+ resource "aws_internet_gateway" "igw" {
  + arn          = (known after apply)
  + id           = (known after apply)
  + owner_id     = (known after apply)
  + region       = "us-central-1"
  + tags         = {
    + "Environment" = "dev"
    + "Name"         = "myproject-igw"
    + "Owner"        = "toshal"
  }
  + tags_all     = {
    + "Environment" = "dev"
    + "Name"         = "myproject-igw"
    + "Owner"        = "toshal"
  }
  + vpc_id       = (known after apply)
}

# aws_route_table.main_rt will be created
+ resource "aws_route_table" "main_rt" {
  + arn          = (known after apply)
  + id           = (known after apply)
  + owner_id     = (known after apply)
  + propagating_vgws = (known after apply)
  + region       = "us-central-1"
  + routes        = [
    + {
      + cidr_block      = "0.0.0.0/0"
      + gateway_id      = (known after apply)
      + (all unchanged attributes hidden)
    }
  ]
  + tags         = {
    + "Environment" = "dev"
    + "Name"         = "myproject-rt"
    + "Owner"        = "toshal"
  }
  + tags_all     = {
    + "Environment" = "dev"
  }
}
```

```
codepaces@codepaces-79c17b: /bin/bash
aws_route_table_association.subnet_assoc: Creating...
aws_route_table_association.subnet_assoc: Creation complete after 1s [id=rtbassoc-06e066178d3fd835]

Apply complete! Resources: 5 added, 0 changed, 0 destroyed.

Outputs:
api_secret_token_output = <sensitive>
compare_collections = {
  "list_example" = tolist([
    "web-1",
    "web-2",
  ])
  "set_example" = toset([
    "us-central-1a",
    "us-central-1b",
  ])
  "tuple_example" = [
    "web-1",
    4,
    true,
  ]
}
is_production = false
monitoring_enabled = true
mutation_comparison = {
  "mutated_tuple" = toset([
    "4",
    "true",
    "web-1",
    "web-2",
  ])
  "original_tuple" = [
    "web-1",
    4,
    true,
  ]
}
optional_tag = "dev"
primary_public_subnet = "subnet-0123456789abcdef0"
resource_name = "myproject-dev"
subnet_cidr_block_output = "10.0.0.0/24"
subnet_count = 1
value_received = {
  "cpu" = 4
  "name" = "server"
}

~/c3-21412-038-aps - /workspaces/Lab11 (note) $
```

Task 9


```

03-22412-038-sys → /workspaces/Lab11 (main) $ terraform init
terraform apply -auto-approve
initializing the backend...
initializing provider plugins...
- Reusing previous version of hashicorp/aws from the dependency lock file
- Using previously-installed hashicorp/aws v6.27.0

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
aws_route_table_association.subnet_assoc: Refreshing state... [id=rtbassoc-08e08b6178d3fd835]
aws_vpc.main_vpc: Refreshing state... [id=vpc-0cd12f95de8556417]
aws_subnet.app_subnet: Refreshing state... [id=subnet-0b1696fda58ee7a71]
aws_internet_gateway.igw: Refreshing state... [id=igw-078d80b905e5f9c60]
aws_route_table.main_rt: Refreshing state... [id=rtb-079ccd3c3c57d9571]

```

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:

- create
- destroy

Terraform will perform the following actions:

```

# aws_default_route_table.main_rt will be created
+ resource "aws_default_route_table" "main_rt" {
+   arn                = (known after apply)
+   default_route_table_id = (known after apply)
+   id                 = (known after apply)
+   owner_id            = (known after apply)
+   region              = "me-central-1"

```

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:

- create
- destroy

Terraform will perform the following actions:

```

# aws_default_route_table.main_rt will be created
+ resource "aws_default_route_table" "main_rt" {
+   arn                = (known after apply)
+   default_route_table_id = (known after apply)
+   id                 = (known after apply)
+   owner_id            = (known after apply)
+   region              = "me-central-1"
+   route               = [
+     {
+       cidr_block      = "0.0.0.0/0"
+       gateway_id       = (known after apply)
+       # (10 unchanged attributes hidden)
+     },
+   ],
+   tags                = {
+     "Name" = "dev-rt"
+   }
+   tags_all             = {
+     "Name" = "dev-rt"
+   }
+   vpc_id               = (known after apply)
}

# aws_default_security_group.myapp_sg will be created
+ resource "aws_default_security_group" "myapp_sg" {
+   arn                = (known after apply)
+   description         = (known after apply)
+   egress              = [
+     {
+       cidr_blocks      = [
+         "0.0.0.0/0",
+       ]
+       from_port         = 0
+       ipv6_cidr_blocks  = []
+       prefix_list_ids   = []
+       protocol          = "-1"
+       security_groups   = []
+       self              = false
+       to_port           = 0
+       # (10 unchanged attributes hidden)

```

```

+ self = false
+ to_port = 0
# (1 unchanged attribute hidden)
},
]
+ id = (known after apply)
+ ingress = [
+ {
+ cidr_blocks = [
+ "0.0.0.0/0",
+ ]
+ from_port = 80
+ ipv6_cidr_blocks = []
+ prefix_list_ids = []
+ protocol = "tcp"
+ security_groups = []
+ self = false
+ to_port = 80
# (1 unchanged attribute hidden)
+ },
+ {
+ cidr_blocks = [
+ "4.240.18.229/32",
+ ]
+ from_port = 22
+ ipv6_cidr_blocks = []
+ prefix_list_ids = []
+ protocol = "tcp"
+ security_groups = []
+ self = false
+ to_port = 22
# (1 unchanged attribute hidden)
+ },
]
+ name = (known after apply)
+ name_prefix = (known after apply)
+ owner_id = (known after apply)
+ region = "me-central-1"
+ revoke_rules_on_delete = false
+ tags = {
+ "Name" = "dev-sg"
+ }
+ tags_all = {
+ "Name" = "dev-sg"
+ }
+ vpc_id = (known after apply)
}

```

```

# aws_internet_gateway.myapp_igw will be created
+ resource "aws_internet_gateway" "myapp_igw" {
+ arn = (known after apply)
+ id = (known after apply)
+ owner_id = (known after apply)
+ region = "me-central-1"
+ tags = {
+ "Name" = "dev-igw"
+ }
+ tags_all = {
+ "Name" = "dev-igw"
+ }
+ vpc_id = (known after apply)
}

# aws_key_pair.ssh_key will be created
+ resource "aws_key_pair" "ssh_key" {
+ arn = (known after apply)
+ fingerprint = (known after apply)
+ id = (known after apply)
+ key_name = "serverkey"
+ key_name_prefix = (known after apply)
+ key_pair_id = (known after apply)
+ key_type = (known after apply)
+ public_key = "ssh-ed25519 AAAAC3NzaC1lZD01INTE$AAAAIPNwu9F8k0FKZ8hsRwTXKX.cwCL0kr5yY861DI/qQ+THD codespace@codespaces-79c519"
+ region = "me-central-1"
+ tags_all = (known after apply)
}

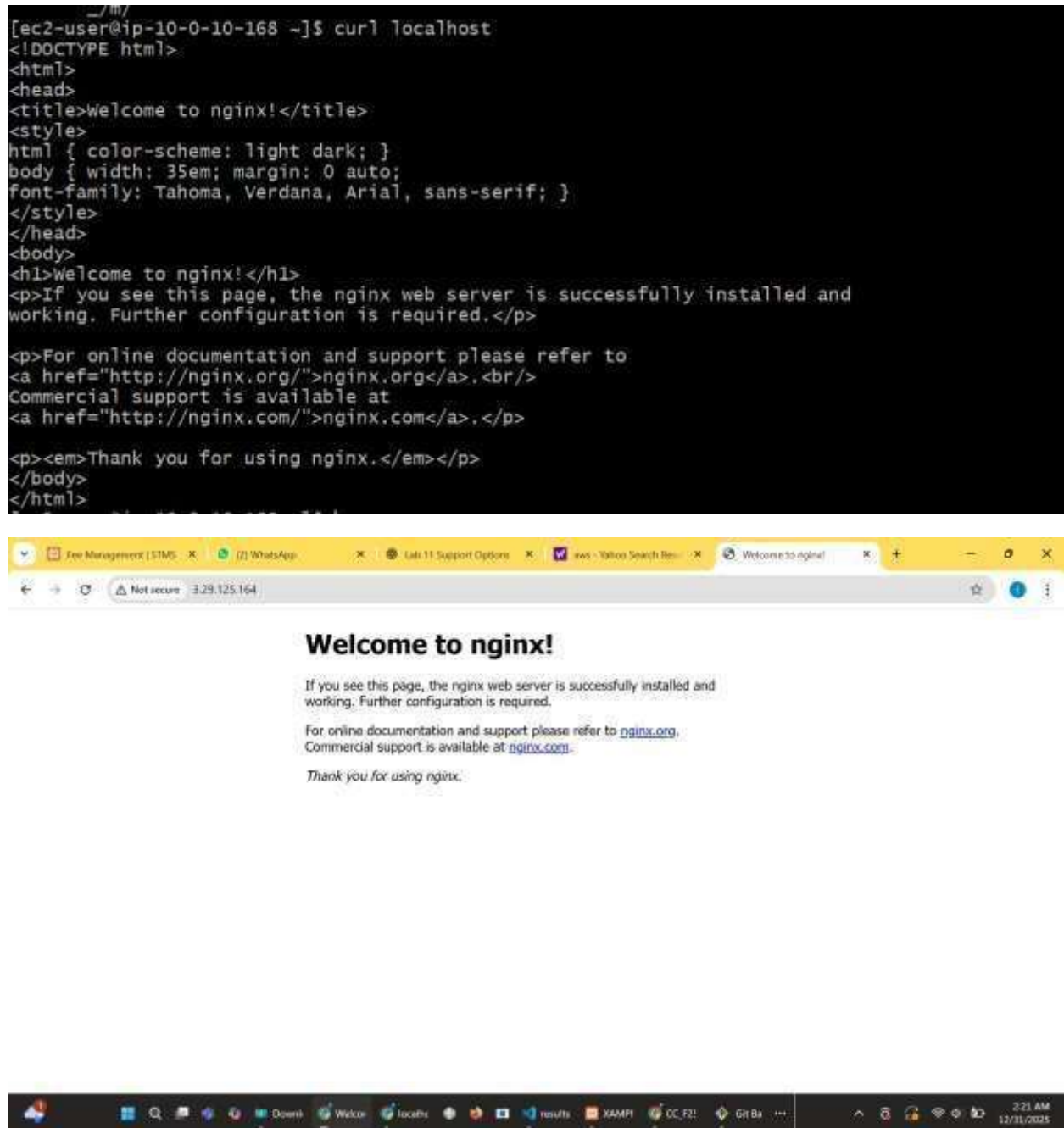
```

Outputs:

```

#
###
#####
###
#
https://aws.amazon.com/linux/amazon-linux-2023
[ec2-user@ip-10-0-10-168 ~]$

```



Cleanup

```

tf1-22412-Q88-SY1 ~ /workspaces/Lab11 (main) % terraform destroy -auto-approve
aws_key_pair.ssh_key: Refreshing state... [id=serverkey]
aws_vpc.myapp_vpc: Refreshing state... [id=vpc-08c0de79b0e26507d]
aws_internet_gateway.myapp_igw: Refreshing state... [id=igw-08deab4a9bafbaac7]
aws_subnet.myapp_subnet_1: Refreshing state... [id=subnet-03e5bda232f8fa581]
aws_default_security_group.myapp_sg: Refreshing state... [id=sg-0a106f65daa2ea6eb]
aws_default_route_table.main_rt: Refreshing state... [id=rtb-0e0e0308f83c69ff2]
aws_instance.myapp_server: Refreshing state... [id=i-097e723716367b59c]

```

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:

- destroy

Terraform will perform the following actions:

```

# aws_default_route_table.main_rt will be destroyed
- resource "aws_default_route_table" "main_rt" {
  - arn              = "arn:aws:ec2:me-central-1:623705167686:route-table/rtb-0e0e0308f83c69ff2" -> null
  - default_route_table_id = "rtb-0e0e0308f83c69ff2" -> null
  - id                  = "rtb-0e0e0308f83c69ff2" -> null
  - owner_id            = "623705167686" -> null
  - propagating_vgw     = [] -> null
  - region              = "me-central-1" -> null
  - route              = [
    - {
      - cidr_block      = "0.0.0.0/0"
      - gateway_id      = "igw-08deab4a9bafbaac7"
      # (10 unchanged attributes hidden)
    }
  ] -> null
  - tags              = {
    - "Name" = "dev-rt"
  } -> null
  - tags_all          = {
    - "Name" = "dev-rt"
  } -> null
  - vpc_id             = "vpc-08c0de79b0e26507d" -> null
}

# aws_default_security_group.myapp_sg will be destroyed
- resource "aws_default_security_group" "myapp_sg" {
  - arn              = "arn:aws:ec2:me-central-1:623705167686:security-group/sg-0a106f65daa2ea6eb" -> null
  - description      = "default VPC security group" -> null
  - egress           = [
    - {
      - cidr_blocks      = [
        - "0.0.0.0/0",
      ]
      - from_port        = 0
      - ipv6_cidr_blocks = []
      - prefix_list_ids  = []
      - protocol         = "-1"
      - security_groups  = []
      - self             = false
      - to_port          = 0
    }
  ]
}

```

```

# aws_default_security_group.myapp_sg will be destroyed
- resource "aws_default_security_group" "myapp_sg" {
  - arn              = "arn:aws:ec2:me-central-1:623705167686:security-group/sg-0a106f65daa2ea6eb" -> null
  - description      = "default VPC security group" -> null
  - egress           = [
    - {
      - cidr_blocks      = [
        - "0.0.0.0/0",
      ]
      - from_port        = 0
      - ipv6_cidr_blocks = []
      - prefix_list_ids  = []
      - protocol         = "-1"
      - security_groups  = []
      - self             = false
      - to_port          = 0
    }
  ]
}

```

```

# aws_instance.myapp_server will be destroyed
- resource "aws_instance" "myapp_server" {
  - ami              = "ami-05524d6658fcf35b6" -> null
  - arn              = "arn:aws:ec2:me-central-1:623705167686:instance/i-097e723716367b59c" -> null
  - associate_public_ip_address = true -> null
  - availability_zone = "me-central-1a" -> null
  - disable_api_stop   = false -> null
  - disable_api_termination = false -> null
  - ebs_optimized      = false -> null
  - force_destroy      = false -> null
  - get_password_data   = false -> null
  - hibernation         = false -> null
  - id                  = "i-097e723716367b59c" -> null
  - instance_initiated_shutdown_behavior = "stop" -> null
  - instance_state      = "running" -> null
  - instance_type       = "t3.micro" -> null
  - ipv6_address_count   = 0 -> null
  - ipv6_addresses      = [] -> null
  - key_name            = "serverkey" -> null
  - monitoring          = false -> null
  - placement_partition_number = 0 -> null
  - primary_network_interface_id = "eni-05c432975c007a7ce" -> null
  - private_dns         = "ip-10-0-10-168.me-central-1.compute.internal" -> null
  - private_ip          = "10.0.10.168" -> null
  - public_ip           = "3.28.126.164" -> null
  - region              = "me-central-1" -> null
  - secondary_private_ips = [] -> null
  - security_groups     = [] -> null
  - source_dest_check    = true -> null
  - subnet_id           = "subnet-03e5bda232f8fa581" -> null
  - tags               = {
    - "Name" = "dev-ec2-instance"
  } -> null
  - tags_all          = {
    - "Name" = "dev-ec2-instance"
  } -> null
  - tenancy            = "default" -> null
  - user_data          = "#!/bin/bash\nyum update -y\nyum install -y nginx\nsystemctl start nginx" -> null
}

```



```

# aws_key_pair.ssh_key will be destroyed
- resource "aws_key_pair" "ssh_key" {
  - arn          = "arn:aws:ec2:me-central-1:623705167686:key-pair/serverkey" -> null
  - fingerprint = "EChX8mPVY3SPkyVYNh9KaRj424WagsWqyFvd/1ztXo=" -> null
  - id          = "serverkey" -> null
  - key_name    = "serverkey" -> null
  - key_pair_id = "key-B45d668074c103868" -> null
  - key_type    = "ed25519" -> null
  - public_key  = "ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIPNhu9F8k0FKzEhsRwTKKLcwtL0Kr5yY8EtDI/aQ+THD_codespace/codespaces-79c519" -> null
  - region      = "me-central-1" -> null
  - tags        = {} -> null
  - tags_all    = {} -> null
}

```

```

# aws_subnet.myapp_subnet_1 will be destroyed
- resource "aws_subnet" "myapp_subnet_1" {
  - arn          = "arn:aws:ec2:me-central-1:623705167686:subnet/subnet-03e5bda232f8fa581" -> null
  - assign_ipv6_address_on_creation = false -> null
  - availability_zone              = "me-central-1a" -> null
  - availability_zone_id          = "mecl-az1" -> null
  - cidr_block                    = "10.0.10.0/24" -> null
  - enable_dns64                  = false -> null
  - enable_inj_at_device_index    = 0 -> null
  - enable_resource_name_dns_a_record_on_launch = false -> null
  - enable_resource_name_dns_aaaa_record_on_launch = false -> null
  - id                          = "subnet-03e5bda232f8fa581" -> null
  - ipv6_native                  = false -> null
  - map_customer_owned_ip_on_launch = false -> null
  - map_public_ip_on_launch      = false -> null
  - owner_id                     = "623705167686" -> null
  - private_dns_hostname_type_on_launch = "ip-name" -> null
  - region                      = "me-central-1" -> null
}

```

Plan: 0 to add, 0 to change, 7 to destroy.

```

Changes to Outputs:
  - ec2_public_ip = "3.29.125.164" -> null
  - subnet_id     = "subnet-03e5bda232f8fa581" -> null
  - vpc_id        = "vpc-08c0de79b0e26507d" -> null
aws_default_route_table.main_rt: Destroying... [id=rtb-0e0e0308f83c69ff2]
aws_instance.myapp_server: Destroying... [id=i-097e723716367b59c]
aws_default_route_table.main_rt: Destruction complete after 0s
aws_internet_gateway.myapp_igw: Destroying... [id=igw-08deab4a9bafbaac7]
aws_instance.myapp_server: Still destroying... [id=i-097e723716367b59c, 00m10s elapsed]
aws_internet_gateway.myapp_igw: Still destroying... [id=igw-08deab4a9bafbaac7, 00m10s elapsed]
aws_instance.myapp_server: Still destroying... [id=i-097e723716367b59c, 00m20s elapsed]
aws_internet_gateway.myapp_igw: Still destroying... [id=igw-08deab4a9bafbaac7, 00m20s elapsed]
aws_internet_gateway.myapp_igw: Destruction complete after 27s
aws_instance.myapp_server: Still destroying... [id=i-097e723716367b59c, 00m30s elapsed]
aws_instance.myapp_server: Still destroying... [id=i-097e723716367b59c, 00m40s elapsed]
aws_instance.myapp_server: Destruction complete after 40s
aws_key_pair.ssh_key: Destroying... [id=serverkey]
aws_subnet.myapp_subnet_1: Destroying... [id=subnet-03e5bda232f8fa581]
aws_default_security_group.myapp_sg: Destroying... [id=sg-0a106f65daa2ea6eb]
aws_default_security_group.myapp_sg: Destruction complete after 0s
aws_key_pair.ssh_key: Destruction complete after 0s
aws_subnet.myapp_subnet_1: Destruction complete after 0s
aws_vpc.myapp_vpc: Destroying... [id=vpc-08c0de79b0e26507d]
aws_vpc.myapp_vpc: Destruction complete after 1s

```

Destroy complete! Resources: 7 destroyed.

@23-22412-038-sys → /workspaces/Lab11 (main) \$ |



FATIMA JINNAH WOMEN UNIVERSITY

Department of Software Engineering

Lab Title:	Experiment # :
Student Name:	Regd. #:
Experiment Name:	

Performance Indicator	Level of Achievements				
	Excellent (5)	Good (4)	Average (3)	Below Average (2)	Poor (1)
Understanding & implementation Problem					
Report and presentation					
Maximum Marks		10	Obtained Marks		
Lab Instructor:					

Lab 12

Task 0

```
A11i@DESKTOP-KQE4P80 MINGW64 ~ (main)
$ gh repo create 23-22412-038-sys/Lab12 --public
✓ Created repository 23-22412-038-sys/Lab12 on github.com
https://github.com/23-22412-038-sys/Lab12
```

```
MINGW64/Users/A11i/Lab12
Cloning into 'Lab12'...
warning: you appear to have cloned an empty repository.

A11i@DESKTOP-KQE4P80 MINGW64 ~/Lab12 (main)
$ echo "# Lab 12 - Terraform & Ngini" > README.md

A11i@DESKTOP-KQE4P80 MINGW64 ~/Lab12 (main)
$ git add README.md
git commit -m "Initial commit"
warning: in the working copy of 'README.md', LF will be replaced by CRLF the next time git touches it
[main (root-commit) 50d9322] initial commit
1 file changed, 1 insertion(+)
create mode 100644 README.md

A11i@DESKTOP-KQE4P80 MINGW64 ~/Lab12 (main)
NAME                DISPLAY NAME     REPOSITORY      BRANCH  STATE    CREATED AT
sturdy-parakeet...  sturdy par...   23-22412-0...   main    Shutdown about 17 d...
obscure-zebra...   obscure zebra... 23-22412-0...   main    Shutdown about 17 d...
reimagined-ba...   reimagined... 23-22412-0...   main    Shutdown about 3 da...
ominous-maffi...   ominous wd... 23-22412-0...   main    Shutdown about 3 da...
didactic-thive...  didactic t... 23-22412-0...   main    Shutdown about 28 h...
animated-spac...   animated s... 23-22412-0...   main    Available less than ...

A11i@DESKTOP-KQE4P80 MINGW64 ~/Lab12 (main)
$
```

```

codespace@codespace-38614f: /bin/bash
all@codespace-top-x024p90: ~$ cd /Lab12 (main)
$ echo '# Lab 12 - Terraform & K8s' > README.md
all@codespace-top-x024p90: ~$ cd /Lab12 (main)
$ git add README.md
git commit -m 'Initial commit'
warning: in the working copy of 'README.md', LF will be replaced by CRLF the next time Git touches it
[main (root-commit) 5093221] Initial commit
1 file changed, 1 insertion(+)
create mode 100644 README.md
all@codespace-top-x024p90: ~$ cd /Lab12 (main)
NAME                                DISPLAY NAME                                STATUS    STATE    CREATED AT
sturdy-parakeet...                  sturdy parakeet...                          23-22412-038-sys/-trahal-nasir-38-  main    Shutdown  about 17 d...
obscure-zebra...                    obscure zebra...                             23-22412-038-sys/-trahal-nasir-38-  main    Shutdown  about 27 d...
reimagined-bassoon...               reimagined bassoon...                       23-22412-038-sys/-trahal-nasir-38-  main    Shutdown  about 3 d...
celinious-waffle...                celinious waffle...                         23-22412-038-sys/-trahal-nasir-38-  main    Shutdown  about 3 d...
didactic-invention...              didactic invention...                       23-22412-038-sys/-trahal-nasir-38-  main    Shutdown  about 23 h...
animated-space-disc...             animated space disc...                      23-22412-038-sys/Lab12              main    Available less than ...
all@codespace-top-x024p90: ~$ cd /Lab12 (main)
$ gh codespace list
NAME                                DISPLAY NAME                                REPOSITORY                                BRANCH    STATE    CREATED AT
sturdy-parakeet-x024p90-7y9p9vdx7j3r1cp  sturdy parakeet...                          23-22412-038-sys/-trahal-nasir-38-  main    Shutdown  about 17 days ago
obscure-zebra-7y9p9vdx7j3r1cp              obscure zebra...                             23-22412-038-sys/-trahal-nasir-38-  main    Shutdown  about 27 days ago
reimagined-bassoon-x024p90-556459nvdxw  reimagined bassoon...                       23-22412-038-sys/-trahal-nasir-38-  main    Shutdown  about 3 days ago
celinious-waffle-556459nvdxw              celinious waffle...                         23-22412-038-sys/-trahal-nasir-38-  main    Shutdown  about 3 days ago
didactic-invention-x024p90-556459nvdxw  didactic invention...                       23-22412-038-sys/Lab12              main    Shutdown  about 23 hours ago
animated-space-disc-x024p90-556459nvdxw  animated space disc...                      23-22412-038-sys/Lab12              main    Available less than a minute ago
all@codespace-top-x024p90: ~$ cd /Lab12 (main)
$ gh codespace ssh -c animated-space-disc-x024p90-556459nvdxw
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-1030-azure x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.
23-22412-038-sys - /workspaces/Lab12 (main) $

```

Task 1

```

@23-22412-038-sys -> /workspaces/Lab12 (main) $ mkdir -p ~/Lab12
cd ~/Lab12

```

```

@23-22412-038-sys -> ~/Lab12 $ touch main.tf variables.tf outputs.tf locals.tf terraform.tfvars entry-script.sh
ls -la
total 12
drwxrwxr-x 2 codespace codespace 4096 Jan  1 06:59 .
drwxr-x--- 1 codespace codespace 4096 Jan  1 06:58 ..
-rw-rw-r-- 1 codespace codespace  0 Jan  1 06:59 entry-script.sh
-rw-rw-r-- 1 codespace codespace  0 Jan  1 06:59 locals.tf
-rw-rw-r-- 1 codespace codespace  0 Jan  1 06:59 main.tf
-rw-rw-r-- 1 codespace codespace  0 Jan  1 06:59 outputs.tf
-rw-rw-r-- 1 codespace codespace  0 Jan  1 06:59 terraform.tfvars
-rw-rw-r-- 1 codespace codespace  0 Jan  1 06:59 variables.tf

```

```
codepace@codepaces-385146: nano variables.tf
GNU nano 2.2
variables.tf
variable "vpc_cidr_block" {}
variable "subnet_cidr_block" {}
variable "availability_zone" {}
variable "env_prefix" {}
variable "instance_type" {}
variable "public_key" {}
variable "private_key" {}
```

```
codepace@codepaces-385146: nano outputs.tf
GNU nano 2.2
outputs.tf
output "aws_instance_public_ip" {
  value = aws_instance.myapp-server.public_ip
}
```

```
codopex@codopexs-385146: nano locals.tf
GNU nano 2.2
locals.tf
locals {
  ey_ip = "${chomp(data.http.ey_ip.response_body)}/32"
}

data "http" "ey_ip" {
  url = "https://icanhazip.com/"
}
```

Help Exit Write Out Read File Where Is Replace Cut Paste Execute Justify Location Go To Line Undo Redo Set Mark Copy To Bracket Where Was Previous Next

11:01 PM 12/31/2025

```
codopex@codopexs-385146: nano terraform.tfvars
GNU nano 2.2
terraform.tfvars
vpc_cidr_block = "10.0.0.0/16"
subnet_cidr_block = "10.0.10.0/24"
availability_zone = "us-central1-a"
env_prefix = "dev"
instance_type = "t3.micro"
public_key = "~/.ssh/id_ed25519.pub"
private_key = "~/.ssh/id_ed25519"
```

Help Exit Write Out Read File Where Is Replace Cut Paste Execute Justify Location Go To Line Undo Redo Set Mark Copy To Bracket Where Was Previous Next

11:02 PM 12/31/2025

```
codespace@codespaces-38814f: nano user_data.sh
GNU nano 2.2.1 user_data.sh
#!/bin/bash
set -e
yum update -y
yum install -y nginx
systemctl start nginx
systemctl enable nginx
```

```
923-22412-038-sys → /workspaces/Lab12 (main) $ ssh-keygen -t ed25519 -f ~/.ssh/id_ed25519 -N ""
Generating public/private ed25519 key pair.
Your identification has been saved in /home/codespace/.ssh/id_ed25519
Your public key has been saved in /home/codespace/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:JeqiSxMUQnNAGWFPkIuyx4r94vSxcKW3vZixqUGiTbg codespace@codespaces-38814f
The key's randomart image is:
+--[ED25519 256]--+
|  =+ |
| ooo |
| ..+.. . x |
| oo. . o |
| o=+.. o S |
| o^=+ + |
| E+8 * + |
| ++o^ = o |
| .++=..+ o |
|          |
+-----+
[ED25519]
```

```
923-22412-038-sys → /workspaces/Lab12 (main) $ terraform init
Initializing the backend...
Initializing provider plugins...
- Finding hashicorp/aws versions matching "~> 5.0"...
- Installing hashicorp/aws v5.100.0...
- Installed hashicorp/aws v5.100.0 (signed by HashiCorp)
Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

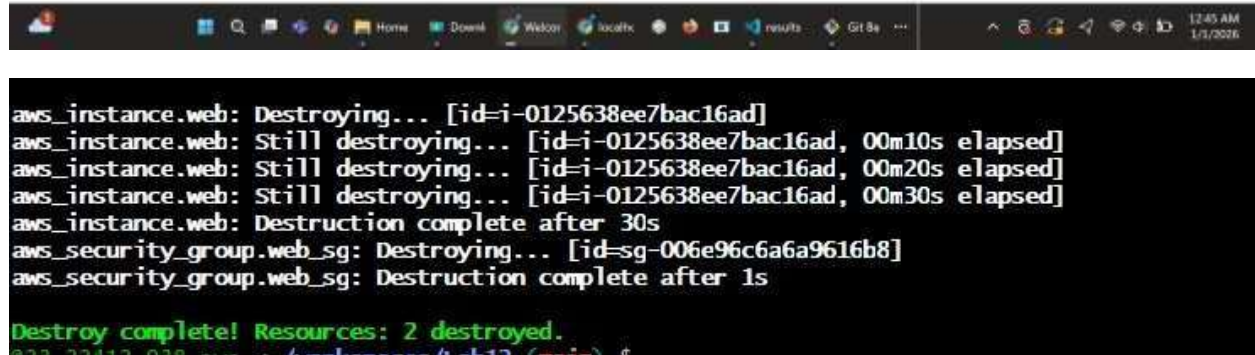
If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
```



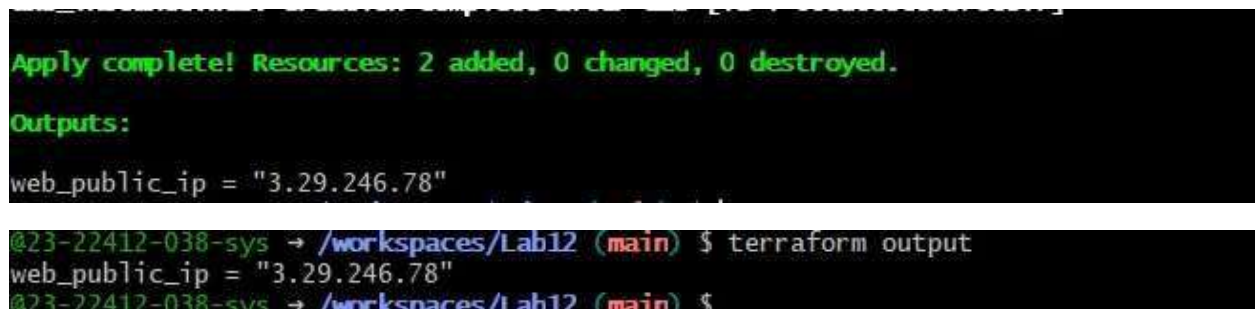
```
Apply complete! Resources: 0 added, 2 changed, 0 destroyed.
```

Outputs:

```
web_public_ip = "40.172.226.85"
```



Task 2



Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.

Task 3

```
cloudspace@cloudspace-386146: /aws$ cat web.yml
apiVersion: "aws-eks-v1"
security_group: ["aws-eks-v1-group-web-sg"]

tags: {}
name: "webserver"

# user data to install nginx
user_data: ["cat <<EOF"]
# EOF
sudo yum update -y
sudo yum install -y nginx
sudo systemctl start nginx
sudo systemctl enable nginx
EOF

# Provisioners
- name: "file"
  # File provisioner: copy a file to the instance
  provisioner: "file"
  source: "nginx.html"
  destination: "/home/ec2-user/nginx.html"

# Remote-exec provisioner: Run commands on the instance
- name: "remote-exec"
  provisioner: "remote-exec"
  scripts:
    - "sudo mv /home/ec2-user/nginx.html /usr/share/nginx/html/nginx.html"
    - "sudo systemctl restart nginx"

connection: {}
type: "ssh"
key: "aws-eks-v1"
host: "ec2-3-29-246-78.us-east-2.compute.amazonaws.com" # path to your private key
host_key: "ssh-rsa" # path to your private key
host_key_fingerprint: "ssh-rsa"

# Local-exec provisioner: Run commands locally after instance creation
- name: "local-exec"
  provisioner: "local-exec"
  command: "echo 'web server is deployed at host ${web_public_ip}'"

# Output
output: "web_public_ip"
action: "aws-eks-v1-web-public-ip"
```

Plan: 0 to add, 2 to change, 0 to destroy.

aws_security_group.web_sg: Modifying... [id=sg-08fca1167275f9469]

aws_security_group.web_sg: Modifications complete after 2s [id=sg-08fca1167275f9469]

aws_instance.web: Modifying... [id=i-008a003c8337863f7]

aws_instance.web: Modifications complete after 2s [id=i-008a003c8337863f7]

Apply complete! Resources: 0 added, 2 changed, 0 destroyed.

Outputs:

web_public_ip = "3.29.246.78"

3.29.246.78

Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.

```
terraform destroy
web.public.ip = "3.29.246.78"
aws_security_group.web_sg: Refreshing state... [id=sg-08fca1167275f9469]
aws_instance.web: Refreshing state... [id=i-008a003c8337863f7]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
  ~ destroy

Terraform will perform the following actions:

# aws_instance.web will be destroyed
+ resource "aws_instance" "web" {
  ~ ami                         = "ami-075212746751066bd" -> null
  ~ arn                        = "arn:aws:ec2:us-central-1:623705187686:instance/i-008a003c8337863f7" -> null
  ~ associate_public_ip_address = true -> null
  ~ availability_zone          = "us-central-1b" -> null
  ~ cpu_core_count             = 2 -> null
  ~ cpu_threads_per_core       = 1 -> null
  ~ disable_api_stop           = false -> null
  ~ disable_api_termination    = false -> null
  ~ ebs_optimized              = false -> null
  ~ get_password_data          = false -> null
  ~ hibernation                = false -> null
  ~ id                         = "i-008a003c8337863f7" -> null
  ~ instance_initiated_shutdown_behavior = "stop" -> null
  ~ instance_state             = "running" -> null
  ~ instance_type              = "t4g.micro" -> null
  ~ ipv6_address_count         = 0 -> null
  ~ ipv6_addresses             = [] -> null
}
```

```
aws_instance.web: Destroying... [id=i-008a003c8337863f7]
aws_instance.web: Still destroying... [id=i-008a003c8337863f7, 00m10s elapsed]
aws_instance.web: Still destroying... [id=i-008a003c8337863f7, 00m20s elapsed]
aws_instance.web: Still destroying... [id=i-008a003c8337863f7, 00m30s elapsed]
aws_instance.web: Still destroying... [id=i-008a003c8337863f7, 00m40s elapsed]
aws_instance.web: Still destroying... [id=i-008a003c8337863f7, 00m50s elapsed]
aws_instance.web: Still destroying... [id=i-008a003c8337863f7, 01m00s elapsed]
aws_instance.web: Still destroying... [id=i-008a003c8337863f7, 01m10s elapsed]
aws_instance.web: Destruction complete after 1m11s
aws_security_group.web_sg: Destroying... [id=sg-08fca1167275f9469]
aws_security_group.web_sg: Destruction complete after 1s
```

Destroy complete! Resources: 2 destroyed.

```
codospex@codospexrv-385146: nano main.tf
protocol = "tcp"
tags_block = ["0.0.0.0/0"]

tags = {}
name = "webapp"

# Key Pair
resource "aws_instance" "web" {
  ami = "ami-07121274710068b4" # replace with your AMI ID
  instance_type = "t3g.micro" # free tier eligible
  key_name = "web-key-pair" # from key name
  subnet_id = aws_subnet.web_subnet_id
  tags = {}
  name = "web-001-vm"
}

# Route Table
resource "aws_route_table" "web" {
  vpc_id = aws_vpc.web_vpc_id
  route_block = [
    {
      cidr_block = "0.0.0.0/0"
      gateway_id = aws_internet_gateway.web_igw_id
    }
  ]
}

# Attach Route Table to Subnet
resource "aws_route_table_association" "web" {
  subnet_id = aws_subnet.web_subnet_id
  route_table_id = aws_route_table.web_id
}
```

Task 4

```
1 variable "vpc_id" {}
2 variable "subnet_cidr_block" {}
3 variable "availability_zone" {}
4 variable "env_prefix" {}
5 variable "default_route_table_id" {}
```

```
1 resource "aws_subnet" "myapp_subnet_1" {
2     vpc_id          = var.vpc_id
3     cidr_block       = var.subnet_cidr_block
4     availability_zone = var.availability_zone
5     map_public_ip_on_launch = true
6     tags = {
7         Name = "${var.env_prefix}-subnet-1"
8     }
9 }
10
11 resource "aws_internet_gateway" "myapp_igw" {
12     vpc_id = var.vpc_id
13     tags = {
14         Name = "${var.env_prefix}-igw"
15     }
16 }
17
18 resource "aws_default_route_table" "main_rt" {
19     default_route_table_id = var.default_route_table_id
20
21     route {
22         cidr_block = "0.0.0.0/0"
23         gateway_id = aws_internet_gateway.myapp_igw.id
24     }
25     tags = {
26         Name = "${var.env_prefix}-rt"
27     }
28 }
```

```
1 output "subnet" {
2     value = aws_subnet.myapp_subnet_1
3 }
```



```

module "myapp-subnet" {
  source           = "./modules/subnet"
  vpc_id           = aws_vpc.myapp_vpc.id
  subnet_cidr_block = var.subnet_cidr_block
  availability_zone  = var.availability_zone
  env_prefix        = var.env_prefix
  default_route_table_id = aws_vpc.myapp_vpc.default_route_table_id
}

```

Initializing the backend...

Initializing modules...

- myapp-subnet in modules/subnet

Initializing provider plugins...

- Reusing previous version of hashicorp/aws from the dependency lock file
- Reusing previous version of hashicorp/http from the dependency lock file
- Using previously-installed hashicorp/aws v6.28.0
- Using previously-installed hashicorp/http v3.5.0

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see any changes that are required for your infrastructure. All Terraform commands should now work.

If you ever set or change modules or backend configuration for Terraform, rerun this command to reinitialize your working directory. If you forget, other commands will detect it and remind you to do so if necessary.

```

module.myapp-subnet.aws_subnet.myapp_subnet_1: Creating...
aws_default_security_group.default_sg: Creating...
module.myapp-subnet.aws_internet_gateway.myapp_igw: Creation complete after 1s [id=igw-0a7894f04dfd9a537]
module.myapp-subnet.aws_default_route_table.main_rt: Creating...
module.myapp-subnet.aws_default_route_table.main_rt: Creation complete after 1s [id=rtb-05b894e5b39dd25a8]
aws_default_security_group.default_sg: Creation complete after 4s [id=sg-0f8b4a66badcdc8ca]
module.myapp-subnet.aws_subnet.myapp_subnet_1: Still creating... [00m10s elapsed]
module.myapp-subnet.aws_subnet.myapp_subnet_1: Creation complete after 11s [id=subnet-0e29387ac22e28b6e]
aws_instance.myapp-server: Creating...
aws_instance.myapp-server: Still creating... [00m10s elapsed]
aws_instance.myapp-server: Creation complete after 14s [id=i-082355d8b3da69cc0]

```

Apply complete! Resources: 7 added, 0 changed, 0 destroyed.

Outputs:

aws_instance_public_ip = "51.112.167.169"

Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.

Task 5

```
$ mkdir -p modules/webserver
$ touch modules/webserver/main.tf modules/webserver/variables.tf modules/webserver/outputs.tf
$ ls -R modules/webserver
```

```
modules > webserver > variables.tf
3  variable "availability_zone" {}
4  variable "public_key" {}
5  variable "my_ip" {}
6  variable "vpc_id" {}
7  variable "subnet_id" {}
8  variable "script_path" {}
9  variable "instance_suffix" {}
```

```

1 resource "aws_security_group" "web_sg" {
2     vpc_id      = var.vpc_id
3     name        = "${var.env_prefix}-web-sg-${var.instance_suffix}"
4     description = "Security group for web server allowing HTTP, HTTPS and SSH"
5
6     ingress {
7         from_port = 22
8         to_port   = 22
9         protocol  = "tcp"
10        cidr_blocks = [var.my_ip]
11    }
12    ingress {
13        from_port = 443
14        to_port   = 443
15        protocol  = "tcp"
16        cidr_blocks = ["0.0.0.0/0"]
17    }
18    ingress {
19        from_port = 80
20        to_port   = 80
21        protocol  = "tcp"
22        cidr_blocks = ["0.0.0.0/0"]
23    }
24    egress {
25        from_port = 0
26        to_port   = 0
27        protocol  = "-1"
28        cidr_blocks = ["0.0.0.0/0"]
29    }
30    tags = { Name = "${var.env_prefix}-web-sg-${var.instance_suffix}" }
31 }
32
33 resource "aws_key_pair" "ssh-key" {

```

```

1 output "aws_instance" {
2     value = aws_instance.myapp-server
3 }

```

```

1 output "webserver_public_ip" {
2     value = module.myapp-webserver.aws_instance.public_ip
3 }

```

```
1  provider "aws" {
2      region = "me-central-1"
3  }
4
5  resource "aws_vpc" "myapp_vpc" {
6      cidr_block = var.vpc_cidr_block
7      tags = { Name = "${var.env_prefix}-vpc" }
8  }
9
10 module "myapp-subnet" {
11     source                = "./modules/subnet"
12     vpc_id                = aws_vpc.myapp_vpc.id
13     subnet_cidr_block    = var.subnet_cidr_block
14     availability_zone     = var.availability_zone
15     env_prefix           = var.env_prefix
16     default_route_table_id = aws_vpc.myapp_vpc.default_route_table_id
17 }
18
19 module "myapp-webserver" {
20     source                = "./modules/webserver"
21     env_prefix           = var.env_prefix
22     instance_type        = var.instance_type
23     availability_zone     = var.availability_zone
24     public_key           = var.public_key
25     my_ip                = local.my_ip
26     vpc_id               = aws_vpc.myapp_vpc.id
27     subnet_id            = module.myapp-subnet.subnet.id
28     script_path          = "./entry-script.sh"
29     instance_suffix      = "0"
30 }
```

Initializing the backend...

Initializing modules...

- myapp-webserver in modules/webserver

Initializing provider plugins...

- Reusing previous version of hashicorp/aws from the dependency lock file
- Reusing previous version of hashicorp/http from the dependency lock file
- Using previously-installed hashicorp/aws v6.28.0
- Using previously-installed hashicorp/http v3.5.0

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see any changes that are required for your infrastructure. All Terraform commands should now work.

If you ever set or change modules or backend configuration for Terraform, rerun this command to reinitialize your working directory. If you forget, other commands will detect it and remind you to do so if necessary.

Apply complete! Resources: 3 added, 0 changed, 3 destroyed.

Outputs:

webserver public ip = "51.112.231.203"

webserver_public_ip = "51.112.231.203"

Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.

Task 6

```
1 #!/bin/bash
2 set -e
3
4 # Install and Start Nginx
5 yum update -y
6 yum install -y nginx
7 systemctl start nginx
8 systemctl enable nginx
9
10 # Create directories for SSL certificates
11 mkdir -p /etc/ssl/private
12 mkdir -p /etc/ssl/certs
13
14 # Get IMDSv2 Token for security
15 TOKEN=$(curl -s -X PUT "http://169.254.169.254/latest/api/token" -H "X-aws-ec2-metadata-token-ttl-seconds: 21600")
16
17 # Get current public IP dynamically
18 PUBLIC_IP=$(curl -s -H "X-aws-ec2-metadata-token: $TOKEN" http://169.254.169.254/latest/meta-data/public-ipv4)
19
20 # Generate self-signed certificate with the Public IP as the Common Name (CN)
21 openssl req -x509 -nodes -days 365 -newkey rsa:2048 \
22   -keyout /etc/ssl/private/selfsigned.key \
23   -out /etc/ssl/certs/selfsigned.crt \
24   -subj "/CN=$PUBLIC_IP" \
25   -addext "subjectAltName=IP:$PUBLIC_IP" \
26   -addext "basicConstraints=CA:FALSE" \
27   -addext "keyUsage=digitalSignature,keyEncipherment" \
28   -addext "extendedKeyUsage=serverAuth"
29
30 # Configure Nginx for SSL and Redirects
31 cat <<EOF > /etc/nginx/nginx.conf
32 user nginx;
33 worker_processes auto;
```

Apply complete! Resources: 7 added, 0 changed, 0 destroyed.

Outputs:

webserver public_ip = "3.28.136.167"

Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.

Task 7

```
2 yum update -y
3 yum install httpd -y
4 systemctl start httpd
5 systemctl enable httpd
6
7 echo "ch$welcome to My Web Servers/h1>" > /var/www/html/index.html
8 hostnamectl set-hostname myapp-subserver
9 echo "ch2$hostname: $(hostname)s/h2>" >> /var/www/html/index.html
10
11 TOKEN=$(curl -s -X PUT "http://169.254.169.254/latest/api/token" -H "X-aws-ec2-metadata-token-ttl-seconds: 21600")
12
13 echo "ch3$Private IP: $(curl -s -H "X-aws-ec2-metadata-token: $TOKEN" http://169.254.169.254/latest/meta-data/local-ipv4)</h2>" >> /var/www/html/index.html
14 echo "ch3$Public IP: $(curl -s -H "X-aws-ec2-metadata-token: $TOKEN" http://169.254.169.254/latest/meta-data/public-ipv4)</h2>" >> /var/www/html/index.html
15 echo "ch3$Public DNS: $(curl -s -H "X-aws-ec2-metadata-token: $TOKEN" http://169.254.169.254/latest/meta-data/public-hostname)</h2>" >> /var/www/html/index.html
16 echo "ch3$Deployed via Terraform/h2>" >> /var/www/html/index.html
```

3. Webserver Module Call - Nginx Proxy Server (Instance Suffix 0)

```
module "myapp-webserver" {
  source                = "./modules/webserver"
  env_prefix            = var.env_prefix
  instance_type         = var.instance_type
  availability_zone     = var.availability_zone
  public_key            = var.public_key
  my_ip                 = local.my_ip
  vpc_id                = aws_vpc.myapp_vpc.id
  subnet_id             = module.myapp-subnet.subnet.id
  script_path           = "./entry-script.sh"
  instance_suffix       = "0"
}
```

4. Webserver Module Call - Apache Backend Server (Instance Suffix 1)

```
module "myapp-web-1" {
  source                = "./modules/webserver"
  env_prefix            = var.env_prefix
  instance_type         = var.instance_type
  availability_zone     = var.availability_zone
  public_key            = var.public_key
  my_ip                 = local.my_ip
  vpc_id                = aws_vpc.myapp_vpc.id
  subnet_id             = module.myapp-subnet.subnet.id
  script_path           = "./apache.sh"
  instance_suffix       = "1"
}
```



```

output "webserver_public_ip" {
  value = module.myapp-webserver.aws_instance.public_ip
}

output "aws_web-1_public_ip" {
  value = module.myapp-web-1.aws_instance.public_ip
}

```

Apply complete! Resources: 3 added, 0 changed, 0 destroyed.

Outputs:

```

aws_web-1_public_ip = "158.252.32.32"
webserver_public_ip = "3.28.136.167"

```

The authenticity of host '3.28.136.167 (3.28.136.167)' can't be established.
ED25519 key fingerprint is
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '3.28.136.167' (ED25519) to the list of known hosts.

A newer release of "Amazon Linux" is available.

Version 2023.10.20260105:

Run "/usr/bin/dnf check-release-update" for full release and version update info

```

#_
~\_ #####_ Amazon Linux 2023
~\_ #####\
~\_ \###|
~\_ \#/_____ https://aws.amazon.com/linux/amazon-linux-2023
~\_ V~' '->
~\_ /
~\_ /
~\_ /
~\_ /m/'

```

```

user nginx;
worker_processes auto;
error_log /var/log/nginx/error.log notice;
pid /run/nginx.pid;

events {
    worker_connections 1024;
}

http {
    include /etc/nginx/mime.types;
    default_type application/octet-stream;

    server {
        listen 443 ssl;
        server_name 3.28.136.167;

        ssl_certificate /etc/ssl/certs/selfsigned.crt;
        ssl_certificate_key /etc/ssl/private/selfsigned.key;

        # --- UPDATED SECTION START ---
        location / {
            proxy_pass http://158.252.32.32:80;
            proxy_set_header Host $host;
            proxy_set_header X-Real-IP $remote_addr;
            proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
            proxy_set_header X-Forwarded-Proto $scheme;
        }
        # --- UPDATED SECTION END ---
    }

    server {
        listen 80;
        server_name _;
        return 301 https://$host$request_uri;
    }
}

```

```

[ec2-user@ip-10-0-10-51 ~]$ sudo systemctl restart nginx
[ec2-user@ip-10-0-10-51 ~]$

```

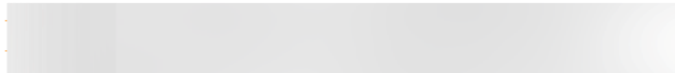
```

[ec2-user@ip-10-0-10-51 ~]$ cat /var/log/nginx/error.log
2026/01/17 19:24:43 [notice] 3012#3012: using the "epoll" event method
2026/01/17 19:24:43 [notice] 3012#3012: nginx/1.28.0
2026/01/17 19:24:43 [notice] 3012#3012: OS: Linux 6.1.158-100.204.el9.aarch64
2026/01/17 19:24:43 [notice] 3012#3012: getrlimit(RLIMIT_NOFILE): 65535-65535
2026/01/17 19:24:43 [notice] 3055#3055: start worker processes
2026/01/17 19:24:43 [notice] 3055#3055: start worker process 3057
2026/01/17 19:24:43 [notice] 3055#3055: start worker process 3058
2026/01/17 19:24:44 [notice] 3055#3055: signal 2 (SIGQUIT) received from 1, shutting down
2026/01/17 19:24:44 [notice] 3057#3057: gracefully shutting down
2026/01/17 19:24:44 [notice] 3057#3057: exiting
2026/01/17 19:24:44 [notice] 3057#3057: exit
2026/01/17 19:24:44 [notice] 3058#3058: gracefully shutting down
2026/01/17 19:24:44 [notice] 3058#3058: exiting
2026/01/17 19:24:44 [notice] 3058#3058: exit
2026/01/17 19:24:44 [notice] 3055#3055: signal 17 (SIGCHLD) received from 3057
2026/01/17 19:24:44 [notice] 3055#3055: worker process 3057 exited with code 0
2026/01/17 19:24:44 [notice] 3055#3055: worker process 3058 exited with code 0
2026/01/17 19:24:44 [notice] 3055#3055: exit
2026/01/17 19:24:44 [warn] 3050#3050: could not build optimal types_hash, you should increase either types_hash_max_size: 1024 or types_hash_bucket_size: 64; ignoring types_h
ash_bucket_size
2026/01/17 19:24:44 [warn] 3004#3004: could not build optimal types_hash, you should increase either types_hash_max_size: 1024 or types_hash_bucket_size: 64; ignoring types_h
ash_bucket_size
2026/01/17 19:24:44 [notice] 3004#3004: using the "epoll" event method

```

Welcome to My Web Server

Hostname: myapp-webserver



Public IP: 158.252.32.32

Public DNS:

Deployed via Terraform

Task 8

```
48 # 5. Webserver Module Call - Second Apache Backend (Instance Suffix 2)
49 module "myapp-web-2" {
50     source          = "./modules/webserver"
51     env_prefix      = var.env_prefix
52     instance_type   = var.instance_type
53     availability_zone = var.availability_zone
54     public_key      = var.public_key
55     my_ip           = local.my_ip
56     vpc_id          = aws_vpc.myapp_vpc.id
57     subnet_id       = module.myapp-subnet.subnet.id
58     script_path     = "./apache.sh"
59     instance_suffix = "2"
60 }
```

```
outputs.tf
1  output "webserver_public_ip" {
2      value = module.myapp-webserver.aws_instance.public_ip
3  }
4
5  output "aws_web-1_public_ip" {
6      value = module.myapp-web-1.aws_instance.public_ip
7  }
8  output "aws_web-2_public_ip" {
9      value = module.myapp-web-2.aws_instance.public_ip
10 }
```

Apply complete! Resources: 3 added, 0 changed, 0 destroyed.

Outputs:

```
aws_web-1_public_ip = "158.252.32.32"
aws_web-2_public_ip = "3.28.44.105"
webserver_public_ip = "3.28.136.167"
```

Welcome to My Web Server

Hostname: myapp-webserver

Public IP: 3.28.44.105

Public DNS:

Deployed via Terraform

Welcome to My Web Server

Hostname: myapp-webserver

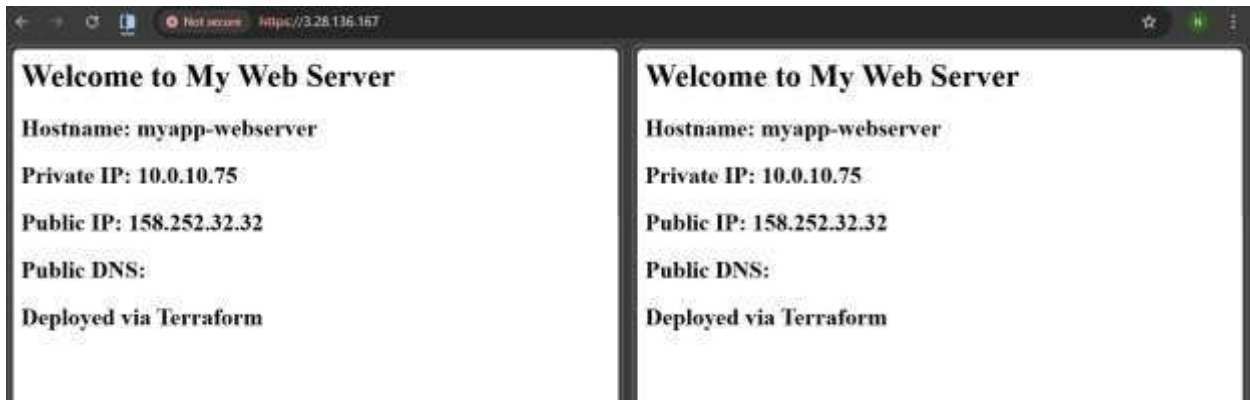
Public IP: 158.252.32.32

Public DNS:

Deployed via Terraform

Task 9

```
# Define the group with Web-2 as BACKUP
upstream backend_servers {
    server 158.252.32.32:80;      # web-1 (Primary)
    server 3.28.44.105:80 backup; # web-2 (Backup)
}
```



```
# Define the group with Web-1 as BACKUP
upstream backend_servers {
    server 158.252.32.32:80 backup;      # web-1 (backup)
    server 3.28.44.105:80 ;             # web-2 (Primary)
}
```



Task 10


```

location / {
    proxy_pass http://backend_servers;

    # 2. Enable caching for this location
    proxy_cache my_cache;
    proxy_cache_valid 200 60m;
    proxy_cache_key "$scheme$request_uri";

    # 3. Add header to see if it's a HIT or MISS in the browser
    add_header X-Cache-Status $upstream_cache_status;

    proxy_set_header Host $host;
    proxy_set_header X-Real-IP $remote_addr;
}

```

```

[ec2-user@ip-10-0-10-51 ~]$ sudo ls -la /var/cache/nginx/
total 0
drwx-----. 2 nginx root   6 Jan 17 20:04 .
drwxr-xr-x.  9 root  root 101 Jan 17 20:04 ..
[ec2-user@ip-10-0-10-51 ~]$

```

Clean up :

```

module.myapp-web-2.aws_key_pair.key: Destruction complete after 0s
module.myapp-subnet.aws_subnet.myapp_subnet_1: Destruction complete after 1s
module.myapp-web-2.aws_security_group.web_sg: Destruction complete after 1s
aws_vpc.myapp_vpc: Destroying... [id=vpc-07f83cc4f9b96672f]
aws_vpc.myapp_vpc: Destruction complete after 1s

Destroy complete! Resources: 13 destroyed.

```

```

{
  "version": 4,
  "terraform_version": "1.14.3",
  "serial": 90,
  "lineage": "2702fe6f-53f9-f4fd-4414-2e9d6d5a13fd",
  "outputs": {},
  "resources": [],
  "check_results": null
}

```

```

total 65420
drwxrwxrwx+ 6 codespace root          4096 Jan 17 20:14 .
drwxr-xrwx+ 5 codespace root          4096 Jan 17 18:11 ..
drwxrwxrwx+ 8 codespace root          4096 Jan 17 18:22 .git
drwxr-xr-x+ 4 codespace codespace      4096 Jan 17 19:10 .terraform
-rw-r--r-- 1 codespace codespace      2422 Jan 17 18:35 .terraform.lock.hcl
-rw-rw-rw- 1 codespace root             11 Jan 17 18:22 README.md
-rw-rw-rw- 1 codespace codespace        926 Jan 17 19:28 apache.sh
drwxr-xr-x+ 3 codespace codespace      4096 Jan 16 19:26 aws
-rw-rw-rw- 1 codespace codespace 66875639 Jan 17 18:37 awscli2.zip
-rw-rw-r-- 1 codespace codespace      1705 Jan 17 19:23 entry-script.sh
-rw-rw-r-- 1 codespace codespace        122 Jan 17 18:26 locals.tf
-rw-rw-r-- 1 codespace codespace      2010 Jan 17 19:47 main.tf
drwxrwxrwx+ 4 codespace codespace      4096 Jan 17 19:13 modules
-rw-rw-r-- 1 codespace codespace        259 Jan 17 19:47 outputs.tf
-rw-rw-rw- 1 codespace codespace        182 Jan 17 20:14 terraform.tfstate
-rw-rw-rw- 1 codespace codespace     41823 Jan 17 20:13 terraform.tfstate.backup
-rw-rw-r-- 1 codespace codespace        219 Jan 17 18:27 terraform.tfvars
-rw-rw-r-- 1 codespace codespace        196 Jan 17 18:26 variables.tf

```



FATIMA JINNAH WOMEN UNIVERSITY

Department of Software Engineering

Lab Title:	Experiment # :
Student Name:	Regd. #:
Experiment Name:	

Performance Indicator	Level of Achievements				
	Excellent (5)	Good (4)	Average (3)	Below Average (2)	Poor (1)
Understanding & implementation Problem					
Report and presentation					
Maximum Marks		10	Obtained Marks		
Lab Instructor:					

Lab 13

Task 0

```
Ali@DESKTOP-KQE4P80 MINGW64 ~/Lab13 (main)
$ gh codespace ssh
? Choose codespace: 23-22412-038-sys/Lab13 [main]: vigilant halibut
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-1030-azure x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.
```

Task 1

```
@23-22412-038-sys →/workspaces/Lab13 (main) $ mkdir -p ~/Lab13
cd ~/Lab13
```

```
@23-22412-038-sys →~/Lab13 $ touch main.tf
ls -la
total 12
drwxr-xr-x 2 codespace codespace 4096 Jan  2 10:08 .
drwxr-x--- 1 codespace codespace 4096 Jan  2 10:07 ..
-rw-r--r-- 1 codespace codespace    0 Jan  2 10:08 main.tf
```

```
GNU nano 7.2 main.tf *
provider "aws" {
  shared_config_files      = ["~/.aws/config"]
  shared_credentials_files = ["~/.aws/credentials"]
}
```

```
@23-22412-038-sys →~/Lab13 $ aws sts get-caller-identity
{
  "UserId": "623705167686",
  "Account": "623705167686",
  "Arn": "arn:aws:iam::623705167686:root"
}
```

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see any changes that are required for your infrastructure. All Terraform commands should now work.

```
@23-22412-038-sys →~/Lab13 $ terraform apply -auto-approve
```

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:

+ create

Terraform will perform the following actions:

```
# aws_iam_group.developers will be created
```

```
@23-22412-038-sys →~/Lab13 $ terraform output
```

```
group_details = {
  "group_arn" = "arn:aws:iam::623705167686:group/groups/developers"
  "group_name" = "developers"
  "unique_id" = "AGPAZCN5ZS5DPX3WSBTHJ"
}
```

```
@23-22412-038-sys →~/Lab13 $ █
```


User groups (1) [Info](#)

A user group is a collection of IAM users. Use groups to specify permissions

 Search

☐	Group name		Users
☐	developers		

Summary

[Edit](#)

User group name
developers

Creation time
January 02, 2026, 02:21 (UTC-08:00)

ARN
 arn:aws:iam::623705167686:group/groups/developers

[Users](#) | [Permissions](#) | [Access Advisor](#)

Items in this group (0)

 [Add users](#) [Add groups](#) [Add roles](#)

Task 2

```
# IAM user: loadbalancer
resource "aws_iam_user" "lb" {
  name      = "loadbalancer"
  path      = "/users/"
}
```

Apply complete! Resources: 2 added, 0 changed, 0 destroyed.

Outputs:

```
group_details = {
  "group_arn" = "arn:aws:iam::623705167686:group/groups/developers"
}
```


Outputs:

```
group_details = {
    "group_arn" = "arn:aws:iam::623705167686:group/groups/developers"
    "group_name" = "developers"
    "unique_id" = "AGPAZCN5ZS5DPX3WSBTHJ"
}
```

```
@23-22412-038-sys → ~/Lab13 $ terraform output
```

```
group_details = {
    "group_arn" = "arn:aws:iam::623705167686:group/groups/developers"
    "group_name" = "developers"
    "unique_id" = "AGPAZCN5ZS5DPX3WSBTHJ"
}
```

```
user_details = {
    "unique_id" = "AIDAZCN5ZS5DOWWNH4ETE"
    "user_arn" = "arn:aws:iam::623705167686:user/users/loadbalancer"
    "user_name" = "loadbalancer"
}
```

loadbalancer [Info](#)

Summary

ARN

 `arn:aws:iam::623705167686:user/users/loadbalancer`

Created

January 02, 2026, 02:28 (UTC-08:00)

Permissions

Groups (1)

Tags (1)

Security

User groups membership

A user group is a collection of IAM users. Use groups to specify

☐ | **Group name**

☐ [developers](#)

Task 3

```
# Attach AmazonEC2FullAccess to developers group
resource "aws_iam_group_policy_attachment" "developer_ec2_fullaccess" {
  group      = aws_iam_group.developers.name
  policy_arn = "arn:aws:iam::aws:policy/AmazonEC2FullAccess"
}
```

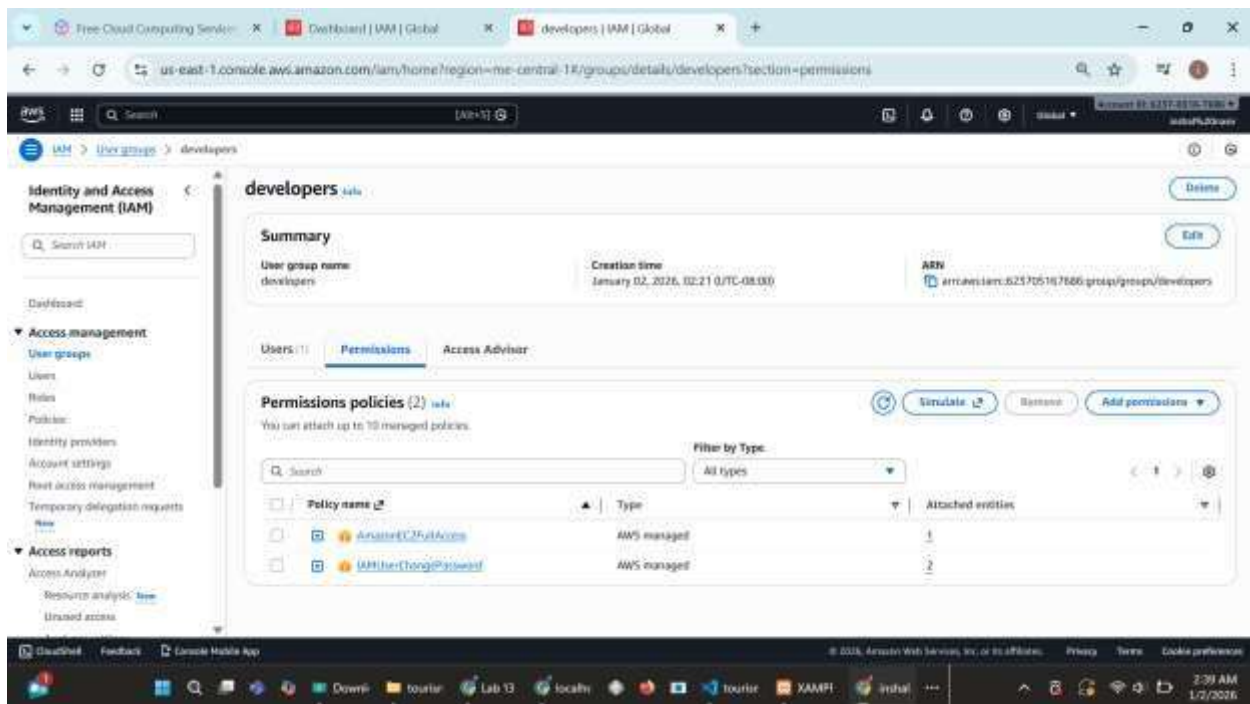
     

Terraform will perform the following actions:

```
# aws_iam_user.lb will be created
+ resource "aws_iam_user" "lb" {
  + arn              = (known after apply)
  + force_destroy    = true
```

```
@23-22412-038-sys →~/Lab13 $ terraform apply -auto-approve
```

```
group_details = {
  "group_arn" = "arn:aws:iam::623705167686:group/groups/developers"
  "group_name" = "developers"
  "unique_id" = "AGPAZCN5ZS5DPX3WSBTHJ"
}
user_details = {
  "unique_id" = "AIDAZCN5ZS5DOWWNH4ETE"
```



Task 4

```

GNU nano 7.2                                variables.tf *
variable "iam_password" {
  description = "Temporary password for the IAM user"
  type        = string
  sensitive    = true
  default      = "1dontKnow"
}

```

```

GNU nano 7.2                                create-login-profile.sh *
#!/usr/bin/env bash
set -euo pipefail

USERNAME="$1"
PASSWORD="$2"

^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location
nano create_login_profile.sh
@23-22412-038-sys →~/Lab13 $ chmod +x create-login-profile.sh
@23-22412-038-sys →~/Lab13 $

```

```

GNU nano 7.2                                main.tf *
resource "null_resource" "create_login_profile" {
  triggers = {
    password_hash = sha256(var.iam_password)
    user          = aws_iam_user.lb.name
  }
}

```

```

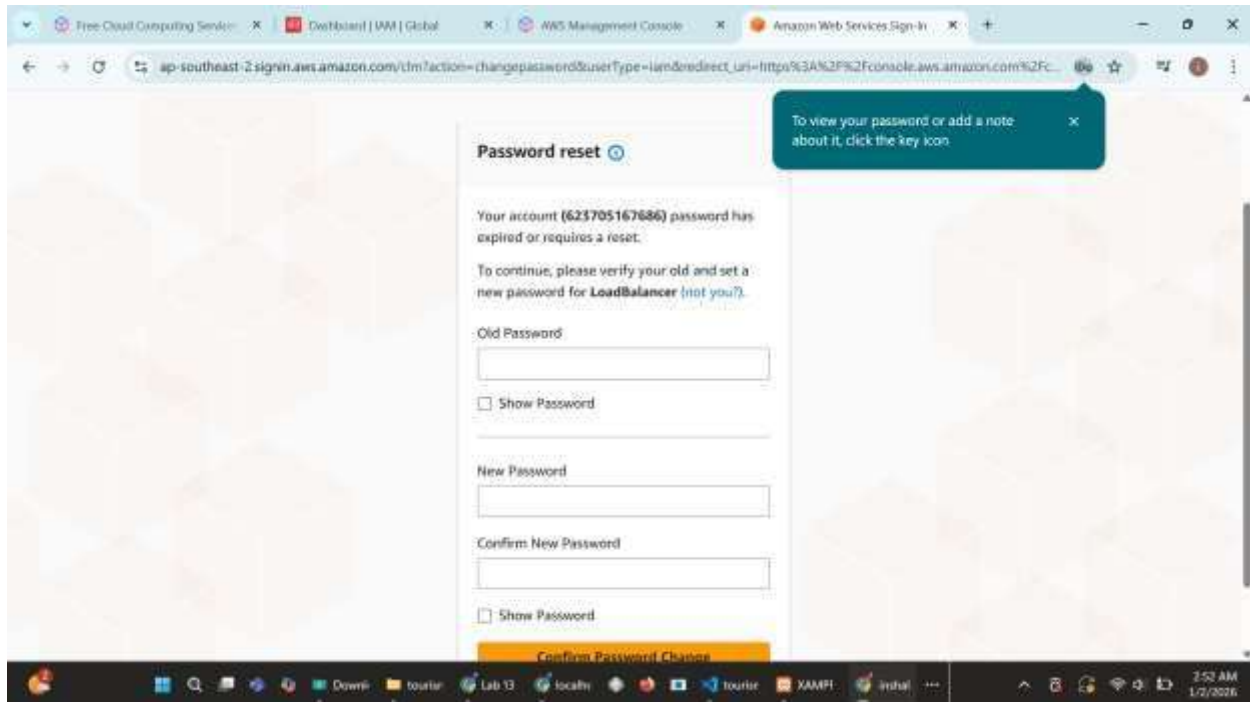
@23-22412-038-sys →~/Lab13 $ terraform output
group_details = {
  "group_arn" = "arn:aws:iam::623705167686:group/groups/developers"
  "group_name" = "developers"
  "unique_id" = "AGPAZCN5ZS5DPX3WSBTHJ"
}

}

user_details = {
  "unique_id" = "AIDAZCN5ZS5DOWWNH4ETE"
  "user_arn" = "arn:aws:iam::623705167686:user/users/loadbalancer"
  "user_name" = "loadbalancer"
}

```

```
@23-22412-038-sys →~/Lab13 $ aws iam get-login-profile --user-name loadbalancer
{
  "LoginProfile": {
    "UserName": "loadbalancer",
    "CreateDate": "2026-01-02T10:49:10+00:00",
    "PasswordResetRequired": true
  }
}
```



Task 5

```
GNU nano 7.2 main.tf
# IAM Access Key for loadbalancer
resource "aws_iam_access_key" "lb_access_key" {
  user = aws_iam_user.lb.name
}

+ access_key_secret = (sensitive value)
aws_iam_access_key.lb_access_key: Creating...
aws_iam_access_key.lb_access_key: Creation complete after 1s [id=AKIAZCN5ZS5DFB02LNNN]

Apply complete! Resources: 1 added, 0 changed, 0 destroyed.
```



```

access_key_id = "AKIAZCN5ZS5DFBO2LNNN"
access_key_secret = <sensitive>
group_details = {
    "group_arn" = "arn:aws:iam::623705167686:group/groups/developers"
    "group_name" = "developers"
    "unique_id" = "AGPAZCN5ZS5DPX3WSBTHJ"
}

```

loadbalancer [Info](#) [Delete](#)

Summary

ARN arn:aws:iam::623705167686:user/users/loadbalancer	Console access Enabled without MFA	Access key 1 AKIAZCN5ZS5DFBO2LNNN - Active ⓘ Never used. Created today.
Created January 02, 2026, 02:28 (UTC-08:00)	Last console sign-in Today	Access key 2 Create access key

[Permissions](#) | [Groups \(1\)](#) | [Tags \(1\)](#) | [Security credentials](#) | [Last Accessed](#)

Console sign-in [Manage console access](#)

Console sign-in link https://623705167686.signin.aws.amazon.com/console	Console password Updated 10 minutes ago (2026-01-02 02:49 PST)
Last console sign-in 7 minutes ago (2026-01-02 02:52 PST)	

Task 6

Free Cloud Computing Services | Dashboard | IAM | Global | AWS Management Console | S3 buckets | S3 | me-central-1 | +

me-central-1 console.aws.amazon.com/s3/buckets?region=me-central-1#

Account ID: 6237-0116-7686 | [Initial name](#)

Amazon S3 > Buckets

Amazon S3

- Buckets**
 - General purpose buckets
 - Directory buckets
 - Table buckets
 - Vector buckets
- Access management and security**
 - Access Points
 - Access Points for FSx
 - Access Grants
 - IAM Access Analyzer
- Storage management and insights**
 - Storage Lens

General purpose buckets [All AWS Regions](#) [Directory buckets](#)

General purpose buckets (0) [Info](#) [Copy ARN](#) [Empty](#) [Delete](#) [Create bucket](#)

Buckets are containers for data stored in S3.

Find buckets by name

Name	AWS Region	Creation date
No buckets		
You don't have any buckets.		

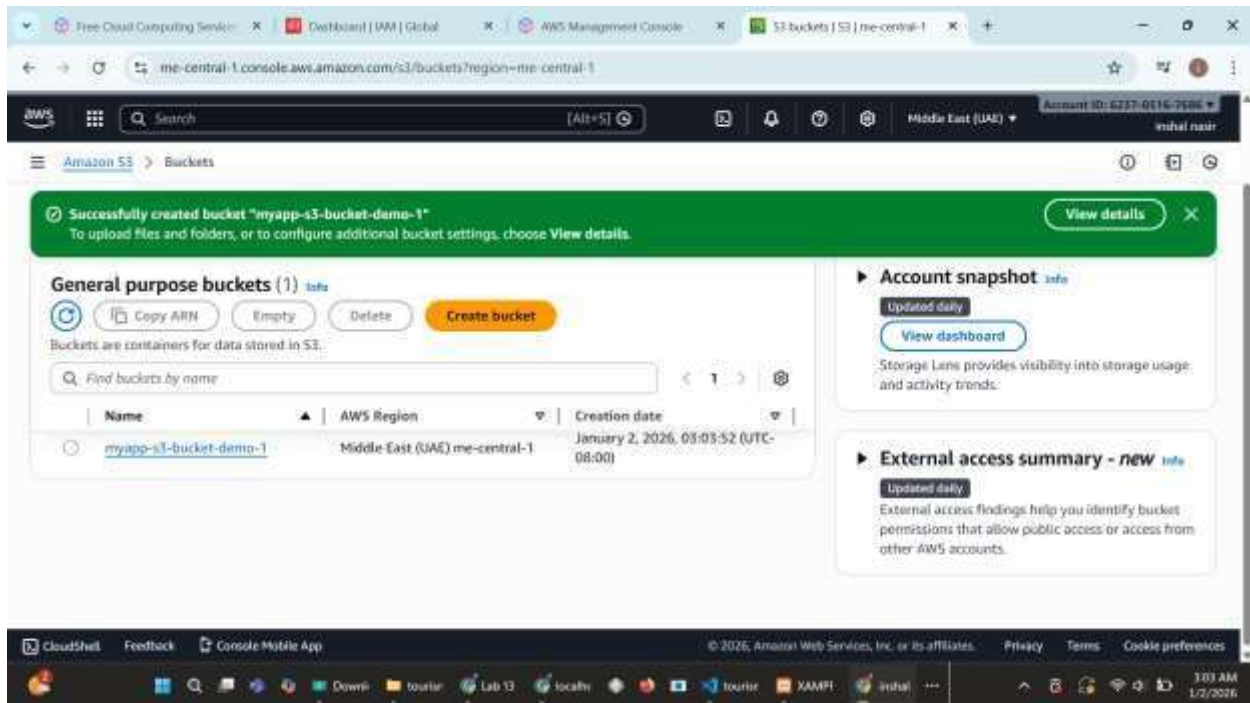
[Create bucket](#)

Account snapshot [Info](#) [View dashboard](#)
Updated daily

External access summary - new [Info](#)
Updated daily

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3:02 AM 1/2/2026



```
@23-22412-038-sys →~/Lab13 $ terraform init -migrate-state
```

- Using previously-installed hashicorp/aws v5.100.0
- Using previously-installed hashicorp/null v3.2.4

Terraform has been successfully initialized!

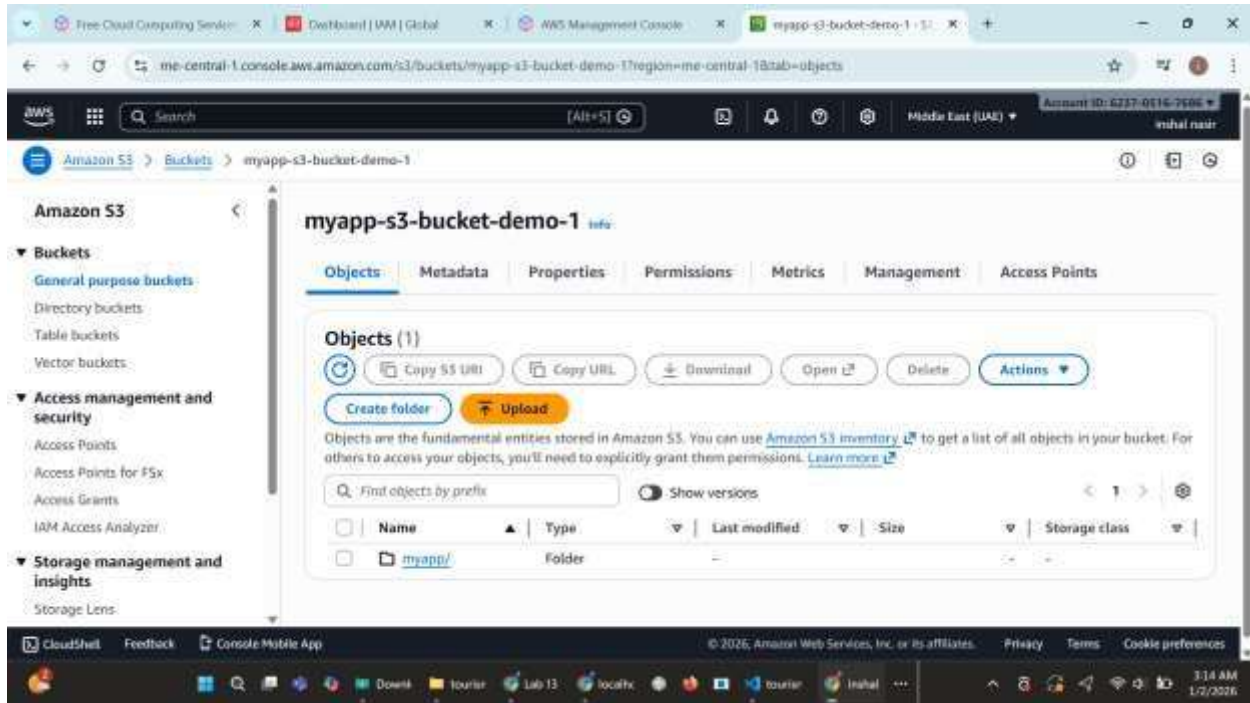
You may now begin working with Terraform. Try running "terraform plan" to see any changes that are required for your infrastructure. All Terraform commands should now work.

```
@23-22412-038-sys →~/Lab13 $ terraform apply -auto-approve -var="iam_password=MySecurePass123!"
```

No changes. Your infrastructure matches the configuration.

Terraform has compared your real infrastructure against your configuration and found no differences, so no changes are needed.

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.

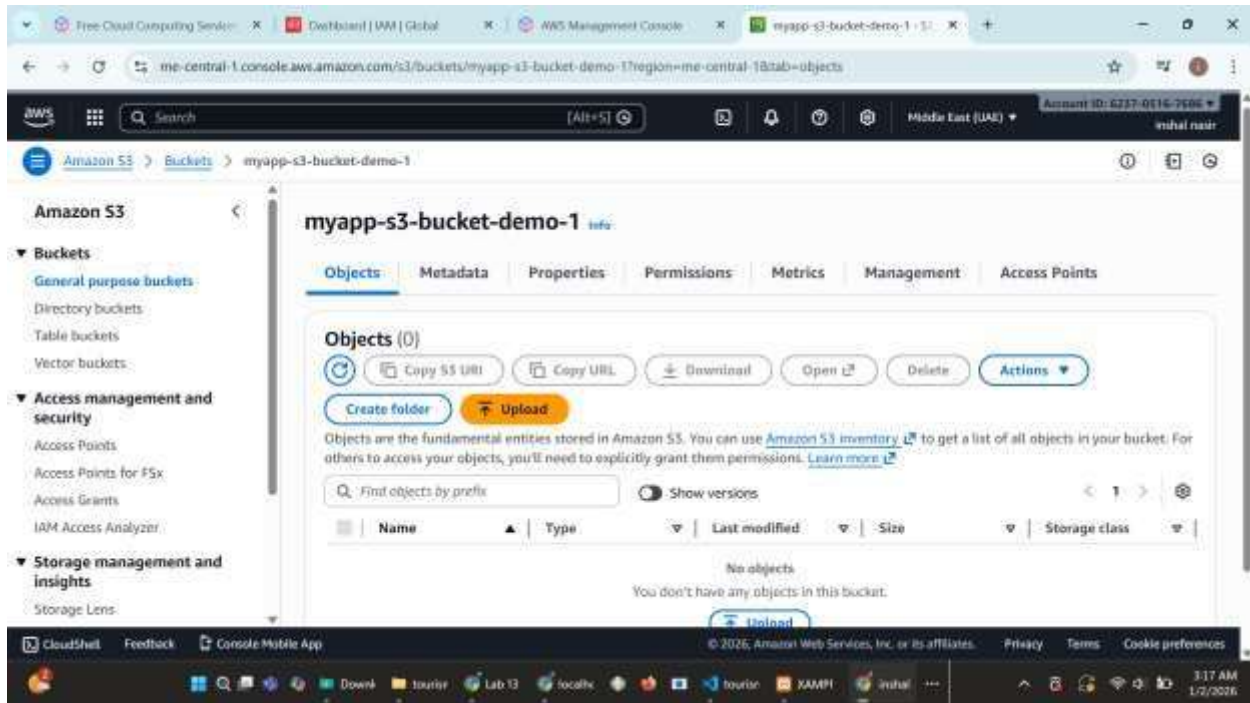


```
@23-22412-038-sys →~/Lab13 $ ls -la terraform.tfstate*  
-rw-r--r-- 1 codespace codespace 0 Jan 2 11:10 terraform.tfstate  
-rw-r--r-- 1 codespace codespace 6882 Jan 2 11:10 terraform.tfstate.backup  
@23-22412-038-sys →~/Lab13 $
```

```
@23-22412-038-sys →~/Lab13 $ terraform destroy -auto-approve  
aws_iam_group_policy_attachment.developer_ec2_fullaccess: Destruction complete af  
ter 1s  
aws_iam_group.developers: Destroying... [id=developers]  
aws_iam_group.developers: Destruction complete after 0s  
aws_iam_user.lb: Destruction complete after 2s
```

Destroy complete! Resources: 7 destroyed.

```
@23-22412-038-sys →~/Lab13 $
```



Task 7

```

GNU nano 7.2      users.csv *
user_name
Michael
Dwight
Jim
Pam

GNU nano 7.2      locals.tf *
locals {
  users = csvdecode(file("users.csv"))
}

```

```
@23-22412-038-sys →~/Lab13 $ terraform init -upgrade
```

- Finding hashicorp/null versions matching "~> 3.2"...
- Finding hashicorp/aws versions matching "~> 5.0"...
- Using previously-installed hashicorp/aws v5.100.0
- Using previously-installed hashicorp/null v3.2.4

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see

PROBLEMS OUTPUT TERMINAL ... bash - Lab13

```
@23-22412-038-sys →~/Lab13 $ terraform apply -auto-approve -var="iam_password=MySecurePass123!"
```

```
    "access_key_id" = "AKIAZCN5ZS5DNSPIMN6L"
    "user_arn"      = "arn:aws:iam::623705167686:user/users/Kevin"
    "user_unique_id" = "AIDAZCN5ZS5DJNJFIRG7D"
  }
  "Meredith" = {
    "access_key_id" = "AKIAZCN5ZS5DE7U7CVIU"
    "user_arn"      = "arn:aws:iam::623705167686:user/users/Meredith"
```

```
@23-22412-038-sys →~/Lab13 $ terraform output
```

```
all_access_key_secrets = <sensitive>
all_users_details = {
  "Andy" = {
    "access_key_id" = "AKIAZCN5ZS5DDMCDTPPG"
    "user_arn"      = "arn:aws:iam::623705167686:user/users/Andy"
    "user_unique_id" = "AIDAZCN5ZS5DDUMIDKGQ3"
  }
}
"Angela" = {
  "access_key_id" = "AKIAZCN5ZS5DED2XGOJO"
  "user_arn"      = "arn:aws:iam::623705167686:user/users/Angela"
  "user_unique_id" = "AIDAZCN5ZS5DL6WDEJWEV"
}
```



```
}
"Charles" = {
  "access_key_id" = "AKIAZCN5ZS5DLQRVIKWQ"
  "user_arn" = "arn:aws:iam::623705167686:user/users/Charles"
  "user_unique_id" = "AIDAZCN5ZS5DGN3KWMLKM"
}
"Clark" = {
}
"Creed" = {
  "access_key_id" = "AKIAZCN5ZS5DEYU5LIIA"
  "user_arn" = "arn:aws:iam::623705167686:user/users/Creed"
  "user_unique_id" = "AIDAZCN5ZS5DGG4SUB7XF"
}
"Darryl" = {
  "access_key_id" = "AKIAZCN5ZS5DLQ2GVWMQ"
  "user_arn" = "arn:aws:iam::623705167686:user/users/Darryl"
  "user_unique_id" = "AIDAZCN5ZS5DBVOPF5RE7"
}
"David" = {
  "access_key_id" = "AKIAZCN5ZS5DC2XVMR32"
  "user_arn" = "arn:aws:iam::623705167686:user/users/David"
  "user_unique_id" = "AIDAZCN5ZS5DEI5V24FT6"
}
"Dwight" = {
  "access_key_id" = "AKIAZCN5ZS5DEPJLNIWD"
  "user_arn" = "arn:aws:iam::623705167686:user/users/Dwight"
  "user_unique_id" = "AIDAZCN5ZS5DKSBBTURE3"
}
"Erin" = {
  "access_key_id" = "AKIAZCN5ZS5DAY3J74W5"
  "user_arn" = "arn:aws:iam::623705167686:user/users/Erin"
  "user_unique_id" = "AIDAZCN5ZS5DDEZR7EQUU"
}
"Gabe" = {
```

```
,
"Gabe" = {
  "access_key_id" = "AKIAZCN5ZS5DOLTKDXDG"
  "user_arn" = "arn:aws:iam::623705167686:user/users/Gabe"
  "user_unique_id" = "AIDAZCN5ZS5DCAKPPGRHA"
}

"Holly" = {
  "access_key_id" = "AKIAZCN5ZS5DPH6QDGGGL"
  "user_arn" = "arn:aws:iam::623705167686:user/users/Holly"
  "user_unique_id" = "AIDAZCN5ZS5DKZLEZQ5B7"
}

}

"Jan" = {
  "access_key_id" = "AKIAZCN5ZS5DE4W367OM"
  "user_arn" = "arn:aws:iam::623705167686:user/users/Jan"
  "user_unique_id" = "AIDAZCN5ZS5DJBXI40AT"
}

}

"Jim" = {
  "access_key_id" = "AKIAZCN5ZS5DNIEWRXZT"
  "user_arn" = "arn:aws:iam::623705167686:user/users/Jim"
  "user_unique_id" = "AIDAZCN5ZS5DHAA4CKHCC"
}

"Jo" = {
  "access_key_id" = "AKIAZCN5ZS5DI2004YNY"
  "user_arn" = "arn:aws:iam::623705167686:user/users/Jo"
}

}

"Kelly" = {
  "access_key_id" = "AKIAZCN5ZS5DLES5QLTE"
  "user_arn" = "arn:aws:iam::623705167686:user/users/Kelly"
  "user_unique_id" = "AIDAZCN5ZS5DEO6HDRJLB"
}

}

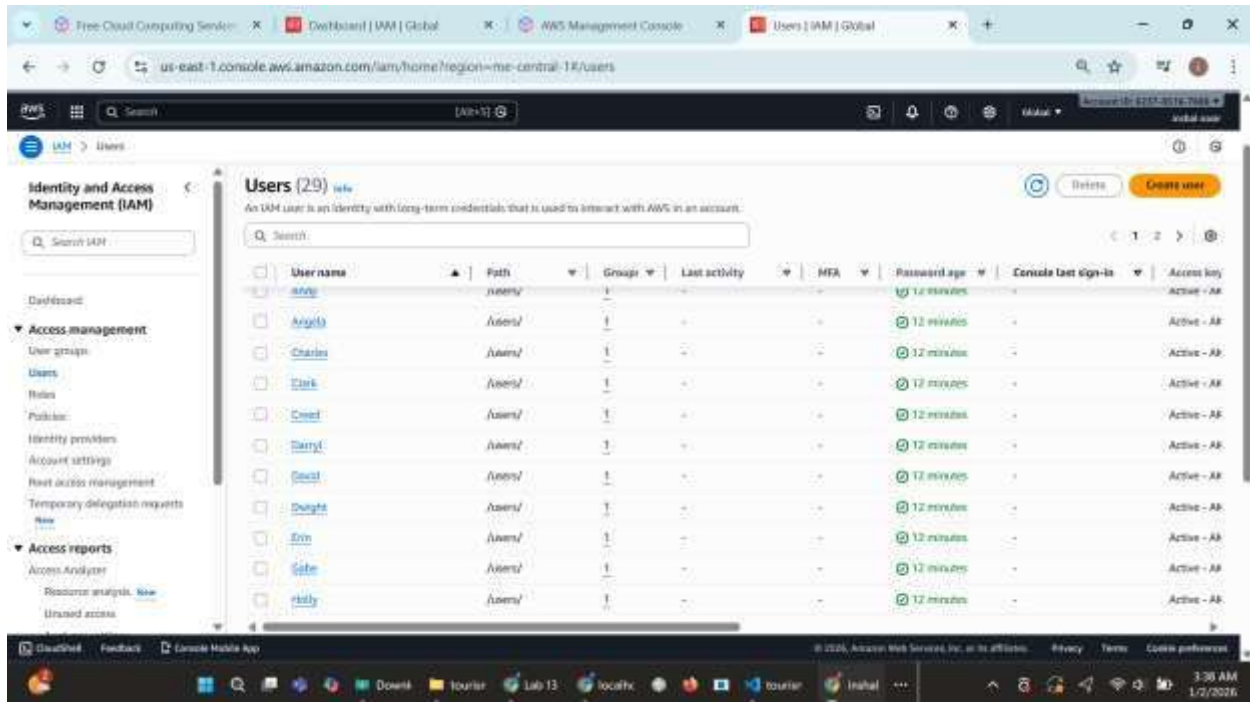
"Kevin" = {
```



```

"Oscar" = {
  "access_key_id" = "AKIAZCN5ZS5DPQUA2PH3"
  "user_arn" = "arn:aws:iam::623705167686:user/users/Oscar"
  "user_unique_id" = "AIDAZCN5ZS5DGRXJH4ZNR"
}
"Pam" = {
  "access_key_id" = "AKIAZCN5ZS5DOGHOFBIH"
  "user_arn" = "arn:aws:iam::623705167686:user/users/Pam"
}
"Robert" = {
  "access_key_id" = "AKIAZCN5ZS5DNYDZS5VH"
  "user_arn" = "arn:aws:iam::623705167686:user/users/Robert"
  "user_unique_id" = "AIDAZCN5ZS5DMKLKZTY6A"
}

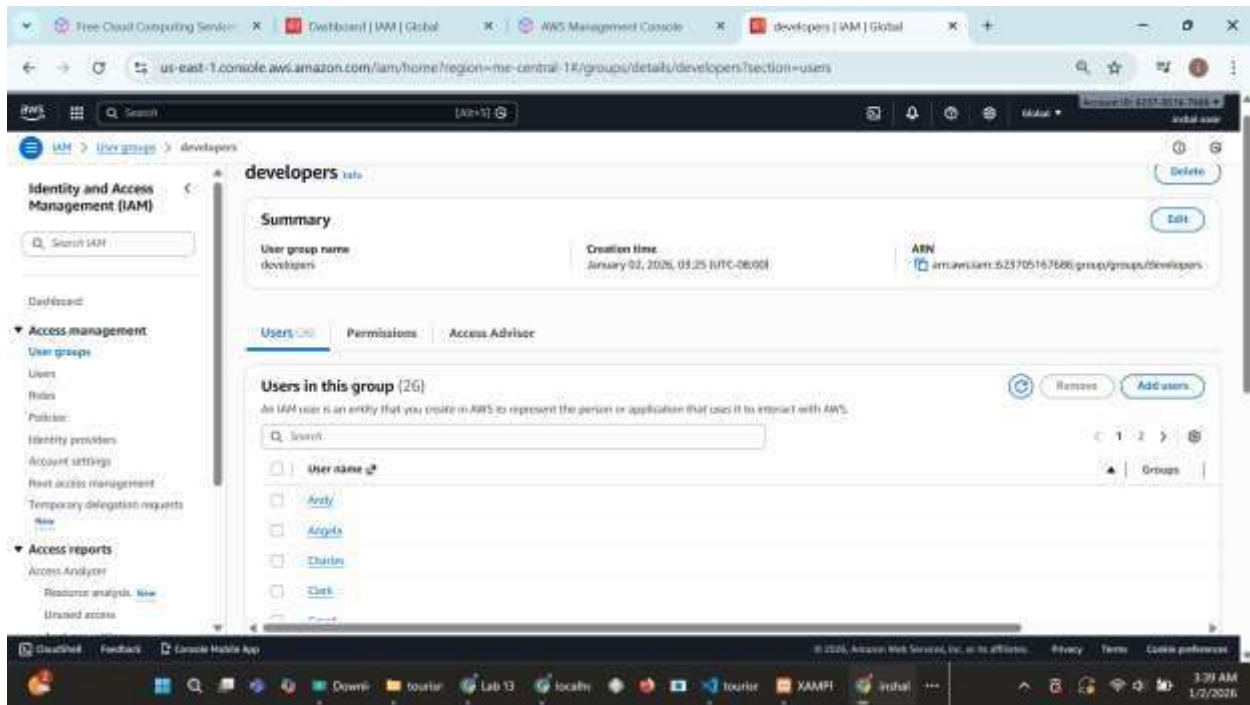
```



The screenshot shows the AWS IAM console interface. The left sidebar contains navigation links for Identity and Access Management (IAM), including Dashboard, Access management, User groups, Roles, Policies, Policies, Identity providers, Account settings, Root access management, Temporary delegation requests, Access reports, Access Analyzer, Resource analysis, and Unshared access. The main content area displays a table of users, with a search bar and pagination controls. The table columns include User name, Path, Groups, Last activity, MFA, Password age, Console last sign-in, and Access key. The table lists 29 users, with the first 10 visible in the screenshot.

User name	Path	Groups	Last activity	MFA	Password age	Console last sign-in	Access key
Alice	/users/	1	-	-	12 minutes	-	Active - AK
Angela	/users/	1	-	-	12 minutes	-	Active - AK
Charles	/users/	1	-	-	12 minutes	-	Active - AK
Clark	/users/	1	-	-	12 minutes	-	Active - AK
Cristi	/users/	1	-	-	12 minutes	-	Active - AK
Darryl	/users/	1	-	-	12 minutes	-	Active - AK
David	/users/	1	-	-	12 minutes	-	Active - AK
Dwayne	/users/	1	-	-	12 minutes	-	Active - AK
Evin	/users/	1	-	-	12 minutes	-	Active - AK
Gabe	/users/	1	-	-	12 minutes	-	Active - AK
Holly	/users/	1	-	-	12 minutes	-	Active - AK

Ivabe	/users/	1	-	-	12 minutes	-	Active - AK
Holly	/users/	1	-	-	12 minutes	-	Active - AK
Jan	/users/	1	-	-	12 minutes	-	Active - AK
Jim	/users/	1	-	-	12 minutes	-	Active - AK
Jo	/users/	1	-	-	12 minutes	-	Active - AK
Kelly	/users/	1	-	-	8 minutes	-	Active - AK
Kevin	/users/	1	-	-	12 minutes	-	Active - AK
Meredith	/users/	1	-	-	12 minutes	-	Active - AK
Michael	/users/	1	-	-	12 minutes	-	Active - AK



☐	Charles	
☐	Clark	
☐	Creed	
☐	Darryl	
☐	David	
☐	Dwight	
☐	Erin	
☐	Gabe	
☐	Holly	
☐	Jan	
☐	Jim	
☐	Jo	
☐	Kelly	
☐	Kevin	
☐	Kelly	
☐	Kevin	
☐	Meredith	
☐	Michael	
☐	Oscar	1
☐	Pam	1

Pam Info Delete

Summary

ARN <code>arn:aws:iam::625705167686:user/users/Pam</code>	Console access Enabled without MFA	Access key 1 AKIAZCN5ZSSDOGHOFBIH - Active Never used. Created today.
Created January 02, 2026, 03:25 (UTC-08:00)	Last console sign-in Never	Access key 2 Create access key

Permissions | Groups (1) | Tags (2) | **Security credentials** | Last Accessed

Console sign-in Manage console access

Console sign-in link: `https://console.aws.amazon.com/iam/home?region=us-east-1#/users/Pam` | Console password: `*****`

Free Cloud Computing | Docker | IAM | Global | AWS Management Console | developers | IAM | Global | myapp-s3-bucket-demo

me-central-1.console.aws.amazon.com/s3/buckets/myapp-s3-bucket-demo-1?prefix=myapp/®ion=me-central-1&tab=objects

Amazon S3 | Buckets | myapp-s3-bucket-demo-1 | myapp/

Amazon S3

- Buckets
 - General purpose buckets
 - Directory buckets
 - Table buckets
 - Vector buckets
- Access management and security
 - Access Points
 - Access Points for FSx
 - Access Grants
 - IAM Access Analyzer
- Storage management and insights
 - Storage Lens

myapp/ Copy S3 URI

Objects | Properties

Objects (1)

[Copy S3 URI](#) [Copy URL](#) [Download](#) [Open](#) [Delete](#) [Actions](#)

[Create folder](#) [Upload](#)

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 Inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix: ☐ Show versions

Name	Type	Last modified	Size	Storage class
terraform.tfstate	tfstate	January 2, 2026, 03:30:18 (UTC-08:00)	95.0 KB	Standard

CloudShell | Feedback | Console Mobile App

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3:42 AM 1/2/2026

Cleanup

```
aws_iam_user.users["Darryl"]: Destruction complete after 2s
aws_iam_user.users["Michael"]: Destruction complete after 2s
aws_iam_user.users["Clark"]: Destruction complete after 2s
aws_iam_user.users["Stanley"]: Destruction complete after 2s
aws_iam_user.users["Jim"]: Destruction complete after 5s
aws_iam_user.users["Peter"]: Destruction complete after 8s
```

Destroy complete! Resources: 107 destroyed.

Free Cloud Computing | Dashboard | IAM | Global | AWS Management Console | developers | IAM | Global | Users | IAM | Global

us-east-1.console.aws.amazon.com/iam/home?region=us-east-1#/users

Identity and Access Management (IAM)

Search IAM

Dashboard

Access management

- User groups
- Users**
- Roles
- Policies
- Identity providers
- Account settings
- Root access management
- Temporary delegation requests

Access reports

- Access Analyzer
- Resource analyzer
- Unshared access

Users (0)

An IAM user is an identity with long-term credentials that is used to interact with AWS in an account.

Search

Username	Path	Groups	Last activity	MFA	Password age	Console last sign-in	Access key ID
No resources to display							

3:46 AM 1/2/2026

Free Cloud Computing | Dashboard | IAM | Global | AWS Management Console | developers | IAM | Global | User groups | IAM | Global

us-east-1.console.aws.amazon.com/iam/home?region=us-east-1#/groups

Identity and Access Management (IAM)

Search IAM

Dashboard

Access management

- User groups**
- Users
- Roles
- Policies
- Identity providers
- Account settings
- Root access management
- Temporary delegation requests

Access reports

- Access Analyzer
- Resource analyzer
- Unshared access

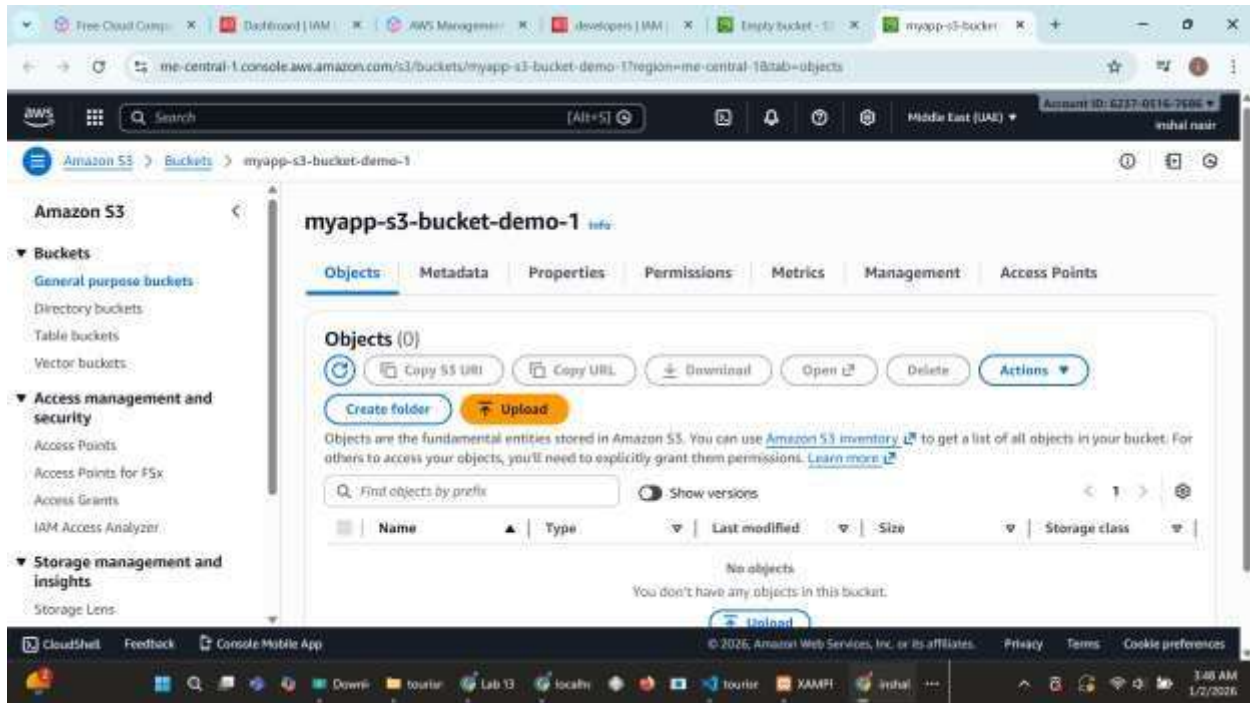
User groups (0)

A user group is a collection of IAM users. Use groups to specify permissions for a collection of users.

Search

Group name	Users	Permissions	Creation time
No resources to display			

3:47 AM 1/2/2026





FATIMA JINNAH WOMEN UNIVERSITY

Department of Software Engineering

Lab Title:	Experiment # :
Student Name:	Regd. #:
Experiment Name:	

Performance Indicator	Level of Achievements				
	Excellent (5)	Good (4)	Average (3)	Below Average (2)	Poor (1)
Understanding & implementation Problem					
Report and presentation					
Maximum Marks		10	Obtained Marks		
Lab Instructor:					

Lab 14

Task 0

```
@23-22412-038-sys → /workspaces/terraform_machine (main) $
```

```
@23-22412-038-sys → /workspaces/terraform_machine (main) $ aws --version  
aws-cli/2.33.2 Python/3.13.11 Linux/6.8.0-1030-azure exe/x86_64.ubuntu.24
```

```
Setting up terraform (1.4.7) ...  
● @23-22412-038-sys → /workspaces/terraform_machine (main) $ terraform --version  
Terraform v1.4.3  
on linux_amd64  
○ @23-22412-038-sys → /workspaces/terraform_machine (main) $
```

```
@23-22412-038-sys → /workspaces/terraform_machine (main) $ ansible --version  
ansible collection location = /home/codespace/.ansible/collections:/usr/share/a  
nsible/collections  
executable location = /usr/bin/ansible  
python version = 3.12.3 (main, Nov 6 2025, 13:44:16) [GCC 13.3.0] (/usr/bin/py  
thon3)  
jinja version = 3.1.6  
libyaml = True  
○ @23-22412-038-sys → /workspaces/terraform_machine (main) $
```

```
● @23-22412-038-sys → /workspaces/terraform_machine (main) $ aws sts get-caller-ide  
ntity  
{  
  "UserId": "623705167686",  
  "Account": "623705167686",  
  "Arn": "arn:aws:iam::623705167686:root"  
}
```

Task 1

```
Cloning into 'terraform_machine'...  
remote: Enumerating objects: 30, done.  
remote: Counting objects: 100% (30/30), done.  
remote: Compressing objects: 100% (26/26), done.  
remote: Total 30 (delta 5), reused 0 (delta 0), pack-reused 0 (from 0)  
Receiving objects: 100% (30/30), 7.03 KiB | 7.03 MiB/s, done.  
Resolving deltas: 100% (5/5), done.
```

```

Generating public/private ed25519 key pair.
Created directory '/home/codespace/.ssh'.
Your identification has been saved in /home/codespace/.ssh/id_ed25519
Your public key has been saved in /home/codespace/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:Uh7T5J+ajLuZeUKDMZB35qBQ2Y8L335i+DJHgXih8bw codespace@codespaces-594742
The key's randomart image is:
+--[ED25519 256]--+
|  ..+  .          |
| o = + o+         |
| B * B+ o         |
| o B =o=o...      |
| . =.XSo o        |
| E *.+o o         |
| o ...+          |
+----+
total 20
drwx----- 2 codespace codespace 4096 Jan  6 07:42 .
drwxr-x--- 1 codespace codespace 4096 Jan  6 07:42 ..
-rw----- 1 codespace codespace 419 Jan  6 07:42 id_ed25519
-rw-r--r-- 1 codespace codespace 109 Jan  6 07:42 id_ed25519.pub

```

```

touch terraform.tfvars
ls -la terraform.tfvars
-rw-rw-rw- 1 codespace codespace 0 Jan  6 07:44 terraform.tfvars

```

```

GNU nano 7.2 terraform.tfvars *
vpc_cidr_block      = "10.0.0.0/16"
subnet_cidr_block   = "10.0.10.0/24"
availability_zone    = "me-central-1a"
env_prefix          = "dev"
instance_type        = "t3.micro"
public_key           = "~/.ssh/id_ed25519.pub"
private_key          = "~/.ssh/id_ed25519"

```

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see any changes that are required for your infrastructure. All Terraform commands should now work.

If you ever set or change modules or backend configuration for Terraform, rerun this command to reinitialize your working directory. If you forget, other commands will detect it and remind you to do so if necessary.

Apply complete! Resources: 10 added, 0 changed, 0 destroyed.

Outputs:

```

webserver_public_ips = [
  "158.252.83.133",
  "3.28.136.221",

```

Task 2

```
@23-22412-038-sys →/workspaces/terraform_machine (main) $ terraform plan
terraform apply
```

```
aws_instance.example: Creating...
aws_instance.example: Still creating... [00m10s elapsed]
aws_instance.example: Creation complete after 13s [id=i-0fd45e8764323df23]
```

Apply complete! Resources: 1 added, 0 changed, 0 destroyed.

```
ansible --version
'ansible-core' already seems to be installed. Not modifying existing installation in '/usr/local/py-utils/venvs/ansible-core'. Pass
'--force' to force installation.
ansible [core 2.20.1]
  config file = None
  configured module search path = ['/home/codespace/.ansible/plugins/modules', '/usr/share/ansible/plugins/modules']
  ansible python module location = /usr/local/py-utils/venvs/ansible-core/lib/python3.12/site-packages/ansible
  ansible collection location = /home/codespace/.ansible/collections:/usr/share/ansible/collections
  executable location = /usr/local/py-utils/bin/ansible
  python version = 3.12.1 (main, Nov 27 2025, 10:47:52) [GCC 13.3.0] (/usr/local/py-utils/venvs/ansible-core/bin/python)
  jinja version = 3.1.6
  pyyaml version = 6.0.3 (with libyaml v0.2.5)
```

```
webserver_public_ips = [
    "158.252.83.133",
    "3.28.136.221",
]
```

```
ls -la hosts
-rw-rw-rw- 1 codespace codespace 0 Jan  6 08:01 hosts
```

```
[Preview] README.md  README.md  hosts  X
workspaces > terraform_machine > hosts
1  158.252.83.133 ansible_user=ec2-user ansible_ssh_private_key_file=~/.ssh/id_ed25519
2  3.28.136.221 ansible_user=ec2-user ansible_ssh_private_key_file=~/.ssh/id_ed25519
```

```
[WARNING]: Host '158.252.83.133' is using the discovered Python interpreter at '/usr/bin/python3.9', but future installation of another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.20/reference_appendices/interpreter_discovery.html for more information.
158.252.83.133 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3.9"
  },
  "changed": false,
  "ping": "pong"
}
[WARNING]: Host '3.28.136.221' is using the discovered Python interpreter at '/usr/bin/python3.9', but future installation of another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.20/reference_appendices/interpreter_discovery.html for more information.
3.28.136.221 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3.9"
  },
  "changed": false,
  "ping": "pong"
}
```

Task 3

[Preview] README.md

main.tf M X

main.tf

```
13 module "myapp-subnet" {
18   env_prefix = var.env_prefix
19   default_route_table_id = aws_vpc.myapp_vpc.default_route_table_id
20 }
21
22 module "myapp-webserver" {
23   source = "./modules/webserver"
24   env_prefix = var.env_prefix
25   instance_type = var.instance_type
26   availability_zone = var.availability_zone
27   public_key = var.public_key
28   my_ip = local.my_ip
29   vpc_id = aws_vpc.myapp_vpc.id
30   subnet_id = module.myapp-subnet.subnet.id
31
32   # Loop count
33   count = 3
34   # Use count.index to differentiate instances
35   instance_suffix = count.index
36 }
```

```
module.myapp-webserver[2].aws_key_pair.ssh-key: Creating...
module.myapp-webserver[2].aws_security_group.web_sg: Creating...
module.myapp-webserver[2].aws_key_pair.ssh-key: Creation complete after 0s [id=dev-serverkey-2]
module.myapp-webserver[2].aws_security_group.web_sg: Creation complete after 3s [id=sg-08a64a1ae3f5df444]
module.myapp-webserver[2].aws_instance.myapp-server: Creating...
module.myapp-webserver[2].aws_instance.myapp-server: Still creating... [00m10s elapsed]
module.myapp-webserver[2].aws_instance.myapp-server: Creation complete after 12s [id=i-01e1fcd5d2b3157e2]
```

Apply complete! Resources: 3 added, 0 changed, 0 destroyed.

Outputs:

```
webserver_public_ips = [
  "158.252.83.133",
  "3.28.136.221",
  "158.252.32.107",
]
```



```
[Preview] README.md  main.tf M  hosts U X

hosts
1  [ec2]
2  158.252.83.133
3  3.28.136.221
4
5  [ec2:vars]
6  ansible_user=ec2-user
7  ansible_ssh_private_key_file=~/.ssh/id_ed25519
8  ansible_ssh_common_args='-o StrictHostKeyChecking=no'
9
10 [droplet]
11 158.252.32.107
12
13 [droplet:vars]
14 ansible_user=ec2-user
15 ansible_ssh_private_key_file=~/.ssh/id_ed25519
16 ansible_ssh_common_args='-o StrictHostKeyChecking=no'
```

```
[WARNING]: Host '158.252.83.133' is using the discovered Python interpreter at '/usr/bin/python3.9', but future installation of another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.20/reference_appendices/interpreter_discovery.html for more information.
158.252.83.133 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3.9"
  },
  "changed": false,
  "ping": "pong"
}
[WARNING]: Host '3.28.136.221' is using the discovered Python interpreter at '/usr/bin/python3.9', but future installation of another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.20/reference_appendices/interpreter_discovery.html for more information.
3.28.136.221 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3.9"
  },
  "changed": false,
  "ping": "pong"
}
```

```
[WARNING]: Host '158.252.83.133' is using the discovered Python interpreter at '/usr/bin/python3.9', but future installation of another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.20/reference_appendices/interpreter_discovery.html for more information.
158.252.83.133 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3.9"
  },
  "changed": false,
  "ping": "pong"
}
```

Task 4

```
1 [defaults]
2 host_key_checking = False
3 interpreter_python = /usr/bin/python3
4 |
```

[Preview] README.md main.tf M hosts U X .ansible.cfg

hosts

```
1 [ec2]
2 158.252.83.133
3 3.28.136.221
4
5 [ec2:vars]
6 ansible_user=ec2-user
7 ansible_ssh_private_key_file=~/.ssh/id_ed25519
8
9 [droplet]
10 158.252.32.107
11
12 [droplet:vars]
13 ansible_user=ec2-user
14 ansible_ssh_private_key_file=~/.ssh/id_ed25519
```

```
158.252.32.107 | SUCCESS => {
  "changed": false,
  "ping": "pong"
}
158.252.83.133 | SUCCESS => {
  "changed": false,
  "ping": "pong"
}
3.28.136.221 | SUCCESS => {
  "changed": false,
  "ping": "pong"
}
```

[Preview] README.md main.tf M hosts U ! my-playbook.yaml U X .ansible.cfg

! my-playbook.yaml

```
1 ---
2 - name: Configure nginx web server
3   hosts: ec2
4   become: true
5   tasks:
6     - name: install nginx and update cache
7       yum:
8         name: nginx
9         state: present
10        update_cache: yes
11
12    - name: start nginx server
13      service:
14        name: nginx
15        state: started
```

```
PLAY [Configure nginx web server] *****

TASK [Gathering Facts] *****
ok: [158.252.83.133]
ok: [3.28.136.221]

TASK [install nginx and update cache] *****
changed: [158.252.83.133]
changed: [3.28.136.221]

TASK [start nginx server] *****
changed: [158.252.83.133]
changed: [3.28.136.221]

PLAY RECAP *****
158.252.83.133      : ok=3    changed=2    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
3.28.136.221      : ok=3    changed=2    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.

[Preview] README.md

main.tf M

hosts U

! my-playbook.yaml U X

ansible.cfg

! my-playbook.yaml

```

1  ---
2  - name: Configure nginx web server
3    hosts: droplet
4    become: true
5    tasks:
6      - name: install nginx and update cache
7        yum:
8          name: nginx
9          state: present
10         update_cache: yes
11
12     - name: start nginx server
13       service:
14         name: nginx
15         state: started

```

Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.

```
ls -la ansible.cfg
-rw-rw-rw- 1 codespace codespace 0 Jan  6 08:52 ansible.cfg
```

[Preview] README.md main.tf M hosts U ! my-playbook.yaml U .ansible.cfg X

home > codespace > .ansible.cfg

```
1 [defaults]
2 host_key_checking=False
3 interpreter_python = /usr/bin/python3
```

[Preview] README.md main.tf M X hosts U ! my-playbook.yaml U .ansible.cfg X

main.tf

```
13 module "myapp-subnet" {
20 }
21
22 module "myapp-webserver" {
23     source = "../modules/webserver"
24     env_prefix = var.env_prefix
25     instance_type = var.instance_type
26     availability_zone = var.availability_zone
27     public_key = var.public_key
28     my_ip = local.my_ip
29     vpc_id = aws_vpc.myapp_vpc.id
30     subnet_id = module.myapp-subnet.subnet.id
31
32     # Loop count
33     count = 1
34     # Use count.index to differentiate instances
35     instance_suffix = count.index
36 }
```

```
webserver_public_ips = [
    "158.252.83.133",
]
```

[WARNING]: Ansible is being run in a world writable directory (/workspaces/terraform_machine), ignoring it as an ansible.cfg
https://docs.ansible.com/ansible/devel/reference_appendices/config.html#cfg-in-world-writable-dir

```
PLAY [Configure nginx web server] *****
TASK [Gathering Facts] *****
ok: [158.252.83.133]

TASK [install nginx and update cache] *****
ok: [158.252.83.133]

TASK [install openssl] *****
ok: [158.252.83.133]

TASK [start nginx server] *****
changed: [158.252.83.133]

PLAY RECAP *****
158.252.83.133      : ok=4    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.

```
ok: [158.252.83.133]

TASK [Create SSL directories] *****
changed: [158.252.83.133] => (item={'path': '/etc/ssl/private', 'mode': '0700'})
changed: [158.252.83.133] => (item={'path': '/etc/ssl/certs', 'mode': '0755'})

TASK [Get IMDSv2 token] *****
ok: [158.252.83.133]

TASK [Get current public IP] *****
ok: [158.252.83.133]

TASK [Show current public IP] *****
ok: [158.252.83.133] => {
  "msg": "Public IP is 158.252.83.133"
}

TASK [Generate self-signed SSL certificate] *****
changed: [158.252.83.133]

PLAY RECAP *****
158.252.83.133      : ok=8    changed=2    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```



```
TASK [Gathering Facts] *****
ok: [158.252.83.133]

TASK [install php-fpm and php-curl] *****
changed: [158.252.83.133]

TASK [Copy website files] *****
changed: [158.252.83.133]

TASK [Copy nginx.conf template] *****
changed: [158.252.83.133]

TASK [Restart nginx] *****
changed: [158.252.83.133]

TASK [Start and enable php-fpm] *****
changed: [158.252.83.133]

PLAY RECAP *****
158.252.83.133      : ok=14   changed=5    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

Not secure <https://158.252.83.133>



Your connection is not private

Attackers might be trying to steal your information from **158.252.83.133** (for example, passwords, messages, or credit cards). [Learn more about this warning](#)

NET::ERR_CERT_AUTHORITY_INVALID



[Turn on enhanced protection](#) to get Chrome's highest level of security

Advanced

Back to safety



Nginx Front End Web Server

Hostnames:	ip-10-0-10-107.mt-central-1.compute.internal
Instance ID:	i-0cafb7153b7a7b7ab
Private IP:	10.0.10.107
Public IP:	158.252.83.133
Public DNS:	https://N/A
Deployed:	2026-01-06 12:04:23
Status:	☒ Active and Running
Managed By:	Terraform + Ansible

Task 8

```
module.myapp-webserver[0].aws_key_pair.ssh-key: Destruction complete after 0s
module.myapp-subnet.aws_subnet.myapp_subnet_1: Destruction complete after 0s
module.myapp-webserver[0].aws_security_group.web_sg: Destruction complete after 1s
aws_vpc.myapp_vpc: Destroying... [id=vpc-0c3680394866245c8]
aws_vpc.myapp_vpc: Destruction complete after 0s

Destroy complete! Resources: 7 destroyed.
```

Task 9

```

compose.yaml
1  - name: Deploy Docker Containers
2    hosts: all
3    become: true
4    user: "{{ normal_user }}"
5    vars_files:
6      - project-vars.yaml
7    tasks:
8      - name: check if docker-compose file exists
9        stat:
10         path: /home/{{ normal_user }}/compose.yaml
11         register: compose_file
12
13      - name: copy docker-compose file
14        copy:
15         src: "{{ docker_compose_file_location }}/compose.yaml"
16         dest: /home/{{ normal_user }}/compose.yaml
17         mode: '0644'
18         when: not compose_file.stat.exists
19
20      - name: deploy containers using docker-compose
21        command: docker compose up -d
22        register: compose_result
23        changed_when: "'Creating' in compose_result.stdout or 'Recreating' in compose_result.stdout"

```

```

TASK [verify docker access] *****
ok: [3.29.5.229]

TASK [display docker ps output] *****
ok: [3.29.5.229] => {
  "docker_ps.stdout": "CONTAINER ID   IMAGE          COMMAND                  CREATED   STATUS    PORTS    NAMES"
}

PLAY [Deploy Docker Containers] *****

TASK [Gathering Facts] *****
ok: [3.29.5.229]

TASK [check if docker-compose file exists] *****
ok: [3.29.5.229]

TASK [copy docker-compose file] *****
changed: [3.29.5.229]

TASK [deploy containers using docker-compose] *****
ok: [3.29.5.229]

PLAY RECAP *****
3.29.5.229          : ok=14   changed=2    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0

```



```

      + cidr_blocks      = [
        + "0.0.0.0/0",
      ]
      + from_port        = 80
      + ipv6_cidr_blocks = []
      + prefix_list_ids  = []
      + protocol         = "tcp"
      + security_groups  = []
      + self             = false
      + to_port          = 80
    },
  ],
  name          = "dev-web-sg-0"
  tags          = {
    "Name" = "dev-default-sg"
  }
  # (9 unchanged attributes hidden)
}

```

Plan: 0 to add, 1 to change, 0 to destroy.

module.myapp-webserver[0].aws_security_group.web_sg: Modifying... [id=sg-01c0dec942001d806]

module.myapp-webserver[0].aws_security_group.web_sg: Modifications complete after 0s [id=sg-01c0dec942001d806]

Apply complete! Resources: 0 added, 1 changed, 0 destroyed.

Outputs:

```

webserver_public_ips = [
  "3.29.5.229",
]

```

3.29.5.229:3000

Gitea HTTP Listen Port *

3000

Port number the Gitea web server will listen on.

Gitea Base URL *

http://3.29.5.229:3000/

Base address for HTTP(S) clone URLs and email notifications.

Log Path *

/data/gitea/log

Log files will be written to this directory.

☐ Enable Update Checker

Checks for new version releases periodically by connecting to gitea.io.

Optional Settings

► Email Settings

► Server and Third-Party Service Settings

► Administrator Account Settings

These configuration options will be written into: `/data/gitea/conf/app.ini` 

Install Gitea

Task 10:

```
main.tf
37 resource "null_resource" "configure_server" {}
38   triggers = {
39     # Is se Terraform ko pata chalega ke naya IP kab aaya
40     webserver_public_ips_for_aws = join(",", module.myapp-webserver.webserver_public_ips)
41   }
42
43   depends_on = [module.myapp-webserver]
44
45   provisioner "local-exec" {
46     command = <<-EOT
47     ansible-playbook -i ${self.triggers.webserver_public_ips_for_aws}, \
48       --private-key "${var.private_key_location}" --user ec2-user \
49       my-playbook.yaml
50     EOT
51   }
52 }
53
```

Destroy complete! Resources: 0 destroyed.

```

null_resource.configure_server (local-exec): skipping: no hosts matched

null_resource.configure_server (local-exec): PLAY [Adding user to docker group] *****
null_resource.configure_server (local-exec): skipping: no hosts matched

null_resource.configure_server (local-exec): PLAY [Deploy Docker Containers] *****
null_resource.configure_server (local-exec): skipping: no hosts matched

null_resource.configure_server (local-exec): PLAY RECAP *****

null_resource.configure_server: Creation complete after 1s [id=1695432964614148986]

Apply complete! Resources: 8 added, 0 changed, 0 destroyed.

```

Outputs:

```

webserver_public_ips = [
  "40.172.121.214",
]

```

40.172.121.214:3000

Gitea Base URL *

http://40.172.121.214:3000/

Base address for HTTP(S) clone URLs and email notifications.

Log Path *

/data/gitea/log

Log files will be written to this directory.

☐ Enable Update Checker

Checks for new version releases periodically by connecting to gitea.io.

Optional Settings

► Email Settings

► Server and Third-Party Service Settings

► Administrator Account Settings

These configuration options will be written into:

/data/gitea/conf/app.ini

📄

Install Gitea

task 11

```

ls -la inventory_aws_ec2.yaml
-rw-rw-rw- 1 codespace codespace 0 Jan  6 13:58 inventory_aws_ec2.yaml

```

Apply complete! Resources: 3 added, 1 changed, 1 destroyed.

Outputs:

```
prod-webserver_public_ips = [
    "3.28.45.153",
]
webserver_public_ips = [
    "40.172.121.214",
]
```

```
prod-webserver_public_ips = [
    "3.28.45.153",
]
webserver_public_ips = [
    "40.172.121.214",
]
```

```
Collecting boto3
  Downloading boto3-1.42.22-py3-none-any.whl.metadata (6.8 kB)
Collecting botocore
  Downloading botocore-1.42.22-py3-none-any.whl.metadata (5.9 kB)
Collecting jmespath<2.0.0,>=0.7.1 (from boto3)
  Downloading jmespath-1.0.1-py3-none-any.whl.metadata (7.6 kB)
Collecting s3transfer<0.17.0,>=0.16.0 (from boto3)
  Downloading s3transfer-0.16.0-py3-none-any.whl.metadata (1.7 kB)
Requirement already satisfied: python-dateutil<3.0.0,>=2.1 in /home/codespace/.local/lib/python3.12/site-packages (from botocore)
Requirement already satisfied: urllib3!=2.2.0,<3,>=1.25.4 in /home/codespace/.local/lib/python3.12/site-packages (from botocore)
Requirement already satisfied: six>=1.5 in /home/codespace/.local/lib/python3.12/site-packages (from python-dateutil<3.0.0,>=2.1)
Downloading boto3-1.42.22-py3-none-any.whl (140 kB)
Downloading botocore-1.42.22-py3-none-any.whl (14.6 MB)
 14.6/14.6 MB 27.8 MB/s 0:00:00
Downloading jmespath-1.0.1-py3-none-any.whl (20 kB)
Downloading s3transfer-0.16.0-py3-none-any.whl (86 kB)
Installing collected packages: jmespath, botocore, s3transfer, boto3
Successfully installed boto3-1.42.22 botocore-1.42.22 jmespath-1.0.1 s3transfer-0.16.0
```

```
[DEPRECATION WARNING]: Importing 'to_text' from 'ansible.module_utils.text' is deprecated. This feature will be removed from ansible-
version 2.24. Use ansible.module_utils.common.text.converters instead.
[DEPRECATION WARNING]: Importing 'to_native' from 'ansible.module_utils.text' is deprecated. This feature will be removed from ansib
re version 2.24. Use ansible.module_utils.common.text.converters instead.
[DEPRECATION WARNING]: Passing 'disable_lookups' to 'template' is deprecated. This feature will be removed from ansible-core version 2
@all:
|--@ungrouped:
|--@aws_ec2:
| |--ip-10-0-1-55.me-central-1.compute.internal
| |--ec2-3-28-45-153.me-central-1.compute.amazonaws.com
| |--ec2-3-28-135-61.me-central-1.compute.amazonaws.com
| |--ec2-40-172-121-214.me-central-1.compute.amazonaws.com
| |--ip-10-0-10-197.me-central-1.compute.internal
| |--ip-10-0-10-21.me-central-1.compute.internal
| |--ip-10-0-1-61.me-central-1.compute.internal
```

Task 12

version 2.24. Use `ansible.module_utils.common.text.converters` instead.
[DEPRECATION WARNING]: Importing 'to_native' from 'ansible.module_utils.text' is deprecated. This feature will be removed in version 2.24. Use `ansible.module_utils.common.text.converters` instead.
[DEPRECATION WARNING]: Passing 'disable_lookups' to 'template' is deprecated. This feature will be removed from ansible

```
@all:
  |--@ungrouped:
  |--@aws_ec2:
  |   |--ip-10-0-1-55.me-central-1.compute.internal
  |   |--ec2-3-28-45-153.me-central-1.compute.amazonaws.com
  |   |--ec2-3-28-135-61.me-central-1.compute.amazonaws.com
  |   |--ec2-40-172-121-214.me-central-1.compute.amazonaws.com
  |   |--ip-10-0-10-197.me-central-1.compute.internal
  |   |--ip-10-0-10-21.me-central-1.compute.internal
  |   |--ip-10-0-1-61.me-central-1.compute.internal
  |--@tag_Name_dev_ec2_instance_2:
  |   |--ip-10-0-1-55.me-central-1.compute.internal
  |--@tag_Name_prod_ec2_instance_0:
  |   |--ec2-3-28-45-153.me-central-1.compute.amazonaws.com
  |--@tag_Name_MyServer:
  |   |--ec2-3-28-135-61.me-central-1.compute.amazonaws.com
  |--@tag_Name_dev_ec2_instance_0:
  |   |--ec2-40-172-121-214.me-central-1.compute.amazonaws.com
  |   |--ip-10-0-10-21.me-central-1.compute.internal
  |--@tag_Name_dev_ec2_instance_1:
  |   |--ip-10-0-10-197.me-central-1.compute.internal
  |   |--ip-10-0-1-61.me-central-1.compute.internal
```

```
! inventory_aws_ec2.yaml
1  ---
2  plugin: aws_ec2
3  regions:
4  - me-central-1
5  keyed_groups:
6  - key: tags
7    prefix: tag
8    separator: "_"
9  - key: instance_type
10   prefix: instance_type
11   separator: "_"
```

```
| |--ec2-3-28-135-61.me-central-1.compute.amazonaws.com
| |--ec2-40-172-121-214.me-central-1.compute.amazonaws.com
| |--ip-10-0-10-197.me-central-1.compute.internal
| |--ip-10-0-10-21.me-central-1.compute.internal
| |--ip-10-0-1-61.me-central-1.compute.internal
|--@tag_Name_prod_ec2_instance_0:
|   |--ec2-3-28-45-153.me-central-1.compute.amazonaws.com
|--@instance_type_t3_nano:
|   |--ec2-3-28-45-153.me-central-1.compute.amazonaws.com
|--@tag_Name_MyServer:
|   |--ec2-3-28-135-61.me-central-1.compute.amazonaws.com
|--@tag_Name_dev_ec2_instance_0:
|   |--ec2-40-172-121-214.me-central-1.compute.amazonaws.com
|   |--ip-10-0-10-21.me-central-1.compute.internal
|--@tag_Name_dev_ec2_instance_1:
|   |--ip-10-0-10-197.me-central-1.compute.internal
|   |--ip-10-0-1-61.me-central-1.compute.internal
```



```

TASK [Save public IP as fact for template] *****
ok: [ec2-3-28-45-153.me-central-1.compute.amazonaws.com]

TASK [Generate self-signed SSL certificate] *****
ok: [ec2-3-28-45-153.me-central-1.compute.amazonaws.com]

PLAY [Deploy Nginx website and configuration files] *****

TASK [Gathering Facts] *****
ok: [ec2-3-28-45-153.me-central-1.compute.amazonaws.com]

TASK [install php-fpm and php-curl] *****
ok: [ec2-3-28-45-153.me-central-1.compute.amazonaws.com]

TASK [Copy website files] *****
ok: [ec2-3-28-45-153.me-central-1.compute.amazonaws.com]

TASK [Copy nginx.conf template] *****
ok: [ec2-3-28-45-153.me-central-1.compute.amazonaws.com]

TASK [Restart nginx] *****
changed: [ec2-3-28-45-153.me-central-1.compute.amazonaws.com]

TASK [Start and enable php-fpm] *****
ok: [ec2-3-28-45-153.me-central-1.compute.amazonaws.com]

PLAY RECAP *****
ec2-3-28-45-153.me-central-1.compute.amazonaws.com : ok=17  changed=1  unreachable=0  failed=0  skipped=0  rescued=0  ignored=0

```

Task 13

```

ansible-galaxy role init nginx
ansible-galaxy role init ssl
ansible-galaxy role init webapp
cd ..
ls -R
- Role nginx was created successfully
- Role ssl was created successfully
- Role webapp was created successfully
.:

```

```
..
ansible.cfg inventory my-playbook.yaml roles
```

```
./inventory:
```

```
./roles:
```

```
nginx ssl webapp
```

```
./roles/nginx:
```

```
README.md defaults files handlers meta tasks templates tests vars
```

```
./roles/nginx/defaults:
```

```
main.yml
```

```
./roles/nginx/files:
```

```
./roles/nginx/handlers:
```

```
main.yml
```

```
./roles/nginx/meta:
```

```
[WARNING]: Ansible is being run in a world-writable directory (/workspaces/terraform-machine/ansible), ignoring it as an ansible.cfg source. For more information see https://docs.ansible.com/ansible/devel/reference\_appendices/config.html#cfg-in-world-writable-dir
```

```
PLAY [Deploy NGINX Web Stack with SSL and PHP] *****
```

```
TASK [Gathering Facts] *****
ok: [40.172.121.214]
```

```
TASK [nginx : Install nginx] *****
ok: [40.172.121.214]
```

```
TASK [nginx : Install openssl] *****
ok: [40.172.121.214]
```

```
TASK [nginx : Start and enable nginx] *****
ok: [40.172.121.214]
```

```
PLAY RECAP *****
40.172.121.214      : ok=4    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

```
TASK [ssl : Generate self-signed certificate] *****
ok: [40.172.121.214]
```

```
TASK [webapp : Install PHP packages] *****
ok: [40.172.121.214]
```

```
TASK [webapp : Copy PHP website] *****
changed: [40.172.121.214]
```

```
TASK [webapp : Deploy nginx config] *****
changed: [40.172.121.214]
```

```
TASK [webapp : Start and enable php-fpm] *****
ok: [40.172.121.214]
```

```
RUNNING HANDLER [webapp : Restart nginx] *****
changed: [40.172.121.214]
```

```
PLAY RECAP *****
40.172.121.214      : ok=15   changed=3    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```



Your connection is not private

Attackers might be trying to steal your information from **40.172.121.214** (for example, passwords, messages, or credit cards). [Learn more about this warning](#)

NET::ERR_CERT_AUTHORITY_INVALID



[Turn on enhanced protection](#) to get Chrome's highest level of security

Advanced

Back to safety

 **Nginx Front End Web Server**

Hostname:	ip-10-0-10-133.mtc-central-1.compute.internal
Instance ID:	i-07f94bc01445483c7
Private IP:	10.0.10.133
Public IP:	40.172.121.214
Public DNS:	https://N/A
Deployed:	2020-01-07 07:36:46
Status:	✔ Active and Running
Managed By:	Terraform + Ansible

Clean up:

Either you have not created any objects yet or the existing objects were already deleted outside of Terraform.

Destroy complete! Resources: 0 destroyed.

```
{  
  "version": 4,  
  "terraform_version": "1.14.3",  
  "serial": 1,  
  "lineage": "0a7b60e6-94d2-a791-71ca-195066bed5e5",  
  "outputs": {},  
  "resources": [],  
  "check_results": null  
}
```



FATIMA JINNAH WOMEN UNIVERSITY

Department of Software Engineering

Lab Title:	Experiment # :
Student Name:	Regd. #:
Experiment Name:	

Performance Indicator	Level of Achievements				
	Excellent (5)	Good (4)	Average (3)	Below Average (2)	Poor (1)
Understanding & implementation Problem					
Report and presentation					
Maximum Marks		10	Obtained Marks		
Lab Instructor:					

Lab 15

Task : 1

```
variables.tf × locals.tf main.tf terraform.tfvars
```

```
1 variable "vpc_cidr_block" {}
2 variable "subnet_cidr_block" {}
3 variable "availability_zone" {}
4 variable "env_prefix" {}
5 variable "instance_type" {}
6 variable "public_key_location" {}
7 variable "private_key_location" {}
```

```
variables.tf locals.tf main.tf terraform.tfvars ×
```

```
1 vpc_cidr_block = "10.0.0.0/16"
2 subnet_cidr_block = "10.0.10.0/24"
3 availability_zone = "us-east-1a"
4 env_prefix = "dev"
5 instance_type = "t2.micro"
6 public_key_location = "~/.ssh/id_rsa.pub"
7 private_key_location = "~/.ssh/id_rsa"
```

```
variables.tf locals.tf × main.tf terraform.tfvars
```

```
1 data "http" "my_ip" {
2   url = "https://icanhazip.com"
3 }
4
5 locals {
6   my_ip = "${chomp(data.http.my_ip.response_body)}/32"
7 }
```

```
variables.tf  locals.tf  main.tf  terraform.tfvars

main.tf
1 resource "aws_vpc" "myapp-vpc" {
2   cidr_block = var.vpc_cidr_block
3   tags = {
4     Name = "${var.env_prefix}-vpc"
5   }
6 }
7
8 resource "aws_subnet" "myapp-subnet-1" {
9   vpc_id      = aws_vpc.myapp-vpc.id
10  cidr_block   = var.subnet_cidr_block
11  availability_zone = var.availability_zone
12  map_public_ip_on_launch = true # Taake servers ko public IP mile
13  tags = {
14    Name = "${var.env_prefix}-subnet-1"
15  }
16 }
17
18 resource "aws_internet_gateway" "myapp-igw" {
19   vpc_id = aws_vpc.myapp-vpc.id
20   tags = {
```

```
Generating public/private rsa key pair.
Created directory '/home/codespace/.ssh'.
Your identification has been saved in /home/codespace/.ssh/id_rsa
Your public key has been saved in /home/codespace/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:BQaFtiP571bv00AekKeD7l6SqtTdXKbgfPZCkdNzYow codespace@codespaces-631a52
The key's randomart image is:
+---[RSA 4096]-----+
|      .++          |
|      o.=          |
|    . o E O..      |
|    . *. * B.+      |
|    oo*.05         |
|    .O +* o        |
|    .+ +O.+ o      |
|    .O o .+.o..    |
|    +o+ o. .O.     |
+---[SHA256]-----+
```

Initializing the backend...

Initializing provider plugins...

- Finding latest version of hashicorp/http...
- Finding latest version of hashicorp/aws...
- Installing hashicorp/http v3.5.0...
- Installed hashicorp/http v3.5.0 (signed by HashiCorp)
- Installing hashicorp/aws v6.28.0...
- Installed hashicorp/aws v6.28.0 (signed by HashiCorp)

Terraform has created a lock file `.terraform.lock.hcl` to record the provider selections it made above. Include this file in your version control repository so that Terraform can guarantee to make the same selections by default when you run "terraform init" in the future.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see any changes that are required for your infrastructure. All Terraform commands should now work.

If you ever set or change modules or backend configuration for Terraform, rerun this command to reinitialize your working directory. If you forget, other commands will detect it and remind you to do so if necessary.

data.http.my_ip: Read complete after 0s [id=https://icanhazip.com]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:

+ create

Terraform will perform the following actions:

```
# aws_internet_gateway.myapp-igw will be created
+ resource "aws_internet_gateway" "myapp-igw" {
  + arn      = (known after apply)
  + id       = (known after apply)
  + owner_id = (known after apply)
  + region   = "us-east-1"
  + tags     = {
    + "Name" = "dev-igw"
  }
  + tags_all = {
    + "Name" = "dev-igw"
  }
  + vpc_id   = (known after apply)
}
```

aws_route_table_association.a-rtb-subnet: Creation complete after 1s [id=rtbassoc-0d76a40660a03d42b]

Apply complete! Resources: 6 added, 0 changed, 0 destroyed.

Task : 2

main.tf

```
72 resource "aws_key_pair" "ssh-key" {
73   key_name   = "server-key"
74   public_key = file(var.public_key_location)
75 }
76
77 data "aws_ami" "latest-amazon-linux-image" {
78   most_recent = true
79   owners      = ["amazon"]
80   filter {
81     name     = "name"
82     values   = ["amzn2-ami-hvm-*-x86_64-gp2"]
83   }
84 }
85
86 resource "aws_instance" "frontend" {
87   ami                    = data.aws_ami.latest-amazon-linux-image.id
88   instance_type         = var.instance_type
89   subnet_id             = aws_subnet.myapp-subnet-1.id
90   vpc_security_group_ids = [aws_security_group.myapp-sg.id]
91   availability_zone      = var.availability_zone
92   associate_public_ip_address = true
93   key_name               = aws_key_pair.ssh-key.key_name
94
95   tags = {
```

variables.tf U

locals.tf U

main.tf U

outputs.tf U X

terraform.tfvars U

outputs.tf

```
1  output "frontend_public_ip" {
2    value = aws_instance.frontend.public_ip
3  }
4
5  output "backend_public_ips" {
6    value = [for b in aws_instance.backend : b.public_ip]
7  }
8
9  output "backend_private_ips" {
10   value = [for b in aws_instance.backend : b.private_ip]
11 }
```

```

XYjZttcLwi/1o3qQc3uLUR/lrOFj14RjflazNqojrZyq5/QzUy1SIEGHxZCD5PI+sCqRaGD/7OuEFg9ij1S4CwG/VG0djbNRT5Hs12CZ1ro7jY+R6ou
Md2faSYHI59uyvbB+vDlnA6Q1CaPpdMM1GX3+IjJIMfVrCw9e6vNYLktIvQ== codespace@codespaces-631a52"
+ region      = "us-east-1"
+ tags_all    = (known after apply)
}

```

Plan: 5 to add, 0 to change, 0 to destroy.

Changes to Outputs:

```

+ backend_private_ips = [
+   (known after apply),
+   (known after apply),
+   (known after apply),
+ ]
+ backend_public_ips = [
+   (known after apply),
+   (known after apply),
+   (known after apply),
+ ]
+ frontend_public_ip = (known after apply)

```

Note: You didn't use the -out option to save this plan, so Terraform can't guarantee to take exactly these actions if you run "terraform apply" now.

Apply complete! Resources: 5 added, 0 changed, 0 destroyed.

Outputs:

```

backend_private_ips = [
  "10.0.10.139",
  "10.0.10.168",
  "10.0.10.7",
]
backend_public_ips = [
  "44.222.93.92",
  "98.92.54.207",
  "34.205.166.198",
]
frontend_public_ip = "44.195.90.205"

```

Task : 3

```

variables.tf U  locals.tf U  main.tf U  outputs.tf U  ansible.cfg U x  terraform.tfvars U
ansible > ansible.cfg
1  [defaults]
2  host_key_checking = False
3  interpreter_python = /usr/bin/python3
4  inventory = ./inventory/hosts

```

```

variables.tf U  locals.tf U  main.tf U  outputs.tf U  ansible.cfg U  hosts U x  terraform.tfvars U
ansible > inventory > hosts
1  [frontend]
2  44.195.90.205
3
4  [backends]
5  44.222.93.92
6  98.92.54.207
7  34.205.166.198
8
9  [all:vars]
10 ansible_user=ec2-user
11 ansible_ssh_private_key_file=~/.ssh/id_rsa

```



```

98.92.54.207 | SUCCESS => {
  "changed": false,
  "ping": "pong"
}
44.222.93.92 | SUCCESS => {
  "changed": false,
  "ping": "pong"
}
44.195.90.205 | SUCCESS => {
  "changed": false,
  "ping": "pong"
}
34.205.166.198 | SUCCESS => {
  "changed": false,
  "ping": "pong"
}
}

```

Task : 4

ansible > roles > backend > tasks > ! main.yml

```

1 ---
2 - name: Install Apache (httpd)
3   yum:
4     name: httpd
5     state: present
6     update_cache: yes
7
8 - name: Enable and start httpd
9   service:
10    name: httpd
11    state: started
12    enabled: true
13
14 - name: Deploy backend index page from template
15   template:
16     src: backend_index.html.j2
17     dest: /var/www/html/index.html
18     owner: apache
19     group: apache
20     mode: '0644'
21   notify: restart httpd

```

main.tf U outputs.tf U ansible.cfg U hosts U ! main.yml U backend_index.html.j2 U ×

ansible > roles > backend > templates > backend_index.html.j2

```

1 <!DOCTYPE html>
2 <html>
3 <head>
4   <title>Backend {{ inventory_hostname }}</title>
5 </head>
6 <body>
7   <h1>Backend server: {{ inventory_hostname }}</h1>
8   <p>Public IP: {{ inventory_hostname }}</p>
9   <p>Private IP: {{ ansible_default_ipv4.address | default('unknown') }}</p>
10  <p>Welcome to the backend cluster!</p>
11 </body>
12 </html>

```

ts.tf U ansible.cfg U hosts U ! main.yml .../tasks U backend_index.html.j2 U ! main.yml .../handlers U ×

ansible > roles > backend > handlers > ! main.yml

```

1 ---
2 - name: restart httpd
3   service:
4     name: httpd
5     state: restarted

```

```
PLAY [Deploy Apache to Backend Servers] *****

TASK [Gathering Facts] *****
ok: [44.222.93.92]
ok: [34.205.166.198]
ok: [98.92.54.207]

TASK [backend : Install Apache (httpd)] *****
changed: [44.222.93.92]
changed: [98.92.54.207]
changed: [34.205.166.198]

TASK [backend : Enable and start httpd] *****
changed: [44.222.93.92]
changed: [34.205.166.198]
changed: [98.92.54.207]

TASK [backend : Deploy backend index page from template] *****
changed: [44.222.93.92]
changed: [34.205.166.198]
changed: [98.92.54.207]

RUNNING HANDLER [backend : restart httpd] *****
changed: [44.222.93.92]
changed: [34.205.166.198]
changed: [98.92.54.207]

PLAY RECAP *****
34.205.166.198      : ok=5    changed=4    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
44.222.93.92       : ok=5    changed=4    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
98.92.54.207       : ok=5    changed=4    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

Backend server: 98.92.54.207

Public IP: 98.92.54.207

Private IP: 10.0.10.168

Welcome to the backend cluster!

Backend server: 34.205.166.198

Public IP: 34.205.166.198

Private IP: 10.0.10.7

Welcome to the backend cluster!

Backend server: 44.222.93.92

Public IP: 44.222.93.92

Private IP: 10.0.10.139

Welcome to the backend cluster!

Task : 5

ansible > roles > frontend > templates > nginx_frontend.conf.j2

```
1 user nginx;
2 worker_processes auto;
3 error_log /var/log/nginx/error.log notice;
4 pid /run/nginx.pid;
5
6 events {
7     worker_connections 1024;
8 }
9
10 http {
11     include /etc/nginx/mime.types;
12     default_type application/octet-stream;
13
14     upstream backend_servers {
15         # Pehle do servers normal challenge
16         server {{ backend1_private_ip }}:80;
```

```
1 ---
2 - name: Install nginx
3   yum:
4     name: nginx
5     state: present
6     update_cache: yes
7
8 - name: Enable and start nginx
9   service:
10    name: nginx
11    state: started
12    enabled: true
13
14 - name: Deploy nginx frontend config
15   template:
16     src: nginx_frontend.conf.j2
17     dest: /etc/nginx/nginx.conf
18   notify: Restart nginx
```

ansible > roles > frontend > handlers > ! main.yml

```
1 ---
2 - name: Restart nginx
3   service:
4     name: nginx
5     state: restarted
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```

```
PLAY [Setup Load Balancer] *****
TASK [Gathering Facts] *****
ok: [44.195.90.205]

TASK [frontend : Enable Nginx from Amazon Linux Extras] *****
changed: [44.195.90.205]

TASK [frontend : Install nginx] *****
changed: [44.195.90.205]

TASK [frontend : Enable and start nginx] *****
changed: [44.195.90.205]

TASK [frontend : Deploy nginx frontend config] *****
changed: [44.195.90.205]

RUNNING HANDLER [frontend : Restart nginx] *****
changed: [44.195.90.205]

PLAY RECAP *****
44.195.90.205      : ok=6    changed=5    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

Backend server: 44.222.93.92

Public IP: 44.222.93.92

Private IP: 10.0.10.139

Welcome to the backend cluster!

Task : 6

```
ansible > roles > trontena > templates > nginx_trontena.conf.j2
1  user nginx;
2  worker_processes auto;
3  error_log /var/log/nginx/error.log notice;
4  pid /run/nginx.pid;
5
6  events {
7      worker_connections 1024;
8  }
9
10 http {
11     include /etc/nginx/mime.types;
12     default_type application/octet-stream;
13
14     upstream backend_servers {
15         # Dynamically picking IPs from inventory
16         server {{ backend1_private_ip }}:80;
17         server {{ backend2_private_ip }}:80;
18         server {{ backup_backend_private_ip }}:80 backup;
19     }
20
21     server {
22         listen 80;
23         server_name _;
```



```
ansible > ! site.yml
```

```
1 ---
2 - name: Configure backend HTTPD servers
3   hosts: backends
4   become: true
5   roles:
6     - backend
7
8 - name: Configure frontend Nginx load balancer
9   hosts: frontend
10  become: true
11  vars:
12    # Requirement 6: Dynamic gathering using hostvars
13    backend1_private_ip: "{{ hostvars[groups['backends'][0]].ansible_default_ipv4.address }}"
14    backend2_private_ip: "{{ hostvars[groups['backends'][1]].ansible_default_ipv4.address }}"
15    backup_backend_private_ip: "{{ hostvars[groups['backends'][2]].ansible_default_ipv4.address }}"
16  roles:
17    - frontend
```

```
changed: [44.195.90.205]
```

```
TASK [frontend : Install nginx] *****
ok: [44.195.90.205]
```

```
TASK [frontend : Enable and start nginx] *****
ok: [44.195.90.205]
```

```
TASK [frontend : Deploy nginx frontend config] *****
changed: [44.195.90.205]
```

```
RUNNING HANDLER [frontend : Restart nginx] *****
changed: [44.195.90.205]
```

```
PLAY RECAP *****
34.205.166.198      : ok=4  changed=0  unreachable=0  failed=0  skipped=0  rescued=0  ignored=0
44.195.90.205      : ok=6  changed=3  unreachable=0  failed=0  skipped=0  rescued=0  ignored=0
44.222.93.92       : ok=4  changed=0  unreachable=0  failed=0  skipped=0  rescued=0  ignored=0
98.92.54.207       : ok=4  changed=0  unreachable=0  failed=0  skipped=0  rescued=0  ignored=0
```

Backend server: 44.222.93.92

Public IP: 44.222.93.92

Private IP: 10.0.10.139

Welcome to the backend cluster!

Backend server: 98.92.54.207

Public IP: 98.92.54.207

Private IP: 10.0.10.168

Welcome to the backend cluster!

Task : 7

```
.terraform > 📄 hosts.tpl
```

```
1  [frontend]
2  ${frontend_ip}
3
4  [backends]
5  %{ for ip in backend_ips ~}
6  ${ip}
7  %{ endfor ~}
8
9  [all:vars]
10 ansible_user=ec2-user
11 ansible_ssh_private_key_file=~/.ssh/id_rsa
12 ansible_python_interpreter=/usr/bin/python3
```

```
📄 main.tf
```

```
114 resource "local_file" "ansible_inventory" {
115   content = templatefile("hosts.tpl", {
116     frontend_ip = aws_instance.frontend.public_ip
117     backend_ips = [for b in aws_instance.backend : b.public_ip]
118   })
119   filename = "ansible/inventory/hosts"
120 }
121
122 resource "null_resource" "ansible_config" {
123   triggers = {
124     frontend_ip = aws_instance.frontend.public_ip
125     backend_ips = join(",", [for b in aws_instance.backend : b.public_ip])
126   }
127
128   depends_on = [
129     aws_instance.frontend,
130     aws_instance.backend,
131     local_file.ansible_inventory
132   ]
133
134   provisioner "local-exec" {
135     command = <<-EOT
136       sleep 30;
137       cd ansible
138       ansible-playbook site.yml
139     EOT
140   }
141 }
```

```

ansible > roles > frontend > tasks > ! main.yml
4  - name: Enable nginx from ansible linux layer
5
6  - name: Install nginx
7    yum:
8      name: nginx
9      state: present
10     update_cache: yes
11
12  - name: Enable and start nginx
13    service:
14      name: nginx
15      state: started
16      enabled: true
17
18  - name: Deploy nginx frontend config
19    template:
20      src: nginx_frontend.conf.j2
21      dest: /etc/nginx/nginx.conf
22      notify: Restart nginx

```

Apply complete! Resources: 2 added, 0 changed, 0 destroyed.

Outputs:

```

backend_private_ips = [
  "10.0.10.139",
  "10.0.10.168",
  "10.0.10.7",
]
backend_public_ips = [
  "44.222.93.92",
  "98.92.54.207",
  "34.205.166.198",
]
frontend_public_ip = "44.195.90.205"

```

Backend server: 44.222.93.92

Public IP: 44.222.93.92

Private IP: 10.0.10.139

Welcome to the backend cluster!

Backend server: 98.92.54.207

Public IP: 98.92.54.207

Private IP: 10.0.10.168

Welcome to the backend cluster!

```
create mode 100644 ansible/site.yml
create mode 100644 hosts.tpl
create mode 100644 locals.tf
create mode 100644 main.tf
create mode 100644 outputs.tf
create mode 100644 variables.tf
```

```
Enumerating objects: 33, done.
Counting objects: 100% (33/33), done.
Delta compression using up to 2 threads
Compressing objects: 100% (25/25), done.
Writing objects: 100% (31/31), 4.96 KiB | 4.96 MiB/s, done.
Total 31 (delta 0), reused 28 (delta 0), pack-reused 0 (from 0)
To https://github.com/23-22411-016-droid/LabProject_FrontendBackend
9a551c6..0a7adac main -> main
```



FATIMA JINNAH WOMEN UNIVERSITY

Department of Software Engineering

Lab Title:	Experiment # :
Student Name:	Regd. #:
Experiment Name:	

Performance Indicator	Level of Achievements				
	Excellent (5)	Good (4)	Average (3)	Below Average (2)	Poor (1)
Understanding & implementation Problem					
Report and presentation					
Maximum Marks		10	Obtained Marks		
Lab Instructor:					